Core-Sparse Monge Construction for Full Optimal Binary Search Trees*

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Abstract

We give an exact construction for optimal binary search trees with arbitrary nonnegative integer weights on keys and on the gaps between them. For total mass $W = O(n)$, a reduction to the core-sparse Monge product of Gawrychowski, Gorbachev, and Kociumaka takes $O(n^{3/2}\log^{3/2}(n+1))$ arithmetic operations and returns the complete tree. Terminal-penalty layers preserve Monge structure; a mixed-weight subtree bound controls the number of layers, and a dyadic boundary argument bounds every intermediate core. Charging individual outcome weights to tiles bounds the sum of all layer cores. Centering around separate constant key and gap baselines gives $O(n(1+\sqrt{R})\log^3(n+1))$ arithmetic work when their total nonnegative deviation R is polynomial in n.

A common lower-bound certificate also gives faster exact construction on specified families. Independently, a common key/gap period b permits $O(nb)$ full construction by cyclic root amortization. A periodic middle of length m with $d = n-m$ arbitrary fringe keys admits $O(mb+nd)$ full construction via Wilber's linear concave least-weight-subsequence primitive. Positive affine translation of key and gap weights also gives $O(nb)$ construction, including geometric scaling and opposing arithmetic drifts. A height bound charging disjoint ancestor increments to exceptional outcomes turns the mixed run-vector construction into an unrestricted exact solver. With a fixed number of bounded positive key runs and exceptional key/gap weights, it takes $O(n^{\log_\varphi 2}\log(n+1))$ arithmetic work, where $\varphi = (1+\sqrt{5})/2$, independently of the exceptional magnitudes. Compressing prefix vectors into valuation-class records removes explicit perfect-tree expansion. With integer mass $W = O(n)$ and a fixed number of normalized key runs and residual gaps, complete unrestricted construction takes $O(n\log^4(n+1))$ arithmetic work, allowing zero-weight runs. A fixed number of arbitrarily large integer outcomes preserves a bound of n times a fixed power of $\log(n+1)$ through their finite reward spectra. We give the certificate once and derive the family guarantees from it. Counterexamples delimit bounded-window relaxations, endpoint-response composition, explicit density representations, and additive separable transfers. Explicit prefix records can be exponential even at the minimum optimal height. Separating each block's first entry from its implicit prefix diagram gives $O(h(D+1))$ arithmetic and dictionary operations per arbitrary mixed update, for height h and input size D. Including traceback and tree repair, complete construction takes $O(h[n+\sum_k D_k])$ such operations. An explicit positive mixed family forces $\Omega(n^3)$ final diagram nodes and $\Omega(n^4)$ distinct nodes in shared prefix history, even when blocks may share under positive affine transformations; the optimum remains a directly certified right chain. A contiguous odd ramp with total mass exactly n has $\Omega(n^{3/2})$ intermediate core entries with explicitly computed densities. Thus the core exponent is necessary, up to a remaining logarithmic factor. A separate positive ruler family has quadratic support at deviation $R = n^2$. An adversarial integer-mass search and independent exact checks are reported in an appendix. The unrestricted subquadratic problem remains open.

1 Model and prior work

Weights are attached to fixed positions in key order; they need not be sorted. Keys have fixed inorder positions $0, \dots, n-1$ and nonnegative weights $p_0, \dots, p_{n-1}$. The $n+1$ gaps have weights $q_0, \dots, q_n$. A key at depth d contributes $p(d+1)$; a gap at depth d contributes qd. Charging an extra comparison at each gap adds the tree-independent constant $\sum q_i$.

For a half-open key interval $[i, j)$, write

$$W(i,j) = \sum_{r=i}^{j-1} p_r + \sum_{r=i}^{j} q_r, \qquad C(i,i) = 0.$$

The exact recurrence is

$$C(i,j) = W(i,j) + \min_{i \le r < j} \{C(i,r) + C(r+1,j)\}.$$

Knuth's monotone-root optimization computes the table in $O(n^2)$ time [2]. The central obstacle here is the number of states, not merely the cost per state. We retain both kinds of weights throughout.

Larmore gives subquadratic exact construction under restrictions on long runs of very low frequency items, together with an approximation tradeoff [4]. Karpinski, Larmore, and Rytter combine exact small intervals with a bounded top-level construction [5]. All running-time bounds below count arithmetic operations and comparisons. They are not bit-complexity bounds for unbounded integer weights.

For uniform key weights a and uniform gap weights b, the external path length equals $I + 2n$, where I is the internal path length. Thus

$$C(T) = (a+b)I + (a+2b)n.$$

Every minimum-internal-path-length tree is optimal. Such a tree fills every level above the last and is constructible in $O(n)$ time. When $a = b = 0$, every tree is optimal. Later baseline arguments use this exact identity.

2 The main Monge construction

Let $n \ge 1$, let all key and gap weights be nonnegative integers, and put $W = \sum p_i + \sum q_i$ and $L = \lceil \log_2(n+1) \rceil$. Write $f(W)$ for the first index with $F_{f(W)} > W$, where $F_0 = 0, F_1 = 1$. Set

$$H = \min\{n, f(W) + L\}, \qquad M = (H+1)W + 1.$$

If every key weight is positive, $H = \min\{n, f(W)\}$ suffices.

Theorem 1 (Integer-mass sparse-product reduction)**.** *Using the core-sparse Monge multiplication and witness algorithms of [14], a full optimal BST can be constructed in*

$$O\Big(H\left[n + \min\{n^2, n\sqrt{WL}\}\right]\log(n+1)\Big)$$

arithmetic operations. Retaining the witness structures uses $O(H[n + \min\{n^2, n\sqrt{WL}\}])$ *words. In particular,* $W = O(n)$ *gives* $O(n^{3/2}\log^{3/2}(n+1))$ *exact construction. Intermediate integers have* $O(\log(n+1) + \log(W+1))$ *bits.*

The dependence on numerical mass is essential to this statement. The linear-mass case fits ordinary $O(\log n)$-bit arithmetic, but arbitrary large integer weights do not give an unrestricted strongly subquadratic bound. Knuth remains a fallback when the displayed bound is unfavorable.

2.1 A full mixed-weight height bound

Lemma 2 (A sufficient mixed-weight subtree bound)**.** *If S is an internal subtree at edge depth d in an optimal full OBST of total outcome mass W, then*

$$W(S) \le W/F_{d+1}, \qquad F_1 = F_2 = 1.$$

Proof. For a subtree Z, $W(Z)$ includes every key and external gap in it; Z may be a single gap. A one-level depth change changes cost by the outcome weight for both kinds of outcomes. Write $v(L, R)$ for a key and its two subtrees. Reflection reduces the straight and bent cases to the following two transformations.

In the straight case, with internal grandchild X, rotate

$$v(y(X, B), D) \longrightarrow y(X, v(B, D)).$$

Key y and all outcomes of X rise one level; key v and all outcomes of D fall one level. Every outcome of B keeps its depth. If $O = p_v + W(D)$, the exact change is

$$O - p_y - W(X).$$

Optimality implies $O \ge p_y + W(X) \ge W(X)$.

For the bent case write $X = x(B, C)$ and perform the double rotation

$$v(y(A, x(B, C)), D) \longrightarrow x(y(A, B), v(C, D)).$$

The changes of edge depth, and hence of comparison length, are:

Outcome or subtree	x	B	C	v	D	y	A
Depth change	-2	-1	-1	$+1$	$+1$	0	0

Every outcome in a listed subtree has the indicated change. Therefore

$$C_{\text{new}} - C_{\text{old}} = p_v + W(D) - 2p_x - W(B) - W(C) = O - p_x - W(X).$$

Optimality again gives $O \ge W(X)$. Both transformations preserve inorder positions and every external gap; all other outcomes keep their depths. These are transformations of complete trees.

Going upward from S, write $T_0 = S, T_1, \ldots, T_d = T$. Nonnegativity gives $W(T_1) \ge W(T_0)$, and the grandchild inequality gives $W(T_i) \ge W(T_{i-1}) + W(T_{i-2})$ for $i \ge 2$. Induction gives $W(T_d) \ge F_{d+1} W(S)$. Weak inequalities include zero weights and ties. □

This lemma concerns internal subtrees. A heavy interior external gap can have nearly all the mass at depth two; zero-key subtrees remain permitted. The Fibonacci approach is prior work [7]; sharpness of this particular sufficient bound is not claimed.

2.2 An upper iteration that finishes in a bounded number of levels

Initialize $A_0(i, j) = M(j - i)$ for $i \le j$, and iterate

$$\begin{aligned} A_h(i, i) &= 0, \\ A_h(i, j) &= W(i, j) + \min_{i \le r < j} \{A_{h-1}(i, r) + A_{h-1}(r + 1, j)\} \quad (i < j). \end{aligned}$$

This is an h-level partial search tree: explicit decisions incur the usual interval mass, while each key still in an unfinished terminal interval incurs penalty M. If t keys are explicit, the cost is

$$M((j-i)-t)+D, \qquad 0 \le D \le hW(i,j).$$

At most $\min\{j-i, 2^h-1\}$ keys can be explicit, and that number is attainable with inorder labels. Since $M > HW$, every minimizer for $h \le H$ uses that maximum number. Writing $c_h = 2^h - 1$ gives

$$A_h(i,j) = M\max\{0, j-i-c_h\} + R_h(i,j), \qquad 0 \le R_h(i,j) \le hW(i,j). \tag{1}$$

The residual R_h is not asserted to be Monge.

The initial bound dominates every interval optimum: any complete tree on $j-i$ keys costs at most $(j-i)W(i,j) \le M(j-i)$. The recurrence is monotone, and $A_1 \le A_0$ because $W(i,j) \le M$. Thus the iterates decrease and remain at least C. Whenever an optimal full tree fits within h levels, it is feasible without a terminal penalty, so $A_h = C$.

The mixed Fibonacci mass lemma applies here to an *actual optimal full tree*. Every positive-mass internal subtree at depth d satisfies $F_{d+1} \le W$. Zero-mass subtrees can be replaced by balanced trees of height at most L without changing cost. Therefore some optimum has height at most $f(W)+L$; the trivial bound n also applies. When all keys have positive integer weight every internal subtree has mass at least one, so the extra L is unnecessary. These arguments apply to every interval, proving $A_H = C$ throughout the upper triangle.

2.3 Finite Monge products, including the diagonal boundary

For a triangular matrix A with zero diagonal, define its interior density by

$$\Delta_A(i,j) = A(i,j+1) + A(i+1,j) - A(i,j) - A(i+1,j+1) \quad (0 \le i < j < n).$$

A *core* lists nonzero densities. To use a finite matrix product, put $P = 4nM + 4W + 1$ and complete each layer by

$$E_h(i,j) = \begin{cases} A_h(i,j), & i \le j, \\ P(i-j), & i > j. \end{cases}$$

We prove by induction that E_h is Monge. This is immediate for E_0. Given the previous completion, multiply its first n columns by its last n rows in the min-plus sense:

$$B_h(i,j) = \min_{0 \le r < n} \{E_{h-1}(i,r) + E_{h-1}(r+1,j)\}.$$

For $i < j$, every invalid root has a term at least P, whereas a valid pair costs at most nM. Thus the finite product gives the restricted root minimum exactly. The product is Monge. Adding $W(i,j)$, a row term plus a column term, leaves density unchanged in squares entirely above the diagonal. The only interior squares affected by resetting a diagonal entry to zero have $j = i+1$. Their new density is

$$\min\{A_{h-1}(i,i+1), A_{h-1}(i+1,i+2)\} - q_{i+1} \ge 0.$$

For the first step both old singleton entries are M; thereafter each is its singleton interval mass. The completion adds exactly the n diagonal densities $P + A_h(i,i+1)$; all densities below them vanish. This proves Monge preservation, including the boundary cases. All entries are integers.

2.4 A bound on every intermediate core

Let K_h be the number of nonzero interior densities of A_h. A direct telescoping identity is

$$\sum_{i<j} \Delta_A(i,j) = A(0,n) - \sum_{i=0}^{n-1} A(i,i+1). \tag{2}$$

For $h \geq L$, a balanced full tree is feasible, so $A_h(0,n) \leq LW$. Nonnegative integrality then gives $K_h \leq LW$. This final-stage argument alone would not bound the early layers, whose terminal penalties are large.

We first make the geometric step explicit, including non-power-of-two sizes.

Lemma 3 (Dyadic boundary count)**.** *In the integer density grid $[0,n)^2$, retain cells with $i<j$ except those on m specified diagonals $j-i=d_t$. Embed the grid in $[0,N)^2$, where N is the least power of two at least n. Subdivide aligned squares into four children, discard squares without retained cells, and accept a square as soon as all its cells are retained. At every dyadic side length s, the accepted squares number at most $8(m+2)n/s$.*

Proof. The enclosing square contains the main diagonal, so it is not accepted. Every accepted square has a parent of side $t=2s$ containing both a retained and a forbidden cell. That parent either crosses $i=n$ or $j=n$, contains a cell of $i=j$, or contains a cell of a removed diagonal. These exhaust the reasons for subdivision. In a fixed row of side-t parent squares, a diagonal's column indices form t consecutive integers, so it meets at most two parents. Thus each of the $m+1$ diagonals meets at most $2N/t$ parents. Each outer boundary crosses at most N/t parents; if it is grid-aligned it crosses none. Each charged parent has at most four accepted children. Their number is at most

$$4\{2(m+1)+2\}N/(2s) \leq 8(m+2)n/s.$$

The boundaries at n account for the padding when n is not a power of two. Forbidden unit cells are discarded. Every retained cell reaches an accepted ancestor or is itself accepted, proving that these squares partition the retained domain. □

For $1 \leq h < L$, the penalty term in (1) has density only on $j-i=c_h$. Account for that line using at most n cells, and apply Lemma 3 with $m=1$ to the remainder. There are at most $24n/s$ accepted tiles of side s.

On a tile $[a,b)\times[c,d)$ the penalty density vanishes. Its density sum telescopes to

$$R_h(a,d) + R_h(b,c) - R_h(a,c) - R_h(b,d) \leq 2hW.$$

All densities in this tile are nonnegative integers. Negative residual densities can occur on the removed line and are not used. The number of nonzero entries in the tile is at most $\min\{s^2, 2hW\}$. Therefore

$$K_h = O\left(n + \sum_{s \text{ dyadic}} \frac{n}{s} \min\{s^2, 2hW\}\right) = O(n + n\sqrt{hW}),$$

capped by $O(n^2)$. Splitting the geometric sum at $s \asymp \sqrt{hW}$ introduces no extra logarithm. For $W=0$, only the removed-line entries can remain. Combining the early and late bounds, including the n completion entries, yields

$$\delta(E_h) = O\Big(n + \min\{n^2, n\sqrt{WL}\}\Big) \qquad (0 \leq h \leq H). \tag{3}$$

For completeness, the early-layer argument gives the explicit bound

$$\delta(E_h) \le 2n + 96n\sqrt{2hW} \qquad (1 \le h < L,\ W > 0),$$

since $\sum_{s \text{ dyadic}} \min(s, t^2/s) \le 4t$ for $t \ge 1$. For $W = 0$ only the exceptional and completion diagonals can contribute. Every layer also has at most $n(n+1)/2$ completed core entries. The constants are conservative; the sharper geometric count is retained in the proof. This controls representation size without charging the large value of M as though it represented many separate core entries.

2.5 Compact computation and full reconstruction

The precise published interface is as follows. Theorem 1.2 of Gawrychowski, Gorbachev, and Kociumaka [14], p. 74:3, states:

> There is a (deterministic) algorithm that, given the condensed representations of a $p \times q$ Monge matrix A and a $q \times r$ Monge matrix B, in time $O(p+q+r+(\delta(A)+\delta(B))\log(1+\delta(A)+\delta(B)))$ computes the condensed representation of $A \otimes B$.

This quotation is reproduced under the paper's CC BY 4.0 license. The following definitions and guarantees specify the interface used here.

- Definition 2.5 uses precisely our density sign and defines the core as triples $(i, j, \Delta_A(i, j))$ with nonzero density. Its size $\delta(A)$ counts triples, regardless of their magnitudes.
- Definition 3.2 stores the topmost row, leftmost column, and core. This is exactly the representation used here.
- Theorem 1.1 bounds the product core by $2(\delta(A) + \delta(B))$.
- Theorem 4.7, p. 74:11, adds within the same construction time a witness structure occupying $O(p + q + r + \delta(A) + \delta(B))$ space, with $O(\log(2 + \delta(A) + \delta(B)))$ query time. Definition 2.3 specifies the smallest minimizing index, so ties are covered.
- Fact 4.1 supplies logarithmic-time entry queries after preprocessing from the condensed representation.

Here the operands have dimensions $(n+1) \times n$ and $n \times (n+1)$. Every entry is a finite integer: the lower-triangle padding is $P(i-j)$, not infinity. Invalid roots have value at least $P > nM$, whereas a valid root pair costs at most nM. Thus the reduction uses neither extended-real entries nor an unstated triangular version of the published theorem.

Use them on the two contiguous submatrices of E_{h-1}. To obtain the next completion, retain the product core strictly above the first superdiagonal, where adding interval mass has zero density. Recompute the $O(n)$ diagonal and first-superdiagonal core entries, discard lower entries, and recover the top row and left column. Filtering and $O(n)$ logarithmic entry queries take $O((n+\delta)\log(n+1))$ work, including coordinate sorting if needed. The product's temporary core is also bounded linearly by the two input cores, by Theorem 1.1 of [14]. No step first enumerates a dense interval table.

Retain the product witness oracle for every layer. Starting from $(0, n, H)$, retrieve a root and recurse on its two intervals at layer $H-1$. There are n nonempty calls. An unfinished nonempty interval at layer zero would incur at least M, but the optimum is at most $LW < M$, since $H \ge L$. Thus none survives; when $H = n$, completeness also follows from depth alone. Witness recovery costs an additional $O(n \log(n+1))$. Equation (3), summed over H layers, proves the general bound in Theorem 1, including dimensions, representation conversion, storage, and reconstruction. Section 2.7 proves its sharper linear-mass clause. Since $H \le n$ and the penalties are polynomial in n, W, the stated bit-length bound follows as well.

2.6 A linear-mass family with $\Omega(n^{3/2})$ intermediate core

A contiguous ramp surrounded by zero-weight keys supplies a lower bound on the representation size. We first identify the exact optimization problem inside its height-limited intervals.

Lemma 4 (Perfect-slot representation for zero gap weights)**.** *If every gap weight is zero and $k \le c = 2^h - 1$, the optimal height-at-most-h key cost equals the minimum over all ordered k-slot subsets of the perfect c-key tree, assigning the input keys to those slots in order. The comparison length of its slot r is $h - \nu_2(r)$, $1 \le r \le c$.*

Proof. Every tree of height at most h occupies an ancestor-closed subset of the perfect tree's slots, proving one inequality. Conversely, start from any k selected slots and delete all unselected keys by ordinary BST deletion. For a node with two children, move its inorder successor to that node and splice out the successor from its old position; otherwise splice out the node directly. Every remaining key keeps or decreases its depth. Inorder order is preserved. With nonnegative key weights and zero gap weights, the resulting feasible tree costs no more than the selected-slot assignment. This proves the opposite inequality. The ruler formula for a perfect tree's inorder depths follows by its recursive halving. □

Let $v_u = \nu_2(u+1)$ for zero-based ruler positions u, and let D bound these coefficients. They alternate between zero and a positive integer. For any consecutive segment of length ℓ,

$$\sum v_u \le \ell + D, \tag{4}$$

because there are at most $\ell/2^d + 1$ multiples of 2^d in the segment, and $v_u = \sum_{d=1}^{D} [2^d \mid u+1]$. Define the odd-ramp subsequence score

$$f_m(a,b) = \max_{a \le t_1 < \cdots < t_m < b} \sum_{r=1}^{m} (2r-1) v_{t_r}.$$

Lemma 5 (Rightmost-zero deletion)**.** *If $b - a = m + s$ and $m > 2s + D$, an optimum for $f_m(a,b)$ deletes the rightmost s zero coefficients and retains every other position.*

Proof. For $s = 0$ every position is retained. For $s > 0$, examine the right endpoint. If its coefficient is zero, some optimum omits it: replace a selected final zero by an earlier omitted position. The inserted coefficient is nonnegative, and the selected coefficients it passes receive larger odd multipliers.

If the final coefficient is positive, every optimum selects it. Otherwise there is a selected zero, since the window has fewer than m positive positions. Let u be the last selected zero. Every zero after u is omitted, and the final positive coefficient uses one more omission. As zeros alternate, the distance from u to the final position is at most $2s - 1$. By (4), the selected coefficients after u have sum at most $2s - 1 + D$. Replace that selected zero by the final positive coefficient. The latter receives multiplier $2m - 1$; each selected coefficient after u loses two in its multiplier. The gain is at least

$$(2m - 1) - 2(2s - 1 + D) > 0,$$

a contradiction. Thus retain the final positive coefficient and omit its preceding zero while omissions remain. Removing this pair changes (m, s) to $(m - 1, s - 1)$ and preserves the required inequality. An initial zero omission only strengthens it. Continue until $s = 0$. □

The rule yields an exact density, rather than only a count. Suppose $b-a=m+s$, $s\geq 1$, b is odd, and $m>2(s+1)+D$, so it applies to all four windows below. Appending the positive coefficient v_b deletes the additional zero at $b-2s-1$. The s retained positive coefficients after that zero each lose two in their multiplier. Hence

$$f_m(a,b+1)-f_m(a,b)=(2m-1)v_b-2\sum_{r=1}^{s}v_{b-2r}.$$

After deleting the left endpoint, the same sum has $s-1$ terms. Subtracting these two increments gives

$$f_m(a,b+1)+f_m(a+1,b)-f_m(a,b)-f_m(a+1,b+1)=-2\nu_2(b-2s+1)<0. \tag{5}$$

Theorem 6 (Linear-mass intermediate-core lower bound)**.** *Let $m=2^t$, $t\geq 5$, $n=m^2$, $z=(n-m)/2$, and $h=2t-1$. Set every gap weight to zero, and set*

$$p_{z+r-1}=2r-1\quad(1\leq r\leq m),\qquad p_i=0\ \textit{otherwise.}$$

Then $W=n$, and the terminal-penalty layer A_h has at least

$$\frac{nm}{16}-\frac{9n}{64}-\frac{m}{4}\ \geq\ \frac{n^{3/2}}{32}$$

strictly positive interior densities. Their values have the explicit formula in (6) below.

Proof. Put $c=2^h-1=n/2-1$. Every interval of at most c keys is fully exposed in layer h, so Lemma 4 applies. An interval $[i,j)$ containing the entire ramp has $z-i$ leading zero keys and $j-z-m$ trailing zero keys. Its positive keys can occupy any m slots in

$$[a,b)=[z-i,\ c-j+z+m).$$

The preceding and following zero keys fit exactly when the positive slots lie in this window. Since the interval's mass is $W=n$,

$$A_h(i,j)=hn-f_m(a,b).$$

Write $s=c-(j-i)$; then $b-a=m+s$. The four original intervals defining a density correspond to the four score windows based at $(a-1,b-1)$. Equation (5) gives, when b is even,

$$\Delta_{A_h}(i,j)=2\nu_2(b-2s)\geq 2. \tag{6}$$

Specifically, take $s=1,\dots,m/4$ and $a=1,\dots,c-m-s-1$ with $b=a+m+s$ even; then set $i=z-a$ and $j=c-b+z+m$. These are valid, distinct density cells, and all four intervals contain the ramp and have at most c keys. The ruler coefficients are bounded by $D=h-1=2t-2$. For $t\geq 5$,

$$m>2(m/4+1)+D,$$

so the deletion rule applies to all the windows used. At each s there are exactly $\lfloor (c-m-s)/2\rfloor$ permitted values of a. With $u=m/4$ even, their sum is

$$\sum_{s=1}^{u}\left\lfloor\frac{c-m-s}{2}\right\rfloor=\frac{u(c-m-1)}{2}-\frac{u^2}{4}=\frac{nm}{16}-\frac{9n}{64}-\frac{m}{4}.$$

For $m\geq 32$ this is at least $nm/32$. □

All counted cells use fully exposed intervals and avoid the penalty diagonal. Their growth is therefore independent of the numerical size of the terminal penalty. Consequently a uniform $O(n \operatorname{polylog} n)$ bound on every intermediate explicit core is impossible even at $W = n$. The polynomial exponent $3/2$ in the upper bound is necessary for this representation; a factor $\sqrt{\log n}$ remains between the core bounds. The lower bound concerns a layer's density list, not the complexity of constructing an optimal tree by a different representation. The original positive ruler family below supplies the complementary quadratic-support result at centered deviation $R = n^2$.

2.7 Charging individual outcomes across all layers

The maximum core size need not be paid at every layer. The exceptional penalty diagonal has distance $c_h = 2^h - 1$ from the main diagonal. Its geometric growth allows a sharper sum of the representation sizes.

Lemma 7 (Per-outcome tile overlap)**.** *Let N be a power of two and $0 < c < N$. Partition the integer cells $0 \le i < j < N$, except $j - i = c$, into maximal dyadic squares avoiding both $j - i = 0$ and $j - i = c$. For a square $T = [a, a + s) \times [b, b + s)$, let $J_T = [a, b + s]$ be its inclusive span of outcome positions. At each side length s there are at most $12N/s$ squares. Each position belongs to at most $60(1 + c/s)$ spans J_T.*

Proof. The whole square meets the main diagonal. Thus every accepted square has a rejected parent of side $2s$, with lower-left coordinates (A, B). The parent cannot lie outside the upper-triangular domain, since it contains the accepted child. Its rejection therefore implies that it meets a line $j - i = d$ for $d \in \{0, c\}$. In integer coordinates,

$$|B - A - d| < 2s, \qquad |b - a - d| < 3s,$$

since each child coordinate differs from its parent's by either 0 or s. For a fixed row coordinate a and a fixed d, at most six multiples of s can be the column coordinate b. There are N/s row coordinates and two lines, giving $12N/s$ squares.

If $x \in J_T$, then $a \le x \le b + s < a + d + 4s$. Hence a is a multiple of s in an interval of length $d + 4s$, with at most $d/s + 5$ choices. Multiplying by six column choices and summing over $d = 0, c$ gives $60 + 6c/s \le 60(1 + c/s)$. Counting a square under both lines only enlarges the bound. Inclusive endpoints cover every gap weight, and also bound the incidences of key weights, whose right endpoint is excluded. □

Theorem 8 (Aggregate core bound)**.** *For the layers of Theorem 1,*

$$S := \sum_{h=0}^{H} \delta(E_h) = O\Big(nH + n\sqrt{WL} + WL^3 + HWL\Big). \tag{7}$$

The complete construction takes $O(S \log(n + 1))$ arithmetic operations, and the retained succinct witness structures occupy $O(S)$ words. In particular, if $W = O(n)$, these bounds are respectively $O(n^{3/2} \log^{3/2}(n + 1))$ and $O(n^{3/2}\sqrt{\log(n + 1)})$.

Proof. For the early layers $1 \le h < L$, pad the input for analysis to $N = 2^{\lceil \log_2 n \rceil}$ keys, where $n \le N < 2n$. Append zero key and gap weights, retaining the original q_n at position n. Keep the same penalty M and the same layer index. Every interval contained in $[0, n]$ has exactly its original recurrence and value. No assertion about the optimal height of the padded full input is needed. The proof of (1) applies on every padded interval: its mass is at most W and $M > hW$. Thus its residual is between zero and h times its own interval mass, and its layer is Monge.

Apply Lemma 7 to the padded grid with $c = c_h$. There is no outer cut through a tile at n: the entire padded tile is used for the bound, even if only part of it lies in the original grid. For a tile T of side s, the two positive corners in the telescoping density sum are residuals on subintervals of J_T. Its density sum is therefore at most $2hW(J_T)$. Its number of positive integer densities is at most both this sum and s^2. Charging each key and gap weight separately to its covering spans gives

$$\sum_{T:\,\mathrm{side}(T)=s} W(J_T) \le 60W(1 + c_h/s).$$

Writing $K_{h,s}$ for the number of nonzero original densities covered by these tiles, we obtain

$$K_{h,s} \le \min\{24ns, 120hW(1 + c_h/s)\}. \tag{8}$$

The original exceptional line and completion diagonal add at most $2n$ entries. There are at most L dyadic scales. Using $\min(x, y + z) \le \min(x, y) + \min(x, z)$ and summing geometrically yields

$$\begin{aligned}\delta(E_h) &\le 2n + 120 \sum_s \min\{ns, hW(1 + c_h/s)\} \\ &= O\Big(n + hWL + \sqrt{nhWc_h}\Big).\end{aligned}$$

Indeed, the hW term contributes at most hWL, while $\sum_s \min\{ns, hWc_h/s\} \le 4\sqrt{nhWc_h}$. If $hWc_h < n$, the latter inequality follows directly from $\sum_s hWc_h/s \le 2hWc_h$; otherwise split at $s = \sqrt{hWc_h/n}$. Zero mass makes both terms zero.

Now $2^L \le 2(n + 1) \le 4n$, and

$$\sum_{h=1}^{L-1} \sqrt{h\,c_h} \le \sqrt{L} \sum_{h=1}^{L-1} 2^{h/2} = O(\sqrt{Ln}).$$

The early layers consequently contribute $O(nL + WL^3 + n\sqrt{WL})$. For $h \ge L$, the earlier balanced-tree argument bounds each completed core by $n + LW$; the initial core has n entries. Since $H \ge L$, summing proves (7).

The product, representation conversion, and witness construction at layer h cost $O((n + \delta(E_{h-1})) \log(n + 1))$. The product core is bounded by its input cores. The n root queries add $O(n \log(n + 1))$, and each completed core already has n completion entries. Hence the total time is $O(S \log(n + 1))$. The per-product succinct witness space is $O(n + \delta(E_{h-1}))$, so retaining all such structures costs $O(S)$. This storage statement is for the published succinct witness structures; it does not assert the same space for an arbitrary range-query implementation.

Finally $W = O(n)$ implies $H = O(L)$. Equation (7) then gives $S = O(n^{3/2}\sqrt{L} + nL^3) = O(n^{3/2}\sqrt{L})$, using $L^{5/2} = O(\sqrt{n})$. This proves the sharper linear-mass clause in Theorem 1. The general maximum-core bound remains available whenever it is smaller. □

2.8 A full mixed-weight slot representation

The key-only ruler subsequence extends to weighted gaps by retaining a pair term. For this subsection, index keys by $1, \dots, n$ and gaps by $0, \dots, n$. Fix a height h with $c = 2^h - 1 \ge n$, put $r = c - n$, and write $v(t) = \nu_2(t)$ for $1 \le t \le c$.

Theorem 9 (Mixed slot representation)**.** *The minimum full mixed cost among BSTs of height at most h equals hW minus the maximum, over all $1 \le t_1 < \cdots < t_n \le c$, of*

$$\sum_{i=1}^{n} p_i v(t_i) + q_0 v(t_1) + q_n v(t_n) + \sum_{i=1}^{n-1} q_i \min\{v(t_i), v(t_{i+1})\}. \tag{9}$$

The selected slots need not be ancestor-closed. A minimizing slot selection can be converted to an optimal height-limited tree in $O(c+n)$ operations. The weights may be arbitrary nonnegative reals in the exact arithmetic model.

Proof. A slot t of the perfect tree has comparison length $\ell(t) = h - v(t)$. A gap between adjacent keys has depth equal to the maximum of their comparison lengths; the endpoint gaps have lengths equal to their neighboring key. The internal-gap identity is Lemma 1 of Becker [8]. Thus every actual height-at-most-h tree embeds into the perfect tree and has cost exactly hW minus (9) for its slots.

Conversely, select any n slots and delete all unselected keys from the perfect BST. Ordinary BST deletion never increases a surviving key's depth: a successor copied into a deleted node moves upward, and any spliced child subtree also moves upward. Inorder order is preserved. The gap-depth identity then shows that every resulting gap depth is at most the maximum of its neighboring selected-slot comparison lengths. Nonnegative weights imply that the repaired tree's mixed cost is at most the slot expression. These two directions prove equality of the minima.

For conversion, store parent and child pointers, inorder threads, and the current node location for every original slot label. Delete unselected labels in increasing order. A two-child deletion obtains its successor from a thread, copies its label, updates that label's location, and splices out the successor. Each deletion changes only constantly many pointers. Building the perfect tree, performing $c - n$ deletions, and writing the output take $O(c + n)$ operations. For an optimal slot sequence, a strict cost decrease during repair would contradict the equality of minima just proved. □

Theorem 10 (Capacity-deficit construction)**.** *With $r = 2^h - 1 - n \geq 0$, an optimal height-at-most-h mixed-weight BST can be constructed in $O(n(r+1))$ arithmetic operations and words, including all preprocessing and full reconstruction.*

Proof. Write $t_i = i + y_i$. The exact feasible sequences are $0 \leq y_1 \leq \cdots \leq y_n \leq r$. Let $D_i(y)$ be the best prefix reward ending at $y_i = y$, including gaps $q_0, \ldots, q_{i-1}$. Then

$$\begin{aligned} D_1(y) &= (p_1 + q_0)v(1+y), \\ D_i(y) &= p_i v(i+y) + \max_{0 \leq z \leq y} \{D_{i-1}(z) + q_{i-1} \min(v(i-1+z), v(i+y))\}. \end{aligned}$$

Finish with $\max_y \{D_n(y) + q_n v(n+y)\}$. There are $n(r+1)$ states.

Within row i, sweep y increasingly and insert predecessor $z = y$. Maintain B_d, the largest inserted predecessor value of valuation d, and $S_e = \max_{d \geq e} B_d$, with attaining predecessors; unpopulated levels are omitted. For $e = v(i+y)$, the transition maximum is

$$\max \left\{ \max_{d<e}(B_d + q_{i-1}d),\ S_e + q_{i-1}e \right\}.$$

Inserting a value of valuation d updates B_d and $S_0, \ldots, S_d$. The query scans the e lower levels and reads S_e once. Generation tags reset these arrays logically between rows, so no length-h clearing pass is required per row. The state costs $O(1 + v(i-1+y) + v(i+y))$ operations.

Across the entire table, a slot index occurs at most $\min(n, r+1)$ times in either valuation sum. Since

$$\sum_{t=1}^{c} v(t) = c - h < c, \qquad \min(n, r+1)c \leq 2n(r+1),$$

all scans and updates take $O(n(r+1))$ operations. The second inequality follows by considering $r+1 \le n$ and $r+1 > n$. Tabulating valuations, initializing the arrays, and processing the first and last rows fit the same bound, since $c = n + r \le n(r+1)$.

Store one predecessor per state and trace back the slots. Apply Theorem 9's $O(c+n)$ repair procedure. All arrays, traceback records, and the reconstructed tree fit in $O(n(r+1))$ words. □

Becker [8] already gives a quadratic construction when the height limit is a fixed additive constant above the minimum. The bound here exposes the capacity deficit: constant r gives linear work, while $r = \Theta(n)$ gives quadratic work. A prescribed height remains a constraint; unrestricted optimality requires a sufficient height bound or a separate certificate. For example, $p = (100, 1, 1, 1)$ and $q = (1, 1, 1, 1, 1)$ have an unrestricted optimum of cost 121: the first key must be the root, since placing it lower already costs at least 200, and its uniform three-key right subtree is balanced. This tree has height three. Every minimum-internal- path-length four-key tree puts that first key below the root, so the height-limited class includes necessary shapes beyond that earlier certificate family.

The complementary penalty regime. For an interval with $n \ge c$, maximum exposure produces a perfect c-key skeleton, and every gap reaches depth h. Consequently, when $M > hW$, its terminal-penalty value is

$$A_h(0, n) = M(n - c) + hW - \max_{1 \le t_1 < \cdots < t_c \le n} \sum_{j=1}^{c} p_{t_j} v(j). \tag{10}$$

The feasibility of selecting the skeleton keys is exactly their increasing order; remaining keys lie in its terminal intervals. At $n = c$ this agrees with (9), since adjacent ruler valuations have minimum zero and both endpoint valuations are zero. In particular, ordinary density cells strictly above $j - i = c$ are independent of all gap weights: the gap contribution $hW(i, j)$ and the penalty's affine length term both have zero density there. Mixed gap/key interactions remain in the fully exposed region and on the boundary.

2.9 Exact gap-only mass and a sharp binary control

The mixed representation separates two sources of core growth. The following identity settles the binary gap-only control case and identifies where its argument uses the absence of key weights. Write $D_h(i, j)$ for the ordinary density of A_h at (i, j).

Lemma 11 (Gap-only core mass)**.** *Let $p_i = 0$, $h \ge 1$, $c = 2^h - 1 \le n$, and $M > hW$. For $0 \le i < n - c$, put*

$$Q_i = \max\{q_{i+1}, q_{i+3}, \ldots, q_{i+c}\}.$$

Densities strictly above $j - i = c$ vanish, and those on that line equal $M - Q_i > 0$. The density mass strictly below the line is exactly

$$S_h = (h-2)W + q_0 + q_n + \sum_{i=0}^{n-c-1} Q_i. \tag{11}$$

For integer gaps, let $X_h = \sum_{j-i<c} \max\{D_h(i, j) - 1, 0\}$. Then

$$\delta(E_h) = 2n - c + S_h - X_h.$$

In particular, binary gaps satisfy

$$\delta(E_h) \le 3n - 2c + (h-2)W + q_0 + q_n. \tag{12}$$

Proof. A length-c interval has the forced perfect tree and cost h times its gap mass. At length $c-1$, Theorem 9 selects all but one perfect slot. Removing an even slot yields no gap reward. Removing an odd slot yields one reward, the corresponding even-indexed gap's weight: the neighboring valuations have minimum one, or the endpoint neighbor has valuation one. Thus

$$A_h(a, a+c-1) = h \sum_{t=a}^{a+c-1} q_t - \max\{q_a, q_{a+2}, \ldots, q_{a+c-1}\}. \tag{13}$$

When $h = 1$, the empty interval gives $q_a - q_a = 0$ directly. Equation (10) gives $M(k-c) + hW(i,j)$ at length $k \geq c$. Four-corner subtraction now gives zero above the line and $M - Q_i$ on it.

For each $i < n - c$, summing below-line densities in that row telescopes to

$$\sum_{j=i+1}^{i+c-1} D_h(i,j) = A_h(i, i+c) - A_h(i+1, i+c) - A_h(i, i+1) = hq_i + Q_i - (q_i + q_{i+1}).$$

The remaining triangular corner has mass

$$A_h(n-c, n) - \sum_{i=n-c}^{n-1} (q_i + q_{i+1}).$$

Its full interval has length c and hence perfect-tree cost. Adding these expressions proves (11), including $c = 1$ and $c = n$. For integer weights, below-line support is $S_h - X_h$; the capacity line contributes $n - c$ entries and the completion contributes n. Finally, $Q_i \leq 1$ proves (12). □

Theorem 12 (Sharp leading term for fixed-mass binary gaps)**.** *Let $n = 2^s$, $s \geq 4$. Over all inputs with zero key weights, binary gap weights, and $W = n/2$, the largest completed core over all layers satisfies*

$$\frac{sn}{2} + 1 \;\leq\; \max_{q,h} \delta(E_h) \;\leq\; \frac{(s+1)n}{2} + 4.$$

In particular, this maximum is $\frac{1}{2} n \log_2 n + O(n)$.

Proof. For $1 \leq h \leq s$, use (12) and $q_0 + q_n \leq 2$. The resulting upper bound $U(h) = 3n - 2(2^h - 1) + (h-2)n/2 + 2$ satisfies $U(h+1) - U(h) = n/2 - 2^{h+1}$. Its maximum occurs at $h = s-2, s-1$ and is $(s+1)n/2 + 4$. Layer zero has core n. For $h \geq s+1$, a perfect $(n-1)$-key skeleton plus one bottom cherry has gap cost at most $sW + 2$. Whole-triangle telescoping gives ordinary density mass $A_h(0,n) - \sum_{i=0}^{n-1}(q_i + q_{i+1})$. Integer densities therefore give $\delta(E_h) \leq n + (s-2)W + q_0 + q_n + 2 \leq sn/2 + 4$.

For the lower bound, put $q_i = 1$ for even $i < n$, and set every other gap weight, including q_n, to zero. This has mass exactly $n/2$. Take $h = s + 1$. An interval avoiding q_n has alternating gap weights. By the parity contraction bound of Lemma 109, its optimum is

$$F(i,k) = tQ + (k + 1 - 2^t), \qquad t = \lfloor \log_2(k+1) \rfloor,$$

where k is its key count and Q its positive-gap count. A minimum-path- length tree attains it within height h. For $k \geq 1$ and $i + k + 1 \leq n - 1$, its four-corner difference is

$$D_h(i, i+k) = \begin{cases} 1, & i \text{ even and } k = 2^d - 1 \text{ for some } d \geq 1, \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

Indeed, away from $k = 2^d - 2, 2^d - 1$, the logarithms agree and both the mass and residual terms cancel. At $k = 2^d - 2$, the longest interval has 2^d alternating gaps, exactly 2^{d-1} positive, which cancels its logarithm jump. At $k = 2^d - 1$, subtraction gives $2^{d-1} - Q_{\text{short}}$, equal to one for even i and zero for odd i. This includes $d = 1$, where the shortest interval is empty with cost zero.

For each $1 \le d \le s - 1$, equation (14) counts $(n - 2^d)/2$ positive entries. Summing gives $(s - 2)n/2 + 1$; adding the n completion entries proves the lower bound. □

This establishes the binary gap-only asymptotic by explicit upper and lower proofs. The fixed-mass search in Appendix E supplies finite controls. Key weights change the saturated perfect-skeleton cost, so the telescoping identity does not extend unchanged to the full mixed objective. The odd-ramp theorem already rules out an $O(n \log n)$ bound for all linear-mass mixed inputs.

2.10 Computing all intervals in a capacity band

The capacity boundary can also be evaluated directly across every interval start. This complements the single-instance slot construction.

Theorem 13 (Exact capacity bands)**.** *Fix an integer $d \ge 0$ and $M > HW$. For $h = 0, \dots, H$, all terminal-penalty values with $|j - i - c_h| \le d$, where $c_h = 2^h - 1$, and minimizing roots for $h \ge 1$ can be computed in $O(n(d+1)(H+1))$ arithmetic operations and words. Every retained fully exposed interval has a complete tree witness that can be reconstructed in time linear in its key count. For $d \ge 1$, these values also give all ordinary densities with $|j - i - c_h| \le d - 1$ within the same bound.*

Proof. At layer h, put $a = c_{h-1}$. For a parent of length k, write the child lengths as u, v, where $u + v = k - 1$. Maximum exposure is forced by $M > hW$. If $k \le c_h$, it requires $u, v \le a$, giving the feasible range

$$\max(0, k - 1 - a) \le u \le \min(a, k - 1).$$

The nonnegative deficits $a - u, a - v$ sum to $c_h - k \le d$, so both children belong to the preceding band. If $k \ge c_h$, maximum exposure instead requires $u, v \ge a$, or $a \le u \le k - 1 - a$. Their excesses sum to $k - c_h \le d$, again keeping both children in the preceding band. Every split in these ranges attains maximum exposure. Thus the restricted recurrence equals the full recurrence on every stored state. Initialize $A_0(i, i + k) = Mk$ for $k \le d$, with empty intervals zero.

At each layer, process lengths from $k_{\min} = \max(1, c_h - d)$ through $k_{\max} = \min(n, c_h + d)$. The first diagonal examines every feasible split, at most $d + 1$ roots per interval, costing $O(n(d + 1))$. For subsequent diagonals, intersect the feasible range with the monotone leftmost-root bounds

$$R_h(i, j - 1) \le r \le R_h(i + 1, j).$$

These roots are already available on the preceding length diagonal. For completeness, if $r < s$, increasing j increases the difference between their candidate costs by the Monge inequality for A_{h-1} on rows $r + 1, s + 1$. Increasing i also increases that difference, by the inequality on columns r, s. Hence the leftmost minimizing root cannot move left when either endpoint increases. The common interval-weight term does not affect root selection. Restricting to maximum-exposure splits preserves the same minimizer.

For a fixed length k, set $b_i = R_h(i, i + k - 1)$. Candidate counts telescope:

$$\sum_i (b_{i+1} - b_i + 1) \le 2n.$$

Feasibility clamping only reduces this work. There are at most $2d$ further diagonals, so the whole layer costs $O(n(d+1))$. Prefix sums, base values, and array initialization fit this bound. Retaining

all values and roots uses $O(n(d+1)(H+1))$ words. Root traceback reconstructs a fully exposed interval in linear output time; band closure supplies every child state. Four-corner differences give the stated densities. □

For $H = O(\log n)$ and $d = 1$, this computes every capacity-line density in $O(n \log n)$ work, with arbitrary mixed weight magnitudes. The result computes a whole family of intervals; the earlier $O(n(r+1))$ slot bound solves one height-limited instance. Unrestricted optimality still requires an adequate height or a separate certificate. The main core bound does not depend on an unproved estimate outside this band.

2.11 Exact capacity densities and their interaction allowance

The capacity line admits a closed form that separates joint optimization from the contributions of individual weight units. Write $P = \sum p_i$, $Q = \sum q_i$, and $v(t) = \nu_2(t)$.

Theorem 14 (Capacity interaction)**.** *Fix $h \geq 1$, $c = 2^h - 1 \leq n$, and $M > hW$. For $0 \leq i < n - c$, define*

$$Z_{i,r} = \sum_{\ell=1}^{r} v(2\ell)(p_{i+2\ell} - p_{i+2\ell-1}), \qquad F_i = \max_{0 \leq r \leq (c-1)/2} (-Z_{i,r}), \qquad G_i = \max_{0 \leq r \leq (c-1)/2} (Z_{i,r} + q_{i+2r+1}).$$

Then $D_h(i, i+c) = M - F_i - G_i$. Let $T_1(w)$ denote total ordinary density mass strictly off this line, e_x a unit on outcome x, and $R_h(w) = A_h(0, n) - M(n - c)$. For arbitrary nonnegative real weights,

$$T_1(w) - \sum_x w_x T_1(e_x) \leq R_h(w) - \sum_x w_x R_h(e_x) \leq (h-1)P. \tag{15}$$

The middle expression is nonnegative. These quantities are independent of the choice of sufficiently large terminal penalties in the unit instances.

Proof. Put $B_i = \sum_{t=1}^{c} p_{i+t-1} v(t)$. The two length-$c$ intervals are perfect. Their inner length-$c-1$ interval has reward $B_i + G_i$, by the one-missing-slot calculation in Lemma 144. The outer length-$c+1$ interval exposes exactly c keys. Omitting its first key gives reward B_{i+1}. Moving the omitted key to index $i + t$ changes this reward by

$$\sum_{k=1}^{t} v(k)(p_{i+k-1} - p_{i+k}) = -Z_{i,\lfloor t/2 \rfloor}.$$

Thus its optimum reward is $B_{i+1} + F_i$. All its gaps have depth h. The four-corner difference cancels the interval masses and perfect rewards, leaving $M - F_i - G_i$. For $h = 1$, the inner interval is empty and the formula holds with $F_i = 0$, $G_i = q_{i+1}$.

The sum of all ordinary densities is the full interval cost minus the singleton costs. Subtracting the capacity densities therefore gives

$$T_1(w) = R_h(w) - P - 2Q + q_0 + q_n + \sum_i (F_i(w) + G_i(w)). \tag{16}$$

Both maxima are positively homogeneous and subadditive in the weights. Apply this to $w = \sum_x w_x e_x$ and cancel the linear terms in (16); this proves the first inequality.

For the second, the full interval's perfect exposed skeleton has reward

$$S_h(p) = \max_{1 \leq t_1 < \cdots < t_c \leq n} \sum_{j=1}^{c} p_{t_j - 1} v(j), \qquad R_h(w) = hW - S_h(p).$$

All gap terms cancel from the interaction. Its value is $\sum_{x=0}^{n-1} p_x S_h(e_x) - S_h(p)$, which is nonnegative because S_h is a maximum of linear forms. It is at most $(h-1)P$ because each unit reward is at most $h-1$ and $S_h(p) \geq 0$. □

Corollary 15 (Sharp key-mass allowance)**.** *No uniform factor smaller than one can multiply* $(h-1)P$ *in* (15)*. In particular an* $O(P)$ *allowance independent of height is impossible.*

Proof. Take $h \geq 2$, $C = 2^h$, $n = 2C$, $p_x = 1$ for even x and zero otherwise, and $q = 0$. Thus $P = C$. Each walk $Z_{i,r}$ is monotone: increasing for even i and decreasing for odd i. The maximizing endpoints of F_i, G_i are common to every positive unit key, so their sums are exactly additive. Equality therefore holds in the first inequality of (15).

The skeleton reward is $c - h$: no reward exceeds $\sum_{j=1}^{c} v(j) = c - h$, and the contiguous subsequence starting at key index one attains it by putting unit keys at every even skeleton rank. Hence $R_h(w) = (h-1)C + h + 1$. For a unit key at one-based position X, feasible skeleton ranks form $[\max(1, X - C - 1), \min(C - 1, X)]$. Among odd positions, $C/2 + 1$ can occupy the root. The leftmost has comparison length h. The remaining positions occur in pairs of groups with comparison length $h - t$ and 2^{t-1} positions per group, for $1 \leq t \leq h - 2$. Consequently

$$\sum_{x \ meven} R_h(e_x) = C/2 + 1 + h + 2 \sum_{t=1}^{h-2} (h-t) 2^{t-1} = 2C - h - 1.$$

The interaction is exactly $(h-3)C + 2h + 2$, whose ratio to $(h-1)C$ tends to one. Scaling all positive weights by two gives $W = n$ with the same ratio. □

2.12 Coupled stability of above-capacity rank rewards

Let $p_0, \ldots, p_{n-1} \geq 0$, $P = \sum_x p_x$, and let $v = (v_1, \ldots, v_c)$ be any real template, with $1 \leq c \leq n$. Set

$$S_v(i,j) = \max_{i \leq t_1 < \cdots < t_c < j} \sum_{r=1}^{c} p_{t_r} v_r \qquad (j - i \geq c).$$

For integer $d \geq 1$, define its upper tail by

$$U_d(v) = \sum_{\substack{0 \leq i < j < n \\ j - i \geq c + d}} [S_v(i,j) + S_v(i+1, j+1) - S_v(i, j+1) - S_v(i+1, j)]. \tag{17}$$

Each summand is nonnegative. Given optima for the outer and inner intervals, their componentwise minimum fits the left interval and their componentwise maximum fits the right interval. The sum of their rewards is unchanged, proving the inequality. For $c = 2^h - 1$ and $v_r = \nu_2(r)$, the above-capacity mixed-slot formula identifies (17) with the ordinary upper density mass at distance at least d. Its terminal-penalty and interval-mass terms cancel in every four-corner difference, including all gap weights.

Theorem 16 (Rank-coordinate tail stability)**.** *For every rank* r *and* $\Delta \geq 0$*,*

$$-2P\Delta \leq U_d(v + \Delta e_r) - U_d(v) \leq P\Delta. \tag{18}$$

Consequently, if $v' \geq v$ *coordinatewise, then* $U_d(v') \leq U_d(v) + P \sum_r (v'_r - v_r)$*. For arbitrary templates the absolute change is at most* $2P\|v' - v\|_1$*.*

Proof. Optima for a fixed interval are closed under componentwise minimum and maximum. Both sequences stay strictly increasing; their total rewards sum to the original total, so both remain optimal. Choose the unique componentwise-leftmost optimum, obtained by taking all these minima. When both interval endpoints move weakly to the right, its selected positions move weakly to the right coordinatewise. Indeed, the minimum of the old and new optima is feasible in the old interval, and their maximum in the new. Reward preservation makes both optimal. Minimality of the old optimum forces it to equal that componentwise minimum.

Put $k = c + d$. If $k \geq n$ the tail is zero. Otherwise $m = n - k \geq 1$ and row telescoping gives

$$U_d(v) = \sum_{i=0}^{m} S_v(i, i+k) - \sum_{i=0}^{m-1} S_v(i+1, i+k) - S_v(0, n). \tag{19}$$

Vary v_r by $\lambda \in [0, \Delta]$. Each score is the maximum of finitely many affine functions, so away from finitely many breakpoints its derivative is the weight at its optimum's rth selected position. Persisting ties have equal slopes. Write a_i for this position in $[i, i+k)$ and b_i for it in $[i+1, i+k)$. Endpoint monotonicity yields

$$a_0 \leq b_0 \leq a_1 \leq b_1 \leq \cdots \leq a_m.$$

The first two sums in (19) therefore have derivative $\sum_i p_{a_i} - \sum_i p_{b_i}$. Equal positions occupy contiguous blocks of this alternating chain. Cancelling each such block leaves a coefficient in $\{-1, 0, 1\}$ at every input position, bounding this derivative between $-P$ and P. The last term contributes $-p_z \in [-P, 0]$. Integrating the resulting almost-everywhere derivative bound proves (18). Successive coordinate changes prove both extensions. □

Corollary 17 (Clipping high ruler rewards)**.** *Let $C = 2^h$, $c = C - 1$, $v_r = \nu_2(r)$, and $v_r^{(t)} = \min(v_r, t)$ for $0 \leq t \leq h - 1$. Then*

$$U_d(v) \leq U_d(v^{(t)}) + P(2^{h-t} - (h-t) - 1). \tag{20}$$

For $t = \min(h-1, \lfloor \log_2 d \rfloor)$ the error is at most $2PC/d$.

Proof. The removed reward sums to

$$\sum_{r=1}^{C-1} [\nu_2(r) - t]_+ = \sum_{s=t+1}^{h-1} (2^{h-s} - 1) = 2^{h-t} - (h-t) - 1.$$

Apply Theorem 16 to the coordinatewise increase from $v^{(t)}$ to v. If $t = h - 1$ the error vanishes; otherwise $2^t > d/2$ gives the stated estimate. □

The remaining template has period 2^t. The corollary controls the high-rank contribution at the desired capacity-over-width scale while retaining joint optimization of all weights. That error scale is necessary, as the following exact family shows. The subsequent alternating-template theorem controls the periodic part at level one. Higher levels remain open; Lemma 147 shows that even level one requires a capacity-dependent term.

Theorem 18 (Exact full-ruler clipping tail)**.** *Let $a \geq 0$, $b \geq 3$, $d = 2^a$, $Q = 2^b$, $C = dQ$, $h = a + b$, $c = C - 1$, and $n = 2C$. Give unit weight to the keys at one-based positions $x = C - 3d$ and $y = C$, and zero weight to every other key. For $v_r = \nu_2(r)$,*

$$U_d(v) = \frac{5Q}{4} - 3b + 2, \qquad U_d(v^{(a)}) = 0. \tag{21}$$

These above-capacity tails are unchanged by arbitrary nonnegative gap weights. Since $P = 2$, the ratio of their difference to the clipping allowance $P(Q - b - 1)$ tends to $5/8$ as $b \to \infty$.

Proof. Put $\ell_r = \min(v_r, a)$ and $z_r = (v_r - a)_+$. The high reward z_r vanishes away from multiples of d, and $z_{dt} = \nu_2(t)$. We first verify compatible optimization on the boundary intervals in (19). This step is needed because optimized scores are not generally additive.

Consider an interval with slack $e = d$ or $d-1$ containing both unit keys, and let r be the first key's relative one-based position. Their selected ranks have the form

$$r - u, \quad r + 3d - w, \qquad 0 \le u \le w \le e,$$

with both ranks in $[1, C-1]$. If $r \bmod d \neq 0$ and the two multiples of d in the feasible rank windows are valid, $u = w = r \bmod d$ selects both. Every other rank has low reward less than a and high reward zero. A selection using one multiple can round the other rank to its multiple without violating $u \le w$, so the same selection maximizes both parts. For slack $d-1$ and $r \bmod d = 0$, the unique feasible pair of multiples has this same property.

For slack d and $r = dR$, the pairs of multiples, after division by d, are

$$(R, R+3), \quad (R-1, R+2), \quad (R, R+2),$$

with out-of-range pairs discarded. The boundary starts with both keys present have $1 \le R \le Q-3$, so the third pair is always valid. If one selected rank is off the multiples, round it while retaining the rank already on a multiple: the extreme shifts force the first two pairs, and the intermediate shifts use the third. A pair with neither rank on a multiple is dominated in low reward by the third pair. Thus a high-score maximizer also has low score $2a$.

The remaining two-key case is $1 \le r < d$. The first key's best low rank lies in $[1, r]$; the second key's unique multiple is $3d$, with reward a and high reward zero. These choices are compatible: the first omission count is less than r, and the second is r. With only one key present, its feasible rank window either contains a multiple of d, giving maximum a plus the high maximum, or contains none and has zero high maximum. On the full interval, two selected ranks are at distance at most $3d$, so at most one is divisible by $4d$. Their reward is at most $(h-1) + (a+1) = h + a$. Ranks $C/2, C/2 + 2d$ attain it, have low score $2a$, and have high score b. They are feasible at x, y because $Q \ge 8$. Consequently every boundary score and the full score satisfy

$$S_v = S_\ell + S_z.$$

On these same intervals the low score is also the sum of the separate one-key low scores. A one-key capped-ruler maximum is the sum, over $1 \le t \le a$, of the indicators that its rank window meets a multiple of 2^t. Its density cells have capacity distance $2^t - 1 < d$, by the consecutive-multiple classification in Section 5.9. Hence $U_d(\ell) = 0$ and $U_d(v) = U_d(z)$.

For the high scores put $k = C + d - 1$ and divide selected multiple ranks by d. The coarse instance has length $2Q$, capacity $Q-1$, unit keys at $Q-3, Q$, and ruler rewards $\nu_2(t)$. If $i = jd$, the long boundary score $S_z(i, i+k)$ maps to the coarse interval $[j, j+Q)$. If $i = jd + r$ with $1 \le r < d$, it maps to $[j+1, j+Q)$. Every short score $S_z(i+1, i+k)$ in this block maps to $[j+1, j+Q)$. These are exactly the unique-pair and three-pair alternatives above. The terminal high score is zero, and the full score maps to the coarse full interval. Cancelling within each block of (19), including the zero endpoint terms, gives $U_d(z) = U_1$ for the coarse instance.

Set $V(r) = \nu_2(r)$ for $1 \le r < Q$ and zero otherwise. In that coarse instance write $R = Q - 3 - i$. The long interval has one omission and the short interval has none. Their score difference is

$$\delta_R = \max\{V(R) + V(R+3),\ V(R) + V(R+2),\ V(R-1) + V(R+2)\} - V(R-1) - V(R+2).$$

The nonzero range is $-2 \le R \le Q-3$. For even R this is $V(R)$, whose sum is $Q - b - 2$. For odd R it is $[V(R+3) - V(R-1)]_+$. The boundary values $R = -1, 1$ contribute $1, 2$; values $R = 3$

(mod 4) contribute zero. Writing $R = 4t + 1$ for the others leaves the positive differences $\nu_2(t + 1)$ for odd t, except the last $t = Q/4 - 1$, where $V(Q) = 0$. Thus the odd sum is

$$3 + \sum_{j=1}^{Q/4-1} \nu_2(j) = Q/4 - b + 4.$$

The full coarse score is b: two ranks within distance three cannot both be divisible by four, and $Q/2, Q/2 + 2$ attain $(b - 1) + 1$. Subtracting this full score gives (21). Corollary 17 removes reward $Q - b - 1$, giving the stated ratio. Gap independence follows from (10). □

This proves necessity of the PC/d scale for full-ruler density mass. Multiplying both key weights by C makes total key mass n and scales the displayed tails by C; the number of positive cells does not scale. The theorem therefore sharpens the tail analysis without changing the current core-cardinality upper bound.

2.13 A width-sensitive bound for the alternating template

The level-one clipped ruler has a fixed-cardinality path-matching representation. Its tail can be bounded while retaining arbitrary nonnegative real key weights and all their interactions.

Theorem 19 (Alternating-template upper tail)**.** *Let $c = 2m + 1 \geq 3$, $n \geq c$, and $v_r = 1$ for even r and zero for odd r. With the scores and tails of (17), for every integer $d \geq 1$,*

$$U_d(v) \leq P \min\left\{m,\, 2 + \frac{c - 3}{d}\right\}. \tag{22}$$

Thus $U_d(v) = O(P(1 + c/d))$.

Proof. The m rewarded positions of a length-c subsequence lie in $[i + 1, j - 2]$, with at least one filler before them, between consecutive positions, and after them. Conversely any m indices in this range separated by at least two admit those fillers. Regard index x as edge $(x, x+1)$ of the path on vertices $i+1, \ldots, j-1$, with weight p_x. Then $S_v(i, j)$ is exactly the maximum weight of a matching with m edges.

Fix $j - i = c + d$ and put $\delta = S_v(i, j) - S_v(i + 1, j)$. If an optimum omits the first edge $i + 1$, it remains feasible on the shorter path and $\delta = 0$. Otherwise remove that edge and the first vertex. The remaining path has $2m + d - 1$ vertices and a matching M of size $m - 1$, hence exactly $d + 1$ unmatched vertices $z_0 < \cdots < z_d$. Between consecutive unmatched vertices every interior vertex is matched. Each of the d intervening segments is therefore an augmenting path: its edges alternate unused and matched, starting and ending unused. Flipping it gives a matching of size m on the shorter path. This is the classical matching augmentation operation [1]; here all candidate paths and their weight accounting are explicit.

These paths can share unmatched endpoints, but their matched edges are disjoint. Their removed weights sum to at most the weight of M. One flip therefore removes at most $(S_v(i, j) - p_{i+1})/d$; its added edges have nonnegative weight. Including the omitted-edge case gives

$$d\delta \leq (d - 1)p_{i+1} + S_v(i, j). \tag{23}$$

For $k = c + d < n$ put $b = n - k$. Equation (19) becomes

$$U_d(v) = \sum_{i=0}^{b-1} (S_v(i, i + k) - S_v(i + 1, i + k)) + S_v(b, n) - S_v(0, n).$$

The last difference is nonpositive by containment. Each score in the sum is at most $\sum_{x=i+1}^{i+k-2} p_x$, and a fixed key appears in at most $k-2$ such intervals. Summing (23) yields

$$dU_d(v) \le (d-1)P + (k-2)P = (2d+c-3)P.$$

For $k \ge n$ the tail is empty. Finally, increasing the m even-rank rewards from zero to one and applying Theorem 16 gives $U_d(v) \le mP$. □

Theorem 20 (Exact weighted cluster tail)**.** *Let $C = 2^h$, $K \ge 3$, $C \ge 4K$, $c = C-1$, $n = 2C$, and $x = C - 3(K-1)$. Place arbitrary nonnegative real weights $w_0, \dots, w_{K-1}$ at $x, x+3, \dots, C$, with other weights zero. For the alternating template, put $\mu = \min_r w_r$ and $P = \sum_r w_r$. Then*

$$U_K = N\mu, \qquad N = \lceil x/2 \rceil - 1. \tag{24}$$

At fixed cluster mass P, the maximum is NP/K, attained exactly by equal weights.

Proof. Put $k = c + K$ and $b = n - k = C - K + 1$. We compute each loss in (19). For $i < x$, the old interval $[i, i+k)$ can reward all K cluster keys. With first local index $R = x - i + 1$, omit $u = 0$ positions before it if R is even, or $u = 1$ if odd, and one in each spacing-three gap. This uses $K - 1 + u \le K$ omissions. The last rewarded rank is $C - K + 2 - i - u$, even and at most $C - 2$. The apparent exception $K = 3, i = 0$ forces $u = 1$ and also fits. Any remaining omission goes after the last cluster key. Thus the old score is P.

The shortened interval has omission budget $K - 1$ and first local index $r = x - i$. If r is even, all cluster keys can again be rewarded. If $r \ge 3$ is odd, rewarding all of them would require one initial omission and at least one in each of $K - 1$ odd gaps: K omissions, too many. The score is therefore at most $P - \mu$.

Every choice of the unrewarded key is feasible. If it is the first, the next local index $r + 3$ is even, and the $K - 2$ remaining odd gaps use $K - 2$ omissions. If it is an interior key, use one initial omission and one in each of $K - 3$ ordinary odd gaps; the gap across the unrewarded key has length six and needs no omission. Again the total is $K - 2$. In both cases the final rewarded rank is $C - K + 2 - i$, even and at most $C - 2$; $K = 3, i = 0$ belongs to the even-r case instead. If the last key is unrewarded, one initial omission and $K - 2$ inter-key omissions give final rewarded rank $C - K - 2 - i$. The unrewarded key can occupy an odd rank, requiring no extra omission. The remaining omissions fit after the final rewarded key. Choosing a minimum weight therefore gives shortened score $P - \mu$.

At $i = x - 1$, the first cluster key is the shortened interval's left endpoint and can only occupy odd rank one. Its loss is w_0, and all remaining cluster keys can be rewarded. For $i \ge x$, each interval has at most $K - 1$ eligible cluster keys. Their first local index is at most four; the initial parity correction and inter-key omissions fit the budget, and their final rewarded rank is at most $2K \le C - 2$. Both intervals reward every eligible key, so their difference is p_{i+1}.

Exactly N starts $i < x - 1$ have $x - i$ odd. The subsequent losses plus $S(b, n)$ equal $P - w_0$, while $S(0, n) = P$. The telescope gives $U_K = N\mu + w_0 + (P - w_0) - P = N\mu$. Finally $\mu \le P/K$, with equality exactly at equal weights, proves the extremal statement. □

Corollary 21 (Both alternating-tail scales are necessary)**.** *The additive P term and the Pc/d term in (22) are both necessary up to constants. The latter remains necessary when d tends to infinity.*

Proof. For the capacity term, take the unit-weight instance of Theorem 20. It gives

$$U_K(v) = \lceil x/2 \rceil - 1 = \left\lceil \frac{C - 3K + 1}{2} \right\rceil. \tag{25}$$

Taking $K = 2^{\lfloor h/2 \rfloor}$ gives $d = K \to \infty$, $P = K$, and $U_K/(Pc/K) \to 1/2$. Scaling each unit by n/K gives integer total mass $W = n$.

For the additive term take $c = 3$, so $S_v(i,j) = \max_{i+1 \le x \le j-2} p_x$. Put $K \ge 2$ unit keys at $1, 1+s, \dots, 1+(K-1)s$, with $s \ge 2$ and $n = (K-1)s + 3$. The positive upper densities connect consecutive unit keys x, y: at $(i,j) = (x-1, y+1)$ the outer and side scores are one and the inner score is zero. There are $K-1$ such unit densities, all at distance $s-1$ from capacity. Hence $U_{s-1} = K-1$ with $P = K$. Letting K, s grow rules out a bound $O(Pc/d)$ alone. □

For higher clipped levels, a matching edge's reward depends on its rank among selected edges. Augmentation can change rewards on retained edges; Section 5.13 gives a small explicit failure of (23) at level two. That additional rank accounting is exact in Section 5.14, which proves that even the best single-augmentation certificate can exceed the desired tail scale by an unbounded factor. Joint rank changes remain necessary.

2.14 A linear above-capacity core for one constant key block

A constant positive key block admits an exact one-dimensional representation of its above-capacity densities, even with arbitrary nonnegative gap weights. The representation retains the joint optimization over ruler ranks. We first record an exchange fact also used in Section 5.19.

Lemma 22 (Binary saturation)**.** *For a binary input word with P ones and arbitrary nonnegative rank rewards $v_1, \dots, v_c$, some maximizing c-subsequence contains* $\min(P, c)$ *ones.*

Proof. If $P \ge c$, selecting c ones attains the upper bound $\sum_r v_r$. Otherwise, if a selected subsequence omits a one, it selects a zero. Choose a selected zero nearest to the omitted one in physical order; all selected entries between them are ones. Insert the omitted one and remove that zero. Shifting the intervening ones changes the selected binary word only by replacing one zero with one, so its reward cannot decrease. Repeat until every positive key is retained. □

Lemma 23 (Range maxima and nonadjacent visibility)**.** *For a sequence $R(0), \dots, R(N-1)$, there are at most* $\max(0, N-2)$ *pairs $a < b-1$ satisfying*

$$\min\{R(a), R(b)\} > \max_{a<r<b} R(r).$$

These pairs and their positive differences can be enumerated in $O(N)$ arithmetic work and space.

Proof. Pairs (a,c) and (b,d) with $a < b < c < d$ cannot both satisfy the inequality: they would require $R(c) > R(b)$ and $R(b) > R(c)$. Place the vertices on a polygon in index order. For $N \ge 3$ the noncrossing proper diagonals number at most $N-3$; the closing edge $(0, N-1)$ adds at most one. For $N \le 2$ there is no nonadjacent pair. This is the usual horizontal visibility geometry [12, 13].

For enumeration, scan left to right with a stack whose heights strictly decrease. On reading a new height x, pop all smaller heights, pairing each popped index with the new index; also pair the remaining top, if any, with the new index. Replace an equal-height top by the new index: that equal intervening height blocks every future strict visibility of the old index. Discard adjacent pairs. The first popped index is the preceding position. For each later candidate the last popped height is exactly the maximum over its interior: previously removed indices are shielded by a retained index of at least their height. Thus each emitted pair satisfies the strict inequality, and the last popped height supplies its positive difference. Conversely, a visible earlier index cannot have been removed by an intervening height at least its own; it is either popped now or is the first remaining top. This enumerates every pair. Each index is pushed once and popped at most once, and there are at most $2N$ pair candidates. □

Theorem 24 (One-block above-capacity core)**.** *Fix $h \geq 1$, $c = 2^h - 1$, and a valid terminal penalty. Suppose keys $z, \dots, z+m-1$ have common weight $\alpha > 0$ and every other key has weight zero. Gap weights may be arbitrary nonnegative reals. The ordinary core strictly above capacity, $j - i > c$, has at most*

$$\max\{0, c - m - 1\}$$

entries. Their coordinates and values can be constructed in $O(n)$ arithmetic work and space. The leading coefficient one is asymptotically sharp.

Proof. Above capacity, the layer cost is an affine terminal penalty plus $hW(i,j)$ minus the optimum ruler reward. The first two terms have zero ordinary density; arbitrary gap weights therefore cancel. Divide the reward by α to obtain a binary input.

Partial-block endpoints. All four intervals of a cell with $j - i > c$ have length at least c. If its removed left key i is positive and its added right key j is inside the block, every interval consists of positive keys and its score saturates at $\sum_{r=1}^{c} \nu_2(r)$. If j is outside the block, its positive keys form a prefix. By Lemma 22, the selected positive keys occupy the first $\min(P, c)$ ranks, where P depends only on the left endpoint. Its score is independent of the right endpoint, so the density is zero again. Reflection proves the same statement when the added right key is positive: $\nu_2(2^h - r) = \nu_2(r)$ makes a positive suffix equivalent to a prefix. Thus every remaining positive cell has all four intervals containing the entire block. If $m > c$, these scores all saturate and the above-capacity core is empty.

Exact range maxima. Assume $m \leq c$. On an interval $[i, j)$ containing the block, an optimum retains all m positive keys. They occupy consecutive selected ranks $r+1, \dots, r+m$. Counting available zeros before and after the block gives precisely

$$\max(0, c + z - j) \leq r \leq \min(c - m, z - i).$$

The normalized score is therefore a range maximum of

$$R_m(r) = \sum_{t=r+1}^{r+m} \nu_2(t) = m + s_2(r) - s_2(r+m), \qquad 0 \leq r \leq c - m, \tag{26}$$

where s_2 counts one bits. Indeed, $\sum_{t=1}^{u} \nu_2(t) = \sum_{a \geq 1} \lfloor u/2^a \rfloor = u - s_2(u)$. Equivalently $R_m(r) = m - s_2(m) + \kappa(r, m)$, where κ counts binary carries, one per column with an outgoing carry, in adding r and m.

Put $l = c + z - j$ and $u = z - i$. If clipping freezes either endpoint of the range, the four-corner difference is zero. Otherwise set $a = l - 1$ and $b = u$, with $0 \leq a < b \leq c - m$. The interior is nonempty because $b - a = j - i - c + 1 \geq 2$. Write $Q = \max_{a<r<b} R_m(r)$. The normalized density is exactly

$$\begin{aligned} D_h(i,j)/\alpha &= \max(Q, R_m(b)) + \max(R_m(a), Q) - \max(R_m(a), Q, R_m(b)) - Q \\ &= \max\{0, \min(R_m(a), R_m(b)) - Q\}. \end{aligned}$$

Thus each positive density is one nonadjacent visibility pair, mapped injectively by

$$i = z - b, \qquad j = c + z - 1 - a. \tag{27}$$

Conversely every such pair with $0 \leq i < j < n$ gives the indicated positive density. Apply Lemma 23 to the $N = c - m + 1$ values in (26) to obtain the bound.

Construction. If $c \geq n$ the region is empty. Otherwise compute the ruler values in $O(c)$ work using $\nu_2(2r) = \nu_2(r) + 1$ and zero on odd arguments. Sliding length-m sums produce R_m in $O(c)$

further work, without assuming a unit-cost population-count primitive. The lemma enumerates all pairs and their density values in $O(c)$ work. Apply (27), discard cells outside the input, and multiply each remaining value by α. Since $c < n$, the work and space are $O(n)$.

Sharp leading coefficient. Take $h \geq 3$, $m = 1$, $C = 2^h$, $n = 2C$, and $z = C + 1$. Every nonadjacent visibility pair of $\nu_2(1), \ldots, \nu_2(C-1)$ maps inside the input. These pairs are exactly consecutive multiples of 2^r, for $1 \leq r < h$: their interior valuations are smaller than r, whereas a nonconsecutive pair has an intervening multiple of the smaller endpoint's power. Their number is

$$\sum_{r=1}^{h-1} (2^{h-r} - 2) = C - 2h.$$

The upper bound is $c - m - 1 = C - 3$, and the ratio tends to one. □

Corollary 25 (A boundary charge for one block)**.** *Under Theorem 24, any $s \times s$ square of ordinary density cells strictly above capacity has at most $2s - 1$ positive entries, independently of the block's length and weight.*

Proof. Equation (27) maps the square to visibility pairs between two ordered disjoint index intervals, each with at most s positions. For distinct left endpoints $a < a'$, every neighbor b of a must be at least every neighbor b' of a'; otherwise $a < a' < b < b'$ would give crossing visibility pairs. Order the nonempty neighbor sets by increasing left endpoint. Their spans are decreasing and have disjoint interiors; two consecutive sets can share at most one endpoint. If k left endpoints and t right endpoints participate, the total number of pairs is at most $t + k - 1 \leq 2s - 1$. An empty support satisfies the same bound. □

The theorem constructs this part of a layer's core, including its values. The below-capacity mixed core and the general local hypothesis in (191) require further arguments.

2.15 The below-capacity core of one constant key block

The complementary side admits an exact classification when the gaps have weight zero. This identifies a source of sparsity inside the region where unequal ramp weights produce the lower bound of Theorem 6.

Theorem 26 (Complete below-capacity plateau core)**.** *Let $p_z = \cdots = p_{e-1} = \alpha > 0$, where $e = z + m$, and let every other key and every gap have weight zero. At height $h \geq 1$, put $c = 2^h - 1$ and*

$$U_h(m) = \sum_{r=1}^{h-1} \max(0, m - 2^r + 1).$$

Every positive ordinary density strictly below capacity equals α. There are at most $2(n-m)+U_h(m)$ such entries, and their coordinates and values can be constructed in $O(n+U_h(m))$ arithmetic work and space. Every dyadically aligned side-s square wholly below capacity and in the ordinary region contains at most $8s$ positive entries. The magnitude of α is arbitrary, and c may exceed n.

Proof. Write $v(t) = \nu_2(t)$ on the one-based perfect-tree slots $1, \ldots, c$. An interval $[i, j)$ containing the complete block, of length at most c, permits its m positive keys to occupy any m slots in $[z - i + 1, c - j + e]$. The remaining slots accommodate its leading and trailing zero keys. The slot-deletion repair therefore makes its exact cost α times hm minus the sum of the m largest values

in this window. For any nonnegative integer word v, denote that sum by $\mathrm{Top}_m(v)$. Threshold counting gives the identity

$$\mathrm{Top}_m(v) = \sum_{r\ge 1} \min\{m, \#\{t : v(t) \ge r\}\}.$$

If the interior of $[u, v]$ has at least m positions, its negative mixed difference is consequently

$$\begin{aligned}&\mathrm{Top}_m([u, v-1]) + \mathrm{Top}_m([u+1, v]) - \mathrm{Top}_m([u, v]) - \mathrm{Top}_m([u+1, v-1])\\ &\quad = \sum_{r\ge 1} \mathbf{1}\{\nu_2(u), \nu_2(v) \ge r,\ \#\{u < t < v : 2^r \mid t\} = m-1\}.\end{aligned}$$

Each clipped count changes only when the two endpoints compete for its last available rank. This is an identity for the top sum, without a separate-optimizer assumption. We classify the changed endpoint keys i, j.

Both outside, with the whole block retained. A nonzero density in this case requires $i < z$ and $j \ge e$. Put $u = z - i$, $v = c - j + e$. The displayed identity gives density α exactly when

$$2^r \mid u, \quad 2^r \mid v, \quad v - u = m2^r \quad \text{for some } r \ge 1. \tag{28}$$

At most one r satisfies the distance equation. The capacity distance is $c - (j - i) = m(2^r - 1)$. Enumerate multiples of $d = 2^r$ in

$$\max(1, c + e - n + 1 - md) \le u \le \min(z, c - md), \qquad i = z - u, \quad j = c + e - u - md,$$

and retain ordinary below-capacity cells. There are at most $\sum_{r\ge 1}\lfloor z/2^r\rfloor < z$ candidates when $z > 0$; reflection also bounds their number by $n - e$.

Positive left endpoint, zero right endpoint. For $z \le i < e \le j$, put $k = e - i$, $b = c + e - j$. The cost is $\alpha(hk - \mathrm{Top}_k([1, b]))$. Its density is α times the increase in the kth largest prefix value when slot b is appended. That increase is one exactly when

$$b = k2^r \quad \text{for some } 1 \le r \le \nu_2(b), \tag{29}$$

and is zero otherwise. Indeed, at level r, the prefix count increases from $k - 1$ to k precisely in this case. Different r give different k. With $t = j - e + 1$, we have $b = 2^h - t$ and $\nu_2(b) = \nu_2(t)$ for $1 \le t < 2^h$. Enumerating those divisors and retaining $1 \le k \le m$ uses at most $n - e$ candidates. The reflected case, a zero left endpoint and a positive right endpoint, uses at most z candidates. Reflection maps a cell to $(n - 1 - j, n - 1 - i)$ and uses $\nu_2(2^h - t) = \nu_2(t)$.

Both endpoints positive. All four intervals have uniform keys. Their minimum-internal-path-length cost has increment $\mathrm{bitlength}(k)$ for the kth key. Thus the density equals α exactly on $j - i = 2^r - 1$, $1 \le r < h$, giving precisely $U_h(m)$ cells. All remaining endpoint cases have zero density. The first three cases contribute at most $z + (n - e) + \min(z, n - e) \le 2(n - m)$. Short ruler prefixes are generated by $\nu_2(2t) = \nu_2(t) + 1$ and zero on odd arguments; no array of length c is needed when $c > n$. Each divisor candidate and each uniform output takes constant arithmetic work, proving the construction bound.

For the local bound, take at most s consecutive positive integers and remove their unique maximum-valuation position. At level r, at most $\lfloor (s-1)/2^r\rfloor$ multiples remain, so

$$\sum \text{remaining valuations} \le \sum_{r\ge 1} \left\lfloor \frac{s-1}{2^r} \right\rfloor \le s - 1. \tag{30}$$

Uniqueness follows because between two equal maximum valuations lies a larger one. In (28), each row has at most $\nu_2(u)$ neighbors. All rows other than that maximum contribute at most $s-1$; the maximum row contributes at most s. Hence this case has at most $2s-1$ cells. Equation (29) gives the same bound using columns, and reflection gives it for the third case.

For the uniform case, write the dyadic square as $[a, a+s) \times [b, b+s)$, $b-a = Ds$, $D \geq 1$. If $s = 1$ the claim is immediate. If $D \geq 2$, eligible powers $(j-i)+1$ lie in $[(D-1)s+2, (D+1)s]$, an interval of width $2s-2$ containing at most one power of two, since each eligible power is at least $2s$. Its diagonal has at most s cells. If $D = 1$, powers $d \leq s$ contribute $d-1$ cells each and $d = 2s$ contributes one, totaling $2s - 1 - \log_2 s$. Cropping to the positive block reduces this count. Summing the four disjoint cases gives at most $8s$ entries. Cropping padded grids to the original non-power-of-two input also only removes cells. □

Together with Corollary 25, this proves a boundary-only local support charge for this key-only family on both sides of capacity, without integrality or a weight-magnitude restriction. The below-capacity formulas require zero gaps; the earlier above-capacity result continues to allow arbitrary nonnegative gaps. Independent verification and the separate fixed-tile excess search are recorded in Appendix R.

2.16 Finite slot-window products and a complete run constructor

The top-sum identity also composes unequal constant runs, if the matrix coordinates are slot boundaries and every input key is retained. This complements the rank transfers of Section 5.22, which select input keys for consecutive ruler ranks.

Lemma 27 (Finite slot-window factor)**.** *Fix $h \geq 1$, $c = 2^h - 1$, and nonnegative key runs (w_ℓ, m_ℓ), $1 \leq \ell \leq r$, with $m_\ell \geq 1$. Put $P = \sum_\ell w_\ell m_\ell$ and $K = 1 + (h-1)P$. For $0 \leq a, b \leq c$,*

$$Q_\ell(a,b) = K \max(m_\ell - b + a, 0) - w_\ell \sum_{t=1}^{h-1} \min\left(m_\ell, \left\lfloor \frac{b}{2^t} \right\rfloor - \left\lfloor \frac{a}{2^t} \right\rfloor\right). \tag{31}$$

This is a finite Monge matrix with fewer than $2c$ core entries and an $O(c)$ condensed construction. If $b - a \geq m_\ell$, it is the negative maximum reward of the run placed in slots $(a, b]$.

Proof. The feasible-entry formula follows by counting valuation thresholds among the m_ℓ largest ruler values. The signed counts in (31) also define all other entries. Write $m = m_\ell$ and $w = w_\ell$. The hinge contributes density K on $b - a = m$. For a threshold $d = 2^t$, a nonzero mixed difference requires both $a+1$ and $b+1$ to be multiples of d. With $x = \lfloor b/d \rfloor - \lfloor a/d \rfloor$, its density is

$$w\,[2\min(m,x) - \min(m, x-1) - \min(m, x+1)].$$

It equals w exactly when $x = m$, and is zero otherwise, including for negative x. Hence the complete density list consists of

$$\begin{aligned} &K && \text{at } (a,b) = (a, a+m), \quad 0 \leq a < c - m, \\ &w && \text{at } (a,b) = (kd-1, (k+m)d-1), \quad 1 \leq k \leq \lfloor c/d \rfloor - m, \quad d = 2^t,\ 1 \leq t < h. \end{aligned}$$

Zero-weight entries are omitted. These supports are disjoint: the second line has $b - a = md$, which distinguishes its levels and the hinge. Thus

$$\delta(Q_\ell) \leq (c-m)_+ + \sum_{t=1}^{h-1} (\lfloor c/2^t \rfloor - m)_+ < 2c.$$

Let $F(a) = \sum_{j=1}^{a} \nu_2(j)$. The left boundary is $Q_\ell(a, 0) = K(m + a) + wF(a)$. To construct the top boundary, sweep b; threshold t changes precisely when $t \le \nu_2(b)$ and $b/2^t \le m$. The total threshold visits are $\sum_{b=1}^{c} \nu_2(b) = c - h < c$. Generate the listed core directly and order it by two bucket passes. All boundaries and core entries therefore take $O(c)$ work and words, even when $m > c$ and no feasible window exists. □

Theorem 28 (Balanced slot-window construction)**.** *For the runs in Lemma 27, let $n = \sum_\ell m_\ell$ and let $w_1, \dots, w_n$ be the expanded weights. The finite Monge product $T = Q_1 \otimes \cdots \otimes Q_r$ satisfies*

$$T(a, b) = -\max_{a < t_1 < \cdots < t_n \le b} \sum_{i=1}^{n} w_i \nu_2(t_i) \qquad (b - a \ge n). \tag{32}$$

Its core has at most $2r2^{\lceil \log_2 r \rceil} c \le 4r^2 c$ entries. The condensed product and an optimal selected sequence for any feasible entry can be constructed in $O(n + r^2 c \log(c + 1))$ arithmetic operations. The ordinary witness arrays used here take $O(r^2 c \log(c + 1))$ auxiliary words.

Proof. A feasible product path partitions $(a, b]$ into consecutive windows, one per run, each of length at least its run length. The best independent selections within these windows concatenate in key order. Conversely, any all-key slot selection admits such a partition: cut after the final selected slot of each run except the last. Thus feasible paths give exactly (32).

An infeasible path has an integer window deficit at least one and incurs at least K penalty. Every threshold minimum in (31) is at most m_ℓ, even for a reversed window. Consequently the reward part of any path is at least $-(h - 1)P$. An infeasible path costs at least $K - (h - 1)P = 1$, whereas every feasible path costs at most zero. This proves strict exclusion, also for zero weights and ties. No infinite entries are used.

Multiply in a balanced tree using the published condensed product and witness primitive [14]. Its bound $\delta(X \otimes Y) \le 2(\delta(X) + \delta(Y))$ charges each leaf by at most $2^{\lceil \log_2 r \rceil}$, proving the final core bound. The sum of core sizes over all nodes is $O(r^2 c)$: at product-tree depth j there are at most 2^j nodes with at most $\lceil r/2^j \rceil$ leaves each, and the resulting quadratic bounds form a geometric sum. Matrix boundaries contribute $O(rc)$ and the product logarithm is $O(\log(c + 1))$. Extracting the runs costs $O(n)$.

Retained witnesses recover each leaf's window in $O(r \log(c + 1))$ work; the penalty argument ensures that every recovered window is feasible. Bucket its valuations into h levels and select the largest m_ℓ, breaking cutoff ties by earliest slot. Disjoint leaf windows partition the outer window, so this costs $O(c + rh)$ altogether. These costs fit the stated bound, including the conservative space for ordinary witness maps. □

Corollary 29 (Equal internal gaps at a prescribed height)**.** *Let $n \ge 2$, let $q_1 = \cdots = q_{n-1} = b \ge 0$, and allow arbitrary nonnegative keys and endpoint gaps. Define*

$$w_0 = p_0 + q_0, \qquad w_i = p_i + b \ (1 \le i \le n - 2), \qquad w_{n-1} = p_{n-1} + q_n. \tag{33}$$

If these weights have r constant runs and $c = 2^h - 1 \ge n$, a complete optimal tree of height at most h can be constructed in $O(n + r^2 c \log(c + 1))$ arithmetic work and $O(r^2 c \log(c + 1))$ auxiliary words. At the minimum feasible height $h = \lceil \log_2(n + 1) \rceil$, the work is $O(n + r^2 n \log(n + 1))$.

Proof. If ℓ_i are key comparison lengths, the total external path length is $E = I + 2n = \sum_i \ell_i + n$. The endpoint gaps have depths ℓ_0 and ℓ_{n-1}. Subtracting their two contributions from bE and restoring the actual endpoint weights proves, for every tree,

$$C_{p,q}(T) = \sum_{i=0}^{n-1} w_i \ell_i + bn.$$

All transformed weights are nonnegative, even when an endpoint gap is smaller than b. Apply Theorem 28 and the $O(c+n)$ slot repair of Theorem 9. The original mixed optimum is $h\sum_i w_i + T(0,c) + bn$, and the repaired tree attains it. For minimum feasible height, $c < 2(n+1)$. For $n = 1$ the unique tree has cost $p_0 + q_0 + q_1$ and needs no product. □

If the original keys have r_0 runs, the transformed sequence has at most $r_0 + 2$. The theorem solves the stated height-constrained problem; unrestricted optimality still requires a sufficient height certificate. Its explicit factor $c = 2^h - 1$ must be retained when applying such a certificate. Unequal internal gaps introduce the signed transition obstruction in Theorem 159. The audit appears in Appendix T.

2.17 Run vectors with arbitrary mixed gaps

A complete tree needs a vector of prefix costs rather than every transfer entry. This removes one factor of the run count and handles unequal gaps without representing their signed transition matrices explicitly.

Theorem 30 (Mixed run-vector construction)**.** *Let $n \geq 2$ and allow arbitrary nonnegative key and gap weights. Put $b = \min_{1\leq i<n} q_i$, define w by (33), and put $g_0 = g_n = 0$, $g_i = q_i - b$ for $1 \leq i < n$. Let r be the number of constant runs in w and $e = |\{i : g_i > 0\}|$. For prescribed h with $c = 2^h - 1 \geq n$, a complete optimal tree of height at most h can be constructed in*

$$O\big(n + (r+e)c\log(c+1)\big) \quad \textit{arithmetic work}, \qquad O\big(n + (r+e)c\big) \quad \textit{words}.$$

The bounds include preprocessing, backpointers and full tree reconstruction. At minimum feasible height the work is $O(n + (r+e)n\log(n+1))$.

Proof. The external-path identity used in Corollary 29, now retaining the residual internal gaps, gives for every tree

$$C_{p,q}(T) = C_{w,g}(T) + bn. \tag{34}$$

All transformed weights are nonnegative. Their total mass is $W - b$. By Theorem 9, it suffices to maximize their mixed slot reward and add bn after reconstruction. Write $v(t) = \nu_2(t)$, set $R = (h-1)(\sum_i w_i + \sum_i g_i)$ and $K = R + 1$. Let $f_k(a)$ be the minimum negative reward of the first k keys ending at slot a. For $k > 0$ retain exactly $k \leq a \leq c$; initially the only state is $f_0(0) = 0$.

Consider a constant run of $m \geq 2$ keys of weight w, with zero residual gaps preceding and inside the run. Force its last key to slot z and let the preceding prefix end at a. Define the finite matrix

$$B_{w,m}(a,z) = K(m - z + a)_+ - wv(z) - w\sum_{t=1}^{h-1} \min\left(m-1, \left\lfloor \frac{z-1}{2^t} \right\rfloor - \left\lfloor \frac{a}{2^t} \right\rfloor\right). \tag{35}$$

On $z - a \geq m$, this chooses the best $m-1$ preceding slots in (a, z). The whole matrix is the window factor for $m-1$ keys, shifted by one column, plus a column potential. Lemma 27's density proof applies for this larger positive K, so it is finite Monge.

To compute $f_{k+m}(z)$, take row minima of $f_k(a) + B_{w,m}(a,z)$, with rows $z = k+m, \ldots, c$ and columns $a = k, \ldots, c-m$ when $k > 0$, or just column 0 when $k = 0$. Transposition and the prefix potential preserve Monge. Each row has a feasible candidate. An infeasible candidate incurs at least K penalty, while the prefix and new-run negative rewards together are at least $-R$, also when $a \geq z$. It therefore costs at least one. A feasible candidate costs at most zero. The finite row minimum consequently gives exactly the true prefix optimum and its attaining predecessor. SMAWK [20] uses

$O(c)$ entry evaluations and additional work; each entry of (35) takes $O(h)$ arithmetic operations. Thus a run pass costs $O(ch)$ work and $O(c)$ working words.

To append a single key of weight w after an arbitrary residual gap g, the exact update is

$$f_{k+1}(z) = -wv(z) + \min_{k \le a < z} \{f_k(a) - g \min(v(a), v(z))\}. \tag{36}$$

For $k = 0$, only $a = 0$ is present, with $v(0) = 0$ and $g = 0$. Sweep z increasingly, inserting $a = z - 1$ when that prefix state exists. Maintain E_d, the minimum prefix value at exact valuation d, and S_d, the minimum at valuation at least d, with leftmost predecessors. For $t = v(z)$ the minimum in (36) is

$$\min \left\{ \min_{d<t}(E_d - gd),\ S_t - gt \right\},$$

with absent states omitted. Inserting valuation d updates $E_d, S_0, \ldots, S_d$; a query scans t lower levels and reads S_t. Every slot is inserted and queried at most once, and $\sum_{a=1}^{c} v(a) = c - h < c$. Array initialization, all updates, all queries, and their predecessors therefore take $O(c + h) = O(c)$ work and $O(c)$ words. This also processes singleton zero-gap runs. No gap-transition matrix or signed density list is constructed.

Make the key after each positive residual gap a singleton stage. Otherwise collect a maximal constant-weight segment with zero residual gaps preceding and inside it. The e exceptional gap positions split the r original runs into at most $r + e$ segments; isolating their first keys adds at most e further stages. Hence there are at most $r + 2e$ stages. If R_2 stages have length at least two and S_1 are singletons, the more precise bounds are

$$O\big(n + c(1 + hR_2 + S_1)\big) \text{ work}, \qquad O\big(n + c(1 + R_2 + S_1)\big) \text{ words}. \tag{37}$$

They imply the theorem because $h = \log_2(c + 1)$.

Retain a predecessor array per stage, choose an attaining final slot for $\min_z f_n(z)$, and trace backward. In each long run choose the $m - 1$ largest valuations in its recovered open window, then append its forced last slot. Valuation buckets and earliest cutoff ties take window length plus $O(h)$ work. The recovered windows are disjoint, so all slot recovery costs $O(c+(r+e)h+n)$. The $O(c+n)$ threaded deletion repair of Theorem 9 produces a complete optimal transformed tree; (34) proves original mixed optimality. For minimum feasible height, $c < 2(n+1)$. The case $n = 1$ is direct. □

For equal internal gaps, $e = 0$, so this improves complete construction in Corollary 29 from quadratic to linear run dependence. The all-entry product and its core bound remain useful separately. For arbitrary weights, $r \le n$ and $e \le n - 2$; if every stage is a singleton, (37) gives $O(nc)$ work. The signed checkerboard in Theorem 159 thus permits fast vector application despite obstructing a sparse density list. The height constraint remains explicit; the next subsection supplies a sufficient height for unrestricted construction. The original interval-core bound and local support question are unchanged. Independent controls appear in Appendix T.

2.18 Unrestricted height from a small exceptional set

The full mixed rotation inequalities also control height after a chosen set of outcomes is set aside in the accounting. Every weight remains in the optimization. Write μ for the smallest positive singleton-interval mass, and z for the longest all-zero interval:

$$\mu = \min\{p_i + q_i + q_{i+1} : p_i + q_i + q_{i+1} > 0\}, \qquad \ell = \lceil \log_2(z + 1) \rceil.$$

Here z is the longest consecutive run of indices with $p_i = q_i = q_{i+1} = 0$. When the total mass is positive, μ exists. Every positive-mass nonempty interval contains a positive singleton interval: for a positive key use that key, and for a positive gap use an adjacent key inside the interval. Thus its mass is at least μ. For $V \geq 0$ define

$$t(V) = \min\{j \geq 0 : F_{j+1}\mu > V\}, \qquad F_1 = F_2 = 1.$$

Lemma 31 (Disjoint charges beyond the light mass)**.** *Mark any k key or gap outcomes and let V be the total mass of the unmarked outcomes. Put*

$$P = \begin{cases} t(W), & k = 0, \\ t(V) + k + 1, & k \geq 1. \end{cases}$$

Every positive-mass internal subtree in every optimal tree has root depth less than P. Some optimal tree has height at most $\min\{n, P + \ell\}$. If $W = 0$, a balanced tree is optimal directly.

Proof. Fix a positive internal subtree at depth d and ascend to the root. Let $A_0, \ldots, A_d$ be the successive masses and $B_i = A_i - A_{i-1}$ the mass of the new ancestor key and its off-path sibling subtree. Lemma 2, including its explicit straight and bent rotations, gives

$$A_j \geq F_{j+1}\mu, \qquad B_i \geq A_{i-2} \quad (2 \leq i \leq d). \tag{38}$$

The outcome sets counted by the B_i are disjoint. For any fixed j, all sets B_i with $i > j$ are also disjoint from the subtree of mass A_j. This partition includes each external gap once.

If $k = 0$, then $A_d \leq W$ implies $d < t(W)$. Suppose $k \geq 1$. When $d < t(V)$ there is nothing to prove. Otherwise let j be the first index with $A_j > V$. Equation (38) ensures $j \leq t(V)$. The subtree of mass A_j contains a marked outcome. For every $i \geq j + 2$, its monotonicity and the same equation give $B_i \geq A_{i-2} \geq A_j > V$, so each such disjoint increment contains another marked outcome. If $d \geq j + 1$, counting these sets gives $k \geq 1 + (d - j - 1) = d - j$; when $d = j$ the same conclusion is immediate. Consequently $d \leq j + k \leq t(V) + k < P$. The increment B_{j+1} is deliberately uncharged: its inequality only involves A_{j-1}, which need not exceed V.

Replace every maximal zero-mass subtree by a balanced tree on its own keys. This changes no cost or positive outcome depth. Such a subtree contains at most z keys and has replacement height at most ℓ. It attaches below a positive internal node, giving height at most $P + \ell$. The trivial height n completes the bound. □

This is a consequence of the sufficient mixed rotation lemma; the Fibonacci method is prior work [7]. No dominance assumption on the marked weights, integrality, or separation of their positions is used. For fixed k, selecting the k largest outcome weights minimizes V and hence $t(V)$. Sorting them and scanning all suffix masses finds the best stated height in $O(n \log(n + 1))$ work and $O(n)$ space. Compute μ, z in one scan. Generate μF_{j+1} only until it exceeds W or $j = n$; the table has $O(n)$ entries and its last entry is at most $2W$. As the suffix mass decreases, the index $\min\{t(V), n\}$ only decreases, so all index updates take $O(n)$ work. Truncation at n cannot falsely improve a bound already capped by the trivial height n.

The same global height works for every contiguous interval. Its marked set and unmarked mass can only decrease, its smallest positive singleton mass can only increase, and its zero interval length is at most z. Use the stronger $k = 0$ case for an interval containing no marked outcome, and balance an all-zero interval directly. Hence this selected budget also suffices for convergence of the original terminal-penalty iteration on the fixed original or normalized instance, with its penalty recomputed for that height.

Theorem 32 (Full construction with a selected sufficient height)**.** *Apply the normalization of Theorem 30, and select height h by Lemma 31 on the normalized weights. Put $c = 2^h - 1$. With r normalized key runs and e positive residual internal gaps, a complete unrestricted optimum can be constructed in*

$$O(n \log(n+1) + (r+e)c \log(c+1)) \text{ work}, \qquad O(n + (r+e)c) \text{ words}.$$

More precisely, if R_2 passes are long runs and S_1 are singletons, choosing between the vector kernel and Knuth before construction gives

$$O\Big(n \log(n+1) + \min\{n^2, c(1 + hR_2 + S_1)\}\Big) \tag{39}$$

arithmetic work and quadratic worst-case space.

Proof. The normalization changes every tree cost by the same constant bn. Thus its sufficient height contains an original unrestricted optimum. Theorem 30 optimizes exactly over that height and constructs the entire tree. Equation (37) supplies the precise kernel work and space. Compare its work proxy with n^2 before allocating slot arrays; the selected kernel has the smaller asymptotic bound. If $h > 2\lceil \log_2(n+1) \rceil$, the slot capacity already exceeds n^2 and can be rejected without constructing it. All-zero normalized input is handled by a balanced tree. □

Corollary 33 (Few exceptional outcomes of arbitrary magnitude)**.** *Let $\varphi = (1+\sqrt{5})/2$ and $\alpha = \log_\varphi 2 \approx 1.44042$. Suppose all normalized keys weigh at least $a > 0$. Apart from s exceptional keys of unrestricted size, their weights lie in $[a, Ca]$ for fixed C. Suppose the background has r_0 constant runs before these exceptional keys are substituted, and there are e positive residual gap weights, also of unrestricted size. For fixed r_0, s, e, a complete unrestricted optimum can be constructed in $O(n^\alpha \log(n+1))$ arithmetic work.*

Proof. Mark the s exceptional keys and e positive residual gaps. The remaining mass is $V \le Cna$, every positive singleton has mass at least a, and $z = 0$. Since $F_{j+1} \ge \varphi^{j-1}$ for $j \ge 1$,

$$t(V) \le 2 + \lceil \log_\varphi(1 + V/\mu) \rceil, \qquad 2^h = O(2^{s+e} n^\alpha).$$

The $k = 0$ case has at least as good a bound. Substituting exceptional keys makes at most $r_0 + 2s$ key runs. Apply Theorem 32; for fixed $s+e$, $h = O(\log(n+1))$. The constants depend on C, r_0, s, e, not on the exceptional magnitudes. □

The earlier dominant-core construction may be faster when its certificate passes. This corollary needs no such certificate, and permits unrelated exceptional magnitudes. Its bound counts exact arithmetic operations; unbounded integer bit costs are additional. Arbitrarily many runs and exceptional outcomes remain valid inputs to the full solver, with the quadratic fallback guarantee. The general arbitrary-weight objective and the original local interval-core support bound remain open.

2.19 Valuation-class records without slot expansion

The prefix vectors admit a second compression: among ending slots of the same valuation, retain only the first feasible slot and strict value decreases. This removes the explicit perfect-tree capacity from an integer-mass bound, including reconstruction of the complete tree.

Lemma 34 (Prefix records and domination)**.** *Fix height h, capacity $c = 2^h - 1$, and normalized weights w, g of total mass M. For each valuation d, the prefix cost $f_k(a)$ is nonincreasing as a increases through feasible slots with $\nu_2(a) = d$. Retaining its first value and strict decreases suffices for every subsequent run or singleton transition. If the weights are integers, the number B_k of retained states satisfies*

$$B_k \le \min\{c,\ h[1 + (h-1)M]\}. \tag{40}$$

More generally, mark t outcomes of arbitrary nonnegative weights, and suppose the unmarked weights are integers of total mass V. Then

$$B_k \le \min\{c,\ h^{t+1}[1 + (h-1)V]\}. \tag{41}$$

The empty prefix has one state, at slot zero.

Proof. Move only the last slot of an attaining prefix to a later slot of the same valuation. Its key reward and preceding gap reward remain unchanged, and order is preserved. This proves monotonicity. If a later state has the same value as an earlier state of its class, replacing it by the earlier one preserves a singleton's preceding-gap reward and feasibility. For a long zero-gap run, it leaves a superset of available inner slots. The earlier state therefore dominates the later state in both transitions. For integer weights, each prefix value is an integer in $[-(h-1)M, 0]$. Each of the h classes has at most $1 + (h-1)M$ retained values, proving (40). For the marked bound, each marked outcome's reward coefficient is one of $0, \dots, h-1$: its slot valuation for a key, the minimum adjacent valuation for an internal gap, and zero for an outcome outside the prefix. Thus there are at most h^t marked coefficient vectors. The unmarked reward is an integer between zero and $(h-1)V$. At most $h^t[1 + (h-1)V]$ distinct costs are possible, regardless of the marked magnitudes or integrality. Multiply by the h ending-slot classes to obtain (41). □

Theorem 35 (Complete construction from prefix records)**.** *Let $s \le r + 2e$ be the number of normalized stages in Theorem 30, and let $B_1, \dots, B_s$ count their retained prefix states, with $B_0 = 1$. For arbitrary nonnegative weights, a complete optimal tree subject to a prescribed feasible height bound h can be constructed in*

$$O\left(h^2\left[n + \sum_{j=0}^{s} B_j\right]\right) \text{ arithmetic work}, \qquad O\left(nh + \sum_{j=0}^{s} B_j\right) \text{ words}. \tag{42}$$

For integer normalized weights of total mass M, these are at most

$$O(nh^2 + s(M+1)h^4) \text{ work}, \qquad O(nh + s(M+1)h^2) \text{ words}. \tag{43}$$

The bounds include all backpointers and the entire output tree.

Proof. For a singleton of weight w following gap g, fix destination valuation d and scan preceding records $(a, f_k(a))$ in increasing a. Its candidate value is $f_k(a) - g\min(\nu_2(a), d) - wd$; it becomes eligible at the first slot of valuation d strictly greater than a. These eligibility positions are nondecreasing. Record each decrease of the running minimum, combining improvements at the same position. Repeating over d costs $O(hB_k)$ work before record organization.

For a long run, restrict the finite Monge factor (35) to preceding records $a \le c - m$. Every row $z \ge k + m$ has a feasible candidate: $a = k$ is the first feasible slot of its valuation class, hence a record (or $a = 0$ for the empty prefix). The same penalty argument excludes infeasible minima. Adding prefix potentials and restricting columns preserve Monge, so SMAWK evaluates any batch of Q destination rows in $O(B_k + Q)$ entry queries. Each query costs $O(h)$ arithmetic work.

The feasible positions of each destination valuation form an arithmetic progression. Query its first and last positions. Equal endpoint values certify a constant interval by Lemma 34. For unequal values, bisect the progression's index interval. Batch all required midpoints across all classes into a single sorted SMAWK query, then continue only on children with unequal endpoint values. Adjacent unequal endpoints identify exactly a strict-decrease record. Retain each class's first position separately.

There are at most h bisection levels. At a fixed level, active intervals within a class are disjoint, and each contains a distinct strict decrease. Thus each level queries $O(B_{k+m})$ rows. All levels together take $O(h(B_k + B_{k+m}))$ entry queries and $O(h^2(B_k + B_{k+m}))$ work. Sorting row batches and organizing records fit this bound since $\log(c+1) = h$. Retain only active endpoints and discovered records, using $O(B_k + B_{k+m} + h)$ working words.

An attaining predecessor stored at a record also works throughout its constant interval in the same valuation class. With that predecessor fixed, the candidate cost cannot increase as the destination moves right, while the global minimum stays equal. The candidate therefore stays equal too. Store these predecessors per stage and recover an attaining final slot and all preceding run windows.

For a recovered window (a, z), let

$$N_t = \left\lfloor \frac{z-1}{2^t} \right\rfloor - \left\lfloor \frac{a}{2^t} \right\rfloor.$$

For $m \geq 2$, find d with $N_{d+1} < m - 1 \leq N_d$. Take all multiples of 2^{d+1} in the window and the earliest required odd multiples of 2^d, then append z. Merge the two sorted arithmetic progressions in $O(m)$ work; the threshold scan costs $O(h)$. All slot recovery costs $O(n + sh)$, without visiting the unused slots.

To repair the slots into a tree, retain only their ancestor paths in the height-h perfect tree, at most nh vertices. If $d = \nu_2(x) < h-1$, the parent of x is $x - 2^d$ when bit $d+1$ of x is set, and $x + 2^d$ otherwise. Stop a path at its first previously retained ancestor. Every absent branch contains no selected key and may be collapsed to an empty tree. On the retained tree, delete each unselected key in increasing label order, using threaded inorder successors. A two-child deletion promotes the successor and splices its possible right child upward; a one-child deletion splices that child upward directly. No surviving key's depth increases. Thus its comparison length is at most $h - \nu_2(t_i)$. Gap depths are the maxima of adjacent key comparison lengths, with the adjacent length at either endpoint. The mixed slot identity and its converse in Theorem 9 prove exact optimality of the resulting full tree.

Deterministic ordered dictionaries or sorted arrays organize these $O(nh)$ vertices in $O(nh^2)$ work and $O(nh)$ words, since the retained vertices are a subset of c slots. Threaded deletions and iterative output then cost linear work in the retained size. Summing stage costs proves (42); inserting (40) proves (43). □

Corollary 36 (Linear integer mass and few mixed runs)**.** *For arbitrary nonnegative integer key and gap weights of total mass $W = O(n)$, suppose the normalization has a fixed number of key runs and positive residual gaps. A complete unrestricted optimum can be constructed in $O(n \log^4(n+1))$ arithmetic work and $O(n \log^2(n+1))$ words. Zero-weight key runs are permitted.*

Proof. Normalization preserves integrality and gives $M \leq W$. When $M > 0$, the smallest positive singleton mass is at least one. The $k = 0$ case of Lemma 31, including balanced zero-mass intervals, supplies $h = O(\log(n+1))$. Selecting a sufficient height costs $O(n \log(n+1))$. Apply (43) with $s = O(1)$. For $M = 0$, construct a balanced tree directly. □

Corollary 37 (Few arbitrarily large integer outcomes)**.** *Suppose the normalized weights are nonnegative integers, with $s = O(1)$ stages. Mark t outcomes of arbitrary magnitude, where t is fixed, and suppose the remaining mass is $V = O(n)$. A complete unrestricted optimum can be constructed in*

$$O(n \log^{t+4}(n+1)) \text{ arithmetic work}, \qquad O(n \log^{t+2}(n+1)) \text{ words}.$$

The constants do not depend on the marked magnitudes; zero weights are allowed.

Proof. Every positive singleton has mass at least one. Lemma 31 gives a sufficient height $h = O(\log(n+1) + t)$, including the balanced zero-mass intervals. Equation (41) gives $B_j = O((V+1)h^{t+2})$ at each stage. Substitution in (42), with fixed s, t, proves both bounds. An all-zero instance is balanced directly. □

All bounds count exact arithmetic, including operations on h-bit slot coordinates. Operand bit costs are additional. Arbitrary nonnegative weights retain the output-sensitive bound (42); its integer score-range specialization is a separate assumption. Choosing Knuth when the integer work proxy exceeds n^2 retains a quadratic worst-case bound. The general arbitrary-weight subquadratic objective and the original interval-core support question remain open. Lemma 165 also gives the sharp universal bound $B_k \le \sum_{r=0}^{\min(k,h)} \binom{h}{r} - k$. Theorem 167 gives exponential explicit record support at necessary optimal height for strictly positive mixed weights. Independent state and full-tree controls appear in Appendix T.

2.20 Implicit prefix functions with additive decision diagrams

The record bound concerns an explicit list of strict changes. A different representation shares dyadic blocks whose value functions differ only by an additive constant. Edge-valued binary decision diagrams are established data structures of Lai and Sastry and subsequent work; see Bryant's account [21]. We use their additive sharing principle with a fixed most-significant-bit-first order on slot coordinates. The new obligations here are the exclusive-prefix operation, the mixed-gap transition, and complete tree recovery. No approximation is used.

Represent a function on $\{0, \dots, 2^d - 1\}$ with values in $\mathbb{R} \cup \{+\infty\}$ by an offset and a normalized binary node. The offset is its minimum finite value. A node stores its level and its two normalized child references, with their offsets. Identical triples are shared. The all-infinite function has a separate sentinel, and the finite level-zero terminal has value zero. Levels are retained even when both children coincide. Induction on the level proves that this representation is canonical up to the chosen node identifiers. Every reference also stores its exact minimum and maximum, allowing $+\infty$ as the maximum when any position is infeasible.

Lemma 38 (Shared exclusive prefix minima)**.** *For a height-h function represented by D reachable nodes, its exclusive prefix-minimum function*

$$(\mathcal{P}F)(b) = \min_{0 \le a < b} F(a), \qquad (\mathcal{P}F)(0) = +\infty,$$

can be constructed with $O(hD)$ arithmetic operations, node constructions, and dictionary operations, creating at most $O(hD)$ nodes.

Proof. Write $F = L \parallel R$ for its two halves. Then

$$\mathcal{P}(L \parallel R) = \mathcal{P}L \parallel \min\{\min L, \mathcal{P}R\}, \tag{44}$$

where $\min L$ denotes the constant function with that value. Also $\mathcal{P}(F+a) = \mathcal{P}F+a$, so a normalized input node needs to be processed only once, regardless of its incoming offset.

Every prefix-minimum function is nonincreasing, including its initial infinite positions. To cap such a function G by a constant t, return G immediately when $\max G \le t$, and return the constant when $t \le \min G$. Otherwise recurse on both halves. At each level at most one half straddles the change from the constant to G, since G is nonincreasing. Thus the cap visits $O(h)$ nodes and creates $O(h)$ nodes. Equation (44) performs one cap per distinct input node. Memoization and normalization give the stated bounds. The all-infinite case and a level-zero input return the infinite function directly. □

Theorem 39 (Implicit mixed-prefix construction)**.** *For arbitrary nonnegative key and gap weights, every feasible height h admits an exact construction of all mixed prefix functions and a complete optimal height-h tree using additive decision diagrams, without enumerating their record sets or their $2^h - 1$ slots. Setting $h = n$, or using a certified sufficient height, gives an unrestricted optimum. For an incoming diagram with D nodes, a zero-residual-gap update uses $O(h^3 D)$ arithmetic and dictionary operations. A general gap update uses $O(h^5 D + X)$ such operations, where X counts the distinct expanded normalized minimum contexts used to combine its h candidate functions. The parameter X includes relative offsets.*

Proof. Use the normalization of Section 2.17, and put $v(b) = \nu_2(b)$ for $b > 0$, with $v(0) = 0$. Initialize $f_0(0) = 0$ and $f_0(b) = +\infty$ for $b > 0$. Let M_d be zero on positions of valuation d and infinite elsewhere; include position zero in M_0. For the next normalized key w_k and its preceding residual gap g_k, form, for every $0 \le d < h$,

$$P_d = \mathcal{P}(f_k + M_d), \qquad Q_d(b) = P_d(b) - g_k \min\{d, v(b)\} - w_k v(b).$$

Then

$$f_{k+1}(b) = \min_{0 \le d < h} Q_d(b). \tag{45}$$

Indeed, the masks partition every feasible predecessor $a < b$ by its valuation, giving exactly the mixed slot recurrence. When $g_k = 0$, use $f_{k+1} = \mathcal{P} f_k - w_k v$ directly.

Masks and functions of valuation have finite shared diagrams, constructed recursively from their binary suffix conditions. Addition and pointwise minimum recurse on corresponding halves and normalize the results. Addition factors out the sum of the incoming offsets. Minimum factors out their common minimum, but must retain the relative offset in its memoization key. Omitting that relative offset would identify different functions. Lemma 38 supplies the remaining operation. Induction on k proves every represented prefix value.

For the work bound, a valuation mask has $O(h)$ nodes, and an arbitrary function of valuation has $O(h^2)$ nodes: at each level its exceptional zero-address value is one of at most h prescribed values. Addition of two diagrams with D_1, D_2 nodes uses $O(D_1 D_2)$ node-pair contexts, since both root offsets factor out. With zero gap, the prefix operation creates $O(hD)$ nodes and adding the valuation reward takes $O(h^3 D)$ work. With a general gap, masking takes $O(hD)$ work per class, prefix minima take $O(h^2 D)$, and the reward addition takes $O(h^4 D)$. Summing over h classes gives $O(h^5 D)$. Combining the candidates costs $O(X + h)$: every newly expanded normalized minimum context makes two recursive calls and a constant amount of arithmetic and node construction.

Keep the prefix roots. Choose an earliest minimum of f_n by following minimum-valued child references. To recover a predecessor of an ending slot b with valuation e, minimize $f_k + M_d$ over $[0, b)$ for each d, subtract $g_k \min(d, e)$, and choose an attaining position. A prefix range decomposes into at most h complete dyadic blocks. Their minima are stored in their offsets, and descending

a minimizing block recovers an attaining position. Repeating this gives every selected slot. Apply the sparse ancestor repair of Theorem 35 to recover the full tree. The mixed-slot identity proves its optimal cost. □

The diagram algorithms operate on shared nodes and normalized operation contexts. Their work need not be proportional to the final live graph: pointwise minima may encounter many relative offsets for the same pair of node identifiers. The audit therefore counts every created intermediate node and every recursive operation, separately from final and history sizes. Exact range pruning is essential to the prefix bound above. The new constructor supports arbitrary mixed weights; a universal subquadratic bound on its intermediate work remains open. Independent prefix and full-tree controls appear in Appendix T.

Lemma 40 (Endpoint dominance and binary skyline)**.** *For every nonempty mixed prefix with nonnegative normalized weights,*

$$1 \le a \le c < 2^h, \quad v(a) \le v(c) \quad \Longrightarrow \quad f_k(c) \le f_k(a).$$

For $1 \le b \le 2^h$, write $b-1 = \sum_{j=1}^{t} 2^{d_j}$ with $d_1 > \cdots > d_t \ge 0$, and let $c_j = \sum_{i=1}^{j} 2^{d_i}$. Define $S(b) = \{0, c_1, \ldots, c_t\}$. For every $g \ge 0$ and $k \ge 0$,

$$\min_{0 \le a < b} \{f_k(a) - g \min(v(a), v(b))\} = \min_{a \in S(b)} \{f_k(a) - g \min(v(a), v(b))\}. \tag{46}$$

Here infeasible values are infinite, $v(0) = 0$, and the endpoint sentinel $b = 2^h$ has $v(b) = h$.

Proof. For a finite $f_k(a)$, move only the final selected key from a to c. All earlier selected slots remain smaller. Its key reward cannot decrease, and its preceding-gap reward cannot decrease because the gap weight is nonnegative and $\min(v(s_{k-1}), v(c)) \ge \min(v(s_{k-1}), v(a))$. Thus the new feasible prefix costs at most $f_k(a)$; taking its optimum proves endpoint dominance. Infinite values make the inequality immediate.

Put $c_0 = 0$. Each interval $(c_{j-1}, c_j]$ has length 2^{d_j}, and c_{j-1} is divisible by 2^{d_j+1}. Its endpoint has valuation d_j and every interior integer has smaller valuation. Therefore each positive predecessor $a < b$ has a representative $c_j \ge a$ with $v(c_j) \ge v(a)$. Endpoint dominance decreases its prefix cost, and $g \ge 0$ ensures that subtracting the new gap reward decreases it again. The representative remains below b. Slot zero is retained separately. These observations prove both inequalities in (46); for $k = 0$, only its retained zero slot is finite. □

Theorem 41 (One-path mixed predecessor queries)**.** *Augment every additive diagram node by the value at its first local address, allowing infinity. This adds constant work and one stored value per created node. On any actual mixed prefix function, the minimum in (46) and an attaining predecessor can then be found in $O(h)$ arithmetic and address operations, with no diagram-node creation. Given the prefix roots from Theorem 39, all optimal selected slots can be recovered in $O(nh)$ such operations and $O(n)$ output space, without creating masked diagrams or invoking pointwise diagram operations.*

Proof. A normalized node's first-address value is its left child's edge offset plus that child's first-address value, or infinity if that address is infeasible. Both are known when the parent is created. An incoming root offset is added on reading this metadata.

Follow the h bits of $b-1$ from most to least significant. Whenever bit d is one, the right child begins at exactly the next skyline position c_j. Its stored first-address value, including the accumulated root offset, is $f_k(c_j)$, and $v(c_j) = d$. Read it in constant work, subtract $g \min(d, v(b))$,

and retain the best feasible candidate. Also consider $f_k(0)$. This is one path walk with at most $h+1$ candidates, so the lemma proves the query and its work bound. Ties may select any attaining skyline candidate; it need not be the earliest minimizer among all slots.

Query f_n with $b = 2^h$ and $g = 0$ to find an optimal last slot. For each preceding key, query its prefix root with the current slot and the relevant residual gap. The current key's reward is constant during this predecessor choice. The mixed recurrence therefore supplies an attaining predecessor at every step. There are exactly n queries. The selected positions are strictly increasing in forward order, and the existing sparse ancestor repair constructs the complete optimal tree from them. □

The new bound concerns slot traceback. The forward construction still has the intermediate-size and minimum-context parameters of Theorem 39. In particular, endpoint dominance has not yet supplied a general subquadratic bound for constructing the prefix roots.

2.21 Scalar-carry forward construction

Endpoint dominance also controls forward construction. It permits a single scalar carry per dyadic block. A reset-path cover bounds all expanded carry states linearly in the input diagram size, with a height factor for the length of each path.

Lemma 42 (Reset paths cover the carry states)**.** *Let V be the reachable nodes of a height-h additive diagram, retaining all levels, and let $d(u)$ be the remaining depth of node u. Consider the following memoized recursion. On a block $R = L \parallel U$ with incoming carry p, prune if R is all infinite or $p \le \min R$; otherwise pass p to L and $\min\{p, \min L, U(0)\}$ to U. Leaves do not expand. Cache by the normalized input node and relative carry, with infinity represented separately. The number Z of expanded states satisfies*

$$Z \le h + \sum_{u \in V} \max\{d(u) - 1, 0\} = O(h|V| + h). \tag{47}$$

Their depth-weighted count satisfies

$$\sum_{s \text{ expanded}} d(s) \le \frac{h(h+1)}{2} + \sum_{u \in V} \frac{d(u)(d(u)-1)}{2}. \tag{48}$$

These bounds permit arbitrary signed finite values and infinite entries; no OBST realizability assumption is needed for the state count.

Proof. Work in the coordinates of a normalized parent input node. Put $\ell = \min L$, $r = \min U$, and $C = \min\{\ell, U(0)\}$, allowing infinity. Classify a right-child edge as a reset when $C \le p$, including ties. Its carry then equals C, so if that child expands its normalized state is $(U \text{ node}, C - r)$. This state depends only on the parent input node, not on p or the parent's external additive offset. An infinite carry remains a separate symbol; no subtraction of two infinities is used. There is consequently at most one possible reset seed per input node.

All left-child edges are inherited. A right-child edge that is not a reset has $p < C \le \ell$, so its left child is pruned. Thus each expanded state has at most one inherited expanded child. The same statement holds when some quantities are infinite: a nonreset inequality forces p to be finite, and an all-infinite child never expands.

Take the root state, when expanded, and every distinct expanded reset seed. Follow inherited edges from each seed. Every expanded state is covered: trace any recursion path from the root, starting a new path after its last reset. Memoization can merge these paths but cannot create an

additional state outside their union. Each edge decreases depth by one. The root path has at most h states and depth sum $h(h+1)/2$. A reset from a depth-d input node starts at depth $d-1$, giving at most $d-1$ states and depth sum $d(d-1)/2$. Summing these allowances, even if paths overlap or some seeds coincide, proves both inequalities. □

Theorem 43 (Polynomial mixed updates on implicit prefixes)**.** *Let $F = f_k$ be a feasible mixed prefix function on 2^h addresses, with D reachable additive-diagram nodes. Appending a key of nonnegative normalized weight w and preceding residual gap $g \geq 0$ constructs the next prefix diagram in $O(h^2 D)$ arithmetic and dictionary operations, without generic pointwise diagram minima. For the empty prefix, use its required initial gap $g = 0$. With incoming diagram sizes $D_0, \ldots, D_{n-1}$, all forward updates take*

$$O\left(h^2 \sum_{k=0}^{n-1} D_k\right)$$

such operations. The stored prefix roots support the complete optimal height-h tree construction and the $O(nh)$ slot traceback of Theorem 41.

Proof. Consider a dyadic block $[B, B + 2^d)$ and put $p_B = \min_{0 \leq a \leq B} F(a)$. For $B > 0$, $v(B) \geq d$. Every $a \leq B$ with $v(a) < d$ and finite value is dominated by B, by Lemma 40; slot zero is infinite for a nonempty prefix. Thus p_B has an attainer of valuation at least d whenever finite. For every interior destination b, $v(b) < d$, so all these outside predecessors contribute exactly $p_B - gv(b)$ after minimization. At $B = 0$, $p_B = +\infty$ for a nonempty prefix. For the empty prefix, $p_B = 0$ and the initial $g = 0$ makes the same formula valid. This separate case is needed because the empty slot has valuation zero.

For a local block function R on 2^d positions define

$$E_g(R) = \min_{0 < t < 2^d} \{R(t) - g\nu_2(t)\}.$$

Exclude the first local address: its global valuation can exceed d. A singleton has summary $+\infty$. If $R = L \parallel U$ and $m_0 = U(0)$, then

$$E_g(R) = \min\{E_g(L), E_g(U), m_0 - g(d-1)\}. \tag{49}$$

The summary commutes with additive offsets, so memoization by normalized input node computes all needed summaries in $O(D)$ work.

The recursive output for a block temporarily has $+\infty$ at its first local address. The parent will supply that boundary's true value. At midpoint $M = B + 2^{d-1}$, the new prefix value and right-child carry are

$$f_{k+1}(M) = \min\{p_B - g(d-1), E_g(L)\} - w(d-1), \tag{50}$$

$$p_M = \min\{p_B, \min L, F(M)\}. \tag{51}$$

Indeed, the internal predecessors in (B, M) have valuation below $d-1$, and the outside contribution has already been established. The carry formula is the definition of the inclusive prefix minimum. Recurse on both children and replace the right output's first address with (50). A persistent replacement follows only the leftmost path, taking $O(d)$ work. The root's first output address correctly remains infinite.

There is a decisive pruning rule. If $p_B \leq \min R$, no internal predecessor can improve the outside one. For every local $t > 0$ the output is simply $p_B - (g + w)\nu_2(t)$, with infinity at $t = 0$. These templates are shared by depth and require $O(h^2)$ total work; an all-infinite input block is handled

by the same rule. Otherwise let $m = \min R$ and memoize the recursive state by the input-node identity and $p_B - m$, representing an infinite carry separately. Translation of both R and its carry translates the result, proving this normalization.

Lemma 42 now gives $O(hD)$ expanded states. For an explicit operational bound, an expanded state at remaining depth d uses one node-assembly call and $d-1$ node-assembly calls along the first-address replacement. Its replacement visits, including the leaf, also number d. Shared templates use at most $h(h+1)/2$ calls of either kind. Thus each of these two operation counts is at most

$$h(h+1) + \sum_{u \in V} \frac{d(u)(d(u)-1)}{2} = O(h^2 D). \tag{52}$$

Here $D \geq 1$ because the input prefix is feasible. There are exactly $2Z+1$ recursive build calls, including memo hits, pruned calls and leaves. Weighted-summary expansions number at most D, with at most $2D+Z$ summary calls. All other scalar and dictionary operations take constant work per call or newly assembled node. This proves the total $O(h^2 D)$ bound, including all intermediate work, and bounds the number of new nodes. Induction over the updates proves the represented mixed prefix functions. The previously proved skyline traceback and sparse repair return their complete optimal tree. Dictionary costs and exact operand bit costs are interpreted as in Theorem 39. □

This removes the uncontrolled generic-minimum parameter X from the per-update guarantee for actual OBST prefixes. A useful bound on the entire sequence D_k, needed for a general subquadratic construction, remains open. The exponential generic-minimum example in Theorem 168 does not contradict this result: this update uses endpoint dominance and the scalar carry recurrence. The height factors in the reset-path accounting are sharp for general diagram inputs by Theorem 169. Its displayed family violates endpoint dominance, leaving sharper actual-prefix bounds open.

2.22 Separating the first entry of each block

The factor h spent on persistent first-address replacement can be removed by representing that address separately. The midpoint value then becomes a scalar stored directly in a node, and a recursive update assembles its result in constant work.

A *punctured diagram* of remaining depth d represents only the local addresses $1, \ldots, 2^d - 1$. At depth zero this domain is empty. At positive depth its decomposition is

$$P = (P_L, c, P_R), \qquad c = P(2^{d-1}), \tag{53}$$

where P_L and P_R represent the interiors of the two half blocks. Normalize by subtracting the minimum finite entry from the two child references and the midpoint scalar. Intern the resulting tuple together with its depth, retaining every nonempty level. An all-infinite interior has a null reference. Thus the incoming offset is exactly the interior minimum, and assembling a node takes constant arithmetic and dictionary work. The first value of a full block is external to this representation.

Lemma 44 (Canonical interiors and conversion)**.** *Two interiors of the same depth, each with a finite entry, have the same normalized punctured node if and only if their functions differ by an additive constant, with identical infinite positions. From an ordinary additive diagram with D reachable nodes, including its terminal, its punctured diagram can be constructed in $O(D)$ work and has at most $D-1$ nodes when the original has a finite entry. An all-infinite original gives a null punctured diagram.*

Proof. For canonicity, induct on remaining depth. The minimum fixes the offset; the left interior, midpoint scalar, and right interior then uniquely fix the normalized tuple by the induction hypothesis. The reverse implication follows by evaluating that tuple. For conversion, first memoize the first value of each ordinary node. An ordinary node $L \parallel U$ maps to $(\text{puncture}(L), U(0), \text{puncture}(U))$. A terminal maps to an empty interior. Offsets commute with this map, so memoizing it creates at most one punctured node per ordinary nonterminal, with constant work per source node. Conversion can merge additional nodes or produce a null reference, which only decreases the size. □

Theorem 45 (Mixed construction with constant boundary assembly)**.** *Let D_k count the punctured nodes of the mixed prefix function f_k on 2^h addresses, with $D_0 = 0$. One arbitrary nonnegative normalized key/gap update takes $O(h(D_k + 1))$ arithmetic and dictionary operations. For $n \geq 1$ and a prescribed feasible height h, a complete optimal mixed BST can be constructed in*

$$O\left(h\left[n + \sum_{k=0}^{n-1} D_k\right]\right) \tag{54}$$

such operations and at most that many stored words. This includes all intermediate nodes, selected-slot traceback, and ancestor repair. Choosing any proved sufficient unrestricted height gives a complete unrestricted optimum with the same bound, plus the height-selection work.

Proof. For a block starting at B, the inclusive carry is $p_B = \min_{a \leq B} f_k(a)$. In particular $p_B \leq f_k(B)$, so the omitted first input entry cannot improve that carry. By the endpoint argument of Theorem 43, outside predecessors contribute $p_B - g\nu_2(t)$ at an interior destination t. For $P = (P_L, c, P_R)$ of remaining depth d, the two recursive carries and the new midpoint value are therefore

$$p_L = p_B, \qquad p_R = \min\{p_B, \min P_L, c\}, \tag{55}$$

$$c' = \min\{p_B - g(d-1), E_g(P_L)\} - w(d-1), \tag{56}$$

where $E_g(P) = \min_{t>0}\{P(t) - g\nu_2(t)\}$ and

$$E_g(P) = \min\{E_g(P_L), c - g(d-1), E_g(P_R)\}.$$

This summary takes $O(D_k)$ work by normalized-node memoization. All minima omit infinite choices, and an empty minimum is infinite. Recurse on the two interiors and assemble (P'_L, c', P'_R) directly. No first-address replacement is performed.

If $p_B \leq \min P$, or the interior is all infinite, the output is the valuation template $p_B - (g+w)\nu_2(t)$ on $t > 0$. A template at depth d is assembled from the depth-$(d-1)$ template on both sides and midpoint $-(g+w)(d-1)$. Thus all needed templates take at most h node-assembly calls. An infinite carry and infinite interior give an infinite output. At the root the nonempty prefix has carry $+\infty$; the empty prefix has no interior, carry zero, and initial gap $g = 0$. This gives the exact first update as well.

Cache each expanded state by its input-node identity and carry relative to the interior minimum. The reset-path proof of Lemma 42 applies with $C = \min\{\min P_L, c\}$. A reset right child has a relative carry fixed by its parent input node. If the right child inherits instead, $p_B < C \leq \min P_L$ prunes the left child. Hence, writing V_k for the input nodes, the number Z of expanded states obeys

$$Z \leq h + \sum_{u \in V_k} (d(u) - 1).$$

Each expanded state now makes exactly one node-assembly call. If $T \le h$ templates are built, the exact number of such calls is $Z + T$, at most

$$2h + \sum_{u \in V_k} (d(u) - 1) = O(h(D_k + 1)). \tag{57}$$

There are $2Z + 1$ recursive build calls, at most D_k summary expansions, and at most $2D_k + Z$ summary calls. Template lookups and remaining scalar operations have the same bound. This accounts for every intermediate operation and created node, including interned nodes that are reused.

Store the prefix roots. The skyline walk of Theorem 41 still follows the bits of $b - 1$. Whenever it enters a right half, the required candidate is exactly the midpoint scalar in (53). Thus each predecessor query takes $O(h)$ work and creates no nodes; all selected slots take $O(nh)$.

For the final repair, collect the union of the selected slots and their perfect-tree ancestors in a dictionary, stopping each upward walk at an already collected ancestor. There are at most nh vertices. This set is ancestor-closed and contains the perfect-tree root. An inorder traversal visiting only collected children enumerates it in increasing slot order in linear work, without comparison sorting. At a collected slot x of valuation $d > 0$, its possible children are $x - 2^{d-1}$ and $x + 2^{d-1}$; check their membership before visiting them. Build the parent/child and inorder-neighbor links, then apply the sparse deletion repair of Theorem 35. Every unwanted ancestor is deleted in constant work using those links, followed by a linear output traversal. The repair therefore takes $O(nh)$ arithmetic/address and dictionary operations. Summing updates, traceback, and repair proves (54). The existing exact mixed-slot identity proves the cost and optimality of the complete tree. As in the preceding diagram theorems, exact operand bit costs are additional. □

The representation changes the cost of an update without imposing a key run structure or restricting the gap weights. The all-stage sum $\sum_k D_k$ remains the obstacle to a general subquadratic guarantee. Corollary 170 gives a matching general-diagram operation family. Theorem 171 proves cubic final diagrams and quartic shared history on actual positive mixed inputs. Further savings require avoiding those explicit canonical representations or exploiting additional input structure.

2.23 Two positive blocks: the whole above-capacity core

The one-block representation extends to two equal positive blocks, including all positive-endpoint cuts. The complete-block interaction is a finite Monge product; the cut regions reduce to visibility pairs and two diagonals.

Lemma 46 (Finite band dominance and a local boundary charge)**.** *For real sequences f_1, f_2 of length $N \ge 1$ and an integer $d \ge 0$, define, for $0 \le A, B < N$,*

$$H(A, B) = \max\{f_1(u) + f_2(v) : u \le A,\ v \ge B,\ 0 \le v - u \le d\}, \qquad 0 \le u, v < N.$$

On its feasible domain $B \le A + d$, $-H$ has a finite Monge completion with core size at most $14(N - 1)$, constructible in $O(N \log(N + 1))$ work. Any $s \times s$ square of density cells with feasible corners has at most $22s$ positive densities.

Proof. Set $K = 1 + \max f_1 - \min f_1 + \max f_2 - \min f_2$ and define

$$\begin{aligned} L(A, u) &= K \max(u - A, 0), \\ Q(u, v) &= K[\max(u - v, 0) + \max(v - u - d, 0)] - f_1(u) - f_2(v), \\ R(v, B) &= K \max(B - v, 0). \end{aligned}$$

All entries are finite. The only densities of L and R are K on the main diagonal. Matrix Q has density K on each of $v-u=0$ and $v-u=d$, adding the two contributions when $d=0$. Its row and column terms do not affect densities. Therefore all three matrices are Monge, with $\delta(L), \delta(R) \le N-1$ and $\delta(Q) \le 2(N-1)$. Every inadmissible pair in $T = L \otimes Q \otimes R$ incurs at least K penalty, so its cost exceeds $-\min f_1 - \min f_2$. A feasible pair has cost at most that quantity. Thus $T = -H$ on the feasible domain. The published inequality $\delta(X \otimes Y) \le 2(\delta(X) + \delta(Y))$ and condensed-product algorithm [14] give

$$\delta(T) \le 2[2((N-1) + 2(N-1)) + (N-1)] = 14(N-1)$$

and the construction bound. The operand cores and boundaries are directly available in $O(N)$ work; no dense conversion is required.

For the local bound let $A \in [a, a']$, $B \in [b, b']$, with at most $s+1$ indices in each range. Discard $u > a'$ and $v < b$. Aggregate $u = 0, \dots, a$ into the first row and keep $a+1, \dots, a'$ as singleton rows. Keep $b, \dots, b'-1$ as singleton columns and aggregate $v = b', \dots, N-1$ into the last column. Let each entry of Q' be the minimum of Q on its row-group/column-group rectangle. Blockwise minima on ordered disjoint groups preserve Monge: take minimizers in two off-diagonal groups and uncross their coordinates using the Monge inequality for Q. Away from the first row and last column, Q' is a consecutive submatrix of Q with at most two density diagonals. These interior diagonals contribute at most $2s$ entries; the first row and last column contribute at most s each. Hence $\delta(Q') \le 4s$. Finite prefix and suffix selectors on the group indices each have at most s core entries. They describe exactly the requested prefixes and suffixes; the same K excludes invalid choices on every feasible entry. Their product therefore equals $-H$ on the rectangle. Applying the core inequality twice gives $2[2(s+4s)+s] = 22s$. A single-row or single-column case has no density. □

Theorem 47 (Complete above-capacity core for two blocks)**.** *Suppose the two blocks $[z, z+m)$ and $[z+m+g, z+m+g+k)$, with $m, k \ge 1$ and $g \ge 0$, have common positive key weight α, and every other key has weight zero. Gap weights may be arbitrary nonnegative reals. At capacity $c = 2^h - 1$ with a valid terminal penalty, the entire ordinary core strictly above capacity has at most $28n$ entries. Any $s \times s$ square strictly above capacity has at most $40s$ entries. All these entries and their values can be constructed in $O(n \log(n+1))$ arithmetic work and space.*

Proof. If $c \ge n$ the region is empty. Normalize α to one and put

$$F(t) = \sum_{r=1}^{t} \nu_2(r), \qquad R_\ell(r) = F(r+\ell) - F(r).$$

Above capacity all gap terms cancel from ordinary densities. By binary saturation, an interval with $P \ge c$ positive keys has score $F(c)$; otherwise an optimum retains every positive key.

Both blocks complete. If $m+k > c$, the score is constant. Otherwise let $N = c-m-k+1$. Let u count selected zeros before the first block and v selected zeros before the second block, excluding the first block's positive keys. For an interval $[i, j)$ containing both whole blocks, feasibility is exactly

$$0 \le u, v < N, \quad 0 \le v - u \le g, \quad u \le A = z - i, \quad v \ge B = c + z + g - j.$$

Its reward is $R_m(u) + R_k(m+v)$, so Lemma 46 applies with $f_1(u) = R_m(u)$, $f_2(v) = R_k(m+v)$, and $d = g$. Clip A from above at $N-1$ and B from below at zero; a frozen endpoint gives zero density. A product-core cell (a, b) maps to

$$i = z - a - 1, \qquad j = c + z + g - b - 1. \tag{58}$$

This region contributes at most $14(N-1) \le 14c$ entries globally and $22s$ in a side-s square.

Separating the endpoint cases. Reserve the at most eight grid lines $i = t-1$ or $j = t-1$ for $t \in \{z, z+m, z+m+g, z+m+g+k\}$, together with

$$j - i = c + g - 1 \quad \text{or} \quad j - i = c + g. \tag{59}$$

They contribute at most $10n$ cells globally and $10s$ locally. Off these lines, the four-corner endpoints lie within fixed pieces of the five-piece input: zero, positive, zero, positive, zero. Cells encountering at most one positive block agree with that one-block input. By Theorem 24 and Corollary 25, the two such cores contribute at most $2c$ entries globally and $4s$ locally.

Exactly one positive endpoint cut. Suppose the left endpoint cuts the first block and the right endpoint lies after the second. Set $a = z + m - i$ and $B = c + z + m + g - j$. For $a + k \le c$, the first a selected ranks are positive. The rank offset t before the second block satisfies precisely

$$\max(a, B) \le t \le \min(a + g, c - k),$$

so the score is

$$F(a) + \max_{\max(a,B) \le t \le \min(a+g, c-k)} R_k(t). \tag{60}$$

At $a + k = c$ this is the saturated score $F(c)$. If $a + k > c$, both left-endpoint choices are saturated, including the case $a-1+k = c$, and their density is zero. For the remaining cells, $B - a = c+g-(j-i)$. Away from (59), either $B \le a$ in all four corners, making the score independent of j, or $B > a$ in all four. In the latter case remove the row term $F(a)$ from (60). The remaining range maximum has endpoints B and $\min(a+g, c-k)$. Frozen upper endpoints give zero; all other positive densities are nonadjacent visibility pairs of R_k. There are at most c globally. In a side-s square their two index intervals have at most s positions each and are ordered and disjoint, since their endpoint difference is $j - i - c + 1 \ge 2$. The noncrossing neighbor-set count in Corollary 25 gives at most $2s$ locally. Reflection, using $\nu_2(2^h - r) = \nu_2(r)$, gives the same bounds for a right endpoint cutting the second block.

Both positive endpoints cut. Let a, b be the remaining positive lengths. For $a + b \ge c$ the score is $F(c)$. Otherwise the interval length supplies enough zeros between them, and the first a and last b selected ranks are positive. Its score is $F(a) + F(b)$. The four scores separate unless $a + b = c - 1$ or c, precisely the two diagonals already reserved in (59). All other densities vanish. Summing the cases proves

$$K_{\text{above}} \le 14c + 2c + 2c + 10n \le 28n, \qquad K_{\text{square}} \le 22s + 4s + 4s + 10s = 40s.$$

Multiplication by α preserves support and scales every density.

Construction, including the reserved cells. The proof enumerates $O(n)$ candidates: the complete-block product core, two one-block visibility cores, two one-cut visibility cores, and the reserved lines and diagonals. Filter to valid cells strictly above capacity and evaluate their four scores. Precompute F, R_m, R_k in $O(c)$ work with the ruler's even/odd recurrence. Segment trees answer range maxima in $O(\log(n+1))$ time. Saturated and partial one-block intervals use $F(c)$ or $F(P)$; two-cut intervals use $F(a) + F(b)$; one-cut intervals use (60) or its reflection. Complete one-block intervals use (26), and complete two-block intervals query the product T above. For these $O(n)$ product queries use its condensed identity

$$T(A, B) = T(A, 0) + T(0, B) - T(0, 0) - \sum_{x < A,\, y < B} D_T(x, y).$$

Sort queries by A, sweep the $O(c)$ core entries in row order, and maintain column prefix sums in a Fenwick tree. Total query work is $O(n \log(n+1))$, with $O(n)$ additional space. The sparse products have the same work bound. The implementation retains ordinary witness maps inside multiplication, so the conservative implemented space bound is $O(n \log(n+1))$; no succinct-array assumption is made. Retain exactly the positive four-corner differences at the enumerated candidates. □

This establishes a boundary charge for interacting equal-weight blocks on the whole above-capacity side. Unequal positive block weights, additional blocks, and the below-capacity mixed core remain open cases of the general size problem.

2.24 Unimodal key weights and a sparse window envelope

A single peak admits a stronger reduction than the constant-block argument. Here unimodality is in the original key order: weights are nondecreasing up to a peak and nonincreasing afterward. No keys are reordered.

Lemma 48 (Contiguous domination)**.** *Every t-element subsequence of a unimodal sequence is componentwise dominated by a contiguous t-element block. Consequently, for every nonnegative reward vector $v_1, \dots, v_t$ and unimodal $x_0, \dots, x_{k-1}$,*

$$\max_{i_1 < \cdots < i_t} \sum_{r=1}^{t} v_r x_{i_r} = \max_{0 \le a \le k-t} \sum_{r=1}^{t} v_r x_{a+r-1}.$$

Proof. Choose a peak z and let b selected indices lie at or before z. Move those indices to $z-b+1, \dots, z$, and the remaining indices to $z+1, \dots, z+t-b$. There are enough positions on each side because the original subsequence already used that many. Each left index moves right inside the nondecreasing part; each right index moves left inside the nonincreasing part. Every selected value weakly increases, in the same order. The new positions are contiguous, including the cases $b = 0$ and $b = t$. The empty subsequence is immediate. □

For arbitrary key weights and capacity $c = 2^h - 1 \le n$, define

$$B_h(a) = \sum_{r=1}^{c} p_{a+r-1} \nu_2(r), \qquad 0 \le a \le n - c, \tag{61}$$

$$\widehat{S}_h(i,j) = \max_{i \le a \le j-c} B_h(a), \qquad j - i \ge c,$$

$$\widehat{A}_h(i,j) = M(j-i-c) + hW(i,j) - \widehat{S}_h(i,j). \tag{62}$$

A contiguous selected block is feasible, so $\widehat{A}_h \ge A_h$. Whenever $p[i:j]$ is unimodal, Lemma 48 gives equality. This identity includes arbitrary nonnegative gap weights through $W(i,j)$.

Theorem 49 (All above-capacity cores for unimodal keys)**.** *With a valid terminal penalty, for arbitrary nonnegative unimodal key weights and arbitrary nonnegative gap weights, a layer's entire ordinary core strictly above capacity has at most $\max(0, n-c-1)$ entries. Every $s \times s$ square wholly above capacity has at most $2s-1$ entries. The global leading coefficient one is asymptotically sharp. All above-capacity cores over every layer with $c \le n$, including their values, can be constructed in $O(n \log(n+1))$ arithmetic work and space. The same preprocessing supports an exact layer-value query and identifies an optimal contiguous selected block for any interval of length at least c, in $O(\log(n+1))$ time.*

Proof. For $j - i > c$, put $a = i$, $b = j - c + 1$, and $Q = \max_{a<t<b} B_h(t)$. The affine terms in (62), including every gap contribution, cancel. Four range maxima give

$$D_{\widehat{A}_h}(i, j) = \max\{0, \min(B_h(a), B_h(b)) - Q\}. \tag{63}$$

These are exactly the nonadjacent strict visibility pairs of the $N = n - c + 1$ window values, with mapping

$$(a, b) \longmapsto (i, j) = (a, c + b - 1).$$

Every such pair maps to a valid above-capacity cell. The noncrossing count in Lemma 23 gives at most $\max(0, N - 2)$ entries. A side-s square maps to two ordered disjoint endpoint ranges, each with at most s indices. The neighbor-span argument of Corollary 25 gives $2s - 1$ entries there. Global unimodality makes $A_h = \widehat{A}_h$ on all four corners, proving both original-core bounds. For sharpness, let $C = 2^h$, $n = 2C$, and put a unit key at zero-based position $C + 1$. The one-key count in Theorem 24 is $C - 2h$, whereas $n - c - 1 = C$. Their ratio tends to one.

For construction initialize $B_0(a) = 0$, $0 \le a \le n$. Writing $d = 2^{h-1}$, the two copies of the preceding ruler word give

$$B_h(a) = B_{h-1}(a) + B_{h-1}(a + d) + (h - 1)p_{a+d-1}. \tag{64}$$

There are $O(n)$ entries at each of $O(\log(n+1))$ relevant layers. The visibility stack constructs every layer's coordinates and densities in $O(n)$ work. It pushes each index once and pops it at most once. Thus all cores take $O(n \log(n + 1))$ work, without an interval-pair scan. For each B_h, a maximum segment tree retains the earliest maximizing window start. Its linear preprocessing preserves the construction bound; a range query costs $O(\log(n + 1))$. Prefix sums give $W(i, j)$ in constant time. The returned start identifies a contiguous optimal selected block; emitting its c keys or perfect partial skeleton takes $O(c)$ additional output work. Keeping all arrays, query trees and cores uses $O(n \log(n + 1))$ space. □

Corollary 50 (Certified exact spans in arbitrary inputs)**.** *For arbitrary nonnegative key and gap weights, all positive above-capacity original densities whose outer key interval is unimodal can be listed in $O(n \log(n + 1))$ work and space over all layers. After $O(n)$ preprocessing, unimodality of an interval can be tested in $O(1)$ time. In particular, an above-capacity tile $[a, a + s) \times [b, b + s)$ whose key span $[a, b + s)$ is unimodal has at most $2s - 1$ original positive densities.*

Proof. Put $\Delta_t = p_{t+1} - p_t$ and let

$$V_t = \max\{u : u < v \le t,\ \Delta_u < 0,\ \Delta_v > 0\},$$

using -1 for an empty set. Maintain the last negative-difference index and update V whenever a positive difference occurs. This computes all V_t in one scan. An interval $[i, j)$ is unimodal exactly when $j - i \le 1$ or $V_{j-2} < i$: a decrease followed by a later increase is precisely a violation, including intervening flat positions.

Construct the global envelope (62) for the arbitrary input. Equation (63) still gives at most $O(n)$ entries per layer. Filter its cell (i, j) by the unimodality of $[i, j + 1)$. On every retained outer span, all four corner intervals have exact envelope values. Conversely, any positive original density with such an outer span must occur in the envelope core. Thus filtering gives the complete claimed set. If a tile's whole key span is unimodal, the same equality holds at every corner in it, and the local visibility count applies. □

This supplies the boundary charge on certified spans even when the full input has many peaks. Positive cells outside this certified part require a nonunimodal outer span. The remaining interactions and the below-capacity mixed region still require separate size bounds.

2.25 Valley weights and logarithmic implicit queries

A valley is nonincreasing up to a minimum and nondecreasing afterward, in the fixed key order. Its above-capacity scores have a different compact representation from the single-peak case. Section 5.24 shows that their explicit cores can nevertheless be quadratic.

Lemma 51 (Universal end-block domination)**.** *For every c, one c-element subsequence of a valley componentwise dominates every other c-element subsequence. It consists of the c largest weights as a prefix and suffix, resolving ties toward the left. It simultaneously maximizes every nonnegative rank-reward vector.*

Proof. Split a valley $x_0, \dots, x_{k-1}$ at a minimum position z, assigning the minimum to the nondecreasing right part. If a subsequence selects t keys from the left part, move them to the first t positions and the remaining keys to the last $c-t$ positions. Each selected value weakly increases within its monotone wing. Let $X(t)$ be this end-block word. Its feasible splits satisfy

$$\max(0, c-(k-z)) \le t \le \min(c, z).$$

The words $X(t)$ and $X(t+1)$ differ only at rank $t+1$, replacing x_{k-c+t} by x_t. On this feasible range, x_t is nonincreasing and x_{k-c+t} is nondecreasing. Their difference is therefore nonincreasing. Move through all nonnegative differences and stop before the first negative one. The stopping word dominates every preceding and subsequent $X(t)$, and hence every original subsequence. It selects the largest c values from the two decreasing end lists. Binary search finds its split in $O(\log(k+1))$ comparisons, including ties and empty wings. □

For arbitrary key weights define the ruler-prefix score

$$P_p(a,t) = \sum_{r=1}^{t} p_{a+r-1}\nu_2(r), \qquad P_p(a,0) = 0.$$

Retain the arrays B_k from (61), with $B_0 = 0$. For $2^k \le t < 2^{k+1}$ the ruler identity gives

$$P_p(a,t) = B_k(a) + kp_{a+2^k-1} + P_p(a+2^k, t-2^k). \tag{65}$$

Scanning decreasing powers evaluates this in $O(\log(n+1))$ arithmetic work. The construction assumes no constant-time population-count operation.

Theorem 52 (Implicit above-capacity queries for valley keys)**.** *For nonnegative valley key weights and arbitrary nonnegative gap weights, $O(n\log(n+1))$ preprocessing and space support an exact layer-value query for any $h \ge 1$, $c = 2^h - 1 \le j - i$, with a valid terminal penalty, in $O(\log(n+1))$ work. A query returns two end blocks identifying an optimal selected subsequence; emitting its keys or perfect partial skeleton costs $O(c)$ additional output work. All layers share the preprocessing.*

Proof. Build the perfect-ruler windows for p and its reversal $\bar{p}$, using (64). Their total construction and storage are $O(n\log(n+1))$. Every interval of a valley is a valley. Find its optimal split t using Lemma 51, clipping the two wings to the interval before binary search. Ruler symmetry $\nu_2(c+1-r) = \nu_2(r)$ yields

$$\begin{aligned} S_h(i,j) &= P_p(i,t) + P_{\bar{p}}(n-j, c-t), \\ A_h(i,j) &= M(j-i-c) + hW(i,j) - S_h(i,j). \end{aligned} \tag{66}$$

The split search and two prefix queries take logarithmic work; prefix mass sums give $W(i,j)$ in constant time. The witness is $[i, i+t)$ followed by $[j-(c-t), j)$. No interval table or explicit core is constructed. □

This gives exact implicit queries even when the original above-capacity core is dense. Below-capacity mixed intervals and efficient composition of these representations remain requirements for a general full-tree algorithm.

2.26 Individual-weight charges for valley cores

The valley query representation also supports a direct charge to individual integer key weights, without the height factor in a density-mass estimate. Let $T_c(i,j)$ be the sum of the c largest key weights in $[i,j)$, and put

$$\Gamma_c(i,j) = T_c(i,j) + T_c(i+1,j+1) - T_c(i,j+1) - T_c(i+1,j) \qquad (j-i>c). \tag{67}$$

For $c \geq 2$, write $x = p_i$, $y = p_j$, and let z, w be the cth and $(c-1)$st largest inner weights in $[i+1,j)$. Only these two smallest retained values can be displaced by adjoining the endpoints, giving

$$\Gamma_c(i,j) = \max\{0, \min(x,y,w) - z\}. \tag{68}$$

For $c = 1$ take $w = +\infty$. Thus $\Gamma_c \geq 0$ for arbitrary input weights.

Lemma 53 (Valley support inclusion)**.** *For nonnegative real valley key weights, arbitrary nonnegative gaps, and $c = 2^h - 1$, every strictly above-capacity cell satisfies*

$$0 \leq D_{A_h}(i,j) \leq (h-1)\Gamma_c(i,j). \tag{69}$$

Proof. At $h = 1$ the ruler score vanishes. Otherwise, Lemma 51 makes a largest-c selection optimal at every corner. All largest-c selections have the same value word: the dominating word has the same sum, so its componentwise inequalities must all be equalities.

If $x \leq z$, adjoining x leaves the inner largest-c sum and value word unchanged. Adjoining y can only raise the cutoff, so x remains ineffective afterward. Both ruler-score changes vanish. Reflection treats $y \leq z$. Suppose now $x, y > z$ but $x < w$. Every inner value on the left wing is at most x, so at most one selected value is on that wing; every retained value except z is at least $w > x$. Adjoining x replaces the first selected value z by x. The endpoint y is at least w, since w is on its wing, and adjoining y raises the cutoff to at least w. Thus the four-corner ruler difference is $(x-z)\nu_2(1) = 0$. Reflection treats $y < w$.

It remains that $x, y \geq w$. Choose adjacent smallest retained values $z \leq w$ in the valley word, choosing within its trough if they tie. Each one-end extension replaces z, and the two-end extension replaces z, w. If z has selected rank r, cancellation gives

$$D_{A_h}(i,j) = (w-z)\nu_2(r). \tag{70}$$

Explicitly, let L_a, R_b be the inner left and reversed-right ruler-prefix scores, $a+b=c$, and let primes adjoin the corresponding endpoint. If w, z are both left, the scores in order (inner, left, right, both) are

$$L_a + R_b, \quad L'_a + R_b, \quad L_{a-1} + R'_{b+1}, \quad L'_{a-1} + R'_{b+1}.$$

Their difference is $(L'_a - L'_{a-1}) + (L_{a-1} - L_a) = (w-z)\nu_2(a)$. If z is left and w right, only the last expression changes to $L'_a + R'_b$. The difference becomes $w\nu_2(b+1) - z\nu_2(a) = (w-z)\nu_2(a)$, since $b+1 = 2^h - a$. Reflection covers the right-minimum cases by ruler symmetry; the expressions also give zero when $w = z$. Now $\nu_2(r) \leq h-1$, and (68) equals $w - z$ in this last case. The affine and gap terms cancel, proving the lemma. □

Theorem 54 (Valley boundary-mass support and construction)**.** *Let the valley key weights be nonnegative integers of mass $P = \sum_i p_i$; gap weights may be arbitrary nonnegative reals. Every layer's strictly above-capacity core has at most P entries. Any rectangle there with row set I and column set J has at most*

$$\min\left\{\sum_{i\in I} p_i,\ \sum_{j\in J} p_j\right\} \tag{71}$$

positive entries. All above-capacity cores over every layer can be constructed in $O((n+P)\log^2(n+1))$ arithmetic work and $O((n+P)\log(n+1))$ space, including their density values.

Proof. Let $z_i(j)$ be the cth largest inner weight in $[i+1, j)$. For arbitrary input weights the endpoint difference is

$$\Delta_i(j) = T_c(i,j) - T_c(i+1,j) = \max\{0, p_i - z_i(j)\}.$$

The cutoff $z_i(j)$ is nondecreasing with j, and $\Gamma_c(i,j) = \Delta_i(j) - \Delta_i(j+1)$. Hence

$$\sum_{j=i+c+1}^{n-1} \Gamma_c(i,j) \le p_i. \tag{72}$$

Reversal gives the column charge $\sum_i \Gamma_c(i,j) \le p_j$. For integer keys every positive Γ_c is at least one. Apply Lemma 53 to obtain the global and rectangle bounds.

Preprocess Theorem 52. Binary search in the two monotone wings also finds a cth-largest cutoff or counts values strictly above a threshold in logarithmic work. In row i, start at $j = i + c + 1$ and compute $z = z_i(j)$. If $z \ge p_i$, no auxiliary density remains. Otherwise let g count inner values greater than z and set $m = c - g$, the multiplicity of the smallest retained value. Find the first future right-wing position k with $p_k > z$. All intervening values are at most z, and all values after k are greater than z. Exactly m such insertions are needed before the cutoff increases; its next event is

$$j' = k + m - 1. \tag{73}$$

If $j' \ge n$, finish the row. Otherwise query $z' = z_i(j'+1) > z$, emit $\Gamma_c(i,j') = \min(p_i, z') - z$, and continue at $j = j' + 1$. If j is still on the decreasing wing, every later key on that wing is at most the current cutoff, justifying the same right-wing search. The multiplicity in (73) skips all zero changes caused by repeated weights. Every positive auxiliary cell is emitted once.

Each event and final row termination take logarithmic work. Evaluate four ruler queries at each emitted cell and retain its original density when positive. Lemma 53 proves completeness; height one has zero ruler core and is skipped. With $L = \lceil \log_2(n+1) \rceil$ and δ total auxiliary events, the work is $O((nL+\delta)L)$ and space is $O(nL+K)$ for K original output entries. Equation (72) gives $K \le \delta \le PL$, proving the bounds. □

This proves the local boundary-plus-mass support target on above-capacity valley spans using only endpoint mass. The normalized rational dense family has fractional auxiliary densities, consistent with the support inclusion and row charge while escaping the integer support count. Below-capacity mixed cores still require a separate argument.

The mass dependence is necessary up to constants. In the integer family (162), $P = n(n+1)/2$ and one layer has at least $n^2/32$ entries. On its displayed square the row-endpoint mass is $3s^2 + s$, the column-endpoint mass is $3s^2$, and the core has $s^2/2$ entries. Thus the local bound (71) is attained within a factor of six by the proved family.

2.27 An exact periodic quotient for the full mixed problem

Periodicity offers a different way to avoid most interval states. Both types of weights are retained, and the construction uses the unrestricted optimum.

Theorem 55 (Cyclic phase construction)**.** *Let $n \geq 1$. If the key and gap weights are restrictions of nonnegative real sequences with common period b, a complete optimal BST can be constructed in $O(nb)$ arithmetic work and space. The dynamic program examines exactly $b(2n-1)$ root candidates. The input length need not be a multiple of b. Given arbitrary explicit weights, their shortest common finite period can be detected in $O(n)$ work; it is at most $n+1$.*

Proof. Extend both period words indefinitely. Translation by b preserves the cost of every ordered tree. Let $C_a(\ell)$ denote the optimum for length ℓ and starting phase $a \in \{0, \ldots, b-1\}$, and let $\rho_a(\ell)$ be its leftmost optimal root relative to the interval start. Then $C_a(0) = 0$ and

$$C_a(\ell) = W_a(\ell) + \min_{0 \leq r < \ell} \Big\{ C_a(r) + C_{(a+r+1) \bmod b}(\ell - r - 1) \Big\}. \tag{74}$$

Process lengths in increasing order. Knuth's monotone-root theorem [2], with consistent leftmost ties, restricts the search at $\ell \geq 2$ to

$$\rho_a(\ell - 1) \leq r \leq 1 + \rho_{(a+1) \bmod b}(\ell - 1). \tag{75}$$

The endpoints lie in $[0, \ell - 1]$. For $a = b - 1$, the right neighbor begins at absolute position b; translation identifies it with phase zero. The total number of candidates at any fixed length $\ell \geq 2$ is therefore

$$\sum_{a=0}^{b-1} \Big(2 + \rho_{(a+1) \bmod b}(\ell - 1) - \rho_a(\ell - 1) \Big) = 2b. \tag{76}$$

Length one uses b candidates, giving exactly $b(2n-1)$ overall. Individual phase ranges can be large; the bound uses the whole cycle. In particular, the algorithm computes all phases at every length, including synthetic intervals extending beyond the original input. Omitting those states would invalidate this cyclic cancellation.

A prefix sum U of $p_t + q_t$ over $n + b$ consecutive positions gives

$$W_a(\ell) = U(a + \ell) - U(a) + q_{(a+\ell) \bmod b}$$

in constant work. Every lookup in (74) uses a shorter length. Thus candidate evaluation, mass queries, and minimizer maintenance take $O(nb)$ work. Cost and relative-root tables each store $(n+1)b$ entries. For reconstruction, interval $[i, j)$ uses absolute root $i + \rho_{i \bmod b}(j - i)$. An explicit stack emits its two child intervals, visiting every key and gap once. This includes the degenerate all-zero case, whose leftmost optimum can have height n.

For period detection, form the signature word

$$(p_i, q_i, q_{i+1}), \qquad 0 \leq i < n. \tag{77}$$

The prefix-function algorithm finds its shortest finite period $d = n - \pi_{n-1}$ in linear work. If $d < n$, signature equality gives every required key and gap equality, including the final gap, so $b = d$. If $d = n$, use $b = n$ when $q_n = q_0$, and $b = n + 1$ otherwise. Every common period below n is a period of (77), proving minimality. For $b = n + 1$, set the unobserved padding key $p_n = 0$ and use period words $(p_0, \ldots, p_{n-1}, 0)$ and $(q_0, \ldots, q_n)$. Their restriction to the input is exact. The empty input is handled separately with cost zero. □

For constant b this is linear construction with arbitrary nonnegative real magnitudes inside the two periods. More generally $b = O(n^\alpha)$, $\alpha < 1$, gives $O(n^{1+\alpha})$ work. A defect in the final gap can invalidate an otherwise repeating pattern; both adjacent gaps in (77) are needed. The method is a direct cyclic application of Knuth's theorem, independent of the integer-mass analysis and the later lower-bound certificate. Its general nonperiodic worst case remains quadratic. Verification is in Appendix X.

2.28 Arbitrary fringes around a periodic middle

An exact central oracle can absorb arbitrary boundary weights without recomputing its central intervals. The needed linear primitive is the concave least-weight-subsequence algorithm of Wilber [3].

Lemma 56 (Exact boundary extension)**.** *Suppose all unrestricted mixed OBST costs $C(i,r)$ for $0 \le i \le r < J$ are available in constant time. The complete new column $C(i,J)$, $0 \le i \le J$, and optimal roots can be computed in $O(J)$ arithmetic work and space. Reflection gives the same bound for a new left row.*

Proof. Write $D(i) = C(i,J)$. The original recurrence is

$$D(J) = 0, \qquad D(i) = W(i,J) + \min_{i \le r < J} \{C(i,r) + D(r+1)\}.$$

Set $F(t) = D(J-t)$ and identify predecessor $s < t$ with root $r = J - s - 1$. This gives $F(0) = 0$ and

$$F(t) = \min_{0 \le s < t} \{F(s) + w(s,t)\}, \qquad w(s,t) = C(J-t, J-s-1) + W(J-t, J). \tag{78}$$

Every old interval has right endpoint at most $J-1$; its value is already available. Prefix masses make each valid arc query constant time. For $s < s' < t < t'$, put $a = J - t'$, $b = J - t$, $c = J - s' - 1$, and $e = J - s - 1$. Then $a < b \le c < e$. The ordinary mixed OBST Monge inequality gives

$$C(a,c) + C(b,e) \le C(a,e) + C(b,c).$$

The mass terms cancel, proving the precise concavity condition

$$w(s,t) + w(s',t') \le w(s,t') + w(s',t). \tag{79}$$

The case $b = c$ uses $C(b,b) = 0$ and is included.

Wilber's published statement is that "the concave least weight subsequence problem can always be solved in $O(n)$ time" [3, p. 418]. Its input is a finite real arc function on $0 \le s < t \le N$ satisfying (79). With constant-time arc access, the algorithm computes every $F(t)$ and predecessor witnesses in $O(N)$ work and space [3, pp. 420–424]; no difference-of-weight condition is required. Here $N = J$, all valid arcs are finite, and (78) supplies exactly this interface. A leftmost predecessor gives a rightmost original root. Reflecting keys and inclusive gaps proves the left-row statement. □

Theorem 57 (Periodic middle with arbitrary fringes)**.** *Let $[l,r)$ be a nonempty middle of $m = r - l$ keys in an n-key input. Suppose its key weights and inclusive gap weights $q_l, \dots, q_r$ have common period $b \le m$. With $d = n - m$ arbitrary fringe keys and arbitrary nonnegative real weights throughout, a complete unrestricted optimal BST and a constant-time oracle for every interval cost and an optimal root can be constructed in $O(mb + nd)$ arithmetic work and space.*

Proof. Build the periodic middle oracle by Theorem 55, using exactly $b(2m-1)$ root candidates. Append the right fringe one key at a time by Lemma 56, then prepend the left fringe by reflection. Extension t creates exactly $m+t$ nonempty states, so

$$S_{\text{boundary}} = \sum_{t=1}^{d}(m+t) = dm + \frac{d(d+1)}{2} = \frac{n(n+1)-m(m+1)}{2} \le nd. \tag{80}$$

Store each completed boundary in an array. An old interval either lies wholly in the middle, ends in a stored right column, or starts in a stored left row. The last case takes precedence when both endpoints are outside. These cases give constant-time lookup, including empty intervals of cost zero. During left extension, every old left endpoint is strictly larger than the new one, so the queried rows already exist.

Explicitly, for a new left row at i, let the reversed endpoint be $N = n - i$. An old reversed interval $[a, u)$ queries $C(n-u, n-a)$; the new interval mass is $W(i, n-a)$; and reversed root u maps to original root $n-u-1$. Thus reflection preserves the inclusive gap indices as well as key order. No unfinished row or column is queried.

Summing the linear boundary costs using (80), and adding the central oracle, gives the claimed bounds. Prefix preprocessing and complete reconstruction cost $O(n)$, absorbed because $mb+nd \ge n$. An explicit stack follows the exact stored roots and visits every key and gap once, with no height assumption. The representation covers every original interval, including those unused by the final tree. □

More generally, any exact constant-time central interval oracle with construction and storage cost T can be extended at cost $O(T+nd+n)$. The periodic quotient is one concrete source of such an oracle. A final-gap defect fits inside a one-key fringe. Neither boundary magnitudes nor total mass enter (80).

Corollary 58 (Automatic middle recognition)**.** *Let b_0 be the shortest whole-input joint period, allowing $b_0 = n+1$. For $1 \le b \le B < n$, let $m_b > b$ be the longest middle with common period b, omitting periods with no such middle. There is an automatic exact construction with work*

$$O\left(n(B+1) + \min\left\{nb_0,\ \min_{1\le b\le B}\left[m_b b + n(n-m_b)\right]\right\}\right). \tag{81}$$

With supplied constant b, d, construction is linear; default recognition with $B = \min\{n-1, \lceil \log_2(n+1)\rceil\}$ adds $O(n\log(n+1))$.

Proof. Use signatures $s_i = (p_i, q_i, q_{i+1})$. A middle $[l, r)$ of length greater than b has joint period b exactly when $s_i = s_{i-b}$ for every $l+b \le i < r$. On a failed comparison at i, an eligible new left endpoint must exceed $i-b$; set it to $i-b+1$. Between failures, the longest current suffix supplies the longest periodic middle. This costs $O(n)$ per b and includes the final gap q_r. For fixed $b < n$, the proxy $mb + n(n-m)$ decreases with m, so a longest middle is best. Compare all these proxies with nb_0, and use the corresponding constructor. If $b_0 \le B$, the whole-period route already costs $O(nB)$ and the scan can be skipped. This proves (81), also with $B = 0$ and an omitted inner minimum. Empty input is handled separately. □

The LWS primitive is established prior work; the boundary reduction and its composition with the central quotient supply the bounds above. A small number of arbitrary interior defects need not leave a long middle, so the general nonperiodic worst case remains quadratic. Independent verification and implementation details are in Appendix Y.

2.29 Positive affine translation of both weight sequences

Literal repetition can be replaced by a transformation that preserves all optimal shapes. The key and gap offsets must have opposite signs.

Lemma 59 (Affine shape-cost identity)**.** *For a tree T on ℓ keys, replacing every key weight by $\lambda p + \alpha$ and every gap weight by $\lambda q - \alpha$ gives*

$$\mathrm{cost}_{\mathrm{new}}(T) = \lambda\, \mathrm{cost}_{\mathrm{old}}(T) - \alpha\ell. \tag{82}$$

For $\lambda > 0$, all minimizing shapes and their leftmost roots are preserved.

Proof. If I is internal path length, key comparison lengths sum to $I + \ell$ and gap depths sum to $I + 2\ell$. The latter identity follows inductively: joining the two subtrees increases the internal sum by $\ell - 1$ and the external sum by $\ell + 1$, giving difference 2ℓ. The offset contribution is therefore $\alpha(I + \ell) - \alpha(I + 2\ell) = -\alpha\ell$. □

Theorem 60 (Finite affine phase construction)**.** *Let $n \geq 1$ and suppose the nonnegative input weights satisfy, for some $1 \leq b \leq n$, $\lambda > 0$, and $\alpha \in \mathbb{R}$,*

$$p_{i+b} = \lambda p_i + \alpha \quad (0 \leq i < n - b), \qquad q_{j+b} = \lambda q_j - \alpha \quad (0 \leq j \leq n - b). \tag{83}$$

A complete unrestricted optimal BST, with constant-time access to every interval cost and leftmost optimal root, can be constructed in $O(nb)$ arithmetic work and space. The smallest admissible b can be found within the same bound. Positivity outside the finite input is not assumed.

Proof. Let $P_k = \lambda^k$ and $G_k = \sum_{t=0}^{k-1} \lambda^t$, with $P_0 = 1, G_0 = 0$. Tabulate these by multiplication and addition. Lemma 59 implies, for $i = a + kb$ and $0 \leq a < b$,

$$C(i, i + \ell) = P_k C(a, a + \ell) - \alpha\ell G_k, \tag{84}$$

whenever the intervals lie in the input. Their relative leftmost roots are equal. All intermediate translations also lie inside the input.

Put $m = n - b + 1$. For each length $1 \leq \ell \leq m$, all b phase intervals $[a, a + \ell)$ are actual input intervals. Compute their costs $C_a(\ell)$ and relative roots $\rho_a(\ell)$ by the original recurrence, using (84) for child costs. Every child has smaller length. Actual prefix masses give each interval weight in constant time. Knuth's theorem [2] restricts roots at $\ell \geq 2$ to

$$\rho_a(\ell - 1) \leq r \leq 1 + \rho_{(a+1) \bmod b}(\ell - 1).$$

For $a = b-1$, the right neighbor begins at b and ends at $b+\ell-1 \leq n$. Its roots equal those of phase zero by (84). Thus every monotonicity argument uses actual nonnegative input weights. Summing the inclusive ranges around all phases gives $2b$ candidates per length, hence exactly $b(2m - 1)$ overall.

For the remaining lengths $\ell = m + 1, \ldots, n$, compute the $t = n - \ell + 1 < b$ actual intervals directly. Ordinary Knuth bounds telescope: if ρ_i is the relative root of $[i, i+\ell-1)$, the total number of candidates at length ℓ is

$$2t + \rho_t - \rho_0 \leq 2t + \ell - 2 = 2n - \ell. \tag{85}$$

Both end roots lie in $[0, \ell-2]$. The last $b-1$ diagonals therefore use at most $(b-1)n+(b-1)(b-2)/2$ candidates and store exactly $b(b-1)/2$ nonempty states. Short child intervals use (84); long children

use the already completed terminal diagonals. The total number of stored nonempty states is exactly

$$bm + \frac{b(b-1)}{2} = nb - \frac{b(b-1)}{2} \le nb. \tag{86}$$

This proves the work and space bounds. Prefixes and power sums cost $O(n)$. An explicit stack follows the stored phase or terminal root, visiting every key and gap once, including degenerate zero-weight chains.

It remains to recognize the law. For a candidate $b < n$, set $X_i = p_i + q_i$ and $Y_i = p_i + q_{i+1}$. Equation (83) implies $X_{i+b} = \lambda X_i$ and $Y_{i+b} = \lambda Y_i$. Conversely, put $e_j = q_{j+b} - \lambda q_j$. These two proportionality conditions give $p_{i+b} - \lambda p_i = -e_i = -e_{i+1}$, so all e_j are constant and $\alpha = -e_0$. A positive source component determines λ by its target/source ratio; zero components require zero targets. Reject a nonpositive ratio and verify all original equations. If all paired components are zero, nonnegativity makes all constrained weights zero; $\lambda = 1, \alpha = 0$ works. This is a complete $O(n)$ test. For $b = n$, $\lambda = 1, \alpha = q_0 - q_n$ always works. Testing $b = 1, 2, \ldots$ until the first success b_* costs $O(nb_*)$, including recognition and construction. The empty input has cost zero separately. □

The finite boundary is essential to this proof. The valid input $p = (1, 2, 2, 3, 3)$, $q = (2, 2, 1, 1, 0, 0)$ has $b = 2, \lambda = 1, \alpha = 1$, but its continued gap sequence becomes negative. The construction uses complete cycles only through $m = 4$ and then computes the last length directly. It never applies nonnegative-weight root monotonicity to that continuation.

For $b = 1$, exactly $2n - 1$ root candidates suffice. This includes pure geometric mixed weights, and the strictly positive opposing arithmetic sequences $p_i = i + 1$, $q_j = n + 1 - j$. The latter have total mass $(n + 1)^2$ and admit $O(n)$ full construction with $O(\log n)$-bit words. Geometric weights can have growing bit lengths; the general theorem counts exact arithmetic operations. Ordinary periodicity is $\lambda = 1, \alpha = 0$. The established boundary-extension lemma also permits an affine-period middle with arbitrary fringes in $O(mb + nd + n)$ work. Arbitrary interior changes and the general nonperiodic subquadratic problem remain open. Verification is in Appendix Z.

2.30 Certified refinement of coarse optimal trees

The integer Monge construction also supplies a structural certificate for nearby arbitrary weights. The coarse optimum may have ties: retain roots shared by every coarse optimum, then resolve the remaining intervals with the original key and gap weights. The resulting work depends on those ambiguous intervals, rather than on the magnitude or precision of the original weights.

Let $(\bar{p}, \bar{q})$ be nonnegative integer weights on the same $n \ge 1$ keys and $n+1$ gaps as the arbitrary nonnegative weights (p, q). For $\delta > 0$, define the total signed-perturbation magnitude

$$E_\delta = \sum_{i=0}^{n-1} |p_i - \delta \bar{p}_i| + \sum_{j=0}^{n} |q_j - \delta \bar{q}_j|. \tag{87}$$

The word coarse refers to the integer base, not to any reordering of outcomes.

Perturbation arguments preserving optimality already occur in search-tree work: Chrobak et al. [22] use infinitesimal changes to resolve ties in two-way comparison trees. Here the integer cost gap gives an explicit finite signed-error certificate, and the common-root partition supplies the construction bound.

Lemma 61 (Preservation of the optimal family)**.** *If*

$$(n-1)E_\delta < \delta, \tag{88}$$

then every optimum for (p, q) is also optimal for $(\bar{p}, \bar{q})$. The errors in (87) may have either sign, and the coarse optimum need not be unique.

Proof. For a nonempty n-key tree, every key comparison length and every gap depth lies in $[1, n]$. Write $C_e(T)$ for the cost of the signed error vector. For any two full trees T, U,

$$|C_e(T) - C_e(U)| \le (n-1)E_\delta.$$

If U is coarse-optimal and T is not, integrality gives $C_{\bar{p},\bar{q}}(T) - C_{\bar{p},\bar{q}}(U) \ge 1$. Therefore

$$C_{p,q}(T) - C_{p,q}(U) \ge \delta - (n-1)E_\delta > 0.$$

Such a T cannot be fine-optimal. This proves the assertion simultaneously for all coarse ties. The empty input has its sole tree separately. □

Lemma 62 (Unrestricted optimal-root extremes)**.** *Let $\bar{S}$ be the aggregate core bound for the integer base in Theorem 8, including its linear boundary terms. In $O(\bar{S}\log(n+1))$ work one can prepare queries for the smallest and largest unrestricted optimal root of any nonempty interval. Each query takes $O(\log(n+1))$ work, without constructing a dense interval table.*

Proof. First compute the final unrestricted cost matrix C^* in condensed form using the Monge construction. Its globally sufficient height applies to every interval. Make one further product whose root-r candidate on $[i, j)$ is $C^*(i, r)+C^*(r+1, j)$, using the same finite padding and boundary conventions as the main construction. The interval mass is independent of r, so the published exact leftmost witness gives the smallest unrestricted optimal root. Repeat on the reversed key and gap sequences; reflection of its smallest root gives the largest original root.

The two extra products cost $O((n+K_H)\log(n+1))$ each, where K_H is the corresponding final core size. Their output sizes are $O(K_H)$ by the product bound. Both are covered by the aggregate preparation bound, and the published witness queries have the stated cost. □

Using unrestricted child values in that last product is essential. A sufficient height preserves some optimum, rather than every tied optimum. For $n = 2^h - 1$ with every coarse weight zero, the only tree of height at most h is perfect, while every full BST is unrestricted-optimal. Two height-limited reconstructions could therefore suggest false uniqueness. The witnesses of Lemma 62 are reused at every subtree depth, with no decreasing height budget.

Theorem 63 (Common-root refinement)**.** *Assume (88). Starting at $[0, n)$, query the two root extremes of each visited nonempty interval. If they coincide, retain that root and recurse on its two children. Otherwise stop at that interval. Let the stopped interval sizes be $m_1, \dots, m_t$, and put $b = \max_j m_j$, or $b = 0$ if $t = 0$.*

A complete optimum for the original weights can be constructed in

$$O\left(\bar{S}\log(n+1) + n + \sum_{j=1}^{t} m_j^2\right) \tag{89}$$

arithmetic and dictionary operations. In particular, if the integer coarse mass $\bar{W} = \sum \bar{p}_i + \sum \bar{q}_j$ is $O(n)$, the bound is

$$O(n^{3/2}\log^{3/2}(n+1) + nb). \tag{90}$$

After preparation, only $O(n)$ data need be retained. Each further certified perturbation of the same base, with any positive scale δ, can be solved completely in $O(n + \sum_j m_j^2)$ work. If the coarse optimum is unique, this is $O(n)$ per original-weight tree.

Proof. At a coincident pair of extreme roots, every coarse-optimal tree has that root. Induction from the full interval shows that every such tree contains the retained skeleton. By Lemma 61, every fine optimum contains it as well. There are at most n visited nonempty intervals. Partition extraction therefore costs $O(n \log(n+1))$; the stopped intervals are disjoint in keys and satisfy $\sum_j m_j \le n$. There are no stopped intervals exactly when the coarse optimum is unique.

Solve each stopped interval by ordinary Knuth on its original key weights and its original gaps, including both endpoint gaps. Attach these subtrees to the retained roots. A subtree's contribution from ancestors outside it is its fixed total outcome mass times its fixed attachment depth. This term is independent of the subtree shape, so all payload optima combine independently. The family with the retained roots contains a global fine optimum; its computed minimum is consequently a global fine optimum.

The payload work is $O(\sum_j m_j^2)$, and writing, attaching, and traversing all original keys and gaps costs $O(n)$. Lemma 62 absorbs partition extraction into the preparation term, proving (89). Since $\sum_j m_j^2 \le nb$, the main linear-mass bound proves (90).

Once the partition is extracted, discard the matrices and witnesses. Retain only the $O(n)$ base weights, skeleton entries, interval endpoints and construction metadata. Each new signed-error certificate costs $O(n)$; the same independent payload argument then proves the reuse bound. The integer coarse optimum being unique gives $t = 0$. □

With the published succinct witnesses, preprocessing uses $O(\bar{S})$ space. The retained implementation uses ordinary maps, which can require $O(\bar{S} \log(n+1))$ peak space. Sequential payload solves add peak $O(b^2 + n)$ space. The arithmetic bounds exclude operand bit costs. No run, period, monotonicity, or integrality condition is imposed on the original fine-weight sequences.

Corollary 64 (An automatic full constructor)**.** *Let X be the largest original weight. For $X > 0$, test $\delta = X/m$ for $m = 1, \dots, 16$, round each original weight to its nearest nonnegative integer multiple of δ, and check (88). Use $\delta = 1$ when $X = 0$. At the first success apply Theorem 63; if none succeeds, use ordinary Knuth. This is an exact full mixed-weight constructor with $O(n^2)$ worst-case arithmetic work. On accepted inputs its work satisfies (90), hence is subquadratic when $b = o(n)$.*

Proof. There are a fixed number of $O(n)$ scale tests. At scale X/m, every rounded coarse weight is in $\{0, \dots, m\}$, so $\bar{W} \le 16(2n+1)$. The certificate and refinement theorem give correctness on acceptance, and Knuth gives correctness on rejection. Use $b \le n$ in (90) for the overall quadratic bound. Half-integer ties may be rounded upward; every comparison and certificate is exact. □

An explicit limitation remains even for arbitrarily small perturbations. For uniform unit coarse weights at $n = 2^h$, $h \ge 1$, start with the perfect $(n-1)$-key tree and expand either its leftmost or rightmost bottom gap. Both trees minimize internal path length, hence coarse cost, and their root ranks differ. The partition stops at the full interval and uses the entire quadratic payload table. The theorem therefore makes its ambiguity parameter explicit. Verification and complete operation counts are in Appendix A.26.

2.31 Adaptive refinement through the reachable optimal grammar

Root ambiguity can occur at the full interval while only a small number of subproblems participate in any coarse optimum. We can retain precisely those subproblems and optimize the original weights over their root choices. The certificate remains Lemma 61; no additional restriction is placed on individual fine-weight errors.

For a fixed integer base, start at $[0, n)$. At every reached nonempty interval $[i, j)$ retain every unrestricted optimal root r, with children $[i, r)$ and $[r + 1, j)$. Deduplicate intervals. Write V for the number of reached nonempty intervals, A for the number of retained triples (i, j, r), and

$$Q = \sum_{[i,j) \text{ reached}} (r_{\max}(i, j) - r_{\min}(i, j) + 1). \tag{91}$$

Thus $V \le A \le Q$. A root strictly between the two extremes need not be optimal, so Q counts unsuccessful candidate tests as well.

Theorem 65 (Adaptive optimal-grammar refinement)**.** *The reachable grammar can be prepared in*

$$O((\bar{S} + V + Q) \log(n + 1)) \tag{92}$$

arithmetic and comparison operations, using the aggregate coarse-core quantity $\bar{S}$ of Lemma 62. After preparation retain $O(n+V+A)$ data. Each arbitrary nonnegative fine-weight input satisfying (88) can then be solved completely in

$$O(n + V + A) \tag{93}$$

arithmetic operations, including certification and full tree reconstruction. Peak preparation space with the retained ordinary-map implementation is $O(\bar{S} \log(n + 1) + n + V + A)$.

Proof. Prepare unrestricted root extremes by Lemma 62. Also index the final condensed cost matrix C^*. For a core entry $(a, b, d_{a,b})$, the condensed representation gives

$$C^*(i, j) = C^*(i, 0) + C^*(0, j) - C^*(0, 0) - \sum_{a<i,\ b<j} d_{a,b}.$$

Maintain a persistent binary sum tree on column coordinates after each core-row prefix. Inserting a density copies $O(\log(n+1))$ nodes. There are $O(n)$ versions and K_H insertions, so this index costs $O(n + K_H \log(n + 1))$ work and space. A column-prefix query in version i evaluates any required entry in $O(\log(n + 1))$ work. In particular, the zero diagonal represents empty-child cost exactly.

Process each reached interval once. Query its root extremes. If they coincide, retain the single rule. Otherwise test every root between them by the exact equality

$$C^*(i, r) + C^*(r + 1, j) + \bar{W}(i, j) = C^*(i, j). \tag{94}$$

Coarse interval masses use prefix sums. Every retained root schedules its nonempty children. A dictionary on endpoint pairs, implemented if desired by a balanced search tree, deduplicates intervals in $O(\log(n+1))$ work per operation. There are $O(V+Q)$ such operations and tests. The final core index is covered by the $\bar{S} \log(n + 1)$ term. This proves (92).

The resulting grammar describes exactly the full coarse-optimal trees. Every coarse-optimal tree uses an optimal root and optimal children at each of its intervals, and hence appears in the grammar by induction. Conversely, combining any retained root with any optimal children attains the parent's coarse optimum by (94). Induction on interval length proves the reverse inclusion, including all combinations of child choices.

Bucket the reached intervals by their integer lengths $1, \dots, n$ in $O(n + V)$ work. Assign array indices in that order, replacing child endpoints by indices and using a sentinel for an empty child. Each child has smaller length. This compilation takes $O(A \log(n+1) + n + V)$ work with the same dictionary and is absorbed by the preparation bound. Discard the cost index, matrices and witness structures.

For each new input check (88), form original weight prefix sums, and compute in increasing length

$$D(i,j) = W(i,j) + \min_{r:\,(i,j,r)\text{ retained}} \{D(i,r) + D(r+1,j)\}, \qquad D(i,i) = 0.$$

Use original gaps at both endpoints. The compiled child indices make each root evaluation constant work with array access; no query-time dictionary is needed. Store an attaining root with its two child indices. The grammar contains every fine optimum by the certificate, so its minimum is the full original optimum. Evaluate V states and A rules, then traverse the selected rules to write all n keys. Every selected child interval is strictly smaller and the two child key sets are disjoint, so full traceback costs $O(n)$. This proves (93) and the retained-space bound. □

Counted expansion and the automatic constructor. Apply the scale tests of Corollary 64. On acceptance, attempt grammar extraction with a budget of $4n\lceil \log_2(n+1) \rceil$ charges, one for each processed interval, candidate root and retained rule. Check the budget before each charge; each accepted rule schedules at most two children. If the budget is exhausted, discard the partial grammar and use ordinary Knuth on the original weights. A certificate rejection also uses Knuth. Thus all failed expansion work is counted and bounded.

The accepted coarse mass is $O(n)$. Coarse preparation, the cost index and the bounded extraction together take $O(n^{3/2}\log^{3/2}(n+1) + n\log^2(n+1))$ work. A completed plan has $V + A = O(n\log(n+1))$, so its first full construction is subquadratic and subsequent certified queries use $O(n+V+A)$ work. The fallback gives an exact $O(n^2)$ worst-case constructor. For a supplied integer base of unrestricted mass, retain the explicit $\bar{S}$ term instead. Operand bit costs are additional. The dense uniform family of Theorem 176 explains why the budget can be necessary.

Theorem 66 (Two distant roots with a linear reachable grammar)**.** *Let $m = 2^k$ with $k \geq 1$, and put $n = 3m-1$, $a = m-1$, $c = 2m-1$. Assign unit weight to every key and gap, then increase the weights of keys a, c to two. The total coarse mass is $2n+3$. There are exactly two coarse-optimal full trees. Their roots are a and c, and their reachable grammar has*

$$V = n+1, \qquad A = n+2, \qquad Q = 4m. \tag{95}$$

The common-root partition has a single ambiguous payload of size n. The adaptive grammar instead gives $O(n)$ work for every certified fine-weight query after generic coarse preparation.

Proof. For unit mixed weights the cost is $2I + 3n$. Every minimum-IPL tree at this size fills the perfect upper skeleton with $2m-1$ keys and expands exactly m of its $2m$ bottom gaps. Its root has rank between a and c. Rank a requires no left-half expansion and all m right-half expansions; rank c requires the reverse. Each extreme therefore determines a unique full shape.

The two extra unit key weights contribute the sum of the comparison lengths of a, c. Distinct keys have comparison-length sum at least three. Both extreme shapes attain three: one marked key is the root, and the other is its child. Any other minimum-IPL shape has neither marked key at the root and contributes at least four. A tree with larger IPL already has a strictly larger baseline cost and cannot offset it with a contribution below three. Consequently the two extreme shapes are exactly the coarse optima.

Let P_0, P_1, P_2 be the perfect $(m-1)$-key trees on $[0,a)$, $[a+1,c)$ and $[c+1,n)$, respectively. The two trees are

$$T_a = a(P_0, c(P_1,P_2)), \qquad T_c = c(a(P_0,P_1),P_2).$$

Their common perfect subtrees contribute $3(m-1)$ distinct nonempty intervals. The full interval and the two different $(2m-1)$-key intervals contribute three more, giving $V = 3m = n+1$. Every

proper reached interval has a unique optimal subtree: an alternative would replace that subtree in a full optimum and create a third full optimum. Thus the full interval has two rules and each of the other $V-1$ intervals has one, yielding $A = V+1 = n+2$. The full root range contains $m+1$ candidates, and every proper interval contributes one. Hence $Q = (m+1)+(3m-1) = 4m$. The different full roots make the common-root partition stop immediately. □

For this explicitly recognized base, even coarse Monge preparation can be omitted. Construct the three perfect subtrees and choose between the two displayed trees. For arbitrary original weights their cost difference is exactly

$$C_{p,q}(T_a) - C_{p,q}(T_c) = p_c + W(c+1,n) - p_a - W(0,a). \tag{96}$$

The middle subtree has the same depth in both trees; the outer subtrees and the two marked keys change depth by one. The certificate, mass sums, choice and complete output therefore take $O(n)$ work in total. The errors at all original key and gap positions may vary independently, subject to nonnegativity and the one total-error certificate. Verification and separate preparation/query counts appear in Appendix A.28.

2.32 Screening roots by coarse cost and height

Coarse weights can also screen roots when the small-error certificate (88) fails. Retain every root whose coarse regret is within a height-based allowance for the original errors. An optimum for the original weights survives this screening. A hierarchy of compressed root-range products enumerates the surviving roots without scanning every position between their extremes.

Lemma 67 (A hierarchy of root-range minima)**.** *Let $n \ge 1$ and let C be a finite $(n+1)\times(n+1)$ Monge matrix in condensed form, with K positive densities, and put $L = \lceil \log_2(n+1) \rceil$. In $O((n+K)L^2)$ arithmetic and comparison operations and space one can index a balanced binary hierarchy of ranges $I = [\ell, u) \subseteq [0,n)$. For every hierarchy node, a query*

$$D_I(i,j) = \min_{r\in I}\{C(i,r) + C(r+1,j)\} \tag{97}$$

takes $O(L)$ work. Given any threshold t and contiguous valid-root range $J \subseteq [0,n)$, all a roots in J satisfying $C(i,r)+C(r+1,j) \le t$ can be enumerated in $O((a+1)L^2)$ work. The bounds include all products and indexing, rather than assuming constant-time access to a condensed entry.

Proof. Split each nonsingleton range at its integer midpoint; this gives $2n-1$ nodes and depth at most $\lceil \log_2 n \rceil$, also when n is not a power of two. Build the persistent entry index used in Theorem 65 for the input C.

At $I = [\ell, u)$, write $d_{a,b}$ for a density of C. The restricted first factor has exactly the densities with $\ell \le b < u-1$; let their number be K_A. The restricted second factor has those with $\ell+1 \le a < u$; let their number be K_B. Keep row coordinates

$$R_I = \{0\} \cup \{a+1 : d_{a,b} > 0,\ \ell \le b < u-1\}$$

in the first factor, and column coordinates

$$T_I = \{0\} \cup \{b+1 : d_{a,b} > 0,\ \ell+1 \le a < u\}$$

in the second. Let $\rho(i)$ and $\kappa(j)$ be the largest retained coordinates no greater than i, j. Telescoping the intervening densities gives, for every $r \in I$,

$$C(i,r) - C(\rho(i),r) = C(i,\ell) - C(\rho(i),\ell),$$
$$C(r+1,j) - C(r+1,\kappa(j)) = C(\ell+1,j) - C(\ell+1,\kappa(j)).$$

For the first identity, a density with row between $\rho(i)$ and $i-1$ and column in $[\ell, u-1)$ would have inserted a retained coordinate in $(\rho(i), i]$, contradicting the choice of $\rho(i)$. Densities outside that column range contribute the same row offset for every $r \in I$. The second identity follows by the transposed argument.

Multiply the compressed factors $C[R_I, I]$ and $C[I+1, T_I]$, retaining their product P_I. There are at most K_A+1 outer rows and K_B+1 outer columns, with $|I|$ inner positions. Increasing-coordinate selection preserves Monge structure. The published product theorem [14] gives at most $K_A + K_B$ product-core entries. Store their persistent entry index and the two offset boundary arrays. Then

$$\begin{aligned} D_I(i,j) = P_I(\rho(i),\kappa(j)) &+ C(i,\ell) - C(\rho(i),\ell) \\ &+ C(\ell+1,j) - C(\ell+1,\kappa(j)). \end{aligned} \tag{98}$$

Here product coordinates are identified with their retained original coordinates. Binary searches locate those coordinates; the two original entry queries and the compressed product query take $O(L)$ work. The other two entries are stored offsets.

At any hierarchy depth, the inner ranges are disjoint and their total length is at most n. Each original density belongs to at most one first-factor range and one second-factor range at that depth. Consequently the sum of $K_A + K_B$ at a depth is at most $2K$. Splitting a range drops densities crossing the new inner boundary; it never copies a density into both children of the same factor. Thus all factor dimensions and core sizes together are $O((n+K)L)$. Sort and deduplicate only the retained coordinates. Obtain factor boundaries from the original index and its top row and left column. The product, coordinate sorting and entry-index costs add at most one further factor L, proving the preparation and space bounds. In particular, no node allocates a fresh full-length outer boundary.

For enumeration, skip a node disjoint from J, and prune a node when its range minimum exceeds t. Otherwise descend, emitting a surviving singleton. Fully contained nodes that survive pruning have a qualifying leaf below them. Their visited descendants can be charged to the at most L ancestors of each emitted root and their pruned siblings. Partially intersecting nodes lie on the two boundary paths of J. There are therefore $O((a+1)L)$ visited nodes, each using at most one $O(L)$ range query. The same bound applies to an interrupted enumeration prefix, with the number of emitted roots plus one. □

Let the coarse nonnegative integer weights be $(\bar{p}, \bar{q})$ and the original arbitrary nonnegative weights be (p, q). Fix any $\delta > 0$. For a nonempty interval $I = [i, j)$, include all its keys and both endpoint gaps when defining

$$E_+(I) = \sum_{o \in I} (x_o - \delta \bar{x}_o)_+, \qquad E_-(I) = \sum_{o \in I} (\delta \bar{x}_o - x_o)_+.$$

Here $y_+ = \max\{y, 0\}$, and the notation $o \in I$ refers to those outcomes. Let H_f bound the height of some full original optimum. Let H_c bound the height of some coarse optimum on every interval. The existing height lemma supplies H_f, with the trivial choice $H_f = n$ always available. The number of completed coarse Monge layers supplies H_c: their final values are unrestricted and are attained within that many layers on every interval.

Lemma 68 (A signed coarse-regret allowance)**.** *For $m = j - i > 0$, put*

$$\eta(I) = (\min\{m, H_c\} - 1)E_+(I) + (\min\{m, H_f\} - 1)E_-(I). \tag{99}$$

There is a full original optimum all of whose subtree roots r satisfy

$$\bar{C}(i, r) + \bar{C}(r + 1, j) + \bar{W}(i, j) \le \bar{C}(i, j) + \eta(I)/\delta, \tag{100}$$

where $\bar{C}$ is the unrestricted coarse optimum table. This conclusion holds for every error magnitude.

Proof. Choose a full original optimum T of height at most H_f. Its subtree T_I on any of its intervals is original-weight optimal there. Compare it with a coarse-optimal tree U_I of height at most $\min\{m, H_c\}$. Every outcome comparison length in a nonempty m-key tree lies in $[1, m]$. Positive errors can increase $C_e(U_I) - C_e(T_I)$ by at most $(\min\{m, H_c\} - 1)E_+(I)$, while negative errors can increase it by at most $(\min\{m, H_f\} - 1)E_-(I)$. Original optimality therefore gives

$$\delta(C_{\bar{p},\bar{q}}(T_I) - \bar{C}(i, j)) \le C_e(U_I) - C_e(T_I) \le \eta(I).$$

The least coarse cost with the same root r is no larger than the coarse cost of T_I. Substituting that root-constrained minimum proves (100) simultaneously at every subtree of T. Only existence of one height-bounded full optimum is required; the screen need not preserve every tall tied optimum. □

Theorem 69 (Full construction through coarse-regret screening)**.** *Prepare the coarse Monge construction, with aggregate core quantity $\bar{S}$, final completed core size K, and H_c as above. In*

$$O(\bar{S}L + (n + K)L^2) \tag{101}$$

work, prepare the root-range index. For any original nonnegative input, start at $[0, n)$, retain precisely the valid roots satisfying (100), and recurse on their children, deduplicating intervals. Let V and A be the reached nonempty interval and retained root counts.

Using the sufficient-height selector of Lemma 31, the complete original optimum can be constructed in

$$O(nL + (V + A)L^2) \tag{102}$$

additional arithmetic and comparison operations. Preparation can be reused for unrelated fine-weight inputs and any positive scale. Retained index and query space is $O((n + K)L^2 + V + A)$; ordinary coarse witness maps add $O(\bar{S}L)$ during preparation.

Proof. Index the finite completed coarse matrix by Lemma 67. For each reached interval, query with valid roots $[i, j)$ and threshold $\bar{C}(i, j) - \bar{W}(i, j) + \eta(I)/\delta$. Prefix sums compute both signed error masses, coarse interval masses and original interval masses. The entry $\bar{C}(i, j)$ costs $O(L)$. Summing the enumeration bound over reached intervals gives $O((V + A)L^2)$ work. A balanced endpoint dictionary adds only $O((V + A)L)$ comparisons.

Every reached interval retains at least a coarse-optimal root because $\eta(I) \ge 0$. Thus the retained acyclic grammar has a feasible completion. By Lemma 68, it also contains the chosen full original optimum: its root survives, then each of its children is reached and its root survives, inductively. Evaluate the original root recurrence on retained rules in increasing interval length and trace an attaining full tree. The result cannot be more expensive than the surviving original optimum, and every output is a feasible original BST, so its cost is exactly optimal. It need not have height at most H_f; that height was used only to prove that an optimum survives. State ordering, recurrence evaluation and traceback fit the stated bound. Height selection costs $O(nL)$; all other input scans are linear. □

A low-mass screening attempt for every input. For a nonzero input set $b = L^3$, $\delta = X/b$, where X is its largest weight, and round each outcome to the nearest nonnegative integer multiple of δ. Then $\bar{W} \le (2n+1)L^3$ regardless of the original magnitudes. Check a budget of $4nL$ charges before each reached state and each retained root. If exhausted, stop expansion and run ordinary Knuth on the original input. Uniform key and gap sequences have their direct linear construction.

This is an exact full constructor on all nonnegative inputs. Its coarse preparation is subquadratic in the arithmetic model. In fact, the final completed core satisfies $K \le n + L\bar{W}$ by (2), while $H = O(L)$ for this polynomially bounded coarse mass. The aggregate theorem and (101) give

$$O(n^{3/2}L^3 + nL^7) = o(n^2) \tag{103}$$

preparation work. The budget bounds a successful or interrupted screening attempt by $O(nL^3)$ additional work, including partial enumeration. Consequently a successful full construction is subquadratic, and the global worst case, including every failed attempt, is $O(n^2)$. The final core bound is used only for preprocessing this index; the early-layer odd-ramp obstruction remains unchanged.

The remaining issue is the size of the screened grammar. A coarse-optimal root has zero regret and always survives, so the dense uniform family can still force quadratic states and cubic rules without a budget. The improvement is that coarse screening now applies to every original weight vector, with an explicit regret allowance, rather than stopping when the small-error certificate fails. All bounds exclude operand bit costs. Independent inclusion, range-product and full-construction checks are reported in Appendix A.29.

2.33 Ordered references without explicit cores

The root-range index in Section 2.32 can be replaced by an implicit cost oracle on a certified ordered class. A second use of the same exchange argument gives a corridor around nonlinear reference profiles. Both constructions retain the original key and gap costs.

Lemma 70 (A uniform baseline forces the shape class)**.** *For nonnegative weights on $n \ge 1$ keys, let*

$$a = \min_i p_i, \qquad b = \min_j q_j, \qquad E_0 = \sum_i (p_i - a) + \sum_j (q_j - b).$$

If

$$a + b > (n-1)E_0, \tag{104}$$

every optimal tree on every interval has minimum internal path length. If $E_0 = 0$, such an optimum also exists, including when all weights vanish. For a fixed root, a best tree has minimum-IPL children. Moreover, whenever two fixed-root classes have different minimum internal path lengths, their best costs have the same strict ordering under (104).

Proof. On an m-key interval, the uniform part has cost $(a+b)I + (a+2b)m$. Every outcome has comparison length in $[1, m]$. The residual-cost difference between any two trees is at most $(m-1)E_0(I) \le (n-1)E_0$ in absolute value. An integer increase of at least one in I therefore cannot improve the total cost. This also applies with a root fixed, and to comparisons between the least-IPL members of two fixed-root classes. For $E_0 = 0$, use the uniform identity directly. The all-zero case permits every tree. □

Theorem 71 (Implicit interval and root-range costs)**.** *Suppose the weights satisfy Lemma 70 and the leftward exchange condition (134), with $h = \lfloor \log_2(n+1) \rfloor$, or its reflection. Put $L = \lceil \log_2(n+1) \rceil$. In $O(nL)$ arithmetic work and space one can prepare an index with the following exact queries:*

1. *the unrestricted interval optimum $C(i,j)$ in $O(L)$ work;*

2. *the minimum of $C(i,r)+C(r+1,j)$ over any nonempty contiguous root range $J \subseteq [i,j)$ in $O(L)$ work;*

3. *all t roots in such a range whose cost is at most a given threshold, in $O(L^2+t)$ work.*

The preparation stores dyadic-stride prefix sums, without constructing an interval table or listing any Monge densities.

Proof. It suffices to treat the leftward condition. It holds on every subinterval: its skeleton height is at most h, and the maximum of d and hd bounds td for every integer $1 \le t \le h$. The exchange theorem and Lemma 70 therefore make the left-filled shape optimal on every interval.

Write $P(u,v) = \sum_{k=u}^{v-1} p_k$ and $Q(u,v) = \sum_{k=u}^{v-1} q_k$. For $m = j-i > 0$ put $h_m = \lfloor \log_2(m+1) \rfloor$, $c = 2^{h_m} - 1$, and $s = m - c$. Define, for positive integer limits,

$$R(o,d;u,v) = \sum_{t=u}^{v} (1+\nu_2(t)) p_{o+dt},$$

with an empty sum equal to zero. The exact full interval cost is

$$\begin{aligned} C(i,j) = {} & (h_m+1)P(i,j) + h_m Q(i,j+1) + Q(i,i+2s) \\ & - R(i-1,2;1,\min\{s,c\}) - R(i+s-1,1;s+1,c). \end{aligned} \tag{105}$$

Indeed, skeleton key t occupies input position $i+t-1+\min\{s,t\}$ and has comparison length $h_m - \nu_2(t)$. The added keys have length h_m+1. The first $2s$ gaps have length h_m+1 and the other gaps length h_m. Subtracting the skeleton's savings from the common key length h_m+1 gives the formula.

For each power-of-two stride $2^d \le n$, store

$$Z_d(k) = \sum_{\substack{t \ge 0 \\ k - t2^d \ge 0}} p_{k-t2^d} \quad (0 \le k < n).$$

These arrays require $O(nL)$ additions. A sum over a contiguous part of one residue class is a difference of two entries. Expand $1+\nu_2(t) = \sum_{d \ge 0} [2^d \mid t]$ in each R query. There are $O(L)$ resulting stride sums, proving the interval-query bound.

For root ranges, let $G(r) = C(i,r) + C(r+1,j)$ and put

$$r_* = i + 2^{h_m - 1} - 1 + \min\{s, 2^{h_m-1}\}. \tag{106}$$

We claim that G is nonincreasing up to r_* and nondecreasing after it. The least IPL with left size $\ell = r - i$ is $m - 1 + I_{\min}(\ell) + I_{\min}(m-1-\ell)$. Since $I_{\min}(k) - I_{\min}(k-1) = \lfloor \log_2 k \rfloor$, this sequence is convex. Its minimum plateau consists precisely of

$$\ell = 2^{h_m-1} - 1 + u, \qquad \max\{0, s - 2^{h_m-1}\} \le u \le \min\{s, 2^{h_m-1}\}.$$

Outside that plateau the strict cost ordering follows from Lemma 70. Within it, the best tree with that root fills the first u bottom slots on the left and the first $s-u$ on the right. Increasing u by one moves one expansion from the right block to the next empty left slot. This transformation is a sequence of swaps $01 \to 10$; each is nonincreasing by (133) and the assumed condition. Thus G is nonincreasing along the plateau, proving the claim. The uniform zero-cost case is immediate.

Clamp r_* to the queried root range. Evaluating G there gives its minimum using two interval queries. If the minimum passes the threshold, binary searches on the two monotone sides find the first and last qualifying roots. All intervening roots qualify. The searches use $O(L)$ interval-query pairs, followed by $O(t)$ enumeration. Reflection reverses the indices and preserves all bounds. □

Corollary 72 (Screening through an implicit ordered reference)**.** *Let coarse weights satisfy Theorem 71. For arbitrary nonnegative original weights and any scale $\delta > 0$, use the allowance of Lemma 68, with $H_c = L$. Preparation takes $O(nL)$ work and space. If screening reaches V nonempty intervals and retains A root rules, complete exact original construction takes*

$$O(nL + VL^2 + AL)$$

additional deterministic arithmetic and comparison operations, and $O(nL + V + A)$ space. The same preparation serves unrelated original inputs.

Proof. Each interval threshold is formed from an exact coarse cost and error prefix sums. The oracle enumerates its retained rules in $O(L^2 + t)$. Balanced endpoint dictionaries account for the AL term during scheduling and evaluation; this term is not treated as constant-time hashing. The inclusion and full-tree correctness proof is exactly that of Theorem 69. A $4nL$ state-and-rule budget bounds a successful or interrupted query by $O(nL^3)$; ordinary Knuth on budget exhaustion gives an exact quadratic worst-case constructor, including all failed work. Preparation is independent of coarse core size and weight magnitudes. Operand bit costs are additional. □

2.34 A nonlinear exchange corridor with linear-time fitting

An exchange margin can also control the whole shape optimization, avoiding interval-graph expansion. This extends the affine corridor of Section 4.13.1 to a reference profile that need not be affine.

Let $h = \lfloor \log_2(n+1) \rfloor$, $S = 2^h$, $s = n - S + 1$, and $z = S - s$. A minimum-IPL shape is a length-S binary expansion word with s ones. Its inversion count is the number of pairs consisting of a zero followed by a one. Suppose every leftward adjacent exchange of a reference input reduces its cost by at least $\gamma > 0$. Let original weights differ from that reference by total absolute key error E_p and gap error E_q.

Lemma 73 (An inversion allowance from exchange margin)**.** *Put $V = hE_p + E_q$ and $D = \lfloor V/\gamma \rfloor$. Every original optimum within the minimum-IPL class has at most D inversions. In particular, every prefix with a zeros already used and b ones still required satisfies $ab \le D$.*

Proof. Sorting an expansion word leftward takes exactly its inversion count in adjacent swaps. Its reference cost therefore exceeds the left-filled reference cost by at least γ times that count. Among minimum-IPL trees, each key length varies by at most h and each gap length by at most one. Original errors can compensate for at most $hE_p + E_q$ of that reference excess. An original class optimum cannot cost more than the left-filled candidate, which proves the inversion bound. Every already used zero precedes every required future one, giving ab inversions. □

Theorem 74 (Nonlinear reference corridor construction)**.** *An original minimum-IPL optimum can be constructed using at most*

$$S + 1 + \min\{sz, D\lceil \log_2(z+1) \rceil\} \tag{107}$$

states and constant work per state, in addition to fitting and linear input/output work. For arbitrary nonnegative original weights, a unit-margin reference can be fitted in $O(n)$ work: set all reference gaps to $\min q$*, and recursively set*

$$\bar{p}_0 = p_0, \qquad \bar{p}_i = \max\{p_i, \bar{p}_{i-1} + 1\}.$$

Compute E_p, E_q directly and use $\gamma = 1$. One may also fit the reflected input and choose the smaller allowance. If the original input passes Lemma 70, the resulting tree is a full unrestricted optimum. Otherwise ordinary Knuth gives an exact fallback. The complete constructor has $O(n^2)$ worst-case arithmetic work and space.

Proof. The fitted key sequence increases by at least one and its gap sequence is constant. Formula (133) makes every leftward exchange improve reference cost by at least one. The fit and error scans are linear.

Retain precisely the feasible expansion-grid states satisfying $ab \le D$. The axes contain $S + 1$ states. For each $1 \le a \le z$, at most $\min\{s, \lfloor D/a \rfloor\}$ positive b survive. Dyadic grouping of a bounds their sum by $D\lceil \log_2(z+1) \rceil$; the full rectangle has only sz such states. This proves (107).

On layer g, after g slots and e expansions, $a = g - e$ and $b = s - e$. Both factors decrease with e, so the retained expansion counts form a single integer interval. Its lower endpoint is nondecreasing with g; a pointer advances at most s times over the whole computation. Every retained state is reachable by placing its used ones first, then its used zeros: all preceding products are at most ab. Completing with the required ones before the remaining zeros also stays in the corridor. Use compact arrays for each layer and one predecessor choice per retained state. There is no dictionary or full-width array per layer.

For the next slot, put $i = g + e$ and $d = h - \nu_2(g+1)$. Choosing no expansion charges hq_i and, if $g < S - 1$, dp_i. Choosing an expansion charges $(h+1)(p_i + q_i + q_{i+1})$ and, if $g < S - 1$, dp_{i+1}. These are the original weights. Each state has at most two outgoing transitions. Every original class optimum survives by Lemma 73; retained extra paths are feasible minimum-IPL trees and cannot invalidate the optimum. Traceback produces the expansion word, its key comparison lengths, and the full tree in linear output work. The baseline lemma supplies global optimality on acceptance. Uniform inputs have their direct linear construction. On rejection only linear checks precede Knuth, and the entire expansion grid itself has $O(n^2)$ states. □

Corollary 75 (Bounded errors around arbitrary increasing profiles)**.** *Let $f_0 = 0$, $f_{i+1} - f_i \ge 1$, and $F = \sum_i f_i$. For arbitrary errors $|u_i|, |v_j| \le U$, let*

$$p_i = B + f_i + u_i, \qquad q_j = B + v_j,$$

with nonnegative resulting weights. If

$$2B - 2U > (n-1)(F + 2U(2n+1)), \tag{108}$$

the automatic constructor of Theorem 74 takes $O(n + Un\log^2(n+1))$ arithmetic work and space, including fitting, certification and full reconstruction. The same holds after reflection. For fixed U this is $O(n\log^2(n+1))$. The profile f is arbitrary subject to its increments; for example $f_i = i + \lfloor i^2/n \rfloor$ is nonlinear. Original perturbations can occur independently on every key and every gap, and need not preserve their ordering.

Proof. The prefix-envelope fit obeys

$$B + f_i - U \le \bar{p}_i = \max_{0 \le k \le i} (p_k + i - k) \le B + f_i + U,$$

because $f_i - f_k \geq i - k$. Since $\bar{p}_i \geq p_i$, this gives $E_p \leq 2Un$. The constant gap fit gives $E_q \leq 2U(n+1)$. Hence $V \leq 2U(hn + n + 1)$, and (107) gives the work bound. The original minimum key and gap weights are at least $B - U$; their total residual mass is at most $F + 2U(2n+1)$. Condition (108) therefore implies (104), completing the unrestricted certificate. The implementation infers its reference from the given weights; it requires neither f, B, nor U as an additional input. □

These two ordered-reference constructions address different costs: the interval oracle supports reusable arbitrary-weight screening, while the inversion corridor supplies a proved small state count for a nonlinear perturbation family. The unrestricted subquadratic objective still requires a representation or certificate covering arbitrary weight profiles. Verification appears in Appendix A.30.

2.35 Range-sensitive shape certification

The total residual mass in Lemma 70 can be replaced by a shortening-capacity bound when only a full-interval shape certificate is required. This gives an alternative baseline requirement for the nonlinear corridor, and an exact universal threshold at perfect-tree sizes.

For $n \geq 1$, put $L = \lceil \log_2(n+1) \rceil$ and define

$$K_L = 2^L - L - 1, \qquad G_L = \begin{cases} 0, & L = 1, \\ 2^{L-1}, & L \geq 2. \end{cases}$$

Write $\ell_i(T)$ for key comparison lengths and $d_j(T)$ for gap depths.

Lemma 76 (Shortening capacities)**.** *For every full n-key tree T,*

$$\sum_i (L - \ell_i(T))_+ \leq K_L, \qquad \sum_j (L - d_j(T))_+ \leq G_L.$$

If $n = 2^h - 1$, the stronger gap bound

$$\sum_j (h - d_j(T))_+ \leq I(T) - I_{\min}(n) \tag{109}$$

holds.

Proof. At most $2^t - 1$ keys have comparison length at most t. Summing this bound for $t = 1, \ldots, L-1$ gives K_L. For an integer $u \geq 1$, $u \leq 2^{u-1}$. Thus each gap with $d_j < L$ contributes at most $2^{L-1}2^{-d_j}$. Kraft equality $\sum_j 2^{-d_j} = 1$ gives the gap bound; for $L = 1$ both gaps have depth one.

At perfect size put $k_j = d_j - h$, and let $A = \sum_j (-k_j)_+$ and $Z = \sum_j (k_j)_+$. Kraft equality, multiplied by 2^h, gives

$$\sum_{k_j < 0} (2^{-k_j} - 1) = \sum_{k_j > 0} (1 - 2^{-k_j}).$$

The left side is at least A, while the right side is at most $Z/2$, using $2^t - 1 \geq t$ and $1 - 2^{-t} \leq t/2$ for positive integers t. Hence $Z \geq 2A$. External path length differs from internal path length by $2n$, so $I(T) - I_{\min} = \sum_j k_j = Z - A \geq A$. □

Corollary 77 (A full shape certificate from weight ranges)**.** *Let $a = \min p$, $b = \min q$, $P = \max p - a$, and $Q = \max q - b$. If*

$$a + b \geq PK_L + QG_L, \tag{110}$$

a minimum-IPL optimum is a full unrestricted optimum. The certificate is computable in $O(n)$ work. It can be combined with the earlier mass certificate by using the smaller sufficient threshold.

Proof. Every minimum-IPL tree S has all key and gap lengths at most L. For any T with excess $e = I(T) - I_{\min} \geq 1$, nonnegative residuals and Lemma 76 imply

$$C(T) - C(S) \geq (a+b)e - PK_L - QG_L \geq 0.$$

Thus optimizing within the class gives a global optimum. Equality in the certificate is sufficient; an outside tree may tie. □

Theorem 78 (Sharp universal box threshold at perfect sizes)**.** *Let $n = 2^h - 1$, $h \geq 2$, and let $a, b, P, Q \geq 0$. The perfect tree is optimal for every fixed-position assignment*

$$p_i \in [a, a+P], \qquad q_j \in [b, b+Q]$$

if and only if

$$a + b \geq (n-h)P + Q. \tag{111}$$

On acceptance it can be constructed in $O(n)$ work and space, including reading the weights and producing the full tree. This is a universal statement about the entire box; particular assignments may remain optimal when the inequality fails. For $n = 1$ there is only one tree.

Proof. The perfect tree S is the unique minimum-IPL shape. For any other T put $e = I(T) - I_{\min} \geq 1$. The key shortening bound is $K_h = n - h$; (109) bounds gap shortening by e. Therefore

$$C(T) - C(S) \geq (a + b - Q)e - P(n-h).$$

Under (111), this is at least $P(n-h)(e-1) \geq 0$. Necessity follows from the explicit one-unit-excess construction in Theorem 139, which attains cost difference $a + b - (n-h)P - Q$. □

Corollary 79 (Nonlinear corridor with a range baseline)**.** *Use the nonlinear profile and error assumptions of Corollary 75, and write $M = f_{n-1}$. Its complete $O(n + Un\log^2(n+1))$ construction bound also holds under*

$$2B - 2U \geq (M + 2U)K_L + 2UG_L. \tag{112}$$

In particular $B = n(M + 4U) + U + 1$ suffices. For the curved profile $f_i = i + \lfloor i^2/n \rfloor$ and fixed U, a baseline $B = O(n^2)$ suffices, whereas the earlier total-mass condition requires order n^3.

Proof. The original baselines are at least $B - U$, and their ranges are at most $M + 2U$ and $2U$. Apply Corollary 77. The same automatic prefix fit still gives $E_p \leq 2Un$ and $E_q \leq 2U(n+1)$, so the corridor state and construction bounds are unchanged. Since $K_L < 2n$ and $G_L \leq n$, the displayed sufficient choice of B satisfies (112). No profile or error bound needs to be supplied to the implementation. □

A full constructor first scans the ranges and the earlier mass threshold. At a perfect size it also tests (111); acceptance returns the perfect tree directly. Otherwise, acceptance permits the original-cost nonlinear corridor, and rejection invokes ordinary Knuth. Every successful and failed route is included in the quadratic worst-case work and space bound. The all-interval root-range oracle retains its separate hypotheses. The exact box threshold does not resolve arbitrary-profile construction; it identifies precisely the worst perturbation scale for this shape class at perfect sizes. Verification is in Appendix A.31.

3 Centering around large constant baselines

The total-mass bound can be pessimistic when large constant key and gap weights dominate smaller variations. We retain the original weights in the solver and center only the analysis of its matrix representation. Let

$$a = \min_i p_i, \quad b = \min_j q_j, \quad u_i = p_i - a, \quad v_j = q_j - b, \quad R = \sum_i u_i + \sum_j v_j.$$

All quantities remain nonnegative integers. The two baselines may differ arbitrarily; the deviations may occur at arbitrary key and gap positions. Write $L = \lceil \log_2(n+1) \rceil$ as before.

Lemma 80 (Normalized height budget)**.** *For $W > 0$, let s be the smallest positive singleton mass $p_i + q_i + q_{i+1}$. Put $z = 0$ if every singleton mass is positive and $z = L$ otherwise. Every interval has an optimal tree with height at most*

$$H = \min\{n, f(\lfloor W/s \rfloor) + z\}.$$

For $W = 0$, use $H = L$.

Proof. Every positive-mass nonempty interval contains a positive-mass singleton: choose a positive key, or an adjacent key within the interval if a gap supplies its positive mass. Thus every positive internal subtree has mass at least s. Lemma 2 gives $W \geq F_{d+1}s$ for such a subtree at depth d in an optimal tree. The positive part has the resulting bounded depth; zero-mass subtrees can be balanced, adding at most L levels without changing the cost. If all singleton masses are positive no nonempty zero subtree exists. The global W, s, L bounds also apply to every subinterval. The trivial height n and the all-zero balanced case complete the argument. □

Use this H, $M = (H+1)W + 1$, and the same finite completion and penalty iteration. All previous preservation arguments apply. Reconstruction is complete because any unfinished nonempty terminal costs at least M, whereas an optimal tree of height at most H costs at most HW.

3.1 The exact baseline layer and its exceptional diagonals

Let $I(k)$ be the minimum unweighted internal path length for k keys. For $k \geq 1$, with $d = \lfloor \log_2 k \rfloor$,

$$I(k) = dk - 2^{d+1} + d + 2, \qquad I(0) = 0.$$

The familiar baseline identity is $(a+b)I(T) + (a+2b)k$ for a full k-key tree. It was already used by the minimum-path-length certificate; the following application concerns every partial-tree layer.

Lemma 81 (Centered partial-tree formula)**.** *With $c_h = 2^h - 1$, define*

$$F_h(k) = \begin{cases} (a+b)I(k) + (a+2b)k, & k \leq c_h, \\ M(k - c_h) + (a+b)h(k+1) - ac_h, & k > c_h. \end{cases}$$

Then the full-weight layer admits

$$A_h(i,j) = F_h(j-i) + U_h(i,j), \qquad 0 \leq U_h(i,j) \leq hR(i,j),$$

where $R(i,j) = \sum_{i \leq r < j} u_r + \sum_{i \leq r \leq j} v_r$. The baseline's density at interval length $k \geq 1$ is

$$F_h(k+1) + F_h(k-1) - 2F_h(k) = \begin{cases} a+b, & k = 2^t - 1, \quad 1 \leq t < h, \\ M - b, & k = 2^h - 1, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. The penalty forces exactly $\min(k, c_h)$ explicit keys. If $k \le c_h$, a full minimum-path-length tree fits in h levels. Otherwise a partial tree exposing c_h keys must fill every position in the first h levels. Its explicit keys have baseline cost $a\sum_{t=0}^{h-1}(t+1)2^t$; unfinished keys cost $ah(k-c_h)$, and all $k+1$ gaps cost $bh(k+1)$. This gives the second branch of F_h. Nonnegative deviations cannot reduce the baseline optimum. Conversely, a baseline-optimal partial tree charges every deviation at most h times, proving the residual bounds. The displayed densities follow by taking second differences of the two branches. In particular, changes in the full-tree slope occur only at $k = 2^t - 1$. □

The residual U_h need not be Monge. The actual layer is Monge, and on any square whose density cells avoid the exceptional diagonals, its density equals the residual density. Baseline magnitudes do not increase the number of exceptional diagonals.

3.2 A weighted packing argument

Remove all density cells on $j - i = 2^t - 1$ within the domain, and the main diagonal, accounting for $O(nL)$ cells. Tile the remainder with maximal aligned dyadic squares that avoid those lines and the outer boundary. For a side-s tile $T = [x, x+s) \times [y, y+s)$, let $J_T = [x, y+s]$ be its spanning input interval, including both gap endpoints.

Lemma 3 gives $O(nL/s)$ tiles at scale s.

Lemma 82 (Per-position packing)**.** *At a fixed side length s, every input position belongs to the spanning intervals of $O(n/s)$ accepted tiles. Consequently*

$$\sum_{T:\,\mathrm{side}(T)=s} R(J_T) = O(nR/s). \tag{113}$$

Proof. Fix an input position and charge each tile to a boundary met by its parent. A tile charged to a diagonal at distance $d \ge s$ lies within $O(s)$ of that diagonal. For its spanning interval to contain the fixed position, its row start lies in an interval of length $O(d+s)$. Row starts are spaced by s, with only $O(1)$ eligible column starts per row near that diagonal. Thus it contributes $O(1 + d/s)$ covers. The distances $d = 2^t - 1$ are geometric, so summing these bounds gives $O(n/s)$. All distances $d < s$ lie together in an $O(s)$-wide strip near the main diagonal and contribute only $O(1)$ covers of the fixed position. Outer-boundary tiles number $O(n/s)$ in total. Charging each nonnegative deviation to its covering intervals proves (113). The outer-boundary case is covered by Lemma 3, including arbitrary n. □

On an ordinary tile the density sum telescopes to

$$U_h(x, y+s) + U_h(x+s, y) - U_h(x, y) - U_h(x+s, y+s) \le 2hR(J_T).$$

The true densities are nonnegative integers. The number of nonzero entries on a tile is therefore bounded both by s^2 and by that density sum. Using the tile count and (113), the number across all tiles of side s is $O(\min\{nLs, hnR/s\})$. Splitting the dyadic geometric sum at $s \asymp \sqrt{hR/L}$ gives $O(n\sqrt{hRL})$; for $R = 0$ every ordinary density vanishes. No cancellation of negative residual densities on the removed lines is assumed.

Theorem 83 (Centered core construction)**.** *For arbitrary nonnegative integer mixed weights, the normalized-height penalty construction has completed layer cores bounded by*

$$K_c = O\Big(n + \min\{n^2, nL + n\sqrt{HRL}\}\Big).$$

Using the published compact product and witness algorithms, a complete optimum can be constructed in $O(HK_c \log(n+1))$ arithmetic operations. Published succinct witnesses use $O(HK_c)$ words; the implemented ordinary maps can require $O(HK_c \log(n+1))$ words.

Proof. Add the $O(nL)$ removed cells and n completion densities to the ordinary tile bound, and use $h \le H$. The initial layer has only $O(n)$ core entries. The quadratic cap is immediate. All products, conversion steps, and root queries operate on the actual condensed matrices, so the already established product reduction applies. The previous $O(n+n\sqrt{WL})$ bound is also available; either bound may be used when smaller. □

Corollary 84 (Arbitrarily large baselines with polynomial deviation)**.** *If R is polynomial in n, exact full construction takes*

$$O\Big(n(1+\sqrt{R})\log^3(n+1)\Big)$$

arithmetic operations, independently of the magnitudes of a, b in the operation count. In particular, $R = O(n)$ gives $O(n^{3/2}\log^3(n+1))$, and $R = O(n^{2-\varepsilon})$ gives $O(n^{2-\varepsilon/2}\log^3(n+1))$ for fixed $0 < \varepsilon \le 2$.

Proof. If $a + b = 0$, then $W = R$, and $s \ge 1$ whenever $W > 0$. Otherwise every singleton has mass at least $a + 2b \ge 1$, and

$$W = na + (n+1)b + R \le n(a+2b) + R, \qquad W/s \le n + R.$$

Thus $H = O(\log(n+R+1)) = O(\log(n+1))$ for polynomial R, including the possible additional L and the all-zero case. Substituting in the theorem gives the result. □

For example, $p_i = n^c + u_i$ and $q_j = n^d + v_j$, with fixed nonnegative integer c, d and bounded nonnegative integer deviations, satisfy the linear-R condition even when W is superlinear. Intermediate integers then still use $O(\log n)$ bits. Arbitrary larger baselines retain a bit-cost dependence on their representation lengths. These families permit nonuniform weights on both keys and gaps. The earlier total-mass result has a smaller logarithmic factor when $W = O(n)$.

4 A common certificate and its constructive corollaries

The following pattern applies to all the exact shortcuts in this section. Every candidate and bound uses the original fixed-position key and gap objective, or a stated tree-independent cost transfer.

Theorem 85 (Certificate pattern)**.** *Let $\mathcal{T}$ be the full set of feasible ordered search trees. Suppose an algorithm constructs a candidate T of cost V and a lower bound $B \le \min_{S\in\mathcal{T}} C(S)$. If $V = B$, the candidate is optimal. More generally, suppose T minimizes cost in a class $\mathcal{F}$ and $B_{\text{out}} \le \min_{S\notin\mathcal{F}} C(S)$. Then $V \le B_{\text{out}}$ certifies optimality. If candidate construction and bound evaluation take $t(n)$ operations including reconstruction, the accepted route takes $O(t(n))$ operations. Exact fallback gives $O(t(n)+n^2)$ worst-case work.*

Proof. In the first case $B \le \text{OPT} \le V = B$. In the second, every tree inside $\mathcal{F}$ costs at least V, and every tree outside costs at least $B_{\text{out}} \ge V$. Adding the work of the two possible stages gives the running-time statement. □

Each family corollary below supplies a candidate construction and proves that one of these comparisons succeeds for every member of the family. A failed comparison leaves an exact fallback available; it does not imply that the candidate itself is suboptimal. The intervening lemmas supply the bounds and construction algorithms used by those corollaries.

4.1 A sharper single-key contraction

A key is a *cherry* when its two children are external nodes. Replacing a cherry with one gap is exact: if the key and its gaps have weights p, x, y, their total is $s = p + x + y$. A cherry rooted at depth d contributes $s(d+1)$, whereas the replacement gap contributes sd.

Theorem 86 (Single-key rule)**.** *Let a key of weight p have adjacent gap weights x, y. If its immediate predecessor and successor keys have weights a, b and*

$$\min(a, b) \;\geq\; 2p + x + y,$$

there is an optimal tree in which this key is a cherry. At the left endpoint, it suffices that the successor weight is at least $p + x$; at the right endpoint, it suffices that the predecessor weight is at least $p + y$. Consequently contraction to a gap of weight $s = p + x + y$ satisfies $\mathrm{OPT}(\textit{original}) = \mathrm{OPT}(\textit{contracted}) + s$.

Proof. Consider the key at depth d in any tree. If its left subtree is nonempty, the immediate predecessor has depth $d + l$, $l \geq 1$. If its right subtree is nonempty, the immediate successor has depth $d + r$, $r \geq 1$. Set the corresponding distance to zero for an empty internal side.

When $l \geq 1$, move the predecessor to the chosen key's position, replacing the predecessor's old position by its left child. Insert the chosen key as a cherry at the leftmost gap of the former right subtree. Inorder order is preserved. All other outcomes either retain their depths or become shallower. The predecessor saves la, and the cost change is at most

$$-la + (r+1)p + (r-l+1)x + y. \tag{114}$$

The mirrored successor exchange has change at most

$$-rb + (l+1)p + x + (l-r+1)y. \tag{115}$$

If both sides are nonempty and $l \geq r$, substituting $a \geq 2p + x + y$ in (114) gives

$$(r + 1 - 2l)(p + x) + (1 - l)y \leq 0.$$

Use (115) when $r \geq l$. If only the left side is nonempty, $r = 0$ and $a \geq p+y$ makes (114) nonpositive. The other case is symmetric. The stated interior condition implies these one-sided conditions. At a global endpoint only the appropriate one-sided case is possible. With both sides empty the key is already a cherry.

Applying the exchange to an optimum proves the existence claim. Contraction subtracts exactly s. Conversely, every contracted tree expands to a valid original tree at additional cost s, proving equality. □

The weaker condition $\min(a, b) \geq p + x + y$ is false. For $p = (24, 14, 24)$ and $q = (1, 0, 9, 0)$, the middle key has local mass 23 and neighbors of weight 24. Contracting it forces cost 142, while the unrestricted optimum costs 130.

Maintain a doubly linked list of remaining keys and weighted gaps. Initially queue every key; after a contraction, only its predecessor and successor need rechecking. At most $3n$ queue entries are processed. Saving the contracted tree at the new gap permits reconstruction. Thus this pass takes $O(n)$ operations and space, and a Knuth solve on the remaining m keys gives total time $O(n + m^2)$. No confluence or minimum-core-size property is claimed.

4.2 Whole-block contraction

Let $B = [i, j)$ be a nonempty proper interval, including its endpoint gaps q_i, q_j. Put $W = W(i, j)$ and $C = C(i, j)$, with costs measured from a stand-alone root of depth zero.

Theorem 87 (Block rule)**.** *If both boundary keys exist and their weights satisfy*

$$p_{i-1} \geq W + C, \qquad p_j \geq W + C,$$

there is an optimum in which all of B forms one subtree. For a prefix or suffix block, it suffices that the sole boundary key has weight at least C. Contracting the block to one gap of weight W then gives

$$\mathrm{OPT}(\textit{original}) = \mathrm{OPT}(\textit{contracted}) + C.$$

Proof. In any tree, let v be the lowest common ancestor of gaps q_i, q_j, at depth d. Its key lies in B, and every outcome of B descends from v. Hence the original contribution of B is at least $(d+1)W$.

Within the subtree rooted at v, delete the keys of B from the inward fringes of its two children. On the left, a deleted key has no retained right child and can be suppressed in favor of its left child; on the right the process is symmetric. Let the retained outside trees be A, D. All their retained outcomes are at depths no greater than before. If A is nonempty, its maximum is boundary key $i - 1$; if D is nonempty, its minimum is boundary key j.

Suppose both are nonempty, and their boundary keys had original depths $d + l, d + r$. If $l \geq r$, remove the maximum key from A, replacing its position by its left child, and make that key the new root at v. Its left child is the remainder of A and its right child is D. Put an optimal tree for B at the leftmost external position of D. This position has depth at most $d + r + 1$. No outside outcome other than the promoted boundary moves deeper, and that boundary saves lp_{i-1}. The increase in the contribution of B is at most

$$C + (d + r + 1)W - (d + 1)W = C + rW.$$

Since $l \geq r \geq 1$ and $p_{i-1} \geq W + C$, the saving is at least $l(W + C) \geq rW + C$. The mirrored construction applies for $r \geq l$.

If only one retained outside tree is nonempty, promote its inward boundary key to v and attach the optimal block as the other child. The block increase is at most C, while that boundary saves at least its own weight. The interior hypothesis suffices, and for a global prefix or suffix the weaker stated hypothesis suffices. If both outside trees are empty, B already occupies the subtree at v and can be optimized in place.

The construction never increases cost. Apply it to an optimum. A block at depth e contributes $C + eW$, while its replacement gap contributes eW. Contraction and expansion now prove the equality. □

A balanced k-key tree has height $h_k = \lceil \log_2(k + 1) \rceil$ and cost at most $h_k W$. Thus the interior test

$$\min(p_{i-1}, p_j) \geq (1 + h_k)W$$

and endpoint test $p_{\mathrm{boundary}} \geq h_k W$ certify a block without first solving it. These are sufficient conditions, not characterizations.

4.3 An adaptive linear selection pass

Build the maximum Cartesian tree of the key-weight sequence, breaking ties deterministically. Its subtree intervals form a laminar family: intervals are disjoint or nested. Compute their sizes and masses by one traversal and prefix sums. Test each proper interval with the balanced-tree bound above. This inspects only $O(n)$ candidates. The restriction to Cartesian intervals is an algorithmic choice, not a claim that all useful blocks occur there.

For a threshold b, select the maximal certified intervals whose sizes are at most b. Different selected intervals have a surviving separator key between them. Contracting one therefore leaves every other selected interval's weights and boundary-key weights unchanged, so the block theorem applies successively. Write their sizes as $k_1, \dots, k_s$, and $m = n - \sum_t k_t$. The exact number of nonempty interval-DP states is

$$S(b) = \tau(m) + \sum_t \tau(k_t), \qquad \tau(u) = \frac{u(u+1)}{2}.$$

We choose a threshold minimizing $S(b)$ without running these DPs. For every certified interval I of size k, find its maximal proper certified descendants J. Immediately before I becomes eligible, these descendants are its selected blocks. Its activation therefore changes the core size and block-state total by

$$\Delta m = -\left(k - \sum_J |J|\right), \qquad \Delta S_{\text{blocks}} = \tau(k) - \sum_J \tau(|J|).$$

These sums are computed bottom-up. Bucket activation events by integer size $1, \dots, n$ and scan the buckets, maintaining m and the block-state total. There are $O(n)$ events and buckets. Equal-size intervals are disjoint; evaluate a threshold after processing its entire bucket. The empty selection $b = 0$ is included.

Corollary 88. *The selected threshold and blocks are found in $O(n)$ arithmetic operations and space. Solving the blocks and surviving core by Knuth's method and expanding their saved trees takes*

$$O\left(n + m^2 + \sum_t k_t^2\right)$$

time and $O(n + m^2 + \max_t k_t^2)$ space when block tables are released after each solve. The chosen state count is no larger than $n(n+1)/2$. Worst-case time remains $O(n^2)$.

The minimization is only over these size thresholds in this candidate hierarchy, not over every possible partition. It does not minimize wall-clock time. Preprocessing, table allocation, root scans, and reconstruction still count as work.

For perspective, a certified partition with $m = O(n^{2/3})$ surviving keys and every $k_t = O(n^{1/3})$ gives $O(n^{4/3})$ time, since $\sum k_t^2 \le (\max k_t)n$. This is a conditional structural consequence, not an unrestricted bound.

4.4 Recursive construction with a bounded preprocessing depth

The recursive constructor combines the proved components. At each call it applies the single-key pass, compares the order-free lower bound of Section 5.1 with a feasible bisection-tree cost, and returns immediately on equality. Otherwise it selects blocks as above. It recurses only if their combined table-state estimate is at most three quarters of the full estimate; small components and unsuccessful decompositions fall back to Knuth's method.

Suppose a component has $m > 8$ keys after the single-key pass and is successfully decomposed. Every child component of size s satisfies $s(s+1) \le \frac{3}{4}m(m+1)$, hence $s < 0.92m$. Child sizes sum to m. Consequently the recursion has logarithmic depth, and the total number of keys processed on any recursion level is at most n. Linear selection and reconstruction cost $O(n \log n)$ in total. The heap-based equality checks cost $O(n \log^2 n)$ over all levels. If the terminal DP sizes are r_t, the construction therefore takes

$$O\left(n \log^2 n + \sum_t r_t^2\right)$$

arithmetic operations. Its worst case is still quadratic. This threshold prevents repeated expensive scans after arbitrarily small reductions. Correctness follows by contraction, exact equality certificates, and induction on the strictly smaller recursive components; reconstruction restores the original key indices and gap subtrees.

4.5 A certified shortcut through nearly complete shapes

Let $I(T)$ be the sum of internal-node depths, with root depth zero. The external path length is $I(T) + 2n$, by induction on the two subtrees. Consequently uniform key weight a and gap weight b give cost

$$(a+b)I(T) + (a+2b)n.$$

For an arbitrary instance set $a = \min_i p_i$, $b = \min_j q_j$, and let H' be the order-free mixed-node Huffman bound of Section 5.1 on residual weights $p_i - a, q_j - b$. These weights remain nonnegative. Write $I_*(n)$ for the minimum possible unweighted internal path length, and let C_* be the best weighted cost among all trees with $I(T) = I_*(n)$.

Corollary 89 (Minimum-path-length certificate)**.** *For $n \ge 1$, if*

$$C_* \;\le\; (a+b)(I_*(n)+1) + (a+2b)n + H',$$

then a tree attaining C_* is a globally optimal full OBST.

Proof. Split each outcome weight into its uniform baseline and residual. Every tree outside the minimum-I class has integer $I(T) \ge I_*(n) + 1$, and its residual cost is at least H'. Its full cost is therefore at least the right-hand side. Within the class, C_* is minimal by definition. Equality is sufficient, including when $a + b = 0$. □

Optimize the class without interval tables. Put $h = \lfloor \log_2(n+1) \rfloor$, $S = 2^h$, $m = S - 1$, and $r = n - m$. Every minimum-I tree has the complete upper skeleton with m keys, and expands exactly r of its S external slots into single-key cherries. Indeed, a missing shallower slot could otherwise receive a deepest internal leaf, strictly reducing I. Inorder relabeling preserves search-tree feasibility in this unweighted shape argument. Thus

$$I_*(n) = (h-2)2^h + 2 + hr.$$

Process skeleton slots $j = 0, \ldots, S-1$ in inorder. State (j, e) records the number of processed slots and expansions, so the next input index is $i = j + e$. An unexpanded slot charges hq_i; an expanded slot charges $(h+1)(p_i + q_i + q_{i+1})$. Except after the last slot, also charge the next skeleton key at comparison length

$$d_j = h - v_2(j+1),$$

using p_i after an unexpanded slot and p_{i+1} after an expansion. Here v_2 counts trailing zero bits. Exactly r expansions are required, so state (j, e) is retained only when

$$\max(0, r - (S - j)) \le e \le \min(r, j).$$

With $s = \min(r, S - r)$, each stage has at most $s + 1$ states and each state has two outgoing choices. This computes C_* and backpointers in $O(n(1 + s))$ time and space. A linear Cartesian-tree pass converts the chosen comparison lengths into the labeled tree. No key or gap weights are discarded in these transitions.

There is an exact early rejection test. Let F be the cost obtained by assigning independently sorted key and gap weights to the shallowest depths of a minimum-I shape. This is feasible for the order-free residual relaxation, so $H' \le F - [(a + b)I_* + (a + 2b)n]$. Therefore $C_* > F + a + b$ proves the certificate cannot succeed. Otherwise the sorted lists are reused in a two-queue Huffman implementation: nonnegative weights and nondecreasing internal weights make the sequence of merged weights nondecreasing. This adds linear work after sorting. The queue optimization is standard and is not claimed as novel.

The implementation attempts this DP only when $s \le \lfloor \log_2 n \rfloor + 1$. Including the residual bound, preprocessing, and reconstruction, a successful attempt costs $O(n \log n)$ arithmetic operations and space. A skipped or failed attempt falls back to Knuth, or to the preceding recursive construction. Both versions remain exact and have quadratic worst-case time. The test is a certificate, not an assumption that near-uniform inputs have balanced optima.

Optimize the baseline rather than fix it. The largest subtractable baseline need not give the strongest certificate. Set $a_0 = \min p_i$, $b_0 = \min q_j$, and use $a = a_0 - \delta$, $b = b_0 - \delta$ for an integer $0 \le \delta \le \lfloor (a_0 + b_0)/2 \rfloor$. The residual weights remain nonnegative and $a + b \ge 0$. Although one baseline term may become negative, it is only an algebraic decomposition; the input probabilities are unchanged. Let H_δ denote the residual order-free optimum. The exclusion bound is

$$B(\delta) = (a_0 + b_0 - 2\delta)(I_* + 1) + (a_0 + 2b_0 - 3\delta)n + H_\delta.$$

Every tree outside the minimum-I class costs at least $B(\delta)$. Thus any sampled price with $C_* \le B(\delta)$ certifies the same candidate.

For a fixed residual tree, increasing δ adds one unit to every key and gap weight, changing its cost at rate $2I + 3n$. Consequently B is a pointwise minimum of affine functions, hence concave. A minimizing residual tree supplies a supergradient

$$g = 2(I - I_* - 1), \qquad B(x) \le B(\delta) + g(x - \delta).$$

If $g > 0$, smaller prices cannot improve the current value; if $g < 0$, larger prices cannot improve it. If $g = 0$, the value is globally maximal. Binary interval elimination, retaining the best sample, therefore finds the maximum over allowed integer prices when run to completion.

The same linear oracle supplies I. Give every original outcome secondary weight1, order queue items lexicographically by residual mass and number of original outcomes, and sum both merge coordinates. The first gives H_δ and the second gives $2I+3n$. This is the Huffman exchange argument with an infinitesimal positive common perturbation, choosing minimum I among residual-cost ties. The weight order is independent of δ, so no additional sorting is needed. Each price costs $O(n)$.

The fixed-profile ceiling extends to all prices: $B(\delta) \le F + a_0 + b_0 - 2\delta$. Therefore an input rejected by the existing profile test cannot be rescued by this search. The implementation tries prices only after the old certificate fails while passing that test, and permits at most $\lfloor \log_2 n \rfloor + 1$

extra oracle calls. This preserves the $O(n \log n)$ attempted-shortcut bound. Reaching the cap does not establish that the best price was found; every sampled certificate is still valid, and failure invokes the exact fallback.

4.6 Exact bounded-excess construction by dyadic intervals

The candidate class can be expanded beyond minimum internal path length. This is necessary even on some mildly nonuniform inputs. For $n = 255$, set all gap weights to 16 and all key weights to 16, except keys $8j + 1$ which have weight 17. The unique minimum-I tree costs 61,680; Knuth gives optimum 61,656, attained with $I = I_* + 2$. The algorithm below constructs and certifies this optimum. It retains both kinds of weight and never assumes that the restricted optimum is globally optimal.

An exact representation. An ordered full binary tree is a tiling of $[0, 1]$ by consecutive aligned dyadic intervals: a gap at depth d has width 2^{-d}. A leaf starting at x must satisfy $x/2^{-d} \in \mathbb{Z}$. Conversely such a tiling is a complete prefix code and determines an ordered full tree. Alignment, not just the Kraft equality, enforces order. On a grid of height H, an interior boundary at integer coordinate Y corresponds to a key with comparison length $H - \nu_2(Y)$. Thus gap and key costs are both local in a left-to-right tiling construction.

Theorem 90 (Bounded-excess construction)**.** *For $n \geq 1$, let $N = n + 1$, $h = \lfloor \log_2 N \rfloor$, $S = 2^h$, $r = N - S$, and $s = \min(r, S - r)$. For any nonnegative integer budget k, the minimum full weighted cost among all trees with $I \leq I_* + k$ can be found and reconstructed in*

$$O\big(n(s + k + 1)(k + 1)^2 2^k\big)$$

arithmetic operations, using $O\big(n(s + k + 1)(k + 1)2^k\big)$ space for backpointers.

Proof. Measure dyadic widths in units of 2^{-h} and put $w = 2^{h-d}$. For integral depth d, the potential

$$f(d) = d - h - 2 + 2^{h+1-d} = d - h - 2 + 2w$$

is nonnegative, with zeros exactly at h and $h + 1$. The minimum external path length is $hN + 2r$. Since external path length equals $I + 2n$ and $\sum_i w_i = S$, every full tree satisfies

$$\sum_i f(d_i) = \sum_i d_i - (h + 2)N + 2S = I - I_*.$$

Every depth in the budgeted class therefore has $f(d) \leq k$. This implies $d \leq H = h + k + 2$, and only $O(k + 1)$ integral depths are possible. For integral d, direct substitution gives

$$\tfrac{1}{2} - \tfrac{1}{2} f(d) \leq w \leq 1 + f(d).$$

Apply these inequalities to a prefix of i leaves and to its remaining suffix, each having potential at most k. The cumulative position y must obey

$$\max\left(\tfrac{i-k}{2}, i - r - k\right) \leq y \leq \min\left(i + k, \tfrac{S-r+i+k}{2}\right).$$

This interval has width at most both $r + 2k$ and $(S - r)/2 + k$, hence $O(s + k)$.

Set $K = 2^{k+2}$. A state after i leaves stores integral drift $u = K(y - i)$ and spent potential $z = K \sum f(d)$. For depth d, let $W = 2^{H-d}$. Require $(iK + u) \bmod W = 0$ and update

$$u' = u + W - K, \qquad z' = z + K(d - h) + 2(W - K).$$

Retain the position band and $0 \le z' \le Kk$. Charge $q_i d$ and, for $i > 0$, charge $p_{i-1}(H - \nu_2(iK + u))$. Start at $(u, z) = (0, 0)$ and finish after N leaves with $u = -rK$. Every retained complete path is an aligned full tiling with excess at most k; every tree in the class passes the same tests. The DP therefore optimizes exactly this class. For fixed u, the congruence $z \equiv 2u \pmod K$ leaves at most $k+1$ possible spent potentials. There are $O((s+k+1)2^k)$ positions per stage and $O(k+1)$ choices per state, proving the bounds. The bands admit direct array indexing. Backpointers recover gap depths; the boundary lengths reconstruct the tree in linear time. □

Perfect-size refinement. When $N = S$, use instead $f_0(d) = d - h + \frac{3}{2}(2^{h-d} - 1)$. It is nonnegative on integer depths, vanishes only at h, and also sums to $I - I_*$. The inequality $|2^{h-d} - 1| \le 2f_0(d)$ and zero total drift imply $|y - i| \le k$. Here $H = h + k + 1$, $K = 2^{k+1}$, and $z = 2K \sum f_0$. Its increment is $2K(d - h) + 3(W - K)$, and $z \equiv 3u \pmod{2K}$. The implementation uses this smaller grid at perfect sizes, with the same asymptotic bound at $s = 0$.

Certify against all other trees. Let C_k be the computed candidate, $a = \min p_i$, $b = \min q_i$, and H' the order-free bound on residual weights. Every excluded tree has $I \ge I_* + k + 1$, so it costs at least

$$B_k = (a + b)(I_* + k + 1) + (a + 2b)n + H'.$$

Thus $C_k \le B_k$ proves global optimality. Sorting and the two-queue bound cost $O(n \log n)$ and are included in the counters. The constructor uses $k = 2$ and attempts the shortcut only when $s \le \text{bit_length}(n)$. Its attempted work is $O(n \log n)$; a failed certificate or skipped size falls back to Knuth. The total worst case remains quadratic. Arbitrary growth of k is exponential, and wide shape spaces remain quadratic even for fixed k.

4.7 A split-height construction covering every input size

The preceding shape band can be wide even at zero excess. We therefore implement the split-height/Monge mechanism already sketched by Karpinski, Larmore, and Rytter in Section 4.4 of their longer preprint [6]. That section explicitly supplies the fixed-height Monge property and the two-part construction. It is different from Section 4.4 of the six-page conference version [5]. No priority claim is made for this mechanism. The present experiment makes the tiling, matrix padding, reconstruction, and separate unrestricted certificate explicit.

Two scales of dyadic intervals. Fix an external height limit $H = L + U$ and write $K = 2^L$, $B = 2^U$. Precompute all relative costs $C_h(i, j)$ for $0 \le h \le L$ and $0 \le j - i \le 2^h - 1$, using

$$C_h(i, j) = W(i, j) + \min_r \{C_{h-1}(i, r) + C_{h-1}(r + 1, j)\}.$$

Empty intervals have zero cost; the root range enforces both child height capacities. Monge inequalities at fixed height give monotone minimizing roots as the endpoints move. Choose leftmost ties and bound each root by the adjacent shorter intervals. Search ranges telescope to $O(n)$ for each interval length. Summing over lengths and heights costs $O(n2^L)$.

Divide the unit dyadic interval into B top-grid blocks. Every tree of height at most H consists of lower subtrees contained in individual blocks, together with shallower gaps occupying aligned groups of blocks. Let $D_b(i)$ be the minimum partial cost after consuming i gaps and reaching grid boundary b. Keys through $i - 2$ have already been charged. For $b > 0$, put

$$R_b(i) = D_b(i) + p_{i-1}(U - v_2(b)); \qquad R_0(0) = 0.$$

The boundary key has exactly this comparison length. There are two types of transition:

$$D_{b+1}(j) \leftarrow \min\{D_{b+1}(j), R_b(i) + C_L(i, j-1) + UW(i, j-1)\}, \qquad i < j \le i + K,$$
$$D_{b+w}(i+1) \leftarrow \min\{D_{b+w}(i+1), R_b(i) + q_i(U - \log_2 w)\}.$$

The second transition requires a power of two $w \ge 2$, $w \mid b$, and $b + w \le B$. Width-one gaps are zero-key lower subtrees in the first transition. Start at $D_0(0) = 0$ and finish at $D_B(n+1)$. Dyadic alignment proves that these paths represent every height-feasible tree and only such trees. Backpointers recover every lower tree and every boundary key.

Linear matrix searches with feasible padding. At fixed b, the first transition is a banded Monge matrix in i, j; row additions and the separable interval mass preserve the inequality. Reachable counts i are contiguous: the coarsest tiling of the prefix has popcount(b) leaves, and successive refinement gives every count through bK, truncated to n before another transition. Search output columns from $i_{\min} + 1$ through $\min(n+1, i_{\max} + K)$, each of which has a feasible input row.

SMAWK requires care outside the band. Let $S = (H+1)(\sum p + \sum q) + 1$, larger than every feasible prefix cost. In the transposed matrix with row j and candidate column i, use

$$A(j,i) = \begin{cases} S + j - K - i, & i < j - K, \\ R_b(i) + C_L(i, j-1) + UW(i, j-1), & j - K \le i < j, \\ S + i - j + 1, & i \ge j. \end{cases}$$

This padding preserves total monotonicity. For $i_1 < i_2$, once i_2 is preferred at a row, it remains preferred as j increases: within the feasible band this is Monge; outside it, distances decrease toward the band and increase away from it, and finite entries beat all padding. When the candidates straddle opposite invalid sides, their cost difference is $i_1 + i_2 + K + 1 - 2j$, also decreasing. Ties preserve the same non-strict implication. Thus SMAWK finds leftmost minima in $O(n)$ queries per grid boundary. Padding by tied infinities does not supply this argument.

Across all boundaries there are $B - 1$ aligned choices of coarse width, so the second transitions and scans also cost $O(nB)$. Choosing $L = \min(H, \lfloor H/2 \rfloor + 1)$ gives

$$O\big(n(2^L + 2^U)\big) = O(n2^{H/2})$$

work and storage including backpointers. Consequently a fixed extra height $H = \lceil \log_2(n+1) \rceil + c$ gives $O(n^{3/2})$ attempted work at *every* size. The prototype uses sparse dictionaries; the bounded integer state coordinates also admit direct array indexing.

Lemma 91 (Minimum path length with a tall path)**.** *For $1 \le R \le n$, the minimum internal path length $J(n, R)$ among n-key trees of external height at least R is obtained by forcing a path of R internal nodes and filling the remaining available positions in increasing order of depth. The excluded class is empty if $R > n$.*

Proof. Every tree in the class contains such a path, contributing $R(R-1)/2$. The remaining nodes form an ancestor-closed subset of its free forest: one side root at each depth $1, \ldots, R-1$, and two roots at depth R. The $n - R$ shallowest positions in this forest are ancestor-closed and minimize the depth sum among all subsets of that size. Their addition constructs a feasible tree attaining the bound. Directions of the forced path do not change the forest's depth counts. □

A level-count implementation computes J in $O(R + \log(n+1))$ work.

Corollary 92 (Certificate against taller trees)**.** *Let C_H be the exact height-restricted optimum above, and let Q be the order-free mixed-node Huffman lower bound for residual weights $p-a, q-b$, where $a = \min p$, $b = \min q$. If $H < n$ and*

$$C_H \le (a+b)J(n, H+1) + (a+2b)n + Q,$$

the constructed tree is globally optimal. If $H \ge n$, there is no height restriction and certification is automatic.

Proof. Every excluded tree has internal path length at least $J(n, H+1)$. Its baseline contribution is $(a+b)I+(a+2b)n$ and its residual cost is at least Q. Within the height class, C_H is optimal. □

Sorting and the residual oracle add $O(n \log n)$ work. Failed tests invoke Knuth, so the full constructor remains quadratic in the worst case. The certificate never assumes success from near-uniform weights.

4.8 A variable interval frontier for exact full construction

The height test can fail even when its candidate is optimal. We now use a class guaranteed in advance to contain an unrestricted optimum. The short-interval/top-skeleton architecture is established work of Larmore and Karpinski–Larmore–Rytter [4, 5]; the experiment compares a uniform width envelope with a variable interval contour. Both use the same safe mass guarantee and matrix infrastructure. This is not a priority claim for the architecture or its restricted subquadratic regime.

The height guarantee follows from Lemma 2.

Store a contour instead of a global width. For fixed d, let $E_d(i)$ be the largest $j > i$ satisfying $F_{d+1}W(i,j) \le W$, or i if there is none. Empty intervals always have stored cost zero. Nonnegative weights imply that $E_d(i)$ is nondecreasing and the domain is closed under taking subintervals. A two-pointer scan computes it in $O(n)$ work. Write

$$S_d = \sum_{i=0}^{n} (E_d(i) - i).$$

By Lemma 2, every internal subtree at depth d of an optimal tree belongs to the domain. For comparison, the uniform version sets $\ell_d = \max_i(E_d(i) - i)$ and stores

$$U_d = \ell_d(n+1) - \ell_d(\ell_d+1)/2 \ \ge S_d$$

intervals. Both classes contain an unrestricted optimum.

Lemma 93 (Root-search work on a hereditary contour)**.** *Computing all unrestricted interval optima in this domain takes $O(n + S_d)$ work, with at most $3S_d$ tested roots.*

Proof. Bucket only existing cells by length in $O(n + S_d)$ work. Knuth's recurrence applies because every child and adjacent shorter interval is present. Let $r(i,j)$ be the leftmost optimal root and put $t(i,j) = r(i,j) - i \ge 0$. For length at least two, the root count is $2 + t(i+1,j) - t(i,j-1)$. Sum over the domain. A positive occurrence $t(a,b)$ cancels its negative occurrence whenever $(a, b+1)$ is present. An unmatched positive occurrence requires $b = E_d(a)$ and $E_d(a-1) = E_d(a)$. There is at most one per row, of size at most $E_d(a) - a$. Discard unmatched negative occurrences. Adding the length-one cells proves the $3S_d$ bound. Scanning nonexistent diagonal cells would lose this saving; the buckets are essential. □

Compose the exact interval costs. Use 2^d top-grid blocks and the preceding dyadic transitions, replacing height-limited lower costs by these unrestricted interval costs. A block starting after i gaps can finish after j exactly when $i < j \le E_d(i) + 1$. It contributes $C(i, j-1) + dW(i, j-1)$ and the starting boundary key cost $p_{i-1}(d - v_2(b))$ at $b > 0$. Coarse gaps retain their aligned power-of-two skip transitions. Every optimal tree is represented, while every reconstructed candidate is a valid full tree. Thus the resulting optimum is unrestricted.

The variable band needs another implementation detail. Reachable input rows, and their union of feasible output columns, may have holes. Scan once per top boundary to retain exactly output columns covered by some reachable row. Put $L(j) = \min\{i : E_d(i) + 1 \ge j\}$, a nondecreasing sequence with $L(j) \le j - 1$. In the transposed matrix, use finite true entries when $L(j) \le i < j$, and padding

$$Z + L(j) - i \quad (i < L(j)), \qquad Z + i - j + 1 \quad (i \ge j), \qquad Z = (n + d + 1)W + 1.$$

All finite prefix costs are smaller than Z. Within the band, exact interval costs are Monge. Across padding, opposite-side differences $i_1 + i_2 + 1 - j - L(j)$ are nonincreasing; same-side preferences also move right. Thus the matrix is totally monotone, even after removing arbitrary rows and columns, and every searched column has a feasible minimum. SMAWK, scans, and coarse transitions together cost $O(n2^d)$. Backpointers recover every key and gap contribution.

Theorem 94 (Exact mass-frontier constructor)**.** *The adaptive implementation constructs a full optimal BST in*

$$O\Big(n \log(n+1) + \min_{d \ge 0}\{S_d + n2^d\}\Big)$$

arithmetic work. The uniform-envelope implementation replaces S_d by U_d. At the selected depth, storage is $O(n + S_d + n2^d)$ or $O(n + U_d + n2^d)$ respectively. Both remain quadratic in the worst case.

Proof. The interval and composition arguments establish exactness at every depth. Start with depth zero and the full Knuth table. Evaluate successive depths while $n2^d$ is smaller than the best work proxy so far; each contour costs $O(n)$. No later depth can improve the proxy because its top term alone is too large. At most $O(\log(n+1))$ depths are evaluated. Depth zero uses Knuth directly, omitting a redundant top pass and its lower-order n proxy term. The preceding work and storage bounds now apply. □

This constructor has no posterior certificate failure followed by a second solve. It may select the full table initially. The Python prototype uses dictionaries; bounded integer top coordinates also allow direct arrays. When each key has weight at least $a > 0$ and $W = O(na)$, we have $S_d = O(n^2/\phi^d)$, where $\phi = (1 + \sqrt{5})/2$. Balancing against $n2^d$ recovers the known exponent $1 + \log_{2\phi} 2 \simeq 1.59023$.

A scoped asymptotic separation. Let n range over fourth powers and $L = n^{3/4}$. Give every key and gap weight $A = n^2$, except a central block of L keys and its $L + 1$ gaps, all of weight one. Every weight is positive and $W = \Theta(nA)$. An admitted interval contains at most $m = O(n/F_{d+1})$ heavy keys. Intervals wholly inside the light block number $O(L^2)$. Other starts contribute $O(nm)$ ordinary endpoints; only $O(m)$ outside starts can reach across a block boundary and gain $O(L)$ endpoints, and starts inside add at most $O(Lm)$ heavy endpoints. Consequently

$$S_d = O(n^2/F_{d+1} + L^2).$$

With $d = \lfloor \log_{2\phi} n \rfloor$, the adaptive work is $O(n^{1+\log_{2\phi} 2} + n^{3/2}) = O(n^{1.59024})$.

Every uniform-envelope plan, however, has proxy $\Omega(nL)$. If $2^d \geq L$, the top term suffices. Otherwise $F_{d+1} = O(L^{\log_2 \phi})$, so W/F_{d+1} exceeds the entire light block's $O(L)$ mass. Hence $\ell_d \geq L$ and the uniform table has at least $nL/2$ cells. This gives $\Omega(n^{1.75})$ at every depth. The separation is within the specified architecture, not against every prior OBST algorithm or combinations of our shortcuts. In particular, this light block also offers an opportunity for certified contraction.

4.9 Dominant-gap separators and local mass frontiers

The global mass bound can admit many unnecessary intervals when a single heavy outcome dominates otherwise ordinary regions. We first split at certified outcomes, then apply the frontier constructor with each terminal component's own total mass. The following exchanges retain all key and gap weights. Their priority is not established.

Lemma 95 (Restriction to an ordered interval)**.** *A BST restricted to a contiguous key interval and its endpoint gaps has no greater depth for any retained outcome. If the old root is outside the retained key interval, every retained outcome loses at least one comparison.*

Proof. Delete each comparison at an outside key and follow the branch containing all retained outcomes. Comparisons at retained keys remain. This gives the required BST, with each retained search path a subsequence of its old path. The argument also applies to an empty key interval containing one gap. □

Fix an interior gap g in a current component. Its adjacent keys are $a = g - 1$ and $b = g$. Let L contain the keys strictly left of a, with its endpoint gaps, and let R contain those strictly right of b with its endpoint gaps. Their masses are W_L, W_R, and the component mass is

$$W = W_L + p_a + q_g + p_b + W_R.$$

Theorem 96 (Interior-gap separator)**.** *If*

$$q_g + \min(p_a, p_b) \;\geq\; W_L + W_R, \tag{116}$$

there exists an optimal tree with gap g at depth two. Its top two keys are a, b, in either orientation, and its remaining subtrees are L, R. Writing C_L, C_R for their unrestricted optimal relative costs gives

$$\mathrm{OPT} = C_L + C_R + W + q_g + \min(p_a + W_L,\; p_b + W_R). \tag{117}$$

At equality, other optimal trees need not have this shape.

Proof. Take any tree in which g has depth at least three. Restrict it to L, R using Lemma 95. If its root lies in L, make a the new root, attach restricted L on its left, and make b its right child, with g on b's left and restricted R on its right. Every L outcome deepens by at most one. Every R outcome lost the old root comparison, then gains two, so also deepens by at most one. Key a saves at least one comparison, key b does not deepen, and gap g saves at least one. The cost change is at most $W_L + W_R - p_a - q_g$. For an old root in R, the mirror exchange costs at most $W_L + W_R - p_b - q_g$.

If the old root is a, keep it and promote b to its right child. The L depths do not increase and R depths increase by at most one. As a, b are adjacent and g was deeper than two, b had depth at least two and saves at least one comparison. The change is at most $W_R - p_b - q_g$. The case with old root b is symmetric. Condition (116) makes each bound nonpositive, so the exchange can be

applied to an optimum. An interior gap cannot have depth below two; at depth two its bounding ancestor keys must be exactly a, b.

Both orientations leave the same two independent optimization problems. With root a, the cost is $C_L + C_R + W + q_g + p_b + W_R$; with root b it is $C_L + C_R + W + q_g + p_a + W_L$. Minimize these two values after solving each outside interval once. This proves (117) without duplicated recursive branches. □

Replacing the smaller neighboring weight in (116) by the larger one is false. For $p = (100, 1, 50)$ and $q = (1, 1, 1, 1)$, the weakened test accepts $g = 1$ since $101 \geq 53$. Yet the full optimum is 212 and forcing this gap to depth two costs 261. The valid test rejects it since $2 < 53$. This strictly positive counterexample is retained.

Endpoint and key certificates. For a component's first key and left endpoint gap, $2(p_i + q_i) \geq W$ suffices to put that key at the root in some optimum; the right endpoint is symmetric. If its original depth is $h \geq 1$, its endpoint gap has depth $h+1$. Promotion saves $h(p_i+q_i)$ while restricting and reattaching the other interval increases cost by at most $W - p_i - q_i$. Likewise, $2p_r \geq W$ certifies an arbitrary key r as root: promote it and attach the two restrictions, increasing every other depth by at most one while saving at least one comparison on r. These are sufficient conditions, not a general largest-weight root rule.

Avoiding quadratic separator scans. Components remain contiguous intervals of the original input, with no weight aggregation. Store the static scores

$$G_j = 2q_j + p_{j-1} + p_j + \min(p_{j-1}, p_j), \qquad K_j = 2p_j.$$

An interior gap in component $[i, k)$ satisfies (116) exactly when $G_j \geq W(i, k)$, for $i < j < k$. Two range-maximum segment trees find qualifying gaps and keys in $O(\log(n+1))$ work per query, after $O(n)$ construction. Prefix sums give component masses in constant work. Each separator removes one or two keys, and terminal intervals are disjoint. Only terminal slices are copied, so a long chain does not cause quadratic copying or rescanning. The implementation tests endpoints first and builds the indexes only when an endpoint test fails on both sides. All-zero components are returned as balanced trees directly.

Theorem 97 (Local-frontier composition)**.** *For terminal components t with m_t keys and local admissible-interval counts $S_{t,d}$, the constructor is exact and uses*

$$O\left(n \log(n+1) + \sum_t \min_d \{S_{t,d} + m_t 2^d\}\right)$$

arithmetic work. If $m_t \leq b$ for every terminal, the bound is $O(n \log(n+1) + nb)$. Sequential reuse of terminal scratch storage gives $O(n + \max_t\{S_{t,d_t} + m_t 2^{d_t}\})$ space at selected depths d_t. The unrestricted worst case remains quadratic.

Proof. The separator identities reduce the optimum to disjoint exact subproblems. There are $O(n)$ separator calls and $\sum_t m_t \leq n$. The local frontier bounds apply separately, including parameter selection since $\sum_t m_t \log(m_t + 1) \leq n \log(n+1)$. Each terminal may select depth zero and use its full Knuth table, so $\sum_t m_t^2 \leq nb$ when all sizes are at most b. Include the reconstructed output and separator stack in $O(n)$ space; completed terminal scratch arrays need not remain allocated. □

Two exact structured regimes. For $p_i = q_i = 2^i$ and $q_n = 2^n$, the total mass is $3 \cdot 2^n - 2$ and the rightmost key/gap pair has mass $3 \cdot 2^{n-1}$. The right endpoint certificate passes, and the remaining prefix has the same form. The optimal chain is constructed with n endpoint decisions, no range index, and no terminal DP: $O(n)$ arithmetic work. At the final single key the implementation selects the equivalent left endpoint rule. Weights have $O(n)$ bits, so this is not a linear bit-complexity assertion.

For a different positive mixed family, recursively plant a heavy central gap separating two outside intervals, stopping below 32 keys. Set each planted weight to four times its component mass after its children have been planted. Each planted gap satisfies the theorem, and all terminal sizes are below 32. This gives $O(n \log n)$ arithmetic work on the family, including varying positive background weights on both keys and gaps. Neither regime is an unrestricted subquadratic result.

A positive obstruction to the combined constructor. For $n \geq 5$, set every key and gap weight to one except $q_1 = q_{n-1} = M = n^3$. Then $W = 2M + 2n - 1$. The largest gap score is $2M + 3 < W$, endpoint scores are four, and key scores are two, so no separator fires. Indeed the frontier planner selects depth zero and the full table. To see this, write $\tau(k) = k(k+1)/2$ and $h = \lfloor (n-1)/2 \rfloor$. Depth one admits all $\tau(n)$ cells. At depth two, an interval containing one heavy gap has mass $M + 2\ell$, so the cutoff $W/2 = M + n - 1/2$ admits it exactly for length $\ell \leq h$. There are four disjoint sets of h such intervals, and $\tau(n-4)$ intervals avoiding both heavy gaps. Thus

$$S_2 + 4n = \tau(n-4) + 4h + 4n = \tau(n) + 6 + 4h > \tau(n).$$

For $d \geq 3$ with $2^d \leq n$, we have $3 \leq F_{d+1} \leq 2^d \leq n$. The cutoff is below M but above every all-unit interval mass, so $S_d = \tau(n-4)$ and

$$S_d + n2^d \geq \tau(n-4) + 8n = \tau(n) + 4n + 6 > \tau(n).$$

If $2^d > n$, the top term alone exceeds $n^2 \geq \tau(n)$. Every positive depth is therefore worse than the direct Knuth plan. This is a quadratic obstruction to the specified constructor, not a general OBST lower bound. An independent contour check rejected the initial assumption that all one-heavy-gap intervals were admitted at depth two; the correct length cutoff above is retained in the test. The family motivates joint treatment of several comparable heavy outcomes. Eight full instances through 256 keys match Knuth and reconstructed costs, with cubic checks through 32 keys. Four larger instances through 65,536 keys check plans only; at that size the selected table has 2,147,516,416 cells, not a completed full solve.

4.10 Joint heavy-outcome cores

The two-gap obstruction motivates selecting several heavy outcomes together. The following certificate uses their necessary comparison keys as an ancestor-closed core. It preserves the complete mixed-weight objective and solves every intervening interval exactly. Priority of the sufficient condition has not been established.

For $n \geq 1$, choose a nonempty set $\mathcal{H}$ of positive-weight key or gap outcomes. Let A contain every selected key and the adjacent key(s) of every selected gap. Write $m = |A|$, $\theta = \min_{x \in \mathcal{H}} w_x$, and let W_0 be the total weight outside $\mathcal{H}$. Both adjacent keys of an interior gap are ancestors of that gap in every BST. Thus each A-key lies on the search path to some selected outcome, possibly itself. Call a tree *core-closed* if no key outside A is an ancestor of an A-key. Then A forms a top BST and the intervals between consecutive A-keys, including both end intervals, are its payloads. Selected gaps remain singleton core gaps.

Lemma 98 (Projection preserving selected depths)**.** *Every BST can be reduced to a BST on A without increasing any selected outcome's depth. If the original tree is not core-closed, at least one selected outcome becomes strictly shallower.*

Proof. Consider an outside key r whose subtree contains an A-key. If just one child contains A-keys, replace r by that child. The other branch contains no selected outcome: any nonempty subtree containing a selected gap must contain one of its neighboring keys, and a selected empty child gap would make r itself an A-key. All retained selected outcomes save a comparison.

If both children contain A-keys, let a be the largest such key in the left subtree. Replace r by a, restrict the old left subtree to keys below a and their endpoint gaps, and retain the old right subtree. No selected outcome lies strictly between a and r: a selected key would belong to A, and a selected gap would put an adjacent key in A before or at r. All A-keys survive. Selected outcomes in the left restriction lose zero or more comparisons by Lemma 95; right depths are unchanged. If a is selected, it moves strictly upward. Otherwise it must border a selected gap on its left. Its right gap cannot be selected, since that would put $a+1$ in A, contradicting either the maximality of a or $r \notin A$. The selected left gap loses the comparison at the old a. Thus some selected depth decreases in either case.

These exchanges may discard light keys, which will be reinserted later. Each exchange reduces the number of outside keys and retains A. Repeat until no outside key has an A-descendant, then discard the remaining anchor-free branches. Selected gaps are preserved throughout. This produces the asserted core tree. Arbitrary deletion of nonretained comparisons would not suffice: a nonretained key can separate retained outcomes on both sides; the explicit promotion above resolves that case. □

A first sufficient condition is $\theta \geq (m+L)W_0$, where $L = \lceil \log_2(n+1) \rceil$. Project an optimum and reinsert balanced payloads: the selected cost falls by at least θ, and the new light cost is at most $(m+L)W_0$. The next theorem substantially sharpens the height term using the earlier Fibonacci lemma.

Corollary 99 (Protected-height core certificate)**.** *Let $W_{\mathcal{H}}$ be the selected total weight and*

$$H = \max\{j \in \{1, \ldots, m\} : F_j\theta \leq W_{\mathcal{H}}\}, \qquad F_1 = F_2 = 1.$$

If

$$\theta \;\geq\; (H+L)W_0, \tag{118}$$

there exists a core-closed optimum. If the payload intervals have masses W_t and unrestricted relative optimum costs C_t, then

$$\mathrm{OPT}(\textit{original}) = \mathrm{OPT}(\textit{original } A\textit{-key weights},\ (W_t)_t) + \sum_t C_t.$$

Equality in the certificate does not force every optimum to be core-closed.

Proof. In a protected-only core instance, keep selected weights and set every other weight to zero. Every internal subtree contains an A-key and therefore a selected outcome of weight at least θ in that subtree. By Lemma 2, a depth-d internal subtree of a protected-only optimum has mass at most $W_{\mathcal{H}}/F_{d+1}$. Its height, including external leaves, is consequently at most H.

Project a non-core-closed full optimum by Lemma 98, saving at least θ in selected weighted cost. Replace the projected core by a protected-only optimum; its selected weighted cost cannot increase, although individual depths may change. Reinsert balanced payloads. Every unselected

key comparison length or gap depth is at most $H + L$, so the new light cost is at most $(H + L)W_0$. The old light cost was nonnegative. Condition (118) makes the exchange nonincreasing, establishing the existence of a core-closed optimum.

At core-gap depth d, an exact payload contributes $dW_t + C_t$. The core and payload relative costs therefore optimize independently, proving the identity. Empty intervals contribute zero relative cost and keep their gap weight. The final full-weight core is optimized without a height restriction: H bounds the protected-only witness, not the output core. □

For k equal selected weights, $H = O(\log(k + 1))$, whereas m can be $2k$. The certificate thus handles much less extreme dominance than the size-based version. It does not remove unselected weights from the final objective.

A threshold sweep with no repeated interval scans. The implementation selects all outcomes of weight at least a threshold θ. Key i is in A precisely when $\max(p_i, q_i, q_{i+1}) \geq \theta$. Sort these activation weights and all outcome weights. Sweep thresholds upward, grouping equal values. Before each threshold, activate as unanchored each key whose activation weight is strictly smaller. A disjoint-set structure maintains the contiguous payload runs and their total triangular state count. A singleton adds one; merging runs of sizes a, b adds ab. Union by size bounds each find by $O(\log n)$, even without using an inverse-Ackermann analysis.

At each certified threshold, evaluate

$$\tau(m) + \sum_t \tau(k_t), \qquad \tau(k) = k(k + 1)/2.$$

Keep the minimum proxy, accepting only a strict improvement on $\tau(n)$. This optimizes a Knuth-state proxy, not actual frontier runtime. Fibonacci values through index n take $O(n)$ additions. Since $W_{\mathcal{H}}/\theta$ is nonincreasing during the upward sweep, a single decreasing Fibonacci index supplies the height bound in $O(n)$ total index decrements; cap it by m. The entire selection uses $O(n \log(n + 1))$ arithmetic work and $O(n)$ arithmetic storage. Failure invokes the preceding separator constructor, adding screening work without a failed quadratic solve.

Kernels, reconstruction, and scoped bounds. Both core and payloads use either exact local frontiers or Knuth. Before each kernel, uniform key weights p and uniform gap weights q are recognized. Such a tree costs $(p + q)I + np + 2nq$, where I is its unweighted internal path length. A recursively balanced tree minimizes I and is built in linear work. This familiar uniform-weight shortcut is not claimed as new.

If $K(s)$ bounds the chosen exact kernel on s keys, successful construction costs

$$O\left(n \log(n + 1) + K(m) + \sum_t K(k_t)\right), \qquad m + \sum_t k_t = n.$$

Sequential scratch reuse retains only the payload outputs between solves. The general work and space bounds remain quadratic. A uniform residual background and k selected outcomes give $m \leq 2k$ and work $O(n \log(n + 1) + k^2)$. If residual weights lie in $[a, Ca]$ for fixed $a > 0$ and C, each payload's local frontier costs $O(s^\alpha)$, where $\alpha = 1 + \log_{2\phi} 2 < 1.59024$. A certified plan then costs

$$O\left(n \log(n + 1) + k^2 + \sum_t k_t^\alpha\right).$$

With payload sizes at most B, the last sum is at most $nB^{\alpha-1}$. These are bounds for the certified plan; minimizing the triangular proxy need not minimize this sharper expression.

For the preceding obstruction, $q_1 = q_{n-1} = n^3$ with unit background, $n \geq 5$, we have $m = 4$, $H = 3$, and $W_0 = 2n - 1$. The certificate passes, so the four-key core and uniform payloads take $O(n \log n)$ total work. The previous separator/frontier constructor used all $n(n+1)/2$ cells. Bounded positive mixed background and sufficiently large equal selected gap weights instead give an $O(n^{1.59024})$ plan. This resolves that scoped obstruction; it is not an unrestricted subquadratic theorem.

4.11 Residual-cost certificates for heavy-outcome cores

The preceding height certificate charges all light search cost against one saved heavy comparison. Every competing tree must itself pay for light searches. Crediting a valid lower bound for that unavoidable cost gives a stronger test for a fixed selected set. This is an additive exclusion certificate built on Lemma 98; it does not assert a new Huffman algorithm or an established priority claim.

Write the original nonnegative weight vector as $w = w_H + w_0$, where w_H retains selected positive outcomes and is zero elsewhere. Let A be the necessary core keys from Section 4.10, let P be the exact protected-only optimum on that core, and let B_0 be any valid lower bound on the full n-key problem with residual weights w_0. In particular, the order-free mixed-node relaxation of Section 5.1 supplies B_0. Residual weights on both keys and gaps are retained.

Corollary 100 (Residual exclusion certificate)**.** *Every tree that is not core-closed has full cost at least*

$$P + \theta + B_0, \qquad \theta = \min_{x \in \mathcal{H}} w_x.$$

Consequently, if a core-closed witness has full cost $U \leq P + \theta + B_0$, some global optimum is core-closed. More strongly, if the exact optimum C_A within the core-closed class satisfies this inequality, then C_A is the unrestricted optimum.

Proof. Project any non-core-closed tree T onto A. Its selected weighted cost decreases by at least θ, and the projected selected cost is at least P. Thus $C_{w_H}(T) \geq P + \theta$. Independently, $C_{w_0}(T) \geq B_0$. Add the inequalities, using linearity of tree cost in the weights. A witness at or below this excluded-class lower bound cannot be worse than every non-core-closed tree. Optimizing within the core-closed class therefore gives a global optimum. Equality allows both classes to contain optima. □

A test before solving the payloads. Solve the protected core, retaining one optimal shape, and attach balanced payloads. This produces a full witness of cost $P + U_0$, where U_0 is its exact residual cost. The inexpensive sufficient condition becomes

$$U_0 - B_0 \leq \theta. \tag{119}$$

Only after it passes does the default constructor solve the full-weight core and all payloads exactly. Thus failed witnesses need no exact payload tables. For the same selected set, this test includes the preceding height certificate: $U_0 \leq (H + L)W_0$ and $B_0 \geq 0$. The implemented threshold selection is not exhaustive, so this logical implication is not a claim that the new selector always chooses the old selector's successful set.

An optional refinement solves one core-closed candidate even after its witness fails and tests $C_A \leq P + \theta + B_0$. For example, $p = (0, 0, 1)$, $q = (0, 4, 1, 1)$, and $\theta = 4$ give $P = 8$, $B_0 = 4$, and excluded-class lower bound 16. The first witness costs 17, while the exact closed candidate costs 14 and is globally optimal. A failed witness is not evidence that the core structure is invalid.

Several candidates without quadratic screening. A single minimum-proxy threshold can include ordinary outcomes whose weights are too small to pay for the exchange. The revised selector keeps two representatives in each dyadic core-size band: the least $\tau(m) + \sum_t \tau(k_t)$ proxy and the highest threshold. It considers only $m \le b = \lfloor \sqrt{n\lceil \log_2(n+1) \rceil} \rfloor$ and only strict proxy improvements over $\tau(n)$. The existing sorted activation sweep finds these representatives in $O(n \log(n+1))$ work. There are $O(\log(n+1))$ candidates, and their squared core sizes sum to $O(b^2)$, because at most two lie in each geometrically increasing band.

Sort the original key and gap profiles once. For a threshold θ, remove the profile entries at least θ and append the same number of zeros. The remaining profile is still sorted. The mixed-node two-queue bound then takes $O(n)$ work per candidate, without another sort. Building and evaluating each balanced witness also takes $O(n + m^2)$ work. All candidate screening therefore costs $O(n \log(n+1))$ arithmetic operations. Candidates are tried in increasing proxy order until a witness succeeds. If none succeeds, the optional mode refines only the first candidate; limiting refinement to one avoids an $O(\log n)$ factor on failed exact payload solves. Failure invokes the preceding exact constructor.

A successful plan retains the bound

$$O\left(n \log(n+1) + K(m) + \sum_t K(k_t)\right).$$

Both modes keep quadratic general worst-case work and storage. The witness screen can use $O(n \log n)$ storage for protected-core tables; the final kernel can still require quadratic space. Ordinary Huffman coding is classical [11]; the mixed-node relaxation and its limitations are already described in Section 5.1. The result here is the certified use of a residual lower bound against a specified excluded class, rather than a relaxation substituted for the full objective.

A linear-weight separation from the height certificate. Let $n = 2^h \ge 8$, put every original key and gap weight equal to one, and replace q_1, q_{n-1} by an equal selected weight $\theta > 1$. The necessary core has four keys. Its protected optimum is $P = 5\theta$: each selected interior gap needs at least two comparisons, and both cannot have depth two because their adjacent-key pairs are disjoint. Depths two and three are attainable.

Use the core whose root is original key 1, left child is key 0, right child is key $n-2$, and right-right child is key $n-1$. The sole nonempty payload contains $n-4$ keys and hangs at depth two. The residual cost of this balanced-payload witness is

$$U_0 = (2h+3)n - 6h - 3.$$

Indeed, the core's residual cost is $4n-1$, while $I_{\min}(n-4) = (h-2)n - 3h + 5$ gives payload cost $2I_{\min}(n-4) + 3(n-4) = (2h-1)n - 6h - 2$.

For the residual problem all key weights are one, all ordinary gap weights are one, and the two selected gaps have weight zero. The identity $E = I + 2n$ rewrites any residual tree cost as $\sum_j (q_j + 1)d_j - n$. The order-free leaf weights are therefore two ones and $n-1$ twos. Huffman first merges the two ones; the resulting $n = 2^h$ weights are all two and have a perfect code of height h. Hence the exact order-free residual lower bound is

$$B_0 = 2nh - n + 2, \qquad U_0 - B_0 = 4n - 6h - 5.$$

Thus $\theta \ge 4n - 6h - 5$ suffices for core closure. In particular, $\theta = 4n$ passes, while the preceding height condition requires $\theta \ge (h+4)(2n-1)$ and rejects it.

At $\theta = 4n$, the described core is also optimal with the full collapsed weights. The four protected-optimal shapes have residual core costs $4n - 1$ or $6n - 9$; every other core shape has selected cost at least $6\theta > 5\theta + 4n - 1$. It follows that the full optimum is

$$\mathrm{OPT} = (2h + 23)n - 6h - 3. \tag{120}$$

The new constructor uses a constant-size core and a uniform payload, for $O(n \log n)$ work and $O(n)$ space on this strictly positive mixed-weight family. These weights and the construction's intermediate values have $O(\log n)$ bits. This is a separation from the preceding height certificate, not a quadratic lower bound for its fallback frontier or for other algorithms.

4.12 Coupled two-gap skeleton certificates

The preceding certificate adds independent protected and residual minima. We now retain enough geometry to couple these costs at the first possible violation. Select internal gaps $a < b$, with weights M_1, M_2, and let $A = \{a - 1, a, b - 1, b\}$, $m = |A| \in \{3, 4\}$, and $\theta = \min(M_1, M_2)$. A tree is A-closed when no outside key is an ancestor of an A-key. All weights below may be zero.

Lemma 101 (First violating skeleton)**.** *In a nonclosed tree the selected gap depth sum D is at least $m + 2$. If $D = m + 2$, the union of the two search paths has exactly $m + 1$ keys: the keys in A and one outside key. The paths share only the root.*

Proof. Each key neighboring a gap is an ancestor of that gap. Thus the path union contains A. Nonclosure puts an outside ancestor of an A-key in the same union. If $L \geq 1$ keys are shared, its size is $D - L$, so $D \geq m + 1 + L \geq m + 2$. Equality forces $L = 1$ and union size $m + 1$. □

The one outside key is called the bridge. Its inorder position is in one of the light intervals between keys of A. Enumerating binary-tree shapes on A and the bridge takes at most $C_5 = 42$ shapes per interval. Only nonclosed shapes with selected depth sum $m + 2$ are retained.

Lemma 102 (Unordered mixed forests)**.** *Let $F_k(P, Q)$ be the minimum mixed cost of a k-root binary forest with roots at depth zero, internal-weight multiset P, and leaf-weight multiset Q, where $|Q| = |P| + k$. Weights may be permuted within each type and empty trees are allowed. Sort P ascending. For each weight in that order, merge the two smallest current leaf weights with it, add the resulting weight to the cost, and reinsert it as a leaf. After all merges the accumulated cost is $F_k(P, Q)$.*

Proof. Choose a deepest internal node in an optimal forest; it is a cherry. Exchange weights to put a smallest internal weight there and the two smallest leaf weights at its children, without increasing cost. Contract the cherry to a leaf of their total weight. Its removed cost is exactly that total. Conversely, expanding this leaf in any feasible contracted forest has that added cost. Induction proves the recurrence, including forests consisting only of leaves. Nonnegative weights ensure the generated merge weights are nondecreasing. Two queues therefore take linear work after sorting, for total $O(n \log n)$ work. □

Delete A from inorder to obtain $m + 1$ intervals. Two are the selected singleton gaps. For every other region s, write its weights as (P_s, Q_s) and mass as W_s. In a retained skeleton, let h_s be its hole depth. An ordinary region contributes the lower bound

$$h_s W_s + F_1(P_s, Q_s).$$

If the bridge is in region t, there are two holes. Lower their depths to their minimum h. If the bridge comparison length is ℓ, then $h \geq \ell$, since both holes descend from it. All other internal slots in this relaxed region have comparison length at least $h+1$. Exchanging internal weights therefore places a largest regional weight x at the bridge. The relaxed contribution is

$$\ell x + h(W_t - x) + F_2(P_t \setminus \{x\}, Q_t). \tag{121}$$

This exchange applies after lowering the two hole depths; it does not identify the bridge key in the original ordered tree. Keep the exact costs of all A-keys and both selected gaps. Minimizing the sum of these costs and the regional bounds over retained skeletons gives B_{first}, a valid lower bound for $D = m + 2$.

Let B_0 be the unordered mixed-tree optimum after zeroing the two selected gap weights, retaining all other weights. Each internal gap has depth at least two, so define

$$B_{\text{basic}} = 2(M_1 + M_2) + (m-2)\theta + B_0,$$
$$B_{\text{tail}} = 2(M_1 + M_2) + (m-1)\theta + B_0.$$

The first bounds every nonclosed tree; the second bounds those with $D \geq m + 3$.

Corollary 103 (Coupled exclusion certificate)**.** *Every nonclosed tree has cost at least*

$$B = \max\{B_{\text{basic}}, \min(B_{\text{first}}, B_{\text{tail}})\}.$$

If the first class is empty, replace the inner minimum by B_{tail}. An A-closed witness of cost $U \leq B$ certifies that some unrestricted optimum is A-closed. Its exact construction then requires independent exact solutions of the intervening intervals and a constant-size core. The screening work is $O(n \log n)$.

Proof. Lemma 101 partitions excluded trees into the first and tail classes. Each has its stated lower bound, and both have B_{basic}. The witness is no more costly than any excluded tree. An optimum of the closed class is consequently globally optimal. There are constantly many regions, forest computations, and skeletons; Lemma 102 gives the screening bound. □

The implementation first uses midpoint payloads and an exact full-weight core as the witness. Optional refinement computes the exact closed candidate when this witness fails. An uncertified attempt falls back to the preceding exact method; the general worst case remains quadratic.

4.12.1 A family with arbitrary nonuniform key positions

Constant key weights require a scope qualification. For $p_i = c$, the internal/external path-length identity $E = I + 2n$ gives

$$C(T; c, q) = \sum_g (q_g + c) d_T(g) - cn.$$

Such inputs reduce to alphabetic leaf-weight problems. The next family instead has a linear number of nonconstant key weights.

Corollary 104 (Nonuniform-key family)**.** *Let $n = 2^h$, $h \geq 5$, $r = n/16$, and $s = r - 1$. All gaps have weight one except $q_1 = q_{n-1} = M = 3n$. All keys have weight one except any s keys in positions $2, \ldots, n-3$, which have weight two. The midpoint witness passes Corollary 103 for every placement of these keys. Combining it with the existing mass-frontier construction takes $O(n^\alpha)$ arithmetic work, where $\alpha = 1 + \log_{2\varphi} 2 < 1.59024$. For $h \geq 6$, the earlier fixed-pair bound B_{basic} cannot certify even the exact closed optimum on this family.*

Proof. For unit keys write $U_0 = (2h+3)n - 6h - 3$. The middle payload has $n-4$ keys, mass $2n-7$, and optimal relative cost $(2h-1)n - 6h - 2$. The minimum four-key core has selected depth sum five, middle hole depth two, and other contribution thirteen, giving cost $5M + 4n - 1$. For completeness, the distinct other triples (selected depth sum, middle depth, other contribution) among its 14 shapes are $(5,3,12), (6,3,13), (6,3,15), (6,4,14), (7,4,13)$. None is cheaper. Thus the unit-key closed optimum is $5M + U_0$. A midpoint payload attains that baseline cost and has maximum relative key comparison length h, so the nonuniform witness satisfies

$$U \le 5M + U_0 + (h+2)s. \tag{122}$$

Let $G_k(t)$ be the minimum sum of comparison lengths of t marked internal nodes in a k-root forest. Filling the capacities $k2^{\ell-1}$ at comparison lengths $\ell = 1, 2, \dots$ gives

$$G_1(s) = (h-5)r + 1, \qquad G_2(s-1) = (h-6)r + 2.$$

In the unit first-violation relaxation the middle two-root forest has $n-5$ internal nodes. Its balanced profile has three leaves at depth $h-2$ and $n-6$ at depth $h-1$. The forest identity $E = I + 2(n-5)$ yields

$$F_2 = (2h-3)n - 6h + 5.$$

There are ten admissible five-key skeleton profiles. Their distinct triples (minimum middle hole depth d, bridge length ℓ, fixed core/endpoint contribution c) are

$$(2,2,16),\ (2,2,15),\ (3,3,13),\ (2,1,15),\ (3,1,14),\ (2,1,16).$$

Their unit costs are $6M + 2d(n-4) + \ell + c + F_2$; the minimum at $(2,1,15)$ is

$$B^{\text{unit}}_{\text{first}} = 6M + (2h+1)n - 6h + 5.$$

With nonuniform keys the bridge takes weight two in the relaxation. The remaining $s-1$ extra units receive hole shift at least two and forest comparison lengths summing to at least $G_2(s-1)$. The bridge adds at least one. Therefore

$$B_{\text{first}} \ge B^{\text{unit}}_{\text{first}} + 2s - 1 + G_2(s-1).$$

The residual unordered optimum is exactly

$$B_0 = 2nh - n + 2 + G_1(s).$$

For the unit baseline, absorbing the key weights leaves two leaf weights one and $n-1$ weights two. Their Huffman tree is a perfect h-level tree with one bottom leaf expanded into the two unit leaves; subtract n to recover mixed cost $2nh - n + 2$. Every tree pays at least $G_1(s)$ for the extra key units. The baseline optimal shape has its top $h-4$ internal levels full, so these same slots attain $G_1(s)$ simultaneously in the unordered problem, establishing equality.

Substitute $M = 3n$ in these bounds and in (122):

$$B_{\text{first}} - U \ge 5n/8 + h + 9 > 0, \qquad B_{\text{tail}} - U \ge 25n/16 + 7h + 8 > 0.$$

The witness certifies every placement. The remaining middle interval has key weights in $[1,2]$ and unit gap weights. Its bounded-positive mass-frontier bound is $O(n^\alpha)$, and the $O(n \log n)$ screening and full reconstruction fit within this bound. This exponent is inherited from that architecture, not a new unrestricted exponent.

Finally, every closed tree costs at least its unit-background closed optimum plus $G_1(s)$. Since $B_{\text{basic}} = 6M + B_0$, its deficit is at least $n - 6h - 5 > 0$ for $h \ge 6$. This is a separation from the earlier certificate for this fixed pair, not a claim about every alternative core its selector could attempt. □

The family's total mass is $O(n)$ and all key weights are at least one. Consequently the earlier global mass frontier already has the same $O(n^\alpha)$ guarantee on the entire instance. Corollary 104 is a certificate separation with arbitrary nonuniform key placements, not an extension of the known subquadratic input regime. Its practical value must be judged by actual construction work and matched measurements.

4.13 Certified pruning around affine weight profiles

The dense ruler example suggests retaining its structure symbolically. Here it bounds the states of a shape DP instead of the entries of a Monge matrix. Perturbations may change the optimal tree; we optimize over all retained shapes and separately certify global optimality. The result covers arbitrary signed perturbations on both keys and gaps, under explicit size conditions. The unrestricted question and priority remain open.

4.13.1 An affine potential and a signed residual bound

Let $h = \lfloor \log_2(n+1) \rfloor$, $S = 2^h$, $s = n - (S-1)$, and $z = S - s$. A minimum internal-path-length shape expands s of S skeleton slots. Let T_0 expand the first s slots. Choose nonnegative slopes λ, μ and reference weights $\bar{p}_i = \lambda i$, $\bar{q}_i = \mu i$. Constant intercepts contribute equally to every shape in this class and can be omitted. Define

$$\gamma = \lambda + 2\mu, \qquad F(g) = (2\lambda + 2\mu)g - \lambda \operatorname{popcount}(g).$$

By (133), moving an expansion one slot right costs $\gamma + \lambda\nu_2(g+1) = F(g+1) - F(g)$. Thus the affine cost of a selected slot set A, relative to T_0, is

$$G(A) = \sum_{g \in A} F(g) - \sum_{g=0}^{s-1} F(g). \tag{123}$$

For $\gamma > 0$, it is at least γ times the number of unexpanded slots preceding expanded slots.

The analytical residuals $\delta_p = p - \bar{p}$, $\delta_q = q - \bar{q}$ may be signed. Actual problem weights remain nonnegative. All minimum-path-length shapes have the same key and gap depth multisets. Assigning the largest residual weights to the shallowest lengths, separately by type, gives a valid rearrangement lower bound even for signed residuals. With descending sorts indicated by $\downarrow$, it is

$$\begin{aligned} L_\delta = {} & \sum_{i=0}^{n-1} \delta_{p,i}^{\downarrow} \operatorname{bitlength}(i+1) + h \sum_i \delta_{q,i} \\ & + \text{sum of the smallest } 2s \text{ gap residuals.} \end{aligned}$$

The last term is zero for $s = 0$. Set

$$V = C_\delta(T_0) - L_\delta \geq 0. \tag{124}$$

For a class optimum T^*, comparison with T_0 gives

$$G(T^*) \leq C_\delta(T_0) - C_\delta(T^*) \leq V.$$

Adding a constant to all residuals of either type changes both terms in (124) equally. The bound is therefore independent of intercepts and takes $O(n \log n)$ work including sorting.

4.13.2 A certified corridor with few states

At a slot-prefix state let a be the number of zeros already used and b the number of ones still needed, with $0 \le a \le z$, $0 \le b \le s$. Increasing F makes the earliest available slots optimal on each side of this prefix. The exact minimum affine regret of any completion through it is

$$K(a,b) = \sum_{t=s-b}^{s-1} \big(F(t+a) - F(t)\big). \tag{125}$$

For $P(t) = \sum_{g=0}^{t-1} F(g)$, this is $P(s+a) - P(s-b+a) - P(s) + P(s-b)$, computable in constant time after a linear prefix pass. Every prefix of T^* satisfies $K(a,b) \le V$. Delete the others and run the existing two-transition shape DP using the *original* weights. The bound preserves a class optimum; unnecessarily retained paths are still feasible original shapes.

Theorem 105 (Affine-regret corridor)**.** *For $\gamma > 0$, put $D = \lfloor V/\gamma \rfloor$. Exact optimization over minimum internal-path-length shapes, including preprocessing and reconstruction, takes*

$$O\Big(n \log(n+1) + \min\{n^2, D \log(n+1)\}\Big)$$

arithmetic work and $O(n + \min\{n^2, D \log(n+1)\})$ auxiliary space.

Proof. Each term in (125) is at least γa, so a retained state must satisfy $ab \le D$. The two boundary segments have $S+1$ states; the total is bounded by

$$S + 1 + \sum_{a=1}^{z} \min\{s, \lfloor D/a \rfloor\} \le S + 1 + \min\{sz, D \operatorname{bitlength}(z)\}. \tag{126}$$

Group the harmonic sum into dyadic blocks for the last inequality. Popcounts and P are computed in a linear pass, each state has at most two transitions, and stored predecessors reconstruct all keys. Sorting for (124) adds $O(n \log n)$ work. □

For $\gamma = 0$ the safe bound is the full shape grid. If $V = 0$, the reference is already class-optimal; a perfect skeleton also bypasses DP. The implementation tries both orientations and uses the smaller certified state bound. The default full solver rejects a bound exceeding $n \operatorname{bitlength}(n)^2$ before starting the shape DP. An unbudgeted class optimizer remains available for the theorem above.

Global optimality still requires the mixed residual-Huffman gate. It also provides an exact preflight rejection: let L_{class} be the same order-free fixed-profile lower bound on the original weights and L_{out} the gate's lower bound for every nonminimum-path-length tree. If $L_{\text{class}} > L_{\text{out}}$, no class candidate can satisfy the gate. The solver goes directly to Knuth before fitting or shape optimization. This rejects an impossible certificate, not the possibility that a class tree is optimal. When preflight passes, its sorted lists and lower bound are reused. An unfiltered mode is retained as a performance control.

4.13.3 Robust fitting and full mixed-weight guarantees

For a list of length N, let $m = \lfloor N/2 \rfloor$. Take the median of the disjoint secants $w_{i+m} - w_i$, $0 \le i < m$, divide by m, round to the nearest integer, and clip at zero. Sorting costs $O(n \log n)$. If $w_i = A + \lambda i + u_i$ with integer $\lambda \ge 0$, the sum of secant errors is at most $E = \sum |u_i|$. Hence its median differs from λm by at most $E/\lceil m/2 \rceil$. The odd unused endpoint remains in every cost and certificate. Fitting quality controls the bound's usefulness; arbitrary fitted slopes still give valid pruning inequalities.

Corollary 106 (Arbitrary perturbations of a positive linear profile)**.** *Let* $n \geq 32$, $B = 4(n+1)^3$, *and*

$$p_i = B + i + u_i, \qquad q_i = B + i + v_i,$$

with integer perturbations. The implemented full constructor has the following guarantees, including inference of slopes, the global certificate, and complete reconstruction:

1. *If* $\sum |u_i| + \sum |v_i| \leq n$, *time and auxiliary space are* $O(n \log^2(n+1))$.
2. *If* $|u_i| \leq 1$ *and* $|v_i| \leq 1$ *independently at every position, time and auxiliary space are* $O(n \log(n+1))$.

Both types of weights are positive and may be nonmonotone. The returned tree is globally optimal in the full mixed-weight problem.

Proof. For the first case the forward fits recover $\lambda = \mu = 1$: $E/(m\lceil m/2 \rceil) < 1/2$ for either list at these sizes. Key lengths range from one to $h+1$ and gap lengths from h to $h+1$, even in the relaxed profile assignment. Writing $E_p = \sum |u_i|$ and $E_q = \sum |v_i|$,

$$V \leq hE_p + E_q \leq hn, \qquad \gamma = 3.$$

Equation (126) gives $O(n \log^2 n)$ states and a bound below the default budget. Choosing the better orientation cannot increase that bound.

For the second case every secant error is at most two, so the forward slopes are again recovered. Let $\rho_p, \rho_q \leq 2$ be residual ranges. The total key depth deficit from $h+1$ is

$$Q = n(h+1) - (I_{\min}(n) + n) = 2S - h - 2.$$

The profile's possible key gain is at most $\rho_p Q$. For gaps it is at most $\rho_q \min\{2s, z\}$, using either the deeper set or its complement. Thus $V < 6S$, $D < 2S$, and (126) gives $O(n \log n)$ states, again below the default budget for $n \geq 32$.

It remains to certify unrestricted optimality in both cases. Put $a_0 = \min p$, $b_0 = \min q$, and $R_0 = \sum(p - a_0) + \sum(q - b_0)$. In the first case,

$$a_0 + b_0 \geq 2B - n, \qquad R_0 \leq n^2 + (n+1)E_p + (n+2)E_q \leq 2n^2 + 2n.$$

In the second, $a_0 + b_0 \geq 2B - 2$ and $R_0 \leq n^2 + 4n + 2 \leq 2n^2 + 2n$. The residual cost of any class candidate is at most

$$(h+1)R_0 \leq 2n(n+1)^2 < 2B - n.$$

Every tree outside the class pays at least one extra unit of internal path length, whose constant-baseline penalty is $a_0 + b_0$. Therefore the existing global gate passes even with zero residual lower bound. In particular, preflight cannot reject these candidates. The computed class optimum is a full optimum. All weights and intermediate costs have $O(\log n)$ bits on these families, so the arithmetic bounds also hold in a word RAM with $O(\log n)$-bit words. □

These guarantees allow perturbations at arbitrary positions and do not require the reference tree to remain optimal. For example, at $n = 24$, $B = 62{,}500$, $p_i = B + i + (-1)^i$, $q_i = B + i$, the left-filled tree costs 13,252,388. Moving its expansion from slot 7 to slot 9 uses two adjacent exchanges and reduces the cost to the exact optimum 13,252,387. The previous one-direction exchange certificate rejects this input; the corridor DP finds and certifies the better tree.

4.14 Exact transfers for joint key/gap signatures

The affine corridor needs a favorable reference slope. The following representation instead compresses constant runs of a local mixed-weight signature, allowing arbitrary values between blocks and no small-residual assumption. Full optimality still requires a separate certificate.

4.14.1 A skeleton word and its exact reward

For $n \geq 1$, let $h = \lfloor \log_2(n+1) \rfloor$, $S = 2^h$, $m = S-1$, and $s = n-m$. Encode a minimum-path-length shape by a word on its inorder keys: a skeleton key is one and an added bottom cherry is zero. Exactly the words with m ones and no adjacent zeros are feasible. Initial and terminal zeros are allowed. Each zero selects a distinct bottom slot, so this is a bijection with the $\binom{S}{s}$ slot subsets.

The skeleton key of rank r among the ones has comparison length $h - \nu_2(r)$; a cherry has length $h+1$. A gap has length $h+1$ exactly when adjacent to a cherry, and otherwise length h. Put $c_i = q_i + q_{i+1}$. Since zeros are nonadjacent, no deep gap is counted twice. The full original cost of a feasible word x is therefore

$$C(x) = A - \sum_{i:x_i=1} \big((1+\nu_2(r_i))p_i + c_i\big), \tag{127}$$
$$A = (h+1)\sum p_i + h\sum q_i + \sum c_i.$$

Both weight types remain in this identity. It does not apply to arbitrary taller shapes.

Partition (p_i, c_i) into its k maximal constant runs. In a block of length L with signature (a, c), define

$$U(r) = (2a+c)r - a\,\mathrm{popcount}(r).$$

The sum of $\nu_2(t)$ through r is $r - \mathrm{popcount}(r)$. Consequently every feasible block increasing the skeleton count from r_0 to r_1 earns exactly $U(r_1) - U(r_0)$. Its internal arrangement affects only feasibility, not reward.

Let t be the previous key's bit and u the final bit of a nonempty block. Its possible number of ones is every integer in $[\ell, b]$:

t	u	ℓ	b
1	1	$\lceil L/2 \rceil$	L
0	1	$\lfloor L/2 \rfloor + 1$	L
1	0	$\lfloor L/2 \rfloor$	$L-1$
0	0	$\lceil L/2 \rceil$	$L-1$

Empty ranges are infeasible. These bounds follow by separating zeros with ones, respecting the first and last bits; every intermediate count is attained. Use virtual initial bit $t = 1$ and either final bit.

4.14.2 Sliding transfers and linear-space reconstruction

For a prefix reward profile $D_t(r_0)$, the exact next profile is

$$D'_u(r) = U(r) + \max_{t \in \{0,1\}} \max_{r-b \leq r_0 \leq r-\ell} \{D_t(r_0) - U(r_0)\}. \tag{128}$$

Bounds ℓ, b depend on t, u. Exclude unreachable and out-of-range entries. The inner maximum is a sliding window; a standard monotone deque evaluates all ranks in linear time. Each candidate is inserted and removed at most once. There are four bit pairs. Backward profiles use the analogous window over $r_1 \in [r+\ell, r+b]$ with values $B'_u(r_1) + U(r_1)$, then subtract $U(r)$. Popcounts are tabulated once.

Theorem 107 (Exact joint-signature optimizer)**.** *For arbitrary nonnegative mixed weights, let k be the number of runs of $(p_i, q_i + q_{i+1})$. Exact optimization over all minimum-path-length shapes, including full reconstruction, takes $O(nk)$ arithmetic operations and $O(n)$ auxiliary space for $n \geq 1$.*

Proof. An initial sweep finds the maximum final reward at rank m. For traceback, split the block list in half and compute its left forward and right backward profiles between fixed endpoint ranks and bits. Their sum selects an optimal middle state. Release both profiles before recursively solving the two halves. A one-block leaf explicitly places its required count of nonadjacent zeros using the boundary rules. The complete word yields all comparison lengths and the existing linear Cartesian reconstruction. This uses Hirschberg's established recomputation principle [15].

At recursion level d, an active subproblem has at most $\lceil k/2^d \rceil$ blocks. The sum of active rank-interval widths is at most $m+2^d$, because endpoint rank spans partition the original span. Summing sweep work gives $O(km + k\log(k+1))$, plus $O(n)$ preprocessing and output. Since $m = \Theta(n)$ and $k \leq n$, this is $O(nk)$. Only a constant number of profiles is retained at once; releasing them before recursion prevents a stack of profiles. All other summaries and the output occupy $O(n)$ space. □

A perfect shape or a single signature block is constructed directly. The potential and transfer use arbitrary block values; they do not assume that an affine approximation is accurate or that the reference tree stays optimal.

4.14.3 Certification and full mixed-weight guarantees

The mixed residual-Huffman gate remains

$$C_* \leq (a_0 + b_0)(I_{\min}(n) + 1) + (a_0 + 2b_0)n + H, \qquad a_0 = \min p, \quad b_0 = \min q.$$

It can also be built from the compressed signatures. Each block contains one key weight. Its gap weights obey $q_{i+1} = c - q_i$, so they alternate between at most two values. Record the block's L keys as one weighted multiplicity and its L left gaps as at most two; finally add q_n once. There are at most k key records and $2k+1$ gap records. Sort these records, expand their multiplicities, and run the two-queue bound. The total work is $O(k\log(k+1) + n)$ and space is $O(n)$. Gaps at block boundaries occur once. Thus an accepted full solve retains the theorem's time and space bounds. Failed gates use exact Knuth; the fixed-profile preflight rejection also remains available.

Corollary 108 (Arbitrary mixed block values above a baseline)**.** *For $n \geq 1$, $M \geq 1$, $0 \leq u_i, v_i \leq M$, set*

$$B = 4(n+1)^2 M, \qquad p_i = B + u_i, \qquad q_i = B + v_i.$$

If $(u_i, v_i + v_{i+1})$ has k runs, the standalone signature constructor returns a full optimal mixed-weight BST in $O(nk)$ arithmetic operations and $O(n)$ auxiliary space. Constant k gives linear construction and $k = o(n)$ gives subquadratic construction.

Proof. The residual mass above a_0, b_0 is at most $(2n+1)M$. Every class candidate has residual cost at most $(h+1)(2n+1)M < 2B \leq a_0 + b_0$. A tree outside the class pays an additional baseline cost of at least $a_0 + b_0$. Hence the gate passes even with $H = 0$, and preflight cannot reject. The exact class optimum is a full optimum. The preceding theorem and compressed certificate give the stated work and space. □

The weights are positive and both types may be nonconstant. Key values can change arbitrarily between blocks, and gap weights can alternate at every position. No small perturbation budget or affine-closeness assumption is imposed on u, v. If M is polynomial in n, all weights and costs

have $O(\log n)$ bits, giving the corresponding word-RAM bound. For arbitrarily large M, bit costs remain additional. The large baseline is a real restriction; few signatures alone do not guarantee the full gate. For example, $p = (100, 1, 1, 1)$ and $q = (1, 1, 1, 1, 1)$ have only two signatures, but their best minimum-path-length shape costs 218 and their full optimum is 121. Direct enumeration of all fourteen trees, cubic DP, and Knuth agree; the separate gate correctly rejects the class candidate.

4.14.4 Bounded routing and its space qualification

Generic transfer sweeps and recomputation have substantial constants when $k = n$. The initial matched experiment exposed roughly tenfold regressions against Knuth in this regime. The standalone kernel remains an explicit control. The implemented portfolio first scans signatures and selects it only when $k \le \max\{16, \lfloor n/32 \rfloor\}$; otherwise it calls the existing affine solver and its exact fallback. The empirical threshold is not a correctness assumption or a claim of optimal routing.

On the preceding positive family, portfolio work is still $O(nk)$: the signature route certifies, while rejected block counts satisfy $k > n/32$, making even quadratic fallback $O(nk)$. The portfolio does not preserve the standalone kernel's linear-space guarantee on every such input; the affine or Knuth route may use more space. The separate standalone constructor retains the theorem's time and space bounds. All scans, certificates, and output reconstruction are included in the implementation and matched measurements.

4.15 Global alternating-gap bounds and linear profile construction

The preceding signature representation can have $k = n$ even on a short alternating pattern. We now avoid its rank table by recognizing attainment of a global key-depth lower bound. The crucial gap inequality removes the large-baseline restriction on the recognized family.

4.15.1 A gap lower bound valid for all tree shapes

For $n \ge 1$, set $h = \lfloor \log_2(n+1) \rfloor$, $S = 2^h$, and $s = n + 1 - S$. Suppose q_i alternates between arbitrary nonnegative values a, b. Every minimum-internal-path-length shape has gap cost

$$G(q) = h \sum_{i=0}^{n} q_i + s(a + b).$$

Lemma 109 (Alternating-gap component bound)**.** *Every BST on these keys, including every taller tree, has gap contribution at least $G(q)$. Every minimum-internal-path-length shape attains equality.*

Proof. Fix one parity of the $n + 1$ inorder gaps. Give its H leaves unit weight and the other $Z = n + 1 - H$ leaves zero weight. Delete a zero leaf and contract its parent into its sibling. The sibling contains a unit-weight leaf: remaining zero leaves are separated in inorder by unit leaves, and a subtree's leaves are contiguous. An all-zero sibling would contradict that separation. Deleting only zero leaves preserves separation. Each contraction decreases the selected depth sum by at least one. After all Z contractions, a full tree with H unit leaves remains. Its external path length is at least $E_{\min}(H)$.

Here $S/2 \le H \le S$, so the standard balanced-leaf formula gives $E_{\min}(H) = (h+1)H - S$, valid at both endpoints. Thus the original selected depth sum is at least

$$E_{\min}(H) + Z = (h+1)H - S + n + 1 - H = hH + s.$$

A minimum-path-length shape has gaps at depths $h, h+1$. Each of its s bottom cherries has two adjacent gaps, one of each parity, at depth $h+1$. Thus its selected depth sum is exactly $hH + s$. Apply the inequality to both parities and multiply by the nonnegative weights a, b. □

Let $p_{(1)} \geq \cdots \geq p_{(n)}$ be the key weights sorted by value, ignoring their fixed positions only for a lower bound. There are at most 2^{d-1} keys at comparison length d. Consequently the sorted lengths of every BST dominate $\text{bitlength}(1), \ldots, \text{bitlength}(n)$ componentwise. Rearrangement yields the globally valid bound

$$K(p) = \sum_{r=1}^{n} p_{(r)} \,\text{bitlength}(r), \qquad \text{OPT}(p, q) \geq K(p) + G(q). \tag{28}$$

An ordered tree attaining both components is globally optimal. This compares with all shapes; it does not assume that an optimum is nearly complete. Finding a tree attaining $K(p)$ is the remaining constructive task.

4.15.2 Two key levels reduce to interlaced intervals

Suppose key weights have two distinct levels $\ell < u$, and let H count the high keys. Put $m = S - 1$ and assume $S/2 - 1 \leq H \leq m$. In a minimum-path-length shape, even skeleton ranks have comparison length less than h, odd skeleton ranks have length h, and cherries have length $h + 1$. Attainment of $K(p)$ is therefore equivalent to retaining every high key in the skeleton and placing high keys at every even skeleton rank. Low keys may occupy odd ranks or cherries. The inorder word must have m ones and no adjacent zeros, as in the preceding section.

Read the high/low pattern from left to right. A high key must be retained; a low key can be retained only when its new rank is odd. At a boundary immediately after a high key, the previous bit is one. Encode reachable even ranks as $2A$ and odd ranks as $2B+1$. The virtual initial boundary has $A = \{0\}$, $B = \varnothing$.

Lemma 110 (Interlaced parity intervals)**.** *The sets A, B are integer intervals, possibly empty. When both are nonempty, their lower endpoint difference and their upper endpoint difference each belong to $\{0, 1\}$. Four endpoints and empty flags thus represent the exact reachable state.*

Proof. Let L low keys precede the next high key. The exact transformations are

L	A'	B'
0	$B + 1$	A
1	$(A \cup B) + 1$	A
2 or 3	$A + 1$	$\varnothing$
≥ 4	$\varnothing$	$\varnothing$

For $L = 1$, remove the low key or retain it at an odd rank. For $L = 2$, the low word is 01 or 10; for $L = 3$ it is 010. These require an even incoming rank. A low-only run can retain at most one key, because two retained lows would occupy consecutive skeleton ranks, one even. At most one removed key can lie on either side, so $L \geq 4$ is impossible.

For $L = 0$, each endpoint difference becomes one minus its previous value. For $L = 1$ with both intervals nonempty, their union is $[\min B, \max A]$. The new lower difference is $1 + \min B - \min A \in \{0, 1\}$ and the upper difference is one. If only A is nonempty, the pair becomes $(A + 1, A)$; if only B is nonempty, only $B + 1$ survives as A'. The remaining cases are immediate. This proves the invariant inductively. □

For a terminal low run, write R for all incoming ranks and E for its even subset. The final ranks are R for $L = 0$, $R \cup (E + 1)$ for $L = 1$, $E + 1$ for $L = 2, 3$, and empty otherwise. Save each high boundary's intervals. After selecting final rank m, trace backward: each step has at

most two candidate predecessor ranks, and the short words above realize the chosen transition. Concatenating the words and applying the comparison-length Cartesian reconstruction returns the full tree in linear time and space. Parity cannot be discarded: the high-key pattern 010100 has reachable final ranks $\{3, 5\}$, so a single ordinary interval would invent rank four.

Corollary 111 (Full linear construction on attaining two-level patterns)**.** *For arbitrary nonnegative two-level key weights with $S/2-1 \le H \le S-1$ and arbitrary nonnegative alternating gap weights, the recognizer decides whether a minimum-path-length shape attains the stated key profile and constructs a witness in $O(n)$ arithmetic operations and $O(n)$ auxiliary space. Every accepted witness is a full optimal mixed-weight BST, without a baseline assumption.*

Proof. The interval transitions and traceback take constant work per high boundary and linear total input/output work. Only two key multiplicity records are ordered, so computing $K(p)$ also takes linear work. The recognized word assigns the high keys to the shortest H comparison lengths, attaining $K(p)$. Its minimum-path-length shape attains $G(q)$. Equality in (28) certifies global optimality, including comparison with every taller tree. □

One key level is handled directly. At perfect sizes $s = 0$, the implementation constructs and evaluates the unique class shape. Equality in (28) certifies it; otherwise the earlier mixed-Huffman gate may certify it. Failure invokes exact Knuth. Rejected recognition does not imply that every minimum-path-length shape is suboptimal. The standalone solver and routed portfolio both remain exact, with quadratic worst-case fallback. The linear-space guarantee applies to the successful route; general fallback can use quadratic space.

4.15.3 An explicit alternating family with no baseline

Corollary 112 (Alternating keys and gaps)**.** *For $n \ge 1$ with $n + 1$ not a power of two, strictly alternating two-level nonnegative key weights and alternating nonnegative gap weights admit full exact $O(n)$-time, $O(n)$-space construction. The relative weight magnitudes are unrestricted. Equal key levels are handled by the uniform route.*

Proof. Here $s > 0$. Let $A = 1$ if the first key is high and $A = 0$ otherwise. If $s \bmod 2 = A$, remove the first s low keys. Otherwise remove the first $s - 1$ lows and the last low. There are enough lows, and their positions are nonadjacent. Every high key remains. Before the initially removed lows end, retained keys are high. After removing t initial lows, a high key's skeleton-rank parity is $A - t$. The suffix therefore has high keys at even ranks when $t \bmod 2 = A$. In the second construction, removing the final low affects at most one subsequent high, which may occupy an odd rank. Every even skeleton rank is high. The high count lies in the theorem's range, so the recognizer succeeds and (28) proves the full claim. □

This family has $k = n$ joint signatures when the key levels differ. The interval representation therefore escapes the preceding $O(nk)$ bound without invoking a small-residual assumption. Nonperiodic accepted patterns are also possible: choose any minimum-path-length shape, mark all even skeleton ranks high, choose the odd skeleton labels freely, and label every cherry low. Recognition receives only the resulting weights, not this witness.

The perfect-size exception matters. For $p = (100, 1, 100)$ and four zero gaps, the unique minimum-path-length tree costs 401 while the full optimum is 303. Cubic DP and Knuth agree. Thus the corollary cannot include every size and orientation. Arbitrary integer magnitudes also retain their bit costs; polynomially bounded weights yield the corresponding word-RAM bounds.

4.16 Threshold runs and a linear Cartesian certificate

We now allow arbitrary nonnegative gap sequences and arbitrary key values in the certificate. The returned tree is optimal for the full mixed objective; failure still requires an exact fallback. This extends the preceding global alternating-gap argument rather than restricting the problem to alphabetic coding.

4.16.1 Binary indicators and arbitrary weighted layers

For $H \geq 1$, write

$$E(H) = (\lfloor \log_2 H \rfloor + 2)H - 2^{\lfloor \log_2 H \rfloor + 1}.$$

This is the minimum external path length of a full tree with H unit leaves; in particular $E(1) = 0$.

Lemma 113. *Give $H > 0$ of N ordered leaves weight one and the others weight zero. Let R be the number of contiguous zero runs, including endpoint runs. The exact minimum weighted external path length is $E(H) + R$. For $H = 0$ the optimum is zero.*

Proof. Remove every maximal zero-only subtree. Each lies within one zero run, and every run contains at least one such subtree. Each removed subtree's parent has a selected descendant in its other child, by maximality. Suppressing the resulting unary nodes therefore decreases the selected depth sum by at least one per removed subtree. What remains is a full H-leaf unit tree, with cost at least $E(H)$. This proves the lower bound.

For attainment, start with any optimal H-leaf unit tree. Assign each interior zero run to the selected leaf immediately to its left; assign an initial run to the first selected leaf's left and a final run to the last selected leaf's right. Each selected leaf has at most one right run, and only the first can also have a left run. Expand each such leaf to a small ordered tree accommodating the required zero-only subtrees. One assigned run increases its selected depth by one; two runs increase it by two. The zero-only subtrees have arbitrary required sizes and zero cost. All original leaves and their order are preserved, with total cost $E(H) + R$. □

For arbitrary $q = (q_0, \ldots, q_{N-1})$, let $Q = \max_i q_i$. At threshold $0 < t \leq Q$, let H_t count weights at least t and R_t count runs of weights below t. Lemma 113 gives

$$\text{gapcost}(T, q) \geq G(q) := \int_0^Q (E(H_t) + R_t)\, dt \qquad \text{for every tree } T. \tag{129}$$

These integrals are exact finite layer sums. The minimum and sum need not commute: $q = (100, 10, 1, 10)$ has $G(q) = 152$, but alphabetic optimum 153.

Let $q_{[r]}$ denote the rth largest weight. Since $E(r) - E(r-1) = \lceil \log_2 r \rceil + 1$ for $r \geq 2$, the bound becomes

$$F(q) = \sum_{r=2}^{N} (\lceil \log_2 r \rceil + 1) q_{[r]}, \tag{130}$$

$$D(q) = Q - q_0 + \sum_{i=1}^{N-1} \max(q_{i-1} - q_i, 0), \tag{131}$$

$$G(q) = F(q) + D(q).$$

For (131), a zero run starts at position zero for $q_0 < t \leq Q$, or at position $i > 0$ for $q_i < t \leq q_{i-1}$. Equivalently,

$$D(q) = Q - \frac{q_0 + q_{N-1}}{2} + \frac{1}{2} \sum_{i=1}^{N-1} |q_i - q_{i-1}|.$$

Variation terms are familiar in alphabetic-code bounds, including upper bounds reviewed by Bruno et al. [18]. The present lower-bound derivation and equality certificate do not by themselves establish priority.

4.16.2 Dyadic selection avoids sorting the gap weights

Theorem 114. *For arbitrary nonnegative weights, $G(q)$ can be computed in $O(N)$ arithmetic and comparison operations and $O(N)$ auxiliary space.*

Proof. The variation term is one scan. For a current multiset of $M > 1$ weights, let k be the largest power of two strictly less than M. Every rank $k+1, \dots, M$ has coefficient $\log_2 k + 2$ in (130). Use deterministic linear selection to retain the k largest values and charge this coefficient times the discarded sum. Repeat on the retained multiset; the last single weight contributes zero. After the first step the retained sizes halve. The total sizes processed, $M, k, k/2, \dots, 2$, sum to less than $3M$. Standard median-of-five selection [16], partitioning, summation, and copying are linear in each processed size. Their total is therefore linear, with linear auxiliary storage. Equal values at a partition boundary may be allocated to either side to obtain exactly the required cardinality. □

The implementation also uses a cheaper scan-only formula when applicable. Put $h = \lfloor \log_2 N \rfloor$, $S = 2^h$, $s = N - S$, and $\beta = \min_i q_i$. If at least $S/2$ gaps have weight Q and at least s have weight β, then

$$F(q) = (h+1)\sum_i q_i - SQ + s\beta. \tag{132}$$

Indeed, for $\beta < t \le Q$, $S/2 \le H_t \le S$, and $E(H_t) = (h+1)H_t - S$. The part below β is $\beta E(N)$, where $E(N) = hN + 2s$; integration gives (132). Uniform weights and $N = 1$ also satisfy the formula. These checked multiplicities choose an evaluation method, not an optimality certificate.

4.16.3 Checking the key profile without sorting

The preceding key capacity bound remains

$$K(p) = \sum_{r=1}^{n} p_{[r]} \lfloor \log_2 r + 1 \rfloor, \qquad \mathrm{OPT}(p, q) \ge K(p) + G(q).$$

Let H be the number of positive keys in a candidate tree and d_i their comparison lengths. Define $M(H) = \sum_{r=1}^{H} \lfloor \log_2 r + 1 \rfloor$.

Lemma 115. *A candidate's key cost equals $K(p)$ if and only if*

1. *$\sum_{i:p_i>0} d_i = M(H)$; and*
2. *every positive weight at a shallower depth is at least every positive weight at a deeper depth.*

Both conditions are testable in linear work and space.

Proof. The sorted positive comparison lengths dominate the capacity profile $\lfloor \log_2 r + 1 \rfloor$ componentwise. All weights being paired with these lengths are strictly positive. Thus any excess length makes even the rearranged cost strictly larger than $K(p)$. Equality of the unweighted sums forces equality of the complete length profiles. The second condition is precisely equality in the rearrangement step: there is no strict weight inversion between different lengths. A traversal collects the length sum and each positive depth's minimum and maximum weight; adjacent nonempty depth levels suffice for the second test. When $H = 0$ the conditions are vacuous. □

Corollary 116. *Construct the max-Cartesian tree on p, with fixed leftmost priority for ties. If its key cost equals $K(p)$ and gap cost equals $G(q)$, it is a globally optimal full mixed-weight BST. Candidate construction, testing, and successful output take $O(n)$ arithmetic work and auxiliary space. If all keys have positive pairwise distinct weights, this test succeeds if and only if some tree attains $K(p) + G(q)$.*

Proof. The two lower bounds hold for every tree, so their simultaneous attainment proves optimality. Standard monotone-stack Cartesian construction [17] is linear; use Lemma 115 for the key test and Theorem 114 for the gap test. Iterative traversal and reverse traversal construct all output nodes in linear work, even when the candidate has linear height.

For distinct positive weights, any attaining tree has weights ordered by depth, so every parent has strictly greater weight than its children. The inorder max-heap with distinct priorities is the unique Cartesian tree. Hence any attaining tree must be the tested candidate. □

With ties, failure does not prove nonattainment. For $p = (1, 1, 1)$ and $q = (1, 1, 1, 1)$ the balanced tree attains cost 13, whereas leftmost Cartesian tie-breaking produces a chain. Even without this candidate issue, simultaneous attainment is not universal: for $p = (2, 1, 1)$ and $q = (100, 10, 1, 10)$, the mixed bound is 158 and full optimum 160. The standalone solver uses exact Knuth fallback. The portfolio first tries the previous profile certificate, then the Cartesian certificate, then the earlier signature/affine portfolio. The unrestricted worst case remains quadratic. All linear claims here count exact arithmetic operations; they are not bit-complexity or instruction-count bounds.

4.16.4 A positive family with many key and gap levels

Corollary 117. *Start with a perfect skeleton of $S - 1$ keys, $S = 2^h$, and expand any s odd-indexed bottom slots to cherries, where $0 \le s \le S/2$. Give every even slot gap weight $Q > 0$, every unexpanded odd slot an arbitrary weight in $[\beta, Q]$, and both gaps of each expanded slot weight β, where $0 < \beta \le Q$. Give all keys positive weights strictly decreasing between depth layers, with arbitrary weights within each layer. The Cartesian certificate constructs a full optimal tree in $O(n)$ arithmetic work and space, where $n = S - 1 + s$.*

Proof. For $\beta < t \le Q$, every even slot is selected. All selected gaps have depth h. If H are selected, zero runs number $S - H$: every zero slot is odd and separated from the others by selected even slots. Expanding a zero slot into two zero gaps does not change this run count. Since $S/2 \le H \le S$, we have $E(H) + (S - H) = hH$, exactly the candidate cost. For $0 < t \le \beta$, all $N = S + s$ gaps are selected and the minimum-path-length shape attains $E(N)$. The candidate therefore attains every threshold optimum and thus $G(q)$. Its key lengths form the capacity profile, and the assigned weights decrease between depths, so it attains $K(p)$. Strict ancestor inequalities make the Cartesian construction recover this tree, even if some same-depth weights coincide. □

For the explicit tested family let $s = S/4$, $\beta = 1$, $Q = n^2 + 2$, give unexpanded odd slot j weight $j + 2$, and assign

$$p_i = (h + 2 - d_i)(n + 1) + i.$$

All n key weights are distinct, and there are $S/4 + 2$ gap levels. Both alphabets have $\Theta(n)$ values, gaps need not alternate, and there is no large common baseline. The weights have $O(\log n)$ bits. Independent positive rescaling of the key and gap vectors preserves the two equalities; the theorem requires no dominance between their masses. The solver receives only p, q, without the reference tree or slot pattern.

4.17 Shifted alphabetic certificates beyond key-capacity equality

Theorem 118. *Let $m = \min p$, with $m = 0$ when $n = 0$. Define*

$$L_m = K(p - m) + A(q + m) - mn.$$

Then $K(p) + G(q) \le K(p) + A(q) \le L_m \le \mathrm{OPT}(p, q)$. Among all admissible uniform transfers, m gives the largest such separate key/exact-gap bound. An alphabetic optimum T for $q+m$ with original full cost $C(T; p, q) = L_m$ is globally optimal. The bound, candidate, certificate, and complete tree construction take $O(n \log(n + 1))$ arithmetic work and $O(n)$ space.

Proof. The transfer identity gives $L(a) = K(p - a) + A(q + a) - an \le \mathrm{OPT}$ for all admissible a. Uniform subtraction preserves ranks and $K(p - a) = K(p) - aM(n)$, so

$$L(a) = K(p) + \min_T \left\{ \sum_i q_i d_i(T) + a \left(\sum_i d_i(T) - E(n + 1) \right) \right\}.$$

Every affine term has nonnegative slope. Their minimum is nondecreasing, so the largest admissible $a = m$ maximizes the bound. Taking $a = 0$ gives the claimed chain of inequalities. Equality of a feasible full cost with a lower bound proves optimality. Equivalently, the alphabetic candidate must attain $K(p - m)$ in the original residual key order.

Implement Hu–Tucker compatible merging using a leftist heap of internal items in each gap between consecutive original leaves, and a priority queue of each group's best compatible pair. Each pair is selected among at most four items: two boundary leaves and two heap minima. Removing a leaf melds its incident groups. There are $O(n)$ groups and events, each meld or priority operation takes $O(\log(n+1))$, and stale events are discarded. Recovering depths, alphabetic stack reconstruction, and full original-cost evaluation are linear. Sorting residual keys costs $O(n \log(n + 1))$. No interval table is used on acceptance. □

The tie convention exactly reproduces infinitesimal original-leaf weights 2^i without storing those large integers. Active items partition the leaves; at equal mass, individual items compare by their largest original index. Two pairs compare lexicographically by the larger, then the smaller, of their items' largest indices. A shared item cancels; otherwise the largest differing leaf determines the symbolic sum order. Only ordinary indices are stored. The implementation separately counts heap-item, priority-entry, and key-sort comparisons, as well as all events, melds, reconstruction, cost visits, and fixed scans. These are arithmetic/logical counts, with bit costs additional.

The six-key obstruction has $L_1 = 213$, and its shifted alphabetic candidate is the full optimum. Thus normalization repairs the bound when the exact alphabetic optimum replaces G, even though $K + G$ is invariant. Tied alphabetic optima can hide an attaining candidate, and rejection does not prove suboptimality or nonattainment. The standalone implementation uses exact Knuth fallback.

4.17.1 A positive nonuniform family with linear capacity deficit

Corollary 119. *Let $r = 2^h$, $n = 7r-1$, and repeat the original seven-gap vector $(18, 2, 13, 3, 17, 1, 2)$ exactly r times. Place the six-key witness tree above into every leaf of a perfect r-leaf skeleton. Give its r deepest keys weight 1, and every other key weight 1 plus a positive residual decreasing between depth layers. The shifted alphabetic certificate constructs a full optimal tree in $O(n \log(n+1))$ work. Its original key contribution exceeds $K(p)$ by exactly r.*

Proof. The $6r-1$ residual-positive keys fill all capacity levels above the last occupied level, with $2r$ keys in that last level. Their weights decrease between levels, so they attain $K(p-1)$. The r residual-zero keys have length $h+4$, one greater than their original unit-weight capacity slots at length $h+3$. This gives the deficit r.

The shifted gap block is $(19,3,14,4,18,2,3)$. In every block compatible merging performs $2+3=5$, $3+14=17$, $17+4=21$, and $18+5=23$. Each stage finishes in all blocks before the next: cross-block pairs do not undercut these costs. Per block remain an original leaf 19 and internal items $21, 23$. For even r, all original leaves pair at 38 before any competing remaining pair, removing all barriers. Then 21 pairs to 42 and 23 to 46, giving $r/2$ copies of twice $(19,21,23)$. Repeat until a single triple remains, with final merges $19+21=40$ and $40+23=63$. For $r=1$ these are already compatible. All same-stage ties give the same depths. Thus the ordered gap depths are $h+(2,4,4,3,2,3,3)$ per block, with alphabetic cost $r(169+63h)$. Reconstruction gives the stated tree, which attains both shifted terms. □

For a concrete integer family, if ℓ_i is its key comparison length, set $p_i=1$ at length $h+4$, and otherwise

$$p_i = 1 + (h+4-\ell_i)(n+1) + i.$$

This gives $6r$ distinct key weights, six distinct gap weights, strictly positive outcomes, common key minimum 1, and $O(\log n)$-bit weights. No key-versus-gap dominance assumption is needed for the equality proof. The family is outside the earlier original-capacity certificate for every r: its full cost exceeds $K(p)+G(q)$ by at least r.

4.18 Endpoint prices and the limit of additive transfers

The endpoint identities $\ell_0=d_0$ and $\ell_{n-1}=d_n$ permit transfers that the preceding uniform normalization does not exploit.

Theorem 120. *For $n\ge 3$, every additive weight change whose cost difference is independent of the tree has the following form. Put $L=p_0+q_0$ and $R=p_{n-1}+q_n$, and choose*

$$p' = (x, p_1-a, \dots, p_{n-2}-a, y),$$
$$q' = (L-x, q_1+a, \dots, q_{n-1}+a, R-y).$$

Then $C(T;p',q') = C(T;p,q)+an$. Nonnegativity is exactly

$$-\min_{1\le j<n} q_j \le a \le \min_{1\le i<n-1} p_i, \qquad 0\le x\le L, \quad 0\le y\le R.$$

Proof. The depth identity and endpoint equalities prove sufficiency. For necessity write $U_i=\delta p_i+\delta q_i$ and $V_i=\delta p_i+\delta q_{i+1}$. A two-key rotation gives $U_i=V_{i+1}$. Compare a rotation on keys $i+1, i+2$ with and without key i in its outer left subtree: subtraction gives $U_i=0$ for $0\le i\le n-3$. The symmetric comparison gives $V_i=0$ for $2\le i<n$. The two-key identities then supply $V_1=0$ and $U_{n-2}=0$. All these rotations can be embedded in full trees on the original keys. Consequently all interior gap changes equal one a, all interior key changes are $-a$, and the endpoint key/gap sums are unchanged. The resulting constant is an. □

This characterizes additive transfers on the given outcome coordinates, not arbitrary encodings or all possible lower bounds.

Theorem 121. *Let $F(a,x,y)=K(p')+A(q')-an$. The maximum of this valid lower bound over the entire admissible box is attained on the face $a=a_*:=\min_{1\le i<n-1} p_i$.*

Proof. Increase a by δ while replacing x, y by $\max(0, x-\delta), \max(0, y-\delta)$. On a segment with $e \in \{0,1,2\}$ endpoint residual keys clamped at zero, the other $n-e$ positive keys decrease uniformly until a_*. Thus K has slope $-M(n-e)$. All gaps except those e endpoints increase uniformly. By Lemma 113, their minimum depth sum is $E(n+1-e)+e = M(n-e)+n$. Every affine tree term in F therefore has nonnegative slope. Their minimum is nondecreasing; continuity joins the segments. Moving any feasible point to the maximal face cannot reduce F. □

4.18.1 An exact bounded two-price search

Fix $a = a_*$. An alphabetic optimum for q' and a decreasing assignment of p' to capacity lengths provide the value $F(x, y)$. If the assigned key lengths are λ_i and the alphabetic gap depths are d_j, they also give a global supporting plane

$$F(X, Y) \le c + (\lambda_0 - d_0)X + (\lambda_{n-1} - d_n)Y.$$

The plane is an upper bound on the concave *bound function*; the sampled $F(x, y)$ is a lower bound on the full optimum. The alphabetic tree evaluated with the original ordered weights gives a feasible upper bound on that optimum. These directions are distinct.

Maintain the planes and maximize their minimum over $[0, L] \times [0, R]$. The implementation enumerates vertices of the three-dimensional hypograph using integer determinants and Cramer's rule. Feasibility and objective comparisons use exact cross-products; prices and bound values are rational. There is no floating-point tolerance. Retain the largest sampled bound and the smallest original tree cost across calls. For integer original weights, equality of the latter with the ceiling of the former certifies full optimality. Equality between the LP upper bound and best sampled bound instead proves completion of the price maximization. An early full certificate or a call-budget stop need not complete that maximization.

With b alphabetic calls, all preprocessing, exact pricing, and candidate construction take

$$O\big(bn \log(n+1) + b^5\big) \text{ arithmetic work}, \qquad O(n+b) \text{ space}.$$

Each LP call checks $O(b^3)$ triples against $O(b)$ constraints. The default $b = \max(2, \text{bitlength}(n))$ gives $O(n \log^2(n+1) + \log^5(n+1))$, asymptotically $O(n \log^2 n)$, for the attempted certificate. Rejection uses Knuth; the unrestricted worst case remains quadratic. Supporting slopes have magnitude $O(n)$ and integer intercepts $O(nW)$, so vertex denominators are $O(n^2)$ and the scaled alphabetic weights use only $O(\log n)$ additional bits. Bit costs remain additional. The uncapped diagnostic has no proved small iteration bound here.

4.18.2 A positive family with distinct gaps and light endpoint keys

Corollary 122. *Let $h \ge 3$, $B = 2^h$, $n = B - 1$, and let $\ell_i = h - v_2(i+1)$ be the perfect tree's key comparison lengths. Initially set $p_i = B + (h+1-\ell_i)B + i$, then replace $p_0 = p_{n-1} = 1$ and $p_2 = B$. Set $q_0 = B$, $q_n = B + n$, and $q_j = j + 1$ for $1 \le j < n$. The first endpoint-price call constructs a full optimum in $O(n \log n)$ arithmetic work and $O(n)$ space, with exact cost*

$$\frac{(8h-11)B^2 + (13-7h)B}{2} - 1.$$

There are $n-1$ distinct key weights and $n+1$ distinct gap weights, all positive and with $O(\log n)$ bits. The preceding uniform-shift bound is at least three below the optimum for every such size.

Proof. At $a_* = B$, $x = y = 0$, exactly three bottom residual keys are zero. The other $n - 3$ keys attain capacity and decrease between depth layers. The transformed gaps are $B + j + 1$, all in $[B + 1, 2B]$. Unrestricted Huffman merging pairs these B outcomes level by level: at each level every newly generated mass exceeds every unmerged same-level mass since the maximum weight is less than twice the minimum. All leaves have depth h. The perfect alphabetic tree attains that unrestricted optimum and both transformed lower bounds, proving full mixed optimality. Summing the perfect lengths, their squares, and the three bottom-key corrections gives the formula.

The preceding uniform shift uses only $\min p = 1$. Its residual key bound is already attained by the perfect tree. In the last four shifted gaps, the two left weights sum to $2B - 3$, while the rightmost weight is $2B$. Changing the balanced four-leaf subtree to the right-heavy alternative saves three in the gap objective. Thus the old bound is strictly below the perfect full optimum, regardless of which candidate it constructs. □

Twelve complete family constructions reach $n = 262143$, with 262142 distinct keys, 262144 distinct gaps, and full cost 4569830391807. The preceding bound is 4561240391642, a deficit of 8590000165. The largest run uses one alphabetic call, 262143 merges, 7111153 priority comparisons, 8924411 heap-item comparisons, and 1482369 key-sort comparisons. It constructs every key with no LP vertices or interval-DP cells. Seven sizes through 511 match Knuth; three also match cubic. The formula and an independently built perfect tree verify all larger sizes. All scans, depth recovery, events, and output visits are retained in the source-hashed results.

4.19 Root/nonroot coupling beyond the transfer barrier

The four-key obstruction proved in Theorem 141 loses the coupling between one residual interior key and the surrounding gap tree. A partition by the identity of the root retains part of that coupling at the cost of two additional alphabetic constructions. It applies to arbitrary mixed input weights; successful certification, rather than the validity of the bound, is conditional.

Theorem 123. *Let $n \geq 2$, let r index a largest key weight $w = p_r$, and put $v = \max_{i \neq r} p_i$, $P = \sum_i p_i$, and $Q = \sum_j q_j$. Define*

$$N_r = K(p) + A(q) + w - v,$$
$$R_r = P + Q + K(p_{[0,r)}) + K(p_{(r,n)}) + A(q_{[0,r]}) + A(q_{[r+1,n]}).$$

Then

$$K(p) + A(q) \;\leq\; \min(N_r, R_r) \;\leq\; \mathrm{OPT}(p, q).$$

Here K is the free key-capacity bound and A the exact alphabetic gap optimum; empty key subvectors have bound zero.

Proof. If the root is not r, let its weight be $u \leq v$, and let the comparison length of r be $\ell \geq 2$. Exchanging those two weights reduces the key cost by $(w - u)(\ell - 1) \geq w - v$. The exchanged cost is at least $K(p)$, so the original key cost is at least $K(p) + w - v$. Adding $A(q)$ proves N_r. The exchange is solely a lower-bound argument; it does not change the ordered input or construct a different search tree.

If the root is r, the recurrence pays $P+Q$ and the two ordered subproblem costs. Bounding their keys and gaps separately gives R_r. These cases partition all trees. Moreover, the root-constrained free key placement costs at least $K(p)$, and the root-constrained alphabetic tree costs at least $A(q)$. Hence each case bound dominates the original separable bound. □

The implementation first applies Theorem 120 with $a = \min_{1 \le i < n-1} p_i$ and endpoint residuals $x = \max(0, p_0 - a)$, $y = \max(0, p_{n-1} - a)$. It subtracts an from transformed bounds and evaluates every candidate using the original mixed weights. One alphabetic call constructs the global gap optimum. A single decreasing key sort supplies K and, by scanning its left and right subsequences, both conditional capacity sums. If the global candidate already meets $K + A$, the attempt ends. Otherwise two alphabetic calls construct the gap optima on either side of r, and a second candidate joins them at root r. The smaller original candidate cost is compared with $\min(N_r, R_r) - an$. Equality certifies full optimality.

At most three alphabetic calls process a total of $2(n+1)$ leaves. Sorting, normalization, rebasing, reconstruction, and cost evaluation give $O(n \log(n+1))$ arithmetic work and $O(n)$ space for the complete attempt. The constant-size $n < 3$ case is solved directly. Rejection uses Knuth and retains the unrestricted quadratic worst case. The fixed endpoint choice need not preserve certificates found at another price; the preceding solvers remain separate and unchanged.

For Theorem 141, normalization gives $p' = (0, 0, 1, 0)$ and $q' = (2, 3, 2, 2, 2)$ with additive constant four. Here $K = 1$, $A = 26$, $r = 2$, and both N_r and R_r equal 28. The new original bound is therefore 24, exactly the full optimum, while the maximum over the entire preceding transfer class is 23.

4.19.1 A positive family outside every separable additive transfer

Corollary 124. *Let $h \ge 3$, $N = 2^h$, $n = N - 1$, and let $\ell_i = h - \nu_2(i+1)$ be the perfect tree's key comparison lengths. Put*

$$r = N/4 - 1, \qquad s = N/2 - 1, \qquad C = 4hN, \qquad w = hN^2/2.$$

Initially set $p_i = 1 + (h - \ell_i)N + i + 1$, then replace

$$p_0 = p_2 = p_{n-1} = 1, \qquad p_r = w + 1.$$

Set $q_j = C + j - 1$ for $0 \le j < N$. The root/nonroot certificate constructs a full optimum in $O(n \log n)$ arithmetic work and $O(n)$ space. Its exact cost is

$$\mathrm{OPT} = \frac{(8h^2 + 6h - 7)N^2 + (12 - 5h)N}{2} + 1 - 3h.$$

Every weight is positive and has $O(\log n)$ bits. There are $n-2$ distinct key weights and $n+1$ distinct gap weights. Nevertheless, the maximum $K+A$ bound over every admissible additive cost-preserving transfer falls short by exactly

$$D = \frac{hN^2 - (2h-1)N}{2} > 0.$$

Proof. The maximal interior transfer is $a = 1$. At endpoint prices zero, residual keys $\widehat{p}_i = p_i - 1$ have largest value w at r, next-largest value $v = (h - \frac{1}{2})N$ at the perfect root s, and three zero bottom keys. All other weights decrease strictly between perfect depth layers. The perfect tree attains the nonroot key bound $K(\widehat{p}) + w - v$: exchanging the weights at r and s attains the free capacity profile.

Transformed gaps are $\widehat{q}_j = C + j$, with maximum less than twice the minimum. Since N is a power of two, unrestricted Huffman merging pairs nodes level by level. Every gap has depth h, and the perfect alphabetic tree attains the unrestricted minimum. Consequently

$$A(\widehat{q}) = h\left(CN + \frac{N(N-1)}{2}\right).$$

If r is the root, its two sides contain $N/4$ and $3N/4$ gaps. The minimum unit-weight total depth, including that root, is $Nh + N/4$. All transformed gap weights are at least C, so the root-r alphabetic cost exceeds $A(\widehat{q})$ by at least

$$\frac{CN}{4} - \frac{hN(N-1)}{2} = w + \frac{hN}{2} > w - v.$$

Its conditional key bound is at least $K(\widehat{p})$. Thus $R_r > N_r$, and the perfect candidate attains N_r, proving full optimality.

For the cost formula, use $\sum_i \ell_i = (h-1)N + 1$, $\sum_i \ell_i^2 = N(h^2 - 2h + 3) - 3$, and $\sum_i (i+1)\ell_i = (N/2)\sum_i \ell_i$. After the three zero-key corrections and the replacement at r, the perfect residual key cost is $(5h-7)N^2/2 + (7-2h)N - 3h$. Adding $A(\widehat{q})$ and subtracting n yields the displayed full cost.

Finally, Theorem 121 reduces all admissible transfers to $a = 1$ and two endpoint prices. At $x = y = 0$, both zero endpoint keys occupy capacity slots of length h, and both endpoint gaps also have depth h. The supporting plane from Section 4.18.1 therefore has both slopes zero and value $K(\widehat{p}) + A(\widehat{q})$. This bounds the entire price objective from above and is attained at zero. The best original separable bound is exactly $K(\widehat{p}) + A(\widehat{q}) - n = \mathrm{OPT} - (w - v)$, giving the stated deficit. □

4.20 A sufficient exchange condition for a wide last level

Let $h = \lfloor \log_2(n+1) \rfloor$, $c = 2^h - 1$, and $s = n - c$. As proved earlier, minimum internal-path-length shapes expand s of the 2^h external slots of a perfect c-key skeleton. Consider adjacent skeleton slots $g, g+1$ containing one expansion. If their first input index is i, moving the expansion from the second slot to the first changes the weighted cost by exactly

$$(1 + \nu_2(g+1))(p_i - p_{i+1}) + q_i - q_{i+2}. \tag{133}$$

Only the intervening skeleton key, added key, and three gaps change their comparison lengths; all later positions are unchanged. The coefficient $1 + \nu_2(g+1)$ lies between one and h.

Theorem 125 (Exchange-certified minimum-path-length shape)**.** *If*

$$\max\{p_i - p_{i+1},\ h(p_i - p_{i+1})\} \le q_{i+2} - q_i \quad (0 \le i < n-1), \tag{134}$$

then filling the last level from the left minimizes weighted cost among all minimum internal-path-length shapes. The reflected condition gives the corresponding right-filled shape. For $s = 0$ the shape is unique without any condition.

Proof. The maximum in (134) bounds the key term in (133), whether its difference is positive or negative. Every adjacent swap of expansion pattern 01 to 10 is therefore nonincreasing in cost. Repeated swaps move every expansion leftward, yielding the claimed shape. Reflection proves the other case. □

In particular, nondecreasing key and gap weights satisfy the condition; both lists nonincreasing satisfy its reflection. Nonmonotone keys are also permitted when the gap increments compensate for their decreases. This proves optimality *within the shape class*. Global optimality still requires the earlier residual-Huffman gate. For candidate value V, minimum weights a, b, and residual mixed-node lower bound H_{res}, the sufficient condition is

$$V \le (a+b)(I_{\min}(n) + 1) + (a+2b)n + H_{\mathrm{res}}.$$

The exchange constructor constructs the candidate, checks this gate, and otherwise falls back to Knuth. Accepted instances take $O(n \log n)$ time and $O(n)$ auxiliary space, including sorting for the bound. If both weight lists are monotone, scans/reversal and the existing two queues make the whole accepted route $O(n)$. No optional-slot width appears. Sorted geometric controls pass the shape condition but fail the global gate, preventing an incorrect general claim about balanced trees.

5 Negative results and representation barriers

These results identify information lost by particular relaxations or costs imposed by particular representations. Each statement specifies its model of failure. None is an unrestricted computational lower bound.

5.1 A useful relaxation that failed as a general solver

Allow arbitrary assignment of key weights to internal positions and gap weights to external positions. In this relaxed problem, repeatedly take the two smallest current external weights a, b and the smallest remaining internal weight p, replace them by $a + b + p$, and add that sum to the cost.

To prove the rule, take a deepest internal node of an optimum. Its children are leaves. Swapping the smallest internal weight into it and the two smallest external weights into its children cannot increase cost. Contract that cherry and induct. Sorting and a heap give $O(n \log n)$ time. This elementary extension of the Huffman exchange argument is not claimed as novel.

The relaxed cost is a valid lower bound for full OBST. Equality with a feasible ordered tree certifies optimality. It agreed with an independent shape-enumeration and rearrangement oracle on 22,143 exhaustive and 2,700 random instances in the basic harness.

However, using the relaxation on demand in an exact sparse DP was poor on random inputs. At $n = 128$ it visited roughly 4,300–4,600 exact intervals instead of Knuth's 8,256, but spent approximately 3.9–4.1 million weight comparisons constructing bounds and inspected over 100,000 candidate roots. Fewer exact states did not mean less computation.

Nor does the relaxation's interval matrix inherit the required quadrangle inequality. For $p = (9, 1, 12)$, $q = (5, 1, 18, 10)$, its costs H satisfy

$$H(1, 2) + H(2, 3) = 60 > 54 = H(1, 3) + H(2, 2).$$

Thus applying Knuth-style matrix optimization directly to this bound would be unjustified.

Two additional exact counterexamples exclude tempting local certificates. With $p = (3, 5, 18)$, $q = (13, 14, 4, 8)$, the tree with root 2 and left subtree $0(\varnothing, 1)$ costs 127, every single-rotation neighbor costs at least 134, yet the balanced tree costs 125. With $p = (18, 8, 18)$, $q = (17, 11, 1, 13)$, the balanced tree costs 164 and every distinct tree obtained by promoting a descendant to the root of an existing subtree costs at least 169; the tree $0(\varnothing, 2(1, \varnothing))$ costs 157. These are barriers to those local certificates, not complexity lower bounds.

5.2 An ordered depth-chain lower bound

Let l_i be the comparison length of key i and g_i the depth of gap i. For every valid BST, adjacent keys have different lengths and

$$g_0 = l_0, \quad g_n = l_{n-1}, \quad g_i = \max(l_{i-1}, l_i) \quad (0 < i < n).$$

This identity is existing work; see Becker [8], Lemma 1. Let K_d count keys of length d and G_d count gaps at depth d. The root and child-count identities are

$$K_1 = 1, \qquad G_d + K_{d+1} - 2K_d = 0 \quad (1 \le d \le n), \qquad K_{n+1} = 0.$$

Permit any sequence in $\{1, \dots, n\}^n$ with unequal adjacent entries. For arbitrary multipliers $\lambda_0, \dots, \lambda_n$, minimize

$$\sum_i p_i l_i + \sum_i q_i g_i + \lambda_0(K_1 - 1) + \sum_{d=1}^{n} \lambda_d(G_d + K_{d+1} - 2K_d). \tag{135}$$

Call the minimum $L(\lambda)$. Every actual tree is permitted and has zero penalty, so $L(\lambda) \le \mathrm{OPT}$ for every multiplier vector. The expression is a chain with key energy $U_i(b) = p_i b + \lambda_{b-1} - 2\lambda_b$, gap energy $V_i(b) = q_i b + \lambda_b$, and constant $-\lambda_0$. For an interior transition into length b, take the smaller of

$$\min_{a<b} F(a) + V_i(b), \qquad \min_{a>b}\{F(a) + V_i(a)\}.$$

Prefix and suffix minima give $O(H)$ work per key for a domain of H lengths. Using $H = n$ would cost quadratic time even for one bound evaluation.

Theorem 126 (Exact affine-tail compression)**.** *Fix $h \ge 1$, arbitrary $\lambda_0, \dots, \lambda_h$, and a slope $0 \le s \le \min_i p_i$. Extend $\lambda_d = \lambda_h + s(d - h)$ for $d > h$. The minimum in (135) is unchanged if lengths are restricted to $1, \dots, \min(n, h + 3)$.*

Proof. For $b \ge h + 1$, the unary key energy has slope $p_i - s \ge 0$ and the gap energy has slope $q_i + s \ge 0$. Both are nondecreasing in this tail. If some $l_i > h + 3$, replace it by a member of $\{h + 1, h + 2, h + 3\}$ unequal to its current neighbors. At most two values are forbidden. The key energy does not increase. Each adjacent maximum is either unchanged or decreases within the affine tail, so neither gap energy increases. Endpoint gaps obey the same argument. Repeat for all large lengths. The restricted domain is a subset of the original domain, and every original sequence has a no-more-expensive restricted sequence; therefore the minima are equal. □

Taking $h = \lceil \log_2(n+1) \rceil + 2$ gives an $O(n \log n)$ oracle. This is a statement about minimization of the relaxed energy, with no height assumption on the unknown optimal BST. Optimizing only the head multipliers requires folding all tail constraint violations into the derivative for λ_h. The implementation uses exact integer multipliers and costs scaled by 1024. Its subgradient ascent is heuristic; the bound is valid at every iteration regardless of convergence.

A stronger oracle requires exactly one length1 key. For each candidate root, the left and right chains then contain only lengths at least2. One forward and one reverse pass supply their closing costs for every root position, giving all forced-root bounds in $O(nH)$ total time. The projection above preserves the unique root, so tail compression remains valid for each marginal. Retaining the maximum bound seen for each root, then taking the minimum over roots, is also a valid lower bound.

A structural gap. The sequence $l = (3, 1, 2, 3, 2)$ has $g = (3, 3, 2, 3, 3, 2)$, a unique root, and all the level-count equalities. It is not a valid BST depth sequence. For

$$p = (3, 27, 21, 16, 12), \qquad q = (18, 22, 27, 22, 12, 22),$$

it costs 470, while the true optimum is 472. Its penalty is zero for every multiplier vector. Hence neither oracle family can ever exceed 470 on this instance. More optimization iterations cannot remove this obstruction.

5.3 Exact depth ordering and an arbitrary-window obstruction

Call a positive integer sequence *ranked* if every nonempty contiguous interval has a unique minimum. Equivalently, equal labels have a smaller label between them. This is vertex ranking on a path with the label order reversed; vertex ranking and bounded-path variants are existing notions [9, 10]. The following results explain the ordering information absent from the preceding relaxation; no novelty claim is made for this characterization.

Theorem 127 (Depth characterization)**.** *A sequence $l \in \{1, \dots, n\}^n$, $n \geq 1$, is the inorder key-comparison-length sequence of a BST if and only if it is ranked, $K_1 = 1$, and every level-flow equality holds, with gaps derived from l as above.*

Proof. Necessity follows from the unique shallowest key in each inorder interval and the two children of every internal node. For sufficiency, delete all labels at most d. Let U_d count the remaining nonempty runs, and let A_d count the deleted labels. If the deleted labels form R runs and include e endpoints, then the number of gaps of length at most d is $A_d - R + e$, whereas $U_d = R + 1 - e$. Thus

$$G_{\leq d} = A_d + 1 - U_d.$$

Summing the level-flow equalities from 1 through d, and using $K_1 = 1$, also gives

$$G_{\leq d} = A_d + 1 - K_{d+1}.$$

Hence $U_d = K_{d+1}$. Ranking permits at most one label $d + 1$ per remaining run, so equality forces exactly one. Starting with the unique length1 root, every remaining nonempty interval therefore receives its root at the next level. This constructs the tree with exactly the supplied lengths. □

Ranking has a linear recognizer: scan with an increasing stack, pop larger labels, reject an equal top, and otherwise push. A rejected pair has no smaller intervening label; conversely every such pair causes rejection. Each entry is pushed once and popped at most once. This establishes cheap recognition, not cheap minimization over all valid sequences.

Theorem 128 (Bounded-window obstruction)**.** *For every integer $w \geq 2$ and $t \geq 2$, there is a strictly positive full-OBST instance with $n = w + 3$ keys and an inadmissible depth sequence satisfying all level-flow equalities, a unique length1 root, and ranking on every interval of at most w keys. Its cost is at most $(7/8 + 1/(8t))\,\mathrm{OPT}$. Consequently every lower bound obtained by pricing the level-flow equalities over this window-constrained domain has the same upper limitation, regardless of the multipliers.*

Proof. Take

$$l = (3, 1, 2, 3, 4, \dots, w + 1, 2).$$

The ascending middle portion has $w - 1$ entries. Its two enclosing length2 labels form the only interval with a repeated minimum, of width $w+1$. Every shorter window passes. Direct substitution into the gap formula verifies $K_1 = 1$ and every level-flow equality. The sequence is invalid: the unique root is key1, but its singleton left subtree has length3, and two keys to its right both claim length2.

Give every gap weight1 and every key weight1 except

$$p_1 = 3B, \qquad p_2 = p_{n-1} = B.$$

The unit-weight outcomes contribute $E = w^2 + 4w + 10$ in the inadmissible sequence, whose cost is $7B + E$. Every actual tree costs at least $8B$. If key1 is the root, the two B-weight keys lie to its

right and their comparison lengths sum to at least5, giving $3B + 5B$. Otherwise key1 has length at least2 and each B-weight key has length at least1, giving $6B + 2B$.

Set $B = tE$. The inadmissible cost satisfies

$$7B + E = \left(\frac{7}{8} + \frac{1}{8t}\right) 8B \le \left(\frac{7}{8} + \frac{1}{8t}\right) \mathrm{OPT}\,.$$

Its level-flow penalty is zero for every multiplier vector, so every priced minimum is at most this cost. □

For $t = 16$ the upper ratio is 113/128, independently of w; as t increases it approaches 7/8. This rules out universal exactness of the specified window-minimum repair, not other local constraints, adaptive nonlocal cuts, or subquadratic full-OBST algorithms. Equivalently, at every $n \ge 5$ the construction passes all windows of up to $n - 3$ keys.

Burying the obstruction below a forced prefix. Let a current instance have mass W and a feasible tree of cost U. Prepend a key of weight $A = W + U + 2$ and a new leftmost gap of weight1. Using the new key as root costs $A + 1 + W + U = 2A - 1$, while every other root costs at least $2A$ from that key alone. Hence the new root is forced and $\mathrm{OPT}_{\mathrm{new}} = A + 1 + W + \mathrm{OPT}_{\mathrm{old}}$. The transformation $l \mapsto (1, l_0 + 1, \ldots, l_{n-1} + 1)$ preserves zero flow and local window ranking, and changes the inadmissible cost by the same $A + 1 + W$. Repeating this D times places the strict gap beneath the true optimal top-D right-chain prefix. Thus fixing any constant number of top levels cannot make the remaining window/flow relaxation exact. The constant relative gap is not claimed after this embedding, because the forcing weights add a potentially large common cost.

5.4 Endpoint responses: insufficiency and representation size

An intact subtree interacts with its ancestors through its relative cost and total mass. If its interior is not yet fixed, one tempting richer summary is the response to added weights at its endpoint gaps:

$$\mathcal{R}(x, y) = \mathrm{OPT}(p, q + xe_0 + ye_n), \qquad x, y \ge 0.$$

For a fixed tree the response is $C(T) + xd_0(T) + yd_n(T)$, so $\mathcal{R}$ is a concave lower envelope of affine planes. We distinguish whether this function contains enough information for composition from how many pieces an explicit representation requires. Neither result below is a general lower bound for optimal-BST construction; priority remains unestablished.

5.4.1 Equal responses need not be interchangeable

Theorem 129 (Insufficient endpoint-value interface)**.** *An interval's endpoint response, length, total mass, and endpoint gap weights do not determine its optimal cost in an arbitrary surrounding search problem. This remains true if its full vector of key weights and its multiset of gap weights are also retained.*

Proof. Consider two four-key intervals with

$$p = (1, 1, 2, 3), \quad q^A = (1, 4, 2, 1, 1), \quad q^B = (1, 4, 1, 2, 1).$$

Both have mass 16 and endpoint gap weights one. Enumerating the 14 four-key shapes gives the following least intercepts for each possible endpoint-depth pair:

Depths	A	B
$(1,2)$	45	45
$(1,3)$	42	42
$(1,4)$	44	45
$(2,1)$	43	43
$(2,2)$	36	36
$(2,3)$	36	37
$(3,1)$	39	39
$(3,2)$	36	36
$(4,1)$	43	42

Each plane with slopes $(1,4), (2,3), (3,2)$, or $(4,1)$ is pointwise dominated on the nonnegative quadrant by a common plane whose slopes and intercept are no larger. Hence both responses are exactly

$$\mathcal{R}(x,y) = \min\{45 + x + 2y,\ 42 + x + 3y,\ 43 + 2x + y,\\ 36 + 2x + 2y,\ 39 + 3x + y\}.$$

This proves equality for all parameters, not just sampled ones.

Prepend keys of weights $(7,1)$, append keys of weights $(1,7)$, and give the four new gaps weight one. The full inputs are

$$P = (7,1,1,1,2,3,1,7),$$
$$Q^A = (1,1,1,4,2,1,1,1,1),$$
$$Q^B = (1,1,1,4,1,2,1,1,1).$$

Using the exact recurrence, the least full costs conditional on each root are

Root	0	1	2	3	4	5	6	7
A	108	108	103	101	101	101	111	110
B	108	108	103	101	100	101	111	110

Thus the optima are 101 and 100. Direct enumeration of all 1,430 eight-key trees independently verifies both rows. Equal summaries with the identical surrounding context therefore have unequal answers. □

In particular, exact prepend/append updates closed on the stated summary cannot exist: identical starting summaries and identical updates would remain indistinguishable. The theorem does not exclude extra shape information or richer interfaces. Section 5.4.2 proves that non-exposed depth states and signed-price responses do retain the original gap vector with the stated metadata. It does not assert a distinction between mixed and alphabetic algorithms; it rejects this summary even on small positive mixed inputs.

The lost information includes internal alternatives with identical boundary depths. Both intervals have minimum cost 36 at depths $(2,2)$. The shape rooted at key 2, with left child 0 (whose right child is 1) and right child 3, also has depths $(2,2)$, but costs 37 in A and 36 in B. The least intercept erases this distinction; the surrounding tree can make the additional alternative valuable.

5.4.2 The legal domain loses information; signed prices recover it

The information boundary can be located more precisely. Allowing all nonnegative endpoint assignments still loses information, while a bounded line of signed assignments determines the original interval.

Theorem 130 (Insufficiency on the full legal endpoint domain)**.** *Even the response to every nonnegative reassignment of the two endpoint gaps, together with the key vector, total mass, original endpoint gaps, and gap multiset, does not determine arbitrary contextual optima.*

Proof. Use $p = (2, 1, 2, 4)$ and interior gaps $(4, 2, 1)$ or $(4, 1, 2)$. With endpoints set to $u, v \geq 0$, both responses equal

$$\min\{45 + u + 2v, 42 + u + 3v, 43 + 2u + v, 36 + 2u + 2v, 39 + 3u + v\}.$$

Indeed, move each unit outer gap in Theorem 129 to its adjacent endpoint key. Their depths agree in every isolated tree, so the same complete intercept table and domination certificate apply. Choose actual endpoints one: both inputs are strictly positive, have mass 18, and have all the listed metadata equal. Prepending keys $(1, 6)$, appending $(4, 3)$, and giving the four new gaps weight one yields keys $(1, 6, 2, 1, 2, 4, 4, 3)$ and full gap vectors $(1, 1, 1, 4, 2, 1, 1, 1, 1)$ and $(1, 1, 1, 4, 1, 2, 1, 1, 1)$. The root-conditioned costs are respectively

$$(128, 106, 103, 101, 101, 101, 109, 120), \qquad (128, 106, 103, 101, 100, 101, 109, 120).$$

Their optima are 101 and 100. Thus equality holds on the entire legal domain, including zero endpoints, yet the same context distinguishes them. □

Lemma 131 (Extreme boundary states determine all gaps)**.** *For nonnegative real weights, the complete endpoint-depth profile determines the original gaps when the key vector p, total mass W, and left gap q_0 are retained. Only its n states with endpoint-depth sum $n + 1$ are needed, and recovery takes $O(n)$ arithmetic operations.*

Proof. If the endpoint depths are a, b, their two paths contain $a + b - 1$ distinct keys. Equality $a + b = n + 1$ forces every key onto these paths. For each root r, there is exactly one such tree: a decreasing left chain and increasing right chain, with endpoint depths $(r + 1, n - r)$. Let its cost be C_r. Every outcome depth changes by at most one between adjacent chain trees, so $|C_r - C_{r-1}| \leq W$. Direct differencing gives

$$C_{r-1} - 2C_r + C_{r+1} = 2p_r + q_r + q_{r+1} \geq 0, \qquad 1 \leq r \leq n - 2, \tag{136}$$

$$C_1 - C_0 = 2(p_0 + q_0) + q_1 - W. \tag{137}$$

For example, key comparison length is $|i - r| + 1$; interior gap j has depth $r - j + 2$ for $j \leq r$ and $j - r + 1$ for $j > r$. The outer gaps have depths $r + 1, n - r$. These formulas prove both identities. For $n \geq 2$, recover

$$\begin{aligned} q_1 &= W + C_1 - C_0 - 2(p_0 + q_0), \\ q_{r+1} &= C_{r-1} - 2C_r + C_{r+1} - 2p_r - q_r, \qquad 1 \leq r \leq n - 2, \\ q_n &= W - \sum_i p_i - \sum_{j=0}^{n-1} q_j. \end{aligned} \tag{138}$$

For $n = 1$, the last mass identity suffices. The chain coefficients themselves are computable in linear work: initialize $C_0 = \sum_i (i + 1)p_i + q_0 + nq_n + \sum_{j=1}^{n-1}(j + 1)q_j$, then use (137) and (136). □

Theorem 132 (Recovery from a bounded signed response)**.** *Define* $F(x,y) = \min_T\{\mathrm{cost}(T) + xd_0(T) + yd_n(T)\}$ *for arbitrary real* x, y*. Given* p, W, q_0*, a bounded line of this signed response determines all gaps. For nonnegative integer input on* $n \geq 2$ *keys, at most* n *exact value-only oracle queries recover them, with* $O(n)$ *additional arithmetic work and space. Query coordinates have magnitude* $O(n(W+2))$*, denominator at most* n^2-1*, and bit length* $O(\log(n(W+2)))$*.*

Proof. Put $M = n+1$ and $K = 4(n+1)(W+2)$. On the line $x = -K+t$, $y = -K$, $|t| \leq W+2$, any non-chain tree has endpoint-depth sum at most n and value at least $-Kn - n|t|$. Any chain has value at most $nW - K(n+1) + n|t|$. Since $K > nW + 2n|t|$, a chain always wins. Hence

$$G(t) := F(-K+t, -K) + K(n+1) = \min_{0 \leq r < n}\{C_r + (r+1)t\}. \tag{139}$$

By (136), C is convex, and its first differences lie in $[-W, W]$. Thus every coefficient is exposed, possibly with ties:

$$C_r = \sup_{|t| \leq W+1}\{G(t) - (r+1)t\}.$$

Together with Lemma 131, this proves determination for real input weights.

For integer input, take a rational $t_0 = A/B$ in lowest terms, $B > 0$, and query at $t = t_0 + 1/(BM)$. Unperturbed scores are multiples of $1/B$; the perturbation changes each pairwise difference by less than $1/B$. It therefore selects the smallest index among the minimizers at t_0. If that index is r, put $a = r+1$. The returned value gives an integer

$$z = BMG(t) = BMC_r + a(AM+1), \qquad z \bmod M = a.$$

Because $1 \leq a \leq n < M$, the residue recovers r, and then $C_r = G(t) - at$ is exact. No minimizer witness or numerical limit is requested.

Queries with $t_0 = W+1$ and $-W-1$ recover the two extreme coefficients. For known endpoints $l < r$ with $r - l \geq 2$, query at the negative chord slope $t_0 = (C_l - C_r)/(r-l)$. If the recovered unperturbed score equals the endpoint score, convexity puts every intermediate coefficient on the chord; fill them by interpolation. Otherwise the recovered index lies strictly inside (l, r); recurse on the two resulting intervals. In the collinear case, the decoded minimizer can lie outside $[l, r]$; score equality, rather than index membership, is the stopping test.

Each nonterminal query discovers at least one previously unknown interior coefficient, and the recursive interiors are disjoint. Including the first two queries, at most n queries suffice. Interpolation, recursion, and (138) take $O(n)$ further work and space. Chord denominators satisfy $B \leq n-1$, so $BM \leq n^2-1$. Their slopes have magnitude at most W; the initial queries have magnitude $W+1$ before perturbation. All requested points satisfy the stated bounds. □

The oracle's signed domain is part of its specification: endpoint weights may become negative. Ordinary nonnegative-weight Knuth monotonicity is not used for those queries. For the legal-domain collision, the absolute assignment $(u, v) = (0, -1)$ already gives distinct costs 33 and 34. The complete endpoint-depth profile is therefore sufficient with the stated metadata, whereas its envelope on legal endpoint weights is not. A known chain sequence supports (139) by binary search in $O(\log(n+1))$ comparisons after linear preprocessing. These are representation and recovery results; an efficient general composition rule still has to be supplied. Verification is in Appendix A.13.

5.4.3 Quadratic explicit envelopes with linear factorizations

For any $n \geq 1$, the endpoint paths share only the root. If their depths are a, b, their union has $a+b-1 \leq n$ keys. At most one least intercept per depth pair is needed, giving at most $n(n+1)/2$ affine pieces. The quadratic order can occur already for a single interval.

Theorem 133 (Explicit-facet obstruction)**.** *There are strictly positive mixed-weight instances on $n = 2m+1$ keys whose endpoint response has at least m^2 affine pieces on the nonnegative quadrant. Consequently, explicitly listing all such pieces can require $\Omega(n^2)$ entries.*

Proof. Set $K = 4m$ and $P_i = K8^i$ for $0 \le i < m$. First consider these m keys with all gaps one and added left endpoint weight x. Write $S_j = \sum_{i=0}^{j} P_i$ and $x_j = 2S_j + 2m$.

An endpoint key and its adjoining outer gap are forced to the root if their combined weight exceeds the mass of all other outcomes. Indeed, each rotation of that key toward the root saves one comparison on both outcomes and adds at most one to each other outcome. The strict inequality makes every rotation improve cost. At x_j, for every $k > j$,

$$P_k + 1 > S_{k-1} + x_j + k + 1,$$

while in the remaining prefix

$$P_0 + 1 + x_j > (S_j - P_0) + (j + 1).$$

The geometric series and $K = 4m$ imply these inequalities. For example, $S_{k-1} < P_k/7$ and $x_j \le 2S_{k-1}+2m$ for $k > j$, whereas $P_k \ge 32m$. Thus every optimum first uses roots $m-1, m-2, \ldots, j+1$, then key 0, giving left endpoint depth $m - j$. All m possible one-sided slopes occur, with strict inequalities persisting in open neighborhoods of the probes.

Mirror this instance about a central key:

$$p = (P_0, \ldots, P_{m-1}, R, P_{m-1}, \ldots, P_0), \quad q = (1, \ldots, 1).$$

Let $X = x_{m-1}$, $S = S_{m-1}$, $B = 2S + 2m + 2 + 2X$, and $R = (2m + 2)B$. For $0 \le x, y \le X$, the mass outside the central key is at most B. A center-root tree costs at most $R + nB < 2R$; any other root charges the central key at least $2R$. Thus the center root is forced. At (x_j, x_k), every optimum has endpoint depths

$$(m - j + 1, m - k + 1).$$

These are m^2 different gradients. Since there are finitely many trees and all minimizing trees at each probe have the same gradient, that probe has an open affine neighborhood. Different gradients require different pieces. □

The weights have $O(n)$ bits each. This is an arithmetic/output-size obstruction to explicit facet enumeration, not a word-RAM lower bound with $O(\log n)$-bit weights, nor a lower bound for a single optimum query. The same example also explains why counting facets cannot rule out implicit representations.

Lemma 134 (Exact hinge factorization)**.** *For the one-sided family above, let*

$$C_0 = \sum_{i=0}^{m-1} (m - i)P_i + m + \frac{m(m+1)}{2}.$$

Its response for every $x \ge 0$ is

$$H_m(x) = C_0 + mx - \sum_{i=1}^{m-1} \max\{0, x - (P_i - P_0)\}.$$

Within the forced-center box the two-sided response equals

$$R + 2S + 2m + 2 + x + y + H_m(x) + H_m(y).$$

Proof. At x_j, the remaining right subtree of key 0 is a decreasing-key chain, by the same strict endpoint condition. Passing from j to $j+1$ rotates key 0 above key $j+1$. At base weights the intercept increases by $P_{j+1} - P_0$, while the endpoint slope decreases by one. At $j = 0$ the intercept is C_0. Every possible endpoint depth occurs among these m trees, and optimality at a probe proves its intercept is minimal for that depth. Their successive intersections are the increasing thresholds $P_i - P_0$, giving the hinge formula. The center-root recurrence gives the two-sided identity. □

The factorization extends to the entire quadrant. Set

$$W = R + 2S + 2m + 2, \qquad V = W - P_0 - 1,$$
$$L = C_0 - mP_0 - m - 1, \qquad J = R + 2(S - P_0) + 2m + 2L.$$

Here L is the fixed cost of the remaining one-sided keys $1, \ldots, m-1$, and J is the fixed inner cost after both outer keys and outer gaps are removed.

Theorem 135 (Full-quadrant factorization)**.** *For the geometric family and every $x, y \geq 0$,*

$$\begin{aligned}\mathcal{R}(x,y) = W + x + y + \min\{&H_m(x) + H_m(y),\\ &V + y + L + H_m(y),\ V + y + J,\\ &V + x + L + H_m(x),\ V + x + J\}.\end{aligned}$$

This function has exactly $m^2 + 2m + 2$ affine pieces. It admits $O(m)$ arithmetic preprocessing and storage, $O(\log m)$ value queries, and $O(m)$ complete tree construction per query.

Proof. Any root other than keys $0, m, 2m$ gives both added endpoint weights depth at least two and the central key comparison length at least two. A center-root candidate with the two outer keys as children also puts both endpoints at depth two, with base cost at most $R + n(W - R) < 2R$. It therefore strictly beats such a root for all x, y. If root 0 is used, its remaining one-sided problem similarly has only the center or the other outer key as a possible root; symmetrically for root $2m$. Expanding these five cases gives the formula. The fixed remaining intervals are geometric chains on either side of a forced central root, giving the stated L, J.

The center-root term supplies m^2 slopes, all exposed by the original probes. For root 0, the same $2R$ argument forces the center in the right problem throughout the original probe range, exposing all m pieces inherited from $H_m(y)$. As y grows unbounded, one final piece appears when the outer key becomes its root. Taking x sufficiently large forces root 0, exposing these $m+1$ pieces globally. Their slopes are $(1, 2), \ldots, (1, m+2)$. Reflection gives another $m+1$ pieces with second slope one. All slopes are distinct. The formula has no further candidate slopes, proving the count.

Prefix sums of the ordered thresholds let two binary searches evaluate $H_m(x)$ and $H_m(y)$. Four comparisons select the outer minimum. The selected case and one-sided chain indices reconstruct exactly $2m + 1$ keys in linear work. □

Thus $2(m-1)$ hinge terms and five alternatives represent a quadratic number of affine regions. Endpoint-dominance shortcuts already solve this family. It is not a hard-query example, and the facet theorem is not an expression-size lower bound. A lossless factorization also does not repair Theorem 129: the value function itself omits information needed by some surrounding contexts.

A favorable bound independent of the added weights

Lemma 136 (Endpoint mass-ratio bound)**.** *Let the original total mass be W, with $q_0, q_n > 0$. The entire response for arbitrary $x, y \geq 0$ has*

$$O(\log(2 + W/q_0) \log(2 + W/q_n))$$

affine pieces, also bounded by $O(n^2)$. Polynomial endpoint mass ratios therefore give $O(\log^2 n)$ pieces.

Proof. If an optimum has left endpoint depth $a \geq 2$, its last internal ancestor is at depth $a - 2$ in the root's left subtree. This subtree is optimal and excludes the opposite endpoint and its added weight y. Its mass is at most $W + x$, while the last ancestor's subtree has mass at least $q_0 + x$. Lemma 2 gives

$$F_{a-1} \leq \frac{W + x}{q_0 + x} \leq \frac{W}{q_0}, \qquad F_1 = F_2 = 1.$$

The symmetric bound is $F_{b-1} \leq W/q_n$ for right depth $b \geq 2$. Fibonacci growth bounds the number of possible slopes in each direction; only a least intercept per pair can be required. □

The added parameters do not occur in this bound. The quadratic example has exponential original mass ratio, so the two results are consistent. This lemma bounds output size, not the work to discover the pieces. Moreover, the composition counterexample has just five pieces: a small response need not be a sufficient interface.

5.5 Ruler patterns and a dense-core representation barrier

The geometric-weight control has a dense core whose entries are constant along each row. This motivates asking whether another finite difference could restore sparse support. The answer is negative even for a smooth, positive, polynomial-weight family that has an exact compact formula. Before proving that barrier, we isolate the ruler pattern underlying both the penalty layers and a new sufficient exchange condition. Priority of these identities and conditions is unestablished.

5.5.1 Saturated layers as weighted subsequences

Write $c_h = 2^h - 1$ and let $\nu_2(t)$ be the exponent of two dividing the positive integer t. Use the penalty recurrence above with $M > HW$. For any interval of length $k = j - i \geq c_h$,

$$A_h(i, j) = M(k - c_h) + hW(i, j) - \max_{i \leq s_1 < \cdots < s_{c_h} < j} \sum_{t=1}^{c_h} \nu_2(t) p_{s_t}. \tag{140}$$

Indeed, the penalty forces exactly c_h exposed keys: losing one exposure costs more than any possible reduction in the ordinary weighted cost. Every position in the first h levels is occupied. The exposed skeleton is perfect, all gaps and unexposed keys pay h comparisons, and its tth inorder key has comparison length $h - \nu_2(t)$. This proves the identity. The coefficient sequence is $0, 1, 0, 2, 0, 1, 0, 3, \ldots$.

At $h = 2$ the maximum is simply $\max_{i+1 \leq r \leq j-2} p_r$. At $h = 3$, greedy selection of the largest eligible root fails: for $p = (0, 9, 0, 9, 10, 9, 0, 0)$ and all nine gap weights equal to one, root 3 permits score 36, while the larger key at root 4 permits only 29. Adding one to all key weights adds four to every feasible score, giving the same failure with strictly positive weights. For nondecreasing p, the rightmost c_h keys maximize the score coordinatewise; for nonincreasing p, the leftmost keys do. Thus their saturated-layer density vanishes whenever $j - i \geq c_h + 1$, independently of the gap weights. This does not discard the gaps or solve the full problem: shorter intervals have nonuniform gap depths, and their band becomes quadratic at large h. The weighted-subsequence reformulation by itself supplies no new efficient min-plus product.

5.5.2 A positive family with dense differences and compact costs

Set

$$B = 4(n+1)^3, \qquad p_r = B + r \ (0 \le r < n), \qquad q_r = B + r \ (0 \le r \le n). \tag{141}$$

All weights use $O(\log n)$ bits and the deviation above their minimum baselines is exactly $R = n^2$.

Theorem 137 (Dense support does not force expensive construction)**.** *For the family (141), the optimal interval-cost matrix has $\Omega(n^2)$ nonzero Monge densities and $\Omega(n^2)$ nonzero mixed fourth differences. It also has quadratically many rowwise density changes. Nevertheless, its exact tree is constructible in $O(n)$ time, and any interval value can be queried using $O(\log n)$ arithmetic operations after linear recognition of the family.*

Proof. First force the shape class. An interval of k keys starting at i has constant baseline $B+i$. A tree of internal path length I has total key and gap comparison length $D = 2I + 3k \le k^2 + 2k$. All relative weights lie between zero and k, so their total contribution is at most $n(n^2+2n)$. Increasing I by one adds at least $2B$, which exceeds this entire bound. Hence every optimum has minimum I. Its adjacent exchange cost (133) is $-3 - \nu_2(g+1) < 0$, so its last level is filled from the left.

Here is an exact value formula. For $k = j - i > 0$ put $h = \lfloor \log_2(k+1) \rfloor$, $c = 2^h - 1$, $s = k - c$, and define

$$\begin{aligned} I(k) &= (h-2)2^h + 2 + hs, & D(k) &= 2I(k) + 3k, \\ J(c) &= \sum_{t=1}^{c} (t-1)\nu_2(t), & T(c,s) &= \sum_{t=1}^{c} \min\{s,t\}\nu_2(t), \\ E(k) &= hc^2 - J(c) + s(2hc + h + 1) \\ &\quad + (h+3)s(s-1) - T(c,s). \end{aligned}$$

Then

$$C(i,j) = (B+i)D(k) + E(k), \qquad C(i,i) = 0. \tag{142}$$

To verify E, the perfect skeleton has linear-weight cost $hc^2 - J(c)$. For an arbitrary set S of expanded slots, each skeleton key at r shifts by the number of selected slots at most r, and each skeleton gap at g shifts by the number strictly below g. Summing these index changes and the new key/gap contributions gives

$$\begin{aligned} E(S) = hc^2 - J(c) &+ \frac{(2h+3)|S|(|S|-1)}{2} \\ &+ \sum_{g \in S} \left(2hc + 3g + h + 1 - \sum_{t=g+1}^{c} \nu_2(t) \right). \end{aligned}$$

Taking $S = \{0, \dots, s-1\}$ gives (142). For each $\ell = 1, \dots, h-1$, let $d = 2^\ell$, $m = \lfloor c/d \rfloor$, and $u = \min\{m, \lfloor s/d \rfloor\}$. The contributions to J and T are, respectively,

$$\frac{dm(m+1)}{2} - m, \qquad \frac{du(u+1)}{2} + (m-u)s.$$

This evaluates the formula in $O(\log n)$ steps, using $\nu_2(t) = \sum_{\ell \ge 1} [2^\ell \mid t]$. The tree itself needs one recursive visit per key: its left subtree has $2^{h-1} - 1 + \min\{s, 2^{h-1}\}$ keys. Recognition scans the given input; the interval oracle then retains only n and B.

Now examine support. Write

$$d(i,j) = C(i,j+1) + C(i+1,j) - C(i,j) - C(i+1,j+1).$$

Inside a fixed h band, for $1 \le s < 2^h$,

$$d(i,j) = 3 + \nu_2(s), \qquad s = (j-i) - (2^h - 1). \tag{143}$$

Indeed, D has slope $2h+3$, while E has second difference $2h+6+\nu_2(s)$. The endpoint at $s = 2^h$ is the next perfect tree, so the calculation remains valid for $s = 2^h - 1$. For $2 \le s \le 2^h - 2$ the mixed fourth difference is therefore

$$\begin{aligned} &d(i,j+1) + d(i+1,j) - d(i,j) - d(i+1,j+1) \\ &\qquad = \nu_2(s+1) + \nu_2(s-1) - 2\nu_2(s). \end{aligned} \tag{144}$$

It is strictly positive for odd s and strictly negative for even s. For $n = 2^m \ge 32$, choose $h = m-2$ and $2 \le s \le n/4 - 2$. There are $n/4 - 3$ lengths and at least $n/2 + 2$ valid placements per length, giving at least $(n/4-3)(n/2+2) = \Omega(n^2)$ nonzeros. For arbitrary sufficiently large n, use $h = \lfloor \log_2 n \rfloor - 2$; both the band width and number of placements remain $\Theta(n)$. The same bands prove quadratic second-difference support and rowwise density changes. □

Consequently any method explicitly listing either support has quadratic output work on this family. This is a representation lower bound; the linear constructor rules out interpreting it as a computational lower bound for the family. The example also reaches the $R = \Theta(n^2)$ boundary of the centered analysis with smooth, very nearly equal relative weights. Symbolically representing the dyadic baseline avoids the dense list, but subtracting it need not leave a Monge residual. Efficient product closure with arbitrary residuals is still missing.

A maximally dense core at minimum height. The following variant makes every interior density positive, while remaining in the implicit-oracle class of Theorem 71. It sharpens the final-core diagnostic; the preceding ruler family already rules out a general sparse final core.

Corollary 138 (Strict density arbitrarily close to uniform weights)**.** *For $n \ge 1$, set $p_i = B + i$, $q_j = B$, where $B = n^3 + 1$ or any larger integer. Every interval has a minimum-IPL optimum, and its unrestricted cost table has a positive density at every cell $0 \le i < j < n$. The finite completion therefore has exactly $n(n+1)/2$ core entries. Writing $m = j - i$, $h = \lfloor \log_2(m+1) \rfloor$, $c = 2^h - 1$, and $s = m - c$, the interior density is exactly*

$$\Delta_C(i,j) = \begin{cases} 2B + i - c + h, & s = 0, \\ 1 + \nu_2(s), & s > 0. \end{cases} \tag{145}$$

After dividing the weights by B, the same support occurs with every weight in $[1, 1+\varepsilon]$, for any prescribed $\varepsilon > 0$ and sufficiently large B. All interval optima have height at most $\lceil \log_2(n+1) \rceil$.

Proof. The residual mass is $n(n-1)/2$, so the baseline lemma applies. The leftward exchange is strictly improving because the key difference is -1 and the gaps are constant. Let $A(m) = I_{\min}(m) + m$, and let $T(m)$ be the weighted key cost of the left-filled shape when its key weights are $0, 1, \ldots, m-1$ and all gaps vanish. Then

$$C(i, i+m) = B(2A(m) + m) + iA(m) + T(m).$$

For $m = c + s$ in a fixed h band, the skeleton positions give

$$T(c+s) = T(c) + hsc + (h/2+1)s(s-1) - \sum_{t=1}^{c} \min\{s,t\}\nu_2(t).$$

For $1 \le s < 2^h$, its second difference is $h + 2 + \nu_2(s)$; the endpoint uses the next perfect tree. Also $A(m) - A(m-1) = \lfloor \log_2 m \rfloor + 1$. The density formula thus gives $1 + \nu_2(s)$ away from perfect lengths. At $m = c$, use

$$A(c) = (h-1)2^h + 1, \qquad T(c+1) - T(c) = A(c), \qquad T(c) - T(c-1) = h(c-1).$$

Substitution gives $2B + i - c + h$. Both cases are positive. There are $n(n-1)/2$ interior cells and n completion diagonal entries. Scaling by a positive constant preserves support, and the shape characterization supplies the height bound. The exact interval and root-range queries remain available without listing this core. □

5.6 A sharp one-unit shape-sensitivity barrier

The gap and key coefficients in Theorem 78 are attained simultaneously by an explicit mixed-weight input.

Theorem 139 (Exact failure below the box threshold)**.** *For $n = 2^h - 1$, $h \ge 2$, let S be the perfect tree and set*

$$p_i = a + P[\, i \ge 2 \text{ and } i \text{ is even}], \qquad q_j = b + Q[j = n].$$

There is a tree T with $I(T) = I(S) + 1$ and

$$C(T) - C(S) = a + b - (n-h)P - Q. \tag{146}$$

Consequently the perfect tree is strictly suboptimal whenever $a + b < (n-h)P + Q$, even though the competing shape has only one unit of excess internal path length.

Proof. Delete the greatest key of S, which is a leaf at internal depth $h-1$. Insert a new least key as the left child of the surviving least key, at internal depth h, and relabel the keys $0, \dots, n-1$ in order. Thus the internal path length increases by exactly one. The resulting comparison lengths are

$$\ell_0(T) = h + 1, \qquad \ell_i(T) = h - \nu_2(i) \quad (1 \le i < n).$$

The first two gaps have depth $h+1$, the last gap has depth $h-1$, and all other gaps have depth h. In S, key i has length $h - \nu_2(i+1)$ and every gap has depth h.

Each even key $i \ge 2$ therefore shortens by $\nu_2(i)$, and the total is

$$\sum_{i=1}^{2^h-2} \nu_2(i) = \sum_{t=1}^{h-1} (2^{h-t} - 1) = 2^h - h - 1 = n - h.$$

These are precisely the keys receiving residual weight P. The only gap receiving residual weight Q shortens by one. The uniform baseline costs $a + b$ for the unit IPL increase, proving (146). □

The witness shows a linear sensitivity to independent key perturbations and constant sensitivity to independent gap perturbations in the universal perfect-tree box. Both statements concern preservation of that shape, not a lower bound for unrestricted optimal-tree algorithms.

5.7 Joint profile coupling and a strict merge obstruction

The separate key and gap minima need not coexist. Let $J(p, q)$ relax only the key positions, keeping the gaps alphabetic. With key weights $p_{[r]}$ decreasing and key comparison lengths $\ell_{(r)}(T)$ increasing,

$$J(p, q) = \min_T \left\{ \sum_r p_{[r]} \ell_{(r)}(T) + \sum_i q_i d_i(T) \right\}.$$

Writing $A(q)$ for the exact alphabetic gap optimum gives $K(p) + G(q) \le K(p) + A(q) \le J(p, q) \le \mathrm{OPT}(p, q)$. If $P = \sum p_i$, S_j sums the j largest key weights, and $a_t(T)$ counts keys of comparison length at most t, then

$$\sum_r p_{[r]} \ell_{(r)}(T) = P + \sum_{t \ge 1} (P - S_{a_t(T)}).$$

Each key is counted once plus once per additional level. The function $P - S_j$ is decreasing and discretely convex. This standard rearrangement and tail-sum identity describes global depth-profile coupling; it does not give a fast minimizer over alphabetic trees.

5.7.1 A precisely specified rejected extension

Start with ordered original gap leaves. Active items are compatible when no unmerged original leaf lies strictly between them. Merge a compatible pair of minimum total mass, adding the smallest unused key weight to its mass; replace its left item and delete its right item. Recover original leaf depths, reconstruct an alphabetic tree, and optimally reassign the key multiset to its internal depths. This is a proposed mixed extension of compatible merging, not the established alphabetic algorithm of Hu and Tucker [19].

For the full ordered instance

$$p = (7, 1, 5, 8, 6, 4), \qquad q = (18, 2, 13, 3, 17, 1, 2),$$

the next-key/pair/new-mass triples are

$$(1; 1, 2; 4),\ (4; 2, 13; 19),\ (5; 3, 17; 25),\ (6; 18, 4; 28),\ (7; 19, 25; 51),\ (8; 28, 51; 87).$$

The margins over the next compatible pair are $12, 1, 1, 1, 3$, followed by the sole remaining pair. Thus all choices are strict, all weights positive, and all key weights distinct. The charges sum to 214.

Quantity	Greedy profile	Better feasible tree
Sorted key lengths	$1, 2, 2, 3, 3, 3$	$1, 2, 2, 3, 3, 4$
Ordered gap depths	$2, 3, 3, 3, 3, 3, 3$	$2, 4, 4, 3, 2, 3, 3$
Optimally assigned key cost	64	65
Gap cost	150	148
Total	214	213

The better tree has root 3, left root 0 with right child 2 and 2's left child 1, and right root 4 with right child 5. Its inorder key lengths are $(2, 4, 3, 1, 2, 3)$, so the displayed ordered weights attain the relaxed assignment. Enumeration of all 132 shapes gives $J = 213$; independent cubic and Knuth recurrences give $\mathrm{OPT} = 213$. Here $K = 64$, $G = 144$, and $A = 148$, leaving deficits even for $K + A = 212$. The value 214 is a relaxed candidate cost: the greedy reconstructed shape with the original ordered p costs 220. Thus the failure persists even after freeing all key positions, and 214 is unsafe as a lower bound.

Theorem 140. *This specified merge extension fails on a positive open box, has arbitrarily large additive error under scaling, and fails for every number of keys $n \geq 6$ with positive weights, distinct keys, and strict pair choices. The latter family uses weights with $O(n)$ bits.*

Proof. Scaling multiplies 214 and 213 by the same factor. Perturb all thirteen weights independently by at most $\epsilon = 1/1000$. Key ranks are unchanged. Each active mass is a sum of original outcomes, so each pair-score difference changes by at most $26\epsilon < 1$. Induction preserves every strict merge. Candidate minus displayed witness cost is $q_4 - q_1 - q_2 - \min p$, at least $1 - 4\epsilon > 0$. No optimality assumption on the perturbed witness is required.

For padding, let an old full instance have mass W and known optimum U. Put $B = W + U + 1$, and prepend a key of weight $2B$ and a gap of weight B. The tree rooted at the new key, with the old tree on the right, costs $3B + W + U = 4B - 1$. Any other root puts the new key at comparison length at least two, costing at least $4B$. Therefore $\mathrm{OPT}' = 3B + W + \mathrm{OPT}$. All old compatible pair masses are at most W, whereas any pair involving the new gap exceeds $B > W$. The new key is largest. All old merges occur first and the final charge is $3B + W$. Alphabetic reconstruction raises all old depths by one, so the relaxed candidate gains the same amount and its excess of one persists. The recurrences $W' = W + 3B$, $U' = 4B - 1$, $B' = W + 7B$ imply $7B \leq B' < 8B$, and hence linear bit length. This proves failure of the rule, not a computational lower bound for the original problem. □

5.7.2 Uniform-transfer invariance and exact controls

Every extended binary tree satisfies $\sum d_i(\text{gap}) - \sum \ell_i(\text{key}) = n$. For $-\min q \leq a \leq \min p$, set $p' = p - a$, $q' = q + a$. Every full tree cost, and also J, increases by an. If $M(n)$ is the minimum unit-key comparison-length sum, $E(n+1) - M(n) = n$. The key bound decreases by $aM(n)$, while G increases by $aE(n+1)$: its variation term is translation invariant and its rank coefficients sum to $E(n+1)$. Consequently $K(p') + G(q') - an = K(p) + G(q)$. For the rejected merge rule every active cluster has one more original gap than consumed key. Its mass increases by a, so all pair scores increase by $2a$, all choices and margins are unchanged, and the sum of charges increases by an. Uniform transfer cannot repair that rule or the $K + G$ deficit. The exact alphabetic bound behaves differently below.

5.8 A four-key barrier for every admissible transfer

Theorem 141. *For $p = (1, 1, 2, 1)$ and $q = (1, 2, 1, 1, 1)$, the full optimum is 24, while the largest $K + A$ lower bound over all nonnegative additive transfers preserving relative tree costs is exactly 23.*

Proof. The all-unit full instance costs at least 20. The two additional units add $\ell_2 + d_1$. If key 2 is root, gap 1 has depth at least three; otherwise $\ell_2 \geq 2$ and the internal gap has depth at least two. The added cost is always at least four. Root 1, left child 0, and right child 2 with right child 3 attains 24.

On the maximal face $a_* = 1$, $L = R = 2$ and

$$p' = (x, 0, 1, y), \qquad q' = (2 - x, 3, 2, 2, 2 - y).$$

The capacity assignment $(2, 3, 1, 2)$ gives $K(p') \leq 2x + 1 + 2y$. The feasible gap depths $(2, 2, 2, 3, 3)$ give $A(q') \leq 26 - 2x - 3y$. Hence $F(x, y) \leq 23 - y \leq 23$ throughout the box. At $x = y = 0$, $K = 1$ and $A = 26$: five gaps of weight two cost at least $2E(5) = 24$, and the additional unit on the internal gap costs at least two; the displayed depths attain equality. Theorems 120 and 121 extend the bound to every admissible transfer. □

Integer rounding does not close this gap. Only one residual interior key has positive weight after normalization. The theorem limits this separable certificate class; it does not rule out a richer joint representation or a faster full algorithm.

The failure persists on a positive open box. Perturb each of the nine weights by at most $\epsilon = 1/100$. The interior minimum remains p_1. The same assignment and gap tree bound the maximum by

$$2p_0 + 2p_1 + p_2 + 3p_3 + 2q_0 + 2q_1 + 2q_2 + 3q_3 + 3q_4 \le 23 + 20\epsilon.$$

Every four-key tree has unit-weight cost at most 24, so every perturbed original tree costs at least $24 - 24\epsilon$. Thus the gap is at least $1 - 44\epsilon > 0$ throughout the box.

A failed transfer of the sharper mass constant. The initial implementation used $2W/F_{d+3}$ as the maximum internal subtree mass at edge depth d, following a literal transfer of the Fibonacci formula in [7]. Its first full-input check failed: the resulting class could be empty. The positive two-key example

$$p = (1, 1), \qquad q = (1, 100, 1)$$

explains the failure. Both trees cost 206; the total mass is 104 and the internal child has mass 102. Thus $102 > 2 \cdot 104/3$ at depth one. Charging an additional comparison at every gap is a tree-independent constant and does not remove this violation. We preserve the counterexample and use the following independently justified, weaker constant in our model. This does not refute the KLR architecture or its asymptotic guarantees.

Partial-tree height caveat. An earlier proposed iteration started from the additive singleton lower bound. Its attempted rotation proof of logarithmic convergence was invalid: a rotation can lift an unfinished interval, requiring extra refinement to restore the prescribed number of levels. Small passing tests do not remove that issue. The upper-penalty proof above avoids that argument.

5.9 A single weight unit obstructs constant per-tile charging

The aggregate argument charges each tile at most $2h$ times the mass of its span. Removing h from this local inequality would strengthen the core bound. The following family rules out that local step for both density mass and support, while retaining a linear global core.

Theorem 142 (Local charging obstruction)**.** *Let $h \ge 4$, $s = 2^{h-2}$, $n = 8s$, $c = 2^h - 1$, and $x = 4s + 1$. Set $p_x = 1$ and every other key and gap weight to zero. At layer h, with terminal penalty $M > h$, the maximal dyadic tile*

$$T = [2s, 3s) \times [7s, 8s)$$

contains exactly $h - 3$ nonzero densities, each equal to one. Its span has mass one. The completed layer core has exactly

$$\delta(E_h) = 2n + 2^h - 2h - 1$$

entries. Thus no universal constant bounds either a tile's density mass or its support by that tile's span mass.

Proof. Write $C = 2^h = c + 1$ and $v(r) = \nu_2(r)$ for $1 \le r \le c$. For an interval $[i, j)$ containing x, put

$$k = j - i, \qquad U = x - i + 1, \qquad V = c + x + 1 - j.$$

Its exact cost is

$$A_h(i,j) = M(k-c)_+ + h - \max_{r\in[\min(U,V),\max(U,V)]\cap[1,c]} v(r). \tag{147}$$

Intervals not containing x have value $M(k-c)_+$. For $k \le c$, the perfect-slot representation permits the positive key in exactly the slots $U,\dots,V$. For $k \ge c$, the c exposed keys form a perfect height-h skeleton. Exposing x at slot r requires $r-1$ preceding keys and $c-r$ following keys; these inequalities give $V \le r \le U$, clipped to $[1,c]$. Leaving x unexposed contributes h, which cannot improve on a feasible slot with nonnegative valuation. The maximum-exposure argument uses $M > hW = h$.

Extend the valuations to virtual endpoints by $v(0) = v(C) = h$. For two endpoints $a < b$, write t for the maximum valuation strictly between them, taking $t = 0$ for an empty interior. A four-window range-maximum difference has value

$$\max\{0, \min(v(a), v(b)) - t\}. \tag{148}$$

This is positive precisely for consecutive multiples of 2^d, $1 \le d < h$, and then equals one. To see this, let d be the smaller endpoint valuation. A separation greater than 2^d contains an interior multiple of 2^d; at separation 2^d the interior maximum is $d-1$. Valuation zero cannot give a positive difference. The pair $(0, C)$ would instead map to $i = j$ and is excluded.

Taking differences in (147), the positive cells below and above the penalty line respectively have coordinates

$$\begin{aligned} j-i<c:&\quad (i,j) = (x-a,\ c+x+1-b), && 0 \le a < b \le C,\\ j-i>c:&\quad (i,j) = (x+1-b,\ c+x-a), && 1 \le a < b \le c. \end{aligned}$$

In each line, (a,b) ranges over the consecutive-multiple pairs just described. Below the line, a virtual endpoint handles the key entering or leaving a contributing interval: its valuation h cancels the formal h cost. Above the line, clipping that prevents an endpoint from moving gives zero density; at a key boundary its comparison length is h in both relevant intervals. Cells whose intervals all miss the key have zero ordinary density.

For the stated n, x, every listed pair lies inside the input grid. The counts below and above the line are

$$\sum_{d=1}^{h-1} 2^{h-d} = 2^h - 2, \qquad \sum_{d=1}^{h-1} (2^{h-d} - 2) = 2^h - 2h.$$

On the penalty line there are $n-c$ positive entries. Their residual density is either zero or minus the absolute difference of adjacent, possibly virtual, valuations, whose magnitude is at most h; adding $M > h$ makes them positive. Including the n completion entries proves the asserted total.

Every cell of T lies above $j - i = c$, while its side-$2s$ parent at $(2s, 6s)$ meets that line. Hence T is a maximal accepted tile. Its four corner slot windows, in telescoping order, are

$$[1, 2s+2], \quad [s+1, s+2], \quad [s+1, 2s+2], \quad [1, s+2].$$

Their maximum valuations are $h-1, 1, h-1, h-2$, so their ordinary costs are $1, h-1, 1, 2$. The affine penalty cancels and the density mass is $h-3$. Explicitly, its positive cells are exactly

$$(i,j) = (3s + 2 - 2^d,\ 7s), \qquad d = 2,\dots,h-2,$$

corresponding to endpoint pairs $(s, s+2^d)$. Each has density one. □

Corollary 143 (One-key global support)**.** *If a single key has positive weight and all other key and gap weights are zero, every completed terminal-penalty layer has at most $3n$ core entries.*

Proof. Scaling the positive weight scales the ordinary densities without changing their support. Input boundaries can only remove pairs from the preceding classification. If $c < n$, the two ordinary regions have at most $c-1$ and $c+1-2h$ entries. Add $n-c$ penalty-line entries and n completion entries to obtain fewer than $3n$. If $n \le c < 2n-1$, only the first region occurs, with at most $c-1$ entries, again giving fewer than $3n$. If $c \ge 2n-1$, every interval can put the weighted key at the root and balance its zero-weight sides within height h. Its cost is the key's weight when it contains that key and zero otherwise, so all interior densities vanish. Layer zero has only n completion entries. □

The local obstruction therefore coexists with a global linear bound. An improved general charging argument must account for how one outcome's contributions are distributed across tiles; it cannot give each tile an independent constant allowance per weight unit.

5.10 Quadratic density mass with bounded positive weights

Counting support through total density mass can lose a polynomial factor, even when every weight is positive and bounded. An exact mixed identity makes that obstruction explicit. Put $P = \sum p_i$, $Q = \sum q_i$, and $v(t) = \nu_2(t)$.

Lemma 144 (Full mixed boundary mass)**.** *Fix $h \ge 1$, $c = 2^h - 1 \le n$, and $M > h(P+Q)$. Define*

$$G_i = \max_{0 \le r \le (c-1)/2} \left\{ q_{i+2r+1} + \sum_{\ell=1}^{r} v(2\ell)(p_{i+2\ell} - p_{i+2\ell-1}) \right\}, \quad 0 \le i < n-c,$$

$$B_{\text{tail}} = \sum_{t=1}^{c} p_{n-c+t-1} v(t).$$

The ordinary density mass strictly below $j - i = c$ is

$$S_h = (h-1)P + (h-2)Q + q_0 + q_n + \sum_{i=0}^{n-c-1} G_i - B_{\text{tail}}. \tag{149}$$

Proof. Write $B_i = \sum_{t=1}^{c} p_{i+t-1} v(t)$. The length-$c$ interval is perfect, with reward B_i. For its final $c-1$ keys, omitting perfect slot one gives the same key reward since $v(1) = 0$. Moving the omitted slot to t changes that reward by

$$\sum_{k=1}^{t-1} p_{i+k}(v(k) - v(k+1)).$$

An even omitted slot cannot improve on the preceding odd one: the additional term is nonpositive, and the odd choice also has a nonnegative gap reward. For odd $t = 2r+1$, grouping consecutive terms gives the sum defining G_i, with gap reward q_{i+2r+1}. The mixed slot representation therefore gives

$$A_h(i+1, i+c) = hW(i+1, i+c) - B_i - G_i.$$

For $h = 1$, this is the empty-interval identity $q_{i+1} - q_{i+1} = 0$. Telescoping row $i < n-c$ below capacity gives $h(p_i + q_i) + G_i - (p_i + q_i + q_{i+1})$. The remaining triangular corner has mass $A_h(n-c, n) - \sum_{i=n-c}^{n-1}(p_i + q_i + q_{i+1})$; its full interval is perfect with reward B_{tail}. Addition proves the formula, including $c = n$. □

Corollary 145 (Mass can be quadratic while support is small)**.** *Let $n = 2^s \geq 16$, $h = s - 1$, $c = n/2 - 1$, and $p_i = 2$ for even i, $p_i = 1$ for odd i, $q_i = 1$ for all gaps. Then $W = 5n/2 + 1$, and this layer has*

$$S_h = \frac{n^2}{8} + \frac{n(h-3)}{4} + 2\big((h-1)n + h - c\big)$$

below-capacity density mass, but only $O(n \log n)$ completed core entries.

Proof. More generally let $p_i = a + 1$ for even i, $p_i = a$ for odd i, and $q_i = b$, with nonnegative integers a, b. Since $\sum_{\ell=1}^{(c-1)/2} v(2\ell) = c - h$, the prefix walk is increasing for even i and decreasing for odd i. Hence $G_i = b + (c-h)$ or b, respectively. The tail starts at odd $n - c$, so $B_{\text{tail}} = (a+1)(c-h)$. Equation (149) gives

$$S_h = \frac{n^2}{8} + \frac{n(h-3)}{4} + (a+b)\big((h-1)n + h - c\big).$$

The diagonal immediately below capacity has the explicit densities

$$D_h(i, i+c-1) = \begin{cases} c+1-h, & i \text{ even}, \\ 0, & i \text{ odd}. \end{cases} \tag{150}$$

To verify this, put $m = 2^{h-1}$ and $K(t) = \sum_{r=1}^{t} \text{bitlength}(r)$. At lengths $c-1, c-2$, Corollary 112 supplies a minimum-path-length tree attaining the sorted high-key profile. Its cost is the uniform (a, b) baseline plus $K(t)$ for the high-key count t. The baseline has zero density at the even length $c-1$. At length c, a high-first perfect interval has incremental high-key cost hm; a low-first interval has $K(m-1)$. Thus an even row gives

$$hm + K(m-2) - 2K(m-1) = 2m - h,$$

using $K(m-1) = (h-2)m + 1$ and $K(m-2) = K(m-1) - (h-1)$. An odd row gives zero, proving (150).

There are $n/4+1$ nonzero entries on that diagonal. Removing their mass $J_h = (n/4+1)(n/2-h)$ leaves

$$S_h - J_h = \frac{n(2h-5)}{4} + h + (a+b)\big((h-1)n + h - c\big).$$

At $a = b = 1$ this is $O(nh)$, so integrality bounds the remaining below-capacity support by $O(nh)$. For lengths at least $c+1$, choose a contiguous c-key skeleton starting with a low key: every even skeleton rank is high, attaining the maximum reward $(a+1)(c-h)$. Four-corner subtraction shows that above-capacity densities vanish except at length $c+1$, where odd rows have density $c - h$. This adds $O(n)$ entries. The removed diagonal, capacity line, and completion also each add $O(n)$ entries, proving the support bound. □

The obstruction concerns a mass-only proof of core size. Its large densities occupy a single diagonal and do not establish a large support lower bound.

5.11 Wider bands need a different interaction argument

For $d \geq 1$, let T_d sum ordinary densities with $|j - i - c| \geq d$. The first inequality of Theorem 14 does not extend to every d. Take four keys of weight zero and gaps $q_2 = q_4 = 1$, all other gaps zero, at $h = 2$, $c = 3$, $d = 2$. Only length-one density cells qualify. The sole positive one is

$$D_h(2,3) = A_h(2,4) - A_h(2,3) - A_h(3,4) = 3 - 1 - 1 = 1.$$

Each gap alone has zero eligible density mass. Yet the full interval's gaps all have depth two, so $R_h(w) = 4 = R_h(e_{q_2}) + R_h(e_{q_4})$. Thus the tail interaction is one while the full-interval interaction is zero. Since $P = 0$, this also defeats any all-width interaction allowance based only on key mass.

The stronger control by full-interval interaction also fails above capacity. Take $n = 10$, $h = 3$, $c = 7$, $d = 2$, and unit keys at indices $3, 6$. The only eligible upper-tail cell is $(0, 9)$. The skeleton rewards for $[0, 10)$, $[0, 9)$, $[1, 10)$, $[1, 9)$ are $3, 3, 3, 1$, respectively, so its density is $3 + 3 - 3 - 1 = 2$. The unit tails are zero, while full-interval interaction is $2 + 2 - 3 = 1$.

Uniform unit keys already force interaction of order hW: with $n = 2^{h+1}$ and $d = 2^{h-1}$, every unit-key tail is zero by the consecutive-power classification in Section 5.9, whereas the combined tail is $(h-1)n + h - c$. The following family shows that even an arbitrary constant multiple of hW is insufficient.

5.12 Two keys defeat every logarithmic additive interaction allowance

Theorem 146 (Sparse interaction barrier)**.** *Let $h \geq 5$, $C = 2^h$, $c = C - 1$, $n = 2C$, and $w = e_{C-5} + e_C$, with unit weights on these two keys and zero weights elsewhere. At width four,*

$$\begin{aligned} T_4(w) &= 21C/16 - 3h + 1, \\ T_4(e_{C-5}) + T_4(e_C) &= C - 4h + 4, \\ T_4(w) - T_4(e_{C-5}) - T_4(e_C) &= 5C/16 + h - 3. \end{aligned}$$

Consequently no absolute constant K gives $T_d(w) \leq \sum_x w_x T_d(e_x) + KhW$ for all inputs and widths. Scaling both positive weights by C gives the same obstruction at total mass $W = n$.

Proof. Put $x = C - 5$, $y = C$ and remove terminal penalties from interval costs, writing $R_w(i, j)$. Define the interval interaction

$$E(i, j) = R_w(i, j) - R_{e_x}(i, j) - R_{e_y}(i, j).$$

It vanishes when an interval omits either key. On the full interval each key alone can occupy the root, with reward $h - 1$. Jointly, selected ranks have distance at most five, so at least one has valuation at most two. Reward $h + 1$ is attained at ranks $C/2, C/2 + 4$. Thus $E(0, n) = h - 3$.

Let $F_k = \sum_{i=0}^{n-k} E(i, i + k)$ and $b_k = E(0, k) + E(n - k, n)$. Summing a density diagonal gives $F_{k+1} - 2F_k + F_{k-1} + b_k - b_{k-1}$. The excluded lengths are $c-3, \ldots, c+3$. Their edge contribution is $b_{c+3} - b_{c-4} = 0$: the short edge intervals omit a key, the left long edge has compatible maximizing rank choices, and the right long edge omits the keys. Singletons have zero interaction. Therefore the tail interaction is exactly

$$h - 3 + F_{c-3} - F_{c-4} + F_{c+3} - F_{c+4}. \tag{151}$$

Write $v(t) = \nu_2(t)$ and $r = x - i + 1$. In a length-$c - a$ interval containing both keys, their perfect slots are $r + u, r + 5 + z$ with $0 \leq u \leq z \leq a$. The local interaction is

$$e_a^-(r) = \max_{0 \leq u \leq a} v(r + u) + \max_{0 \leq z \leq a} v(r + 5 + z) - \max_{0 \leq u \leq z \leq a} \{v(r + u) + v(r + 5 + z)\}.$$

For length $c + a$, replace the two slots by $r - u, r + 5 - z$, keeping $0 \leq u \leq z \leq a$ and clipping to $1 \leq r - u < r + 5 - z \leq c$; call this interaction $e_a^+(r)$. For $a \leq 4$, any feasible rank of one positive key can be supplemented by a rank of the other, without reducing reward. Indeed, given u, choose $z = \max(u, r + 5 - c)$; given z, choose $u = \min(z, r - 1)$. The distance remains positive since $a < 5$.

The rank constraints also leave enough zero keys to fill the other slots. Thus omission of a positive key cannot improve the displayed joint maximum. The interval starts give

$$F_{c-a} = \sum_{r=1}^{C-6-a} e_a^-(r), \qquad F_{c+a} = \sum_{r=1}^{C-4} e_a^+(r), \qquad a = 3, 4.$$

Here are the finite ruler cases needed to sum these expressions. For the lower diagonals, the lists starting at residue one are periodic:

$$e_3^- : (1, 1, 2, 0, 2, 1, 1, 0),$$
$$e_4^- : (1, 1, 1, 1, 1, 1, 0, 0, 1, 1, 2, 2, 1, 1, 0, 0).$$

They follow by taking the two independent maxima in the four- or five-position windows and enforcing $u \le z$. Only valuations $0, 1, 2, 3, \ge 4$ are needed; larger values cancel or the lower competing slot determines the loss. The period sums are eight and fourteen, and the terminal partial periods sum to eight and six. Consequently

$$F_{c-3} = C - 8, \qquad F_{c-4} = 7C/8 - 8.$$

For e_3^+, only residues $3, 7$ modulo eight contribute. At $r = 8k + 3$, the independent high ranks are $8k, 8k + 8$ with valuations A, B. They cannot both be used. Keeping either forces the other's reward to zero; using neither gives at most two. Hence the defect is $\min(A, B) = 3$ away from the edges. At $r = 8k + 7$ the independent maxima are both two, but the best compatible sum is two, giving defect two. Other residues have compatible maxima. Clipping lowers the defects at $r = 3, C - 5$ from three to one. There are $C/8$ cells of residue three and $C/8 - 1$ of residue seven, giving $F_{c+3} = 5C/8 - 6$.

For e_4^+, extend $v(0) = v(C) = h$ and put $A = v(8k)$, $B = v(8k + 8)$. Only residues three and four contribute. Before edge corrections,

$$e_4^+(8k + 3) = \min(A, B - 2), \qquad e_4^+(8k + 4) = \min(A - 2, B).$$

For the first case the best compatible reward is $\max(A + 2, B)$; for the second it is $\max(A, B + 2)$. Keeping one high rank either permits the intervening valuation-two slot or forces the other reward to zero. All other pairs have smaller reward. At $r = 4, C - 5$, clipping lowers the formal value from three to zero, a total correction of six. Put $m = C/8 \ge 4$. For each $k = 0, \dots, m - 1$, one of $k, k+1$ is odd. The pair of displayed defects sums to three when its even neighbor has valuation one, and four otherwise. Exactly $m/2$ indices have each type, including the formal endpoints. Hence $F_{c+4} = 7C/16 - 6$.

Substitution in (151) gives $5C/16 + h - 3$. Separately telescoping below and above the excluded band gives excesses $C/8$ and $3C/16 + h - 3$, respectively; each side alone defeats an $O(hW)$ allowance. For each unit key, all classified pairs at distances $2^t - 1$, $3 \le t < h$, fit inside the input. There are $C/2^t$ below and $C/2^t - 2$ above. Their sum is $C/2 - 2h + 2$ per key, proving the remaining formulas. Since $W = 2$, the excess divided by hW tends to infinity. Homogeneity gives the linear-total-mass scaling. □

The family leaves the broader target $T_d = O(W(L + c/d))$ open: its mass and that target are both of order C at $d = 4$. It shows that the c/d term must cover joint optimization effects as well as independent unit tails.

5.13 The clipped periodic part retains capacity-scale mass

Lemma 147 (Two-key periodic obstruction)**.** *Let $h \geq 3$, $C = 2^h$, $c = C - 1$, $n = 2C$, with unit key weights at $C - 5, C$ and zero elsewhere. For the clipped template $v_r = \min(\nu_2(r), 1)$, the tail in (17) satisfies*

$$U_2(v) = C/2 - 4.$$

Thus no absolute constant multiple of tP bounds $U_d(v^{(t)})$ for all $d \geq 2^t$, even with $t = 1$ and $P = 2$.

Proof. The template rewards even ranks by one and odd ranks by zero. Use (19) with $k = C + 1$. Its first sum has starts $0 \leq i \leq C - 1$, each with two positions omitted from a length-$C + 1$ interval. For $1 \leq i \leq C - 6$, write $r = C - 4 - i$ for the first unit key's local position. Omit $u = 0$ positions before it if r is even, or $u = 1$ if r is odd, and one further position between the two unit keys. Their selected ranks $r - u, r + 4 - u$ are even and lie in $[1, c]$. Remaining omissions can follow the second key. The reward is two. At $i = 0$ the second key is last and, if selected, has odd rank c; at $i = C - 5$ the first key has odd rank one. Each has reward one. The four remaining intervals contain only the second unit key, which can occupy an even rank. This boundary sum is $2C - 6$.

The second sum has starts $1 \leq j \leq C - 1$ and one omission per length-C interval. When both keys are present, their separation five requires that omission to fall between them for both ranks to be even. The reward is two precisely when $r = C - 4 - j$ is even with $2 \leq r \leq C - 6$. There are $C/2 - 3$ such starts; every other interval has reward one. This sum is $3C/2 - 4$. Both keys can occupy even ranks on the full interval, so its reward is two. Subtracting gives $C/2 - 4$. □

This lemma concerns the auxiliary clipped template. It identifies an obstruction to finishing the clipping argument with a height-only allowance. Scaling both weights by C gives total mass $W = n$ and preserves the unbounded ratio to tP.

Retained ranks obstruct an unchanged matching bound. The endpoint inequality (23) cannot be reused at clipping level two. Take $c = 7$, $d = 3$, $n = 10$, one unit key at position three, and template $(0, 1, 0, 2, 0, 1, 0)$. On $[0, 10)$ that key can have rank four, giving score two. On $[1, 10)$ its feasible ranks are one through three, giving score one. Thus $\delta = 1$ and $p_1 = 0$, but $d\delta = 3 > S_v(0, 10) = 2$. Counting only removed edge weights misses the changed reward on a retained key.

5.14 One-augmentation certificates can exceed the tail scale

The matching representation extends to any template with $v_{2r} = u_r \geq 0$ and $v_{2r-1} = 0$, where $c = 2m + 1$. Its score is

$$S(i, j) = \max \left\{ \sum_{r=1}^{m} p_{a_r} u_r : i + 1 \leq a_1 < \cdots < a_m \leq j - 2, \quad a_{r+1} - a_r \geq 2 \right\}.$$

For the clipped ruler, $u_r = \min(1 + \nu_2(r), t)$. Take an optimal matching A on an interval of length $c + d$, $d \geq 1$. If it avoids edge $i + 1$, it remains feasible after shortening the interval on the left. Otherwise delete edge $i + 1$ and vertex $i + 1$. The remaining matching has $m - 1$ edges and $d + 1$ unmatched vertices. As in Theorem 19, each consecutive unmatched pair bounds an augmenting path; flipping it gives a feasible matching B of size m. For removed, added, and retained edge sets

R, N, E, respectively, the exact score loss is

$$F(A) - F(B) = p_{i+1}u_1 + \sum_{x\in R} p_x u_{\mathrm{rank}_A(x)} - \sum_{x\in N} p_x u_{\mathrm{rank}_B(x)} + \sum_{x\in E} p_x\big(u_{\mathrm{rank}_A(x)} - u_{\mathrm{rank}_B(x)}\big).$$

The signed retained-rank term survives even when removed-edge costs are disjoint across the candidate paths.

Put $\delta(i,j) = S(i,j) - S(i+1,j)$. Let $\widehat{\delta}(i,j)$ be the smallest loss over all these single-path repairs and over all optimal starting matchings. An optimum avoiding $i+1$ is allowed unchanged. Then $\widehat{\delta} \geq \delta \geq 0$. For $k = c + d < n$ and $b = n - k$, define

$$\widehat{U}_d = \sum_{i=0}^{b-1} \widehat{\delta}(i, i+k) + S(b,n) - S(0,n) \;\geq\; U_d. \tag{152}$$

This is the strongest upper certificate obtained by choosing one such augmentation independently at each endpoint.

Lemma 148 (Rightmost witnesses recognize zero endpoint loss)**.** *Optimal matching tuples are closed under componentwise minimum and maximum. Their unique componentwise-rightmost optimum A^+ satisfies*

$$S(i,j) = S(i+1,j) \quad\Longleftrightarrow\quad a_1^+ > i+1.$$

Proof. Componentwise minimum and maximum preserve gaps of at least two. At each rank they merely exchange the two chosen positions, preserving the sum of the two objective values. If both original tuples are optimal, both new tuples must be optimal. Taking maxima over the finite set gives A^+, also its lexicographically largest member. When the two scores are equal, a shortened optimum is an old optimum avoiding $i+1$; A^+ is at least that tuple coordinatewise. The converse follows by feasibility. □

Lemma 149 (An unbounded local repair ratio with a unique optimum)**.** *For every integer $B \geq 1$, take $c = 7$, $n = 10$, $u = (1,2,1)$, and $p_2 = p_3 = 2B$, $p_4 = B$, $p_6 = 2B - 1$, with other weights zero. Then*

$$\delta(1,10) = 1, \qquad \widehat{\delta}(1,10) = B + 1.$$

The old optimal matching is unique.

Proof. Of the ten feasible old matchings, $(2,4,6)$ alone has score $6B - 1$. All others have score among $4B, 2B, 6B-2, 5B-2$, each at most $6B - 2$. The four shortened matchings have scores $2B, 2B, 6B-2, 5B-2$; $(3,6,8)$ attains $6B-2$. After deleting edge two, the unmatched vertices are $3, 8, 9$. The two augmentations give $(3,5,7)$ and $(4,6,8)$, with scores $2B$ and $5B - 2$. Their best score is $5B - 2$, proving the claim. □

Theorem 150 (A one-key obstruction to the repair certificate)**.** *Let $h \geq 3$, $C = 2^h$, $c = C - 1$, $m = C/2 - 1$, $n = 4C$, and $d = C$. Put one unit key at $x = C/2 + 1$ and use the full ruler $u_r = 1 + \nu_2(r)$, with maximum reward $t = h - 1$. Then*

$$U_C = 0, \qquad \widehat{U}_C = \frac{(h-2)(h-3)}{2}. \tag{153}$$

Rightmost optimal starts attain this minimum certificate. Consequently no absolute constant bounds $\widehat{U}_d/[P(t + (c+1)/d)]$ over all inputs.

Proof. Put $k = 2C - 1$ and $b = 2C + 1$. For $i < x$, the positive key's feasible matching ranks on $[i, i + k)$ are exactly $1, \dots, R_i$, where

$$R_i = \min\left(m, \left\lfloor \frac{x - i + 1}{2} \right\rfloor\right).$$

The long suffix leaves room for every such rank. The unique rank with maximum ruler reward is $r = 2^{\lfloor \log_2 R_i \rfloor}$. A rightmost optimum is

$$a_s = \begin{cases} x - 2(r - s), & 1 \le s \le r, \\ i + k - 2 - 2(m - s), & r < s \le m. \end{cases}$$

For $i \ge x$, pack all m zero-weight edges against the right boundary. The first edge is used precisely at the thresholds

$$i = x - 2r + 1, \qquad r = 1, 2, 4, \dots, 2^{h-2}.$$

At every other start this witness survives, so both endpoint losses vanish. At a threshold, every optimum puts the positive key at rank r, and its prefix is forced: exactly enough vertices precede the key for that many matching edges. For $r = 1$, deleting the first edge deletes the positive key, and both losses are one.

For $r \ge 2$, the shortened interval's largest feasible rank is $r - 1$; its unique best ruler rank is $r/2$. The reward falls from $1 + \nu_2(r)$ to $\nu_2(r)$, giving true loss one. In the old remaining matching, only its leftmost vertex is unmatched before the positive key. No augmenting path can therefore lie entirely before that key. A path crossing the key removes it, giving score zero; every other path lies after it and retains it at odd rank $r - 1$, giving score one. There is an after-key path because only one of the $d + 1$ unmatched vertices lies before the key, while $d = C \ge 8$. Thus every optimal starting matching, regardless of its suffix, has best repair loss $\nu_2(r)$.

The full score is $h - 1$ and $S(b, n) = 0$. The true endpoint losses sum to $h - 1$, proving $U_C = 0$. The best repair losses sum to

$$1 + \sum_{a=1}^{h-2} a = 1 + \frac{(h - 2)(h - 1)}{2}.$$

Subtracting $h - 1$ gives (153). Since $P = 1$ and $t + (c + 1)/d = h$, the ratio tends to infinity. Multiplying the key weight by n gives integer total mass $W = n$ with the same ratio. □

Corollary 151 (An optimum can require many alternating components)**.** *In Theorem 150, use the threshold $i = 2$, where the old positive key has rank $r = C/4$, and delete the first matched edge. Relative to the remaining matching, every shortened optimum has at least $C/8 - 1$ cardinality-decreasing alternating components and at least $C/8$ cardinality-increasing components. Both bounds are attained for the rightmost old optimum.*

Proof. The positive edge is common to the remaining matching M and every new optimum B. Its new rank must be $r/2$. Before that common edge, M has $r - 2$ edges and B has $r/2 - 1$. Their symmetric-difference components cannot cross the common edge; each changes cardinality by at most one. At least $r/2 - 1$ components must decrease cardinality before the key. Total cardinality rises by one, requiring at least $r/2$ increasing components overall.

For attainment, keep edges $a_{r/2+1}, \dots, a_m$ of the rightmost old optimum and insert $x + 2, x + 4, \dots, x + r$. These fit between the old prefix and suffix. The resulting matching has m edges with the key at rank $r/2$, hence is optimal. Relative to M, its removed edges $a_2, \dots, a_{r/2}$ and added edges are pairwise nonadjacent. They are exactly the claimed decreasing and increasing single-edge components. □

The true tail in this family is zero. The obstruction concerns the specified local repair certificate; its many zero-weight edge changes can potentially be represented together. Continuing the matching route requires joint rank changes rather than a bound on one augmentation.

5.15 Aggregate tail bounds do not imply uniform key sensitivities

The weighted formula in Theorem 20 determines all directional derivatives within its cluster. If $I = \{r : w_r = \min_s w_s\}$ and a direction z preserves nonnegative weights for sufficiently small positive perturbations, then

$$D_z U_K = N \min_{r \in I} z_r.$$

Weights outside I remain above the minimum for a sufficiently small perturbation, which proves the formula. A unique minimum has individual right derivative N. At a tied minimum, every individual coordinate increase has derivative zero, while increasing all cluster weights by ϵ increases the tail by $N\epsilon$. Such directional responses need not be additive.

Corollary 152 (Unbounded pointwise sensitivity at linear mass)**.** *For the alternating template, no absolute constant bounds the right derivative of U_d with respect to an individual key weight by $h + (c+1)/d$, even at nonnegative integer inputs of total mass $W = n$.*

Proof. Take $K = 2^r \geq 4$, $C = K^2$, $h = 2r$, $c = C - 1$, $n = 2C$, $d = K$, and the spacing-three cluster of Theorem 20 with weights $(K, 3K, 2K, \ldots, 2K)$. Their total is exactly n, and the first weight is the unique minimum. Therefore

$$N = \frac{K^2 - 3K + 2}{2}, \qquad U_K = NK, \qquad \frac{\partial_+ U_K}{\partial w_0} = N.$$

The ratio $N/(h + (c+1)/d) = N/(2\log_2 K + K)$ tends to infinity. □

The proved aggregate estimate $U_d = O(P(1 + c/d))$ holds throughout this family. Its exact charge $N \min_r w_r$ is bounded using the mass of all K interacting keys. Thus uniform pointwise sensitivity is a stronger requirement than the aggregate bound. The corollary concerns the auxiliary alternating template; the corresponding full-ruler sensitivity question remains open.

5.16 Nested binary threshold tails can all vanish

Grouping weights by threshold does not by itself preserve a tail bound. For $B_a = \{i : p_i \geq a\}$, even a constant-factor transfer from $\int_0^\infty U_d(\mathbf{1}_{B_a})\, da$ can fail.

Lemma 153 (Zero threshold tails with a positive weighted tail)**.** *There is no finite constant A such that*

$$U_d(p) \leq A \int_0^\infty U_d(\mathbf{1}_{B_a})\, da$$

for every nonnegative input, even for the full height-three ruler template. The same holds for its alternating clipping.

Proof. Take $n = 10$, $c = 7$, $d = 1$, and

$$p = (0, 0, 0, 0, 2, 1, 1, 2, 1, 0).$$

Its two nonempty threshold vectors are $b = (0, 0, 0, 0, 1, 1, 1, 1, 1, 0)$ and $e = (0, 0, 0, 0, 1, 0, 0, 1, 0, 0)$, each over a threshold interval of length one. For $v = (0, 1, 0, 2, 0, 1, 0)$ the scores are as follows.

Input	$S(0,8)$	$S(1,9)$	$S(2,10)$	$S(1,8)$	$S(2,9)$	$S(0,10)$	U_1
p	5	6	5	5	4	6	1
b	3	3	4	3	3	4	0
e	2	3	1	2	1	3	0

Each entry is the maximum of $p_{a_1} + 2p_{a_2} + p_{a_3}$ over $i+1 \le a_1 < a_2 < a_3 \le j-2$ with consecutive indices separated by at least two. Equation (19) subtracts the last three scores from the first three. Both threshold tails vanish, while the weighted tail is one, proving the claim for every A. For the alternating template the three score rows become $(3,4,4;3,3,4)$, $(2,2,3;2,2,3)$, and $(1,2,1;1,1,2)$, again giving tails $1,0,0$. Scaling the weights scales the positive tail and preserves the zero threshold integral. □

5.17 Sparse rank rewards defeat an average-reward tail bound

The full-ruler capacity term cannot be replaced by average template reward in a theorem for arbitrary nonnegative rank templates.

Lemma 154 (Sparse periodic rank obstruction)**.** *There is no absolute constant A such that*

$$U_d(v) \le AP\left(\max_r v_r + \frac{1}{d}\sum_r v_r\right)$$

for every nonnegative template and input, even for periodic binary templates and two unit keys.

Proof. For integers $d \ge 1$ and $m \ge 2$, put $C = 2d(m+1)$, $c = C-1$, and $n = 2C$. Give unit weight to one-based input positions $x = C - 3d, y = C$. Let $v_r = 1$ exactly at ranks $2d, 4d, \dots, 2dm$. We show $U_d(v) = m-1$.

At slack d, both keys receive reward one precisely when the first relative rank is $2dt$, $1 \le t \le m-1$. Their separation $3d$ requires all d omissions between them to reach consecutive rewarding ranks. There are $m-1$ such starts. At slack $d-1$, a simultaneous reward is impossible: feasible rank gaps lie in $[2d+1, 3d]$, containing no positive multiple of $2d$.

For a key at X, long starts giving reward one lie in $[X - 2dt - d, X - 2dt]$, $1 \le t \le m$. Taking the union for x, y within the boundary start range gives

$$[0, C-4d] \;\cup\; [C-3d, C-2d],$$

containing $C-3d+2$ starts. For short starts the union is $[1, C-4d] \cup [C-3d+1, C-2d]$, containing $C-3d$ starts. The two boundary sums in (19) are therefore $C-3d+m+1$ and $C-3d$. The full score is two, so $U_d = m-1$. Taking $m = d \ge 2$ gives ratio $(d-1)/4$ to the proposed right-hand expression without A, which is unbounded. □

This sparse auxiliary template differs from the full ruler. The obstruction preserves the full-ruler target $O(P(h + C/d))$; its capacity term accommodates the explicit family.

5.18 Interactions survive removal of every unit-outcome support

One possible repair to constant-per-weight charging is to first remove every cell attainable by a single positive outcome. The geometric envelope $\mathcal{E}_h$ in Lemma 180 contains all such cells, for every key and gap position. Even removing this whole envelope does not make the remaining support bounded by a constant times local mass.

Theorem 155 (Off-envelope interaction support)**.** *Let $s = 2^t$, $t \geq 3$, $n = 8s$, $C = 4s$, $c = C - 1$, and $h = t + 2$. Put unit keys at one-based positions $X = 3s + 3$ and $Y = 4s + 2$, with all other weights zero. The accepted maximal tile $T = [0, s) \times [5s, 6s)$ has local mass two and at least $\log_2 s - 2$ positive densities outside $\mathcal{E}_h$. In particular, no absolute constant bounds off-envelope support by that constant times local mass.*

Proof. The tile has minimum interval length $4s + 1 > c$; its parent $[0, 2s) \times [4s, 6s)$ meets capacity. For each $r = 2, \ldots, t - 1$, consider

$$i = s + 3 - 2^r, \qquad j = 5s + 2^{r-1}.$$

These are distinct cells in T. Their distance above capacity is $g = 3 \cdot 2^{r-1} - 2$. Since $g + 1$ is not a power of two, they lie outside $\mathcal{E}_h$.

For an above-capacity interval $[a, b)$ containing both keys, feasible exposed ranks $u < v$ are characterized by

$$1 \leq u < v \leq c, \quad u \leq X - a, \quad v \geq Y + c - b, \quad v - u \leq Y - X = s - 1.$$

These conditions give enough zero keys before, between, and after the two positive keys to fill the remaining selected ranks. Write $X - a = 2s + x$ and $Y + c - b = 3s - y$. At all four corners of each witness, $3 \leq x \leq s/2$ and $1 \leq y \leq s/4$. If $v < 3s$, replacing u by $2s$ is feasible and does not lower its valuation: $v > 2s$, $v - 2s \leq s - 1$, and $2s \leq X - a$. If $v \geq 3s$, replacing v by $3s$ is feasible and does not lower its valuation: the original separation forces $u \geq 2s + 1$, while $u \leq X - a < 3s$. The valuations of $2s$ and $3s$ are $t + 1$ and t. The largest valuation on $[3s - y, 3s - 1]$ is $\lfloor \log_2 y \rfloor$, and on $[2s + 1, 2s + x]$ it is $\lfloor \log_2 x \rfloor$. Both resulting pairs are feasible, so the exact maximum score is

$$S(a, b) = t + F(x, y), \qquad F(x, y) = \max\{\lfloor \log_2 x \rfloor, 1 + \lfloor \log_2 y \rfloor\}.$$

Omitting one positive key gives score at most $t + 1$, and omitting both gives zero; neither improves this formula.

At the witness cell, $x = 2^r$ and $y = 2^{r-1} - 1$. The affine penalty and the constant $2h$ cancel in the cost density, leaving

$$F(x, y) + F(x - 1, y + 1) - F(x, y + 1) - F(x - 1, y) = r + r - r - (r - 1) = 1.$$

There are $t - 2$ such cells and the local mass is two. Their ratio to local mass is unbounded as t grows. □

The theorem rules out a mass-only charge after removing the entire unit-outcome envelope. It is compatible with the boundary-plus-mass hypothesis below, whose side term also covers these interaction cells.

5.19 Dyadic block replication preserves interior support

A natural amplification attempt replaces each positive key by a block of unit keys and scales all positions. For binary key weights this has an exact coarse representation on intervals retaining complete positive blocks. It preserves interior core size instead of multiplying it. Keys in this subsection are numbered $1, \ldots, n$.

Theorem 156 (Exact binary-block dilation)**.** *Let $p_1, \dots, p_n \in \{0,1\}$, $C = 2^h$, $c = C - 1$, and $h \geq 1$. Write $S_h(A, B)$ for the maximum ruler reward of a c-subsequence on keys $A+1, \dots, B$, whenever $B - A \geq c$. Replace each input key by $d = 2^\ell$ identical consecutive copies. The fine height is $h + \ell$, its capacity is $c_d = dC - 1$, and its ruler score is S_d. Consider a fine interval $[a, b)$ with $b - a \geq c_d$ that cuts no positive block: each replicated positive block is wholly contained or wholly excluded. Put*

$$A = \lceil a/d \rceil, \qquad B = \lfloor b/d \rfloor, \qquad m = \sum_{k=A+1}^{B} p_k.$$

If $m \leq c$, then $B - A \geq c$ and

$$S_d(a, b) = m(d-1) + S_h(A, B). \tag{154}$$

Endpoints may lie inside zero blocks.

Proof. The length bound gives $B - A \geq C - 1 = c$ by integer rounding. A block of d consecutive selected ones starting after rank r has reward

$$\sum_{t=r+1}^{r+d} \nu_2(t) = d - 1 + \nu_2(\lfloor r/d \rfloor + 1). \tag{155}$$

Every residue modulo d occurs once. The valuations capped at ℓ sum to $\sum_{j=1}^{\ell} d/2^j = d - 1$; the unique multiple of d contributes the remaining coarse valuation. A legal start $0 \leq r \leq c_d - d$ gives coarse rank $1 \leq \lfloor r/d \rfloor + 1 \leq c$.

For $m = 0$ both scores vanish, so assume $m \geq 1$. Let $X_1 < \cdots < X_m$ be the positive coarse positions in the interval. By Lemma 22, all $md \leq dc \leq c_d$ positive fine keys can be retained in an optimum. Their block starts r_j satisfy exactly

$$\begin{aligned} r_1 &\leq R = d(X_1 - 1) - a, \\ r_m &\geq Q = d(X_m + C - 1) - b - 1, \\ d &\leq r_{j+1} - r_j \leq d(X_{j+1} - X_j), \qquad 0 \leq r_j \leq c_d - d. \end{aligned}$$

These inequalities count the available zeros before, between, and after the complete positive blocks. Write $r_j = d(u_j - 1) + \rho_j$, where $0 \leq \rho_j < d$. Taking floors gives

$$\begin{gathered} 1 \leq u_1 < \cdots < u_m \leq c, \qquad u_1 \leq X_1 - A, \qquad u_m \geq X_m + c - B, \\ u_{j+1} - u_j \leq X_{j+1} - X_j. \end{gathered} \tag{156}$$

These are precisely the coarse feasibility conditions when all m positive keys are retained.

Conversely, fix any coarse ranks satisfying (156). A tight first bound imposes $\rho_1 \leq R \bmod d$; otherwise its upper phase bound is $d - 1$. A tight last bound imposes $\rho_m \geq Q \bmod d$; otherwise its lower phase bound is zero. If these endpoint ranges intersect, a common phase satisfies every fine inequality. If they do not, the last phase must exceed the first. Put $D_j = X_{j+1} - X_j$ and $e_j = u_{j+1} - u_j$. If some $e_j < D_j$, increase the phase once across that edge and keep it constant elsewhere. Its upper gap bound has at least d units of slack, whereas the increase is at most $d - 1$; its lower gap bound remains valid even when $e_j = 1$. If every $e_j = D_j$, both endpoint bounds must be tight and

$$(Q \bmod d) - (R \bmod d) = Q - R - d(u_m - u_1) = c_d - (b - a) \leq 0,$$

contradicting the required increase. Thus every coarse choice lifts. The same endpoint equation covers $m = 1$ directly. Summing (155) and applying coarse binary saturation proves (154). □

Corollary 157 (Exact transport of interior densities)**.** *Let $T = [a_0, a_0 + s) \times [b_0, b_0 + s)$ have zero key weights in its two endpoint fringes, so all four-corner intervals contain the same $m \le c$ positive keys. Assume $b_0 - a_0 - s \ge C$. Then T and its dilated square dT lie strictly above their respective capacity lines, and*

$$D_{\text{fine}}(di, d(j+1) - 1) = D_{\text{coarse}}(i, j) \qquad ((i, j) \in T). \tag{157}$$

Every other density in dT vanishes. Thus core cardinality and density mass on these squares are invariant under dilation. Arbitrary nonnegative gap weights are allowed on both inputs. For zero gap weights, the ratio $K(T)/(s + W(J_T))$ decreases by the factor d.

Proof. All relevant fine intervals retain the same complete positive blocks. Above capacity, cost is an affine terminal penalty plus height times interval weight minus ruler score. The first two terms have zero ordinary density, including arbitrary gap weights. In (154), $m(d-1)$ is constant on the rectangle. The rounded left coordinate changes from i to $i+1$ only when $d \mid i$; the rounded right coordinate changes from j to $j+1$ only when $j \equiv d - 1 \pmod d$. At these intersections the four coarse scores give exactly (157). For zero gaps, local mass and tile side both multiply by d. □

If $d \ge 2$, every transported positive cell lies outside the unit-outcome envelope of Lemma 180: its capacity distance is $d(j - i - c) > 0$, so adding one gives an odd integer greater than one. For the two-key family of Theorem 155, the dilated tile stays maximal because its parent meets capacity. Replicating the two keys creates no additional core cell there, although every positive fine density becomes an off-envelope interaction.

The restrictions in Theorem 156 are necessary. For coarse input $(1, 0, 0)$, $h = 1$, and $d = 4$, the fine interval $[1, 9)$ cuts a positive block and has score one. Its rounded coarse interval $[1, 2)$ has no positive key, so the proposed expression would be zero. For coarse input $(1, 1)$, $h = 1$, $d = 2$, and fine interval $[0, 4)$, the fine score is one but $m(d-1) + S_h(0, 2) = 2$, with $m = 2 > c = 1$.

5.20 Binary rank thresholds require incompatible subsequences

The identity $\nu_2(r) = \sum_{a=1}^{h-1} \mathbf{1}\{\nu_2(r) \ge a\}$ does not permit separate optimization at each rank threshold. Here the input weights are already binary; the thresholds act on rank rewards. This differs from the input-weight threshold-tail obstruction in Lemma 153.

Theorem 158 (An asymptotic factor-two incompatibility)**.** *Let $h \ge 5$, $C = 2^h$, $c = C - 1$, $n = C$, and $m = h - 1$. Put unit weight at one-based positions*

$$X = C/2 + 1, \qquad Y_r = C/2 + 4r + 2 \quad (0 \le r < m),$$

and zero at every other key. Let S be the optimum c-subsequence reward for $\nu_2(1), \dots, \nu_2(c)$, and let T_a be the optimum for binary rank rewards $\mathbf{1}\{\nu_2(r) \ge a\}$. Then

$$S = h - 1, \qquad T_1 = h - 1, \qquad T_a = 1 \ (2 \le a < h), \qquad \sum_{a=1}^{h-1} T_a = 2h - 3.$$

Consequently $(\sum_a T_a)/S = 2 - 1/(h-1)$ tends to two.

Proof. The positions fit strictly below C for $h \ge 5$. Every c-subsequence deletes exactly one key t. If $t < X$, key X moves to selected rank $C/2$, of valuation $h - 1$, and every Y_r moves to an odd rank. The full reward is $h - 1$ and every threshold reward is one. If $t = X$, that key disappears and every Y_r moves to an odd rank, so all rewards are zero. If $t > X$, key X stays odd. Each $Y_r < t$ retains its rank, congruent to two modulo four, contributing exactly one to the full reward

and first threshold. A key $Y_r = t$ is deleted; every $Y_r > t$ moves to an odd rank and contributes zero. Thus the full and first-threshold rewards equal the number of $Y_r < t$, at most m, and all higher threshold rewards vanish. Deleting a key after the last Y_r attains m. These cases exhaust the feasible subsequences and prove every claimed optimum. □

No universal constant less than two can bound this separate-level upper estimate by that constant times the full reward. The theorem does not supply a universal factor-two upper bound. It concerns reward optimization, not an approximation barrier for the full OBST cost or a core-size lower bound. Arbitrary gap weights leave the above-capacity reward optimization unchanged.

A smaller example has $h = 4$, $n = 16$, and unit keys at positions $9, 12, 14$. Its full reward is three, while its threshold optima are $(2, 1, 1)$. Deleting before position nine gives the single positive rank reward three. Deleting positions nine through twelve gives zero; deleting thirteen or fourteen gives reward two from key twelve; deleting fifteen or sixteen gives rewards two and one from keys twelve and fourteen. These cases directly prove the four optima.

5.21 A single internal gap and signed slot densities

In the mixed slot representation, the transition between consecutive selected slots $u < v$ contributes $B_q(u, v) = -q \min(\nu_2(u), \nu_2(v))$ for an internal gap of weight q. This natural transition fails the finite Monge requirement even at unit weight, before any products are taken.

Theorem 159 (Dense signed gap-transition squares)**.** *Put $e(u) = u + (u \bmod 2)$. For $q > 0$ and every cell with $u + 1 < v$ and $v + 1 \le c = 2^h - 1$, the exact density is*

$$D_{B_q}(u, v) = (-1)^{u+v} q \min\{\nu_2(e(u)), \nu_2(e(v))\}. \tag{158}$$

For any power of two $s \ge 2$, choose $h = \log_2 s + 2$, hence $c = 4s - 1$. The square $1 \le u \le s$, $2s + 1 \le v \le 3s$ has exactly s^2 nonzero densities, half positive and half negative. Every corner is feasible.

Proof. Each adjacent pair of slots has one odd valuation-zero member and one even positive-valuation member. Precisely one of the four entries of B_q is nonzero. Its coefficient in the density is negative when u, v have equal parity and positive otherwise; since that entry itself is negative, this gives (158). The minimum is always positive. The stated square has $u + 1 < v$ and $v + 1 \le c$; each row has equally many columns of either parity. □

No Monge or anti-Monge completion preserving these feasible entries can exist, and adding row or column potentials leaves the obstruction unchanged. A representation explicitly listing its nonzero signed second differences requires quadratic size already for $q = 1$. These are consecutive-slot coordinates. The result gives no original interval-core lower bound and no barrier to every representation; the original single-gap interval core has its separate sparse classification. Indeed, Corollary 29 removes equal internal-gap terms by a global tree identity. The general interval-tile hypothesis remains open.

5.22 Arbitrary-key rank transfers and dense local cores

The finite-product representation extends to arbitrary key weights by retaining both the starting and ending selected ranks. This preserves the coupled optimization, but its transfer coordinates can have dense cores. They are different coordinates from the original interval endpoints.

Lemma 160 (Finite transfer through constant-weight runs)**.** *Let $c \ge 1$, and partition n nonnegative key weights into r constant runs (w_ℓ, m_ℓ), with $m_\ell \ge 1$. Put*

$$F(t) = \sum_{j=1}^{t} \nu_2(j), \qquad K = 1 + F(c) \sum_{\ell=1}^{r} w_\ell.$$

For $0 \le u, v \le c$ define the finite matrices

$$Q_\ell(u,v) = K\big[\max(u-v,0) + \max(v-u-m_\ell,0)\big] + w_\ell F(u) - w_\ell F(v). \tag{159}$$

Their product $T = Q_1 \otimes \cdots \otimes Q_r$ equals the negative maximum reward of a $(v-u)$-key subsequence assigned consecutive ruler ranks $u+1, \dots, v$, whenever $0 \le v-u \le n$. It is Monge, has core size at most $2r2^{\lceil \log_2 r \rceil} c \le 4r^2 c$, and has a condensed construction using $O(n + r^2 c \log(c+1))$ arithmetic work and $O(r^2 c \log(c+1))$ auxiliary space.

Proof. The row and column potentials in (159) do not change densities. Each factor has density K on $v-u=0$ and $v-u=m_\ell$, when the latter is in range, and no other density. Thus its core has at most $2c$ entries and is directly available with its boundaries in $O(c)$ work. A feasible path chooses $t_\ell \in [0, m_\ell]$ keys from each run and advances its rank by t_ℓ. Its cost is the negative sum of precisely those consecutive ruler rewards. Every subsequence has such a path, and any counts define a subsequence because keys in a run have equal weight. Every infeasible path incurs at least K penalty. Even allowing arbitrary rank decreases, the negative part of its potentials has magnitude at most $F(c)\sum_\ell w_\ell$. Its cost is therefore at least one, whereas any feasible path has cost at most zero. The finite completion consequently preserves every feasible optimum.

Multiply in a balanced tree. The published core-product bound [14] charges each input core by at most $2^{\lceil \log_2 r \rceil}$, proving the stated final bound. The sum of core sizes throughout the tree is $O(r^2 c)$. Each matrix has dimension $c+1$, so the logarithm in the published multiplication bound is $O(\log(c+1))$. Extracting the runs costs $O(n)$. The implementation retains ordinary witness arrays while multiplying; the stated auxiliary-space bound includes their possible logarithmic factor. □

For $c = 2^h - 1$ and $j - i \ge c$, this yields the original layer identity

$$A_h(i,j) = M(j-i-c) + hW(i,j) + T_{p[i:j]}(0,c). \tag{160}$$

All key weights may be arbitrary nonnegative reals, and gap weights enter through $W(i,j)$. The identity preserves their full above-capacity scores; it does not by itself construct all interval transfers efficiently.

Theorem 161 (Dense feasible rank-transfer squares)**.** *For r keys with weights $p_j = j$, $1 \le j \le r$, every completion agreeing with the exact feasible transfers has*

$$D_T(u,v) = \nu_2(u+1) \quad \text{when } 0 \le u, v < c, \quad 1 \le v-u \le r-1.$$

If $r = 2s$ and $c \ge 2s$, the square $0 \le u < s$, $s \le v < 2s$ has exactly $s\lfloor s/2 \rfloor$ positive entries. If $c = 2r-1$ and r is even, the whole strict feasible strip has exactly $3r^2/4 - 3r/2 + 1$ positive entries.

Proof. For $t = v-u$, the kth selected input index is at most $r - t + k$. Since ruler rewards are nonnegative, selecting the last t keys maximizes every summand simultaneously. With $G(t) = \sum_{j=1}^{t} j\nu_2(j)$, the exact transfer is

$$T(u,v) = -\big[(r-v)(F(v)-F(u)) + G(v) - G(u)\big]. \tag{161}$$

Its only term involving both variables is $(r-v)F(u)$. Four-corner differencing gives $F(u+1)-F(u) = \nu_2(u+1)$. The stated strict strip is precisely where these four corners are feasible. In the square, every odd row has s positive entries and every even row has none. For the full strip the count is

$$\sum_{\substack{0\le u\le c-2\\ u \text{ odd}}} \min(r-1, c-1-u) = \frac{r}{2}(r-1) + \left(\frac{r}{2}-1\right)^2 = \frac{3r^2}{4} - \frac{3r}{2} + 1.$$

All counted densities use only feasible entries, so the choice of completion cannot remove them. □

This rules out a uniform $O(s)$ local bound or an $O(c+r)$ total bound for these rank transfers. The family has integer weight $W = r(r+1)/2$, so its density count is consistent with a mass-dependent bound. It does not contradict the original interval-tile hypothesis (191).

Lemma 162 (Balanced chains of finite bands)**.** *For $N \ge 1$ and $r \ge 2$ real functions f_ℓ on $\{0,\dots,N-1\}$ and integers $g_\ell \ge 0$, let*

$$H(A,B) = \max\left\{\sum_{\ell=1}^{r} f_\ell(u_\ell) : u_1 \le A,\ u_r \ge B,\ 0 \le u_{\ell+1} - u_\ell \le g_\ell\right\}.$$

Here all states and endpoints lie in $\{0,\dots,N-1\}$. On the feasible domain $B \le A + \sum_\ell g_\ell$, $-H$ has a finite Monge completion with at most $2r2^{\lceil \log_2(r+1)\rceil}(N-1)$ core entries, constructible in $O(r^2 N \log(N+1))$ work.

Proof. Set $K = 1 + \sum_\ell(\max f_\ell - \min f_\ell)$. Use the two selectors of Lemma 46 and $r-1$ band factors. Charge $-f_\ell(u_\ell)$ in its outgoing band, and charge $-f_r(u_r)$ in the final band. Every unary term occurs exactly once. The sum of input core sizes is at most $2r(N-1)$. Every invalid path pays at least K, making it more expensive than any feasible optimum by the same range argument as in Lemma 46. A balanced product tree has height $\lceil \log_2(r+1)\rceil$; the core bound and construction follow as above. □

For unit-weight positive blocks of lengths $m_1,\dots,m_r$ with total $P \le c$, put $N = c - P + 1$, $P_\ell = \sum_{t<\ell} m_t$, and $f_\ell(u) = F(P_\ell + u + m_\ell) - F(P_\ell + u)$. This represents the complete-block interior, with $A = z - i$ and $B = c + z + G - j$ when the first block begins at z and the total intervening zero length is G. Clipping frozen endpoints and mapping $i = z - a - 1$, $j = c + z + G - b - 1$ constructs that part of the original core. This extension does not include endpoints cutting a positive block.

Theorem 163 (Width-one chains with dense output squares)**.** *There are chains of $r = 2s+1$ functions on $N = 2s+1$ states, with every band width equal to one, whose output has density one in every cell of an $s \times s$ square.*

Proof. Take $f_1(u) = \cdots = f_{r-1}(u) = u$ and $f_r(u) = -(r-1)u$. For $e_\ell = u_{\ell+1} - u_\ell \in \{0,1\}$ the reward telescopes to $-\sum_{\ell=1}^{r-1} \ell e_\ell$. If $B > A$, at least $d = B - A$ increments are necessary. Choosing the first d increments, starting at A, attains the optimum $-d(d+1)/2$. If $B \le A$, a constant path gives zero. Thus, with $d = \max(B-A, 0)$,

$$-H(A,B) = d(d+1)/2.$$

Every output is feasible because $N = r$. Its four-corner difference is one throughout $0 \le A < s$, $s \le B < 2s$. □

The functions in this last theorem are arbitrary unary rewards, not the special ruler block rewards. Therefore a boundary charge uniform in the number of binary blocks must use further ruler structure; narrow bands and finite Monge factors alone do not establish it.

5.23 The window-envelope correction has both density signs

The envelope in Corollary 50 has linear core size per layer, but its remaining correction does not inherit a Monge sign. Let $E_h = S_h - \widehat{S}_h = \widehat{A}_h - A_h \geq 0$. At $h = 3$, $c = 7$, consider the binary key words

$$p^+ = (0,0,0,1,0,0,0,1,0), \qquad p^- = (0,0,0,1,0,0,1,0,0,0).$$

Use cell $(0,8)$ for p^+ and $(0,9)$ for p^-. In corner order $(i,j),(i,j+1),(i+1,j),(i+1,j+1)$, their exact scores are

	S_h	$\widehat{S}_h$	E_h	D_{E_h}
p^+	$(2,3,0,2)$	$(2,2,0,2)$	$(0,1,0,0)$	1
p^-	$(3,3,1,3)$	$(2,2,1,2)$	$(1,1,0,1)$	-1

To verify the table directly, the reward word is $(0,1,0,2,0,1,0)$, so two unit keys score at most three. For p^+ the full interval attains three at selected ranks $(4,6)$; its seven-key middle interval forces $(3,7)$ and scores zero. Its two eight-key intervals attain two, and neither can give a reward-two rank to one key while also rewarding the other. For p^-, the two left intervals attain three at $(4,6)$, and the right nine-key interval attains three at $(2,4)$. In its right eight-key interval, deleting one key leaves positive ranks only in $(2,5)$, $(3,5)$, $(3,6)$, or deletes a positive key, so the maximum is one. Direct contiguous-window evaluation gives $\widehat{S}_h$.

Concatenating the two words gives a single binary input with both signs, at cells $(0,8)$ and $(9,18)$, by interval locality. Arbitrary gap weights cancel from the correction. Thus the remaining correction can be neither Monge nor anti-Monge, despite its nonnegative values. A general construction must control these signed interactions rather than apply the published Monge product to the correction without an additional argument.

The same p^+ also shows that Lemma 53 cannot extend to arbitrary inputs. At its cell $(0,8)$, the displayed ruler scores give original density one, whereas all four largest-seven sums equal two. Thus $\Gamma_7(0,8) = 0$: the auxiliary support misses an actual two-peak interaction at integer key mass two. The score proof above verifies this failure directly.

5.24 A single valley has a quadratic original interval core

Two monotone pieces do not by themselves guarantee a sparse above-capacity core. Unlike the rank-transfer example, the following family uses the original interval coordinates.

Theorem 164 (Dense valley square)**.** *Let $s = 2^t \geq 4$, $n = 4s$, $m = 2s$, and $c = m - 1 = 2^h - 1$. Set*

$$p_k = \begin{cases} 2(m-k), & 0 \leq k < m, \\ 2(k-m)+1, & m \leq k < 2m. \end{cases} \tag{162}$$

With arbitrary nonnegative gap weights and a valid terminal penalty, the above-capacity square $0 \leq i < s$, $3s \leq j < 4s$ satisfies

$$D_{A_h}(i,j) = \nu_2\left(3s - \left\lceil \frac{i+j}{2} \right\rceil\right). \tag{163}$$

It has exactly $s^2/2 = n^2/32$ positive densities.

Proof. The key weights are $1,\dots,n$ arranged as decreasing evens followed by increasing odds. In the inner interval $[i+1,j)$ write

$$d = j - i - m, \qquad a = m - i - 1 - \lfloor d/2 \rfloor, \qquad b = c - a.$$

Its largest c keys omit the d globally smallest values, occupying $[m-\lfloor d/2\rfloor, m+\lceil d/2\rceil)$. The chosen square places this interval and the next two smallest weights strictly inside the relevant endpoints. The two smallest retained weights are $z = d+1$ and $w = d+2$. There is at least one selected key on each wing even after both are removed. Lemma 51 applies at every corner. Let L_t, R_t be ruler-prefix scores of the inner interval's left and reversed right wings. Write L'_t, R'_t for the same prefixes after adjoining the new left or right endpoint, respectively. The four scores are

corner	d even	d odd
$[i+1, j)$	$L_a + R_b$	$L_a + R_b$
$[i, j)$	$L'_{a+1} + R_{b-1}$	$L'_a + R_b$
$[i+1, j+1)$	$L_a + R'_b$	$L_{a-1} + R'_{b+1}$
$[i, j+1)$	$L'_a + R'_b$	$L'_a + R'_b$.

Ruler symmetry justifies the reversed prefixes, as in (66). For even d, z is the first retained right key and w the last retained left key. The score difference $S_h(i,j)+S_h(i+1,j+1)-S_h(i,j+1)-S_h(i+1,j)$ is exactly

$$(L'_{a+1} - L'_a) + (R_{b-1} - R_b) = w\nu_2(a+1) - z\nu_2(b) = \nu_2(a+1),$$

because $b = m-(a+1)$ and $w - z = 1$. For odd d, z is the last retained left key and w the first retained right key, giving

$$(L_{a-1} - L_a) + (R'_{b+1} - R'_b) = w\nu_2(b+1) - z\nu_2(a) = \nu_2(a),$$

because $b+1 = m-a$. Substituting a gives (163). The affine terms of (66), including all gap contributions, cancel from the original density.

The argument of ν_2 lies between $s/2+1$ and $3s/2$. It is even exactly when $i+j$ is 0 or 3 modulo four. Each row has s columns, of which exactly $s/2$ meet these residues, proving the count. □

The dense square itself has a short density representation:

$$D_{A_h}(i,j) = \sum_{\ell=1}^{\log_2 s} \mathbf{1}\{i+j \equiv 6s \text{ or } 6s-1 \pmod{2^{\ell+1}}\}. \tag{164}$$

This follows by expressing $\nu_2(r)$ as the number of powers of two dividing r. It describes the density by logarithmically many modular diagonal patterns; it does not assert a fast product algorithm for them.

With zero gap weights, the integer family has $W = n(n+1)/2$. Dividing every key weight by n gives positive rational weights of total $W = (n+1)/2$, with $O(\log n)$-bit numerators and denominators. Every density scales by $1/n$, so the same quadratic support remains. Consequently the integral-mass core bound cannot be extended to normalized real mass alone. Theorem 52 still supplies logarithmic exact interval queries for both versions. This separates explicit-core size from query complexity and gives no lower bound for computing a full optimal BST.

5.25 Exponential prefix records at necessary optimal height

The prefix-record construction has an exact limitation on arbitrary magnitudes. Its record support can be exponential even when every original key and gap weight is positive and the prescribed height is the smallest height of any optimum. Write $\mathrm{pc}(u)$ for the number of one bits in the binary expansion of u.

Lemma 165 (Universal binary envelope)**.** *For arbitrary nonnegative mixed weights, a record of the k-key prefix at height h must end at a slot b with* $\mathrm{pc}(b) \le k$*. Consequently, for* $1 \le k \le 2^h - 1$*,*

$$B_k \le E_h(k) := \sum_{r=0}^{\min(k,h)} \binom{h}{r} - k. \tag{165}$$

Proof. Given any selected slot sequence, replace each slot successively by the first position after its replaced predecessor having the same valuation. Every replaced position is at most its original position, by induction, and the whole valuation vector is unchanged. Hence all key rewards and all adjacent-gap rewards are unchanged.

One replacement increases popcount by at most one. To see this, write the previous position as $a = Q2^{d+1} + r$, $0 \le r < 2^{d+1}$, where d is the next required valuation. If $r < 2^d$, the next position is $Q2^{d+1} + 2^d$, with popcount at most $\mathrm{pc}(a) + 1$. Otherwise it is $(Q+1)2^{d+1} + 2^d$. In that case $\mathrm{pc}(Q+1) \le \mathrm{pc}(Q) + 1$ and the old remainder already had its dth bit set, giving the same bound. Starting at zero, the earliest ending position after k replacements has at most k one bits.

If an optimal prefix ended at b with $\mathrm{pc}(b) > k$, this replacement would end strictly earlier in the same valuation class with the same reward. Lemma 34's monotonicity then gives equal prefix values, so b cannot be a record. Count all positive h-bit integers with at most k one bits and remove the $k-1$ positions below k; each of those also satisfies the popcount condition. This gives (165). □

Lemma 166 (Lexicographic ruler endpoints)**.** *Let $m \ge 1$, and suppose positive weights satisfy $w_i > (h-1)\sum_{j>i} w_j$. For $m \le U < 2^h$, let*

$$G_m(U) = \max_{1 \le t_1 < \cdots < t_m \le U} \sum_{i=1}^{m} w_{i-1}\nu_2(t_i).$$

The maximizing tuple is unique. Its last slot is U exactly when $\mathrm{pc}(U) \le m$*. Consequently, for $U > m$,*

$$G_m(U) > G_m(U-1) \quad \Longleftrightarrow \quad \mathrm{pc}(U) \le m.$$

The first feasible window $U = m$ is an initial record.

Proof. All valuations lie in $\{0, \ldots, h-1\}$. The first differing coordinate therefore determines the weighted comparison, so optimization is lexicographic in key order. With r slots still required after the previous slot a, choose the largest valuation in $[a+1, U-r+1]$. That maximum is unique: two positions of equal maximal valuation would enclose a position of larger valuation. Repeating this feasible greedy choice gives the unique maximizing tuple.

Write $U = P + s$, where P is its largest power of two and $0 \le s < P$. If $m \le s+1$, the first greedy choice is P. For $m = 1$, it ends at U exactly when $s = 0$, equivalently $\mathrm{pc}(U) = 1$. For $m > 1$, the suffix is the same problem on $[1, s]$ with $m-1$ slots, since $\nu_2(P+x) = \nu_2(x)$ for $1 \le x \le s$. Induction and $\mathrm{pc}(U) = 1 + \mathrm{pc}(s)$ prove the claim. If $m > s+1$, every choice while $r > s+1$ lies below P, because its feasibility upper endpoint $U - r + 1$ is below P. When $r = s+1$, the slot P becomes feasible and is selected. All following slots through U are then required. Here $\mathrm{pc}(U) \le s + 1 < m$, proving the claim in this case as well.

If the unique optimum ends at U, no equally good tuple was feasible at $U-1$, so the value rises strictly. Otherwise the same optimum was already feasible there. The initial window contains only one tuple. □

Theorem 167 (Exact exponential record support)**.** *For $n \geq 2$, put $A = n+1$ and set*

$$p_i = A^{n-i} - 1 \quad (0 \leq i < n), \qquad q_j = 1 \quad (0 \leq j \leq n). \tag{166}$$

All weights are positive integers. The unique optimal tree is the right chain, so its necessary height is $h = n$. At this height, the records of its normalized k-key prefix are exactly

$$\{b : k \leq b \leq 2^n - 1,\ \mathrm{pc}(b) \leq k\}. \tag{167}$$

Their exact count and total across all prefixes are

$$B_k = \sum_{r=0}^{k} \binom{n}{r} - k, \qquad B_n = 2^n - n, \qquad \sum_{k=1}^{n} B_k = (n+2)2^{n-1} - 1 - \frac{n(n+1)}{2}. \tag{168}$$

Proof. Common-gap normalization gives $w_i = A^{n-i}$, zero residual gaps, and the additive cost n for every tree. For every suffix,

$$n \sum_{j>i} w_j = w_i - A < w_i.$$

A right chain on that suffix costs less than $2w_i$. Any tree with its first key below the root costs at least $2w_i$ from that key alone. The first key must therefore be the root; induction forces the unique right chain. Its original cost is $n + \sum_i (i+1)w_i$, and its height is n.

The same weights satisfy the strict dominance assumption of Lemma 166. For $m = k-1$, a prefix forced to end at b has negative reward

$$f_k(b) = -G_m(b-1) - w_m \nu_2(b),$$

where G_0 is constantly zero. Fix valuation d and write $b = (2j+1)2^d$. Between consecutive feasible slots of this valuation, the argument of G_m passes through

$$[b - 2^{d+1},\ b - 1].$$

The minimum popcount in this interval is $\mathrm{pc}(j)$. Indeed, its right half contains $j2^{d+1}$ and has that minimum; its left half has minimum $\mathrm{pc}(j-1)+1$, which is at least $\mathrm{pc}(j)$. Thus f_k decreases strictly at b exactly when $\mathrm{pc}(j) \leq m$. As $\mathrm{pc}(b) = \mathrm{pc}(j) + 1$, this is precisely $\mathrm{pc}(b) \leq k$.

The first feasible slot of each class obeys the same test. Its index is $j = \max\{0, \lceil (k-2^d)/2^{d+1} \rceil\} \leq k-1$, so $\mathrm{pc}(b) = 1 + \mathrm{pc}(j) \leq k$. For $k = 1$, only the first slot of each valuation class is retained, consistent with $G_0 = 0$. This proves (167). There are $\sum_{r=0}^{k} \binom{n}{r} - 1$ positive n-bit integers with at most k one bits. Every positive integer below k satisfies that test; removing those $k-1$ positions proves B_k. Summing the binomial coefficients over k gives the last identity in (168). □

The support calculation requires only lexicographic dominance. With any prescribed height h, base $A = h+1$, and at most $2^h - 1$ keys, the same argument attains $E_h(k)$ at every prefix. Thus Lemma 165 is sharp.

The largest input integer has $O(n \log(n+1))$ bits. Thus a polynomial-size input forces exponential explicit prefix lists, even at the minimum optimal height. Retaining every stage's predecessors requires $\Theta(n2^n)$ records on this family. Its support nevertheless has the succinct binary description (167). Moreover, the original mixed weights satisfy period-one affine translation with $\lambda = 1/A$ and $\alpha = 1/A - 1$. Theorem 60 therefore constructs their full optimum using exactly $2n-1$ root candidates. This is a barrier to enumerating prefix records; the general OBST problem and implicit representations remain separate. The integer-mass and fixed-exception guarantees of Section 2.19 are unaffected. Independent controls appear in Appendix T.

5.26 Monotone functions can require exponential diagram minima

The prefix operation of Lemma 38 has a bound in the input diagram size. General pointwise minimum has no such polynomial bound, even for strictly decreasing inputs in the same address order.

Theorem 168 (Exponential minimum of two monotone diagrams)**.** *Let $m \geq 1$, $N = 2^m$, and write a $2m$-bit address as $b = NX + Y$, where $0 \leq X, Y < N$. The two strictly decreasing functions*

$$F(b) = -2b + X, \qquad G(b) = -2b + Y$$

each have an additive decision diagram with $2m+1$ nodes, including its terminal, in most-significant-bit-first address order. Their pointwise minimum has exactly $3 \cdot 2^m - 2$ nodes in that same order, including 2^m distinct normalized nodes at level m.

Proof. Both functions are weighted sums of individual address bits. A single shared node per bit, with the corresponding coefficient on one edge, represents each function; the terminal supplies the extra node. Increasing b by one changes F by -2 or -1. It changes G by -1 within a low-bit block and by $-(N+1)$ across its boundary. Thus both are strictly decreasing.

Their minimum is

$$H(b) = -2b + \min(X, Y).$$

Fixing the high m bits to $X = x$ gives a low-bit residual function $-2Y + \min(x, Y)$, up to the constant $-2Nx$. Its minimum is at $Y = N - 1$. After subtracting that minimum the normalized residual is

$$H_x(Y) = 2(N-1) - 2Y + \min(x, Y) - x.$$

For $0 \leq Y < N - 1$ its consecutive difference is -1 when $Y < x$ and -2 when $Y \geq x$. These N functions are pairwise distinct, including $x = 0$ and $x = N - 1$. No two can share the same normalized node. Every high-bit assignment must therefore reach a different level-m node.

For the exact total, consider a remaining low-bit level $d \leq m$. After fixing the preceding bits, the residual has the form $-2t + \min(x - z, t)$ up to a constant, where z is the fixed low-bit block's starting address. Normalization depends only on the clipped threshold $\max\{0, \min\{x - z, 2^d - 1\}\}$. There are exactly 2^d distinct thresholds, and all occur by taking $z = 0$ and x in that range. Thus levels $1, \ldots, m$ contain $2^{m+1} - 2$ nodes in total. At level $m + r$, $1 \leq r \leq m$, there are 2^{m-r} high-bit prefixes. They remain distinct up to an additive constant: setting the remaining r bits of X to zero gives different thresholds $X = a2^r$ in the low-bit function. Hence those upper levels contain $2^m - 1$ nodes. Adding the single terminal gives $3 \cdot 2^m - 2$. □

The theorem concerns the fixed address order used by the prefix algorithm. It shows why a polynomial bound for a generic minimum operation cannot follow from small operand diagrams and monotonicity alone. The displayed functions have not been realized as mixed-OBST prefix transitions, so this is a barrier to that generic complexity argument. Bounding the specific functions generated by (45) remains a separate structural question. Independent exact checks appear in Appendix T.

5.27 Sharp reset-path work on general diagram inputs

The reset-path lemma bounds the mechanical recursion even outside the OBST domain. The following family makes both its state bound and the persistent-replacement work sharp. Its failure of endpoint dominance is explicit, so the result does not assert tightness on actual OBST prefixes.

Theorem 169 (A compact input with sharp carry and replacement counts)**.** *Let $m \geq 1$, $N = 2^m$, $h = 2m$, and $A(X) = 2X - \mathrm{popcount}(X)$. On $0 \leq X, Y < N$ define*

$$F(NX + Y) = -\frac{A(X)}{m+1} - Y.$$

Its additive diagram has exactly $D = 2m + 1$ nodes, including the terminal. Run the mechanical recursion in Theorem 43, initialized with carry $F(0)$, using key and gap parameters $w = g = 1$. The number of expanded states is

$$Z = \frac{3m(m+1)}{2} = \Theta(hD).$$

With the stated persistent first-address replacement and shared templates, the number of node-assembly calls, and separately the number of replacement visits including leaves, is

$$M = \frac{m(7m^2 + 15m + 2)}{6} = \Theta(h^2 D). \tag{169}$$

This input cannot be a mixed OBST prefix.

Proof. The identity $A(X) = \sum_{j=0}^{m-1}(2^{j+1} - 1)x_j$ shows that F is a weighted sum of individual bits. Every residual of the same length differs only by an additive constant, so there is exactly one normalized node at each level and one terminal. For $X > 0$, setting $s = \nu_2(X)$ gives

$$A(X) - A(X-1) = s + 1, \qquad \delta_s = \frac{s+1}{m+1} \in (0, 1).$$

Within each X block the values decrease by one, and its last value is δ_s below the preceding block's last value. Thus the incoming inclusive carry at the start of block $X > 0$ is that preceding minimum. Relative to the current block minimum, it is exactly δ_s. Every position before $Y = N - 1$ has larger value than that carry. Only the rightmost path expands inside this block, with the same relative carry δ_s at each remaining low-bit depth $d = 1, \ldots, m$. Each $s = 0, \ldots, m - 1$ occurs, by taking $X = 2^s$.

In the first block $X = 0$, every depth-d subblock starts with its own first value as inclusive carry, giving relative carry $2^d - 1$. This is distinct from every $\delta_s < 1$. All low-bit contexts have therefore been enumerated: at each depth $1 \leq d \leq m$ there are exactly $m + 1$ states.

At an upper depth $m + r$, $1 \leq r \leq m$, a block starts at an X divisible by 2^r. For $X > 0$, its valuation s ranges over $r, \ldots, m - 1$. The block minimum is at its last address, and its relative carry is

$$\frac{A(2^r - 1) + s + 1}{m+1}, \qquad A(2^r - 1) = 2^{r+1} - r - 2.$$

These $m - r$ values are distinct and all occur. The block starting at zero has relative carry $A(2^r - 1)/(m + 1) + N - 1$, a further distinct value. All these carries are positive, so every such context expands. There are exactly $m - r + 1$ contexts at depth $m + r$. Summing the low and upper levels gives

$$Z = m(m+1) + \sum_{r=1}^{m}(m - r + 1) = \frac{3m(m+1)}{2}.$$

The expanded-state depth sum is

$$S = (m+1)\sum_{d=1}^{m} d + \sum_{r=1}^{m}(m + r)(m - r + 1) = \frac{m(m+1)(7m+5)}{6}.$$

Every upper block expands. Inside each later X block the left child of each rightmost-path state is pruned, with largest depth $m-1$. Thus templates of precisely depths $1,\dots,m-1$ are assembled once; for $m=1$ this set is empty. Their assembly work and replacement visits both sum to $m(m-1)/2$. Adding this to S proves (169), counting all assembly calls even when interning reuses an existing node.

Finally,

$$F(N) = -\frac{1}{m+1} > -(N-1) = F(N-1), \qquad \nu_2(N) = m > \nu_2(N-1) = 0.$$

This contradicts the endpoint-dominance condition of Lemma 40. Hence no actual mixed prefix has this function. The construction proves sharpness of the general recursion's accounting; it supplies no lower bound for the full OBST problem. □

Corollary 170 (Sharp punctured-diagram update count)**.** *Puncturing the input in Theorem 169 gives exactly $D = 2m$ nodes at height $h = 2m$. The recursion of Theorem 45 expands $Z = 3m(m+1)/2$ states and makes exactly*

$$Z + m - 1 = \frac{3m(m+1)}{2} + m - 1 = \Theta(hD)$$

node-assembly calls. Thus its general-diagram work bound is sharp.

Proof. At each positive depth the input remains a weighted sum of residual bits, restricted to nonzero addresses, so there is exactly one normalized node per level and no terminal. Every nontrivial input block has its minimum at its last address; omitting its first address preserves that minimum. The omitted value is at least the inclusive carry, so removing it from the right-carry formula changes no carry. Consequently the expanded relative carries and every level count are precisely those computed in Theorem 169. The same depths $1,\dots,m-1$ need pruning templates, now costing one assembly call each. Every expanded state also costs one assembly call, proving the formula. The earlier endpoint-dominance violation still excludes this family from OBST prefixes. □

5.28 Cubic actual-prefix diagrams and quartic shared history

The generic operation family above violates endpoint dominance. The next family consists of actual mixed OBST prefixes and gives a size obstruction for the canonical representation itself. Its optimum has necessary height n, and all original outcome weights are strictly positive.

Theorem 171 (Actual-prefix diagram size)**.** *For $n \ge 2$, put $A = 4n+1$, $w_i = A^{n-i}$, and*

$$p_i = w_i - 1 \quad (0 \le i < n), \qquad q_j = 1 \quad (0 \le j \le n).$$

At the necessary optimal height $h = n$, let D_k count the canonical punctured nodes of the k-key prefix, and let $\mathcal{H}$ count distinct nodes in the union of all prefix diagrams, allowing full sharing across stages. Define

$$Q = 2\lceil n/2 \rceil, \qquad d_0 = \lceil \log_2(Q+1) \rceil + 2, \qquad N = \max\{0, n - d_0\}.$$

Then

$$D_k \geq L_k := \sum_{t=1}^{N} (N - t + 1) \min\{t, (k-3)_+\}, \tag{170}$$

$$D_n \geq \binom{N+2}{3} = \Omega(n^3), \tag{171}$$

$$\mathcal{H} \geq (n-2)\binom{N+2}{3} - \binom{N+3}{4} = \Omega(n^4). \tag{172}$$

The unique optimum is the right chain. Its cost is $n + \sum_{i=0}^{n-1}(i+1)w_i$*, and the complete tree has a linear-work construction and certificate.*

Proof. Common-gap normalization gives w, zero residual gaps, and additive cost n. The inequality $n\sum_{j>i} w_j < w_i$ forces the first key to be the root in every suffix, by the dominance argument in Theorem 167. This proves the chain, cost, and necessary height assertions. It also gives lexicographic optimization of slot valuations, as in Lemma 166.

We first record why scalar differences preserve valuation information. Two differences of valuation vectors have coordinates in $[-(n-1), n-1]$. If they disagree first at coordinate i, their weighted difference has leading magnitude at least w_i, whereas its remaining magnitude is at most

$$2(n-1)\sum_{j>i} w_j < \frac{2(n-1)}{A-1} w_i < w_i.$$

Thus equality of their exact scalar scores implies equality of every coefficient. Padding shorter vectors with zeros preserves this statement.

For each quadruple (d, s, r, k) with

$$d_0 \leq d \leq s < n, \qquad 1 \leq r \leq \min\{n-s, k-3\}, \tag{173}$$

choose a block boundary B with exactly r one bits, its lowest at position s. One choice uses the top $r-1$ bit positions and bit s. This is a multiple of 2^d, and its whole punctured block $B+t$, $1 \leq t < 2^d$, lies within the height-n slots. Use three local addresses

$$t_* = 1, \qquad t_- = 2^{d-2} + Q, \qquad t_+ = 2^{d-1} + Q. \tag{174}$$

Since $2^{d-2} > Q \geq n$, all are interior and all corresponding prefix entries are finite. Let $v(t)$ be the valuation vector of the unique optimal k-slot sequence forced to end at $B+t$. Its first $k-1$ slots are obtained by successive feasible greedy choices, and $f_k(B+t) = -\sum_i w_i v_i(t)$.

At t_- and t_+, the first r choices are the successive partial sums of the high bits of B. Each is the unique highest-valuation feasible position. Feasibility holds because at least n positions remain after B, and still more remain after each earlier partial sum. There remain $m = k-1-r \geq 2$ predecessor slots after B. Their first feasible upper endpoint, relative to B, is $t - m$. As $0 < Q - m < 2^{d-2}$, their next greedy choice has valuation $d-2$ at t_- and $d-1$ at t_+. Therefore

$$\min\{i : v_i(t_+) \neq v_i(t_-)\} = r, \qquad v_r(t_+) - v_r(t_-) = 1. \tag{175}$$

At $t_* = 1$, the first $r-1$ partial sums remain feasible and are still greedy maxima. Put $P = B - 2^s$. There are $j = k - r \geq 3$ predecessor slots left in $(P, B]$. The first two greedy choices are

$$P + 2^{s-1}, \qquad P + 3 \cdot 2^{s-2},$$

with valuations $s-1$ and $s-2$. For the first choice B is infeasible because $j>1$, while the midpoint is feasible because $2^{s-1}\geq j-1$. For the second, B is still infeasible because $j>2$, while the quarter point is feasible because $2^{s-2}\geq j-2$. These margins follow from $2^{s-2}\geq 2^{d-2}>Q\geq n$. Hence

$$v_r(t_*)-v_r(t_-)=(s-2)-(d-2)=s-d. \tag{176}$$

The forced ending key has valuation zero at t_* and $\nu_2(Q)\geq 1$ at t_-. The latter follows from $0<Q<2^{d-2}$. After zero padding, this gives

$$\max\{i : v_i(t_*)\neq v_i(t_-)\}=k-1. \tag{177}$$

If two same-depth blocks shared a normalized punctured node, their scalar value differences at the three addresses would agree. Coefficient uniqueness would then give the same vector differences. Equation (177) recovers k, (175) recovers r, and (176) recovers s. Thus every quadruple in (173) gives a different canonical node, even across prefix stages. Different depths are distinct by the definition of the representation.

For fixed s, write $t=n-s$. There are $N-t+1$ eligible depths and $\min\{t,(k-3)_+\}$ choices of r, proving (170). At $k=n$, summing gives $\binom{N+2}{3}$. For each r, eligible stages are $k=r+3,\dots,n$, numbering $n-r-2$. Counting their distinct shared nodes gives

$$\sum_{t=1}^{N}(N-t+1)\sum_{r=1}^{t}(n-r-2)=(n-2)\binom{N+2}{3}-\binom{N+3}{4}.$$

Finally $d_0=O(\log n)$, so the two formulas have leading terms $n^3/6$ and $n^4/8$, respectively. This proves the lower bounds. □

Theorem 45's full work bound remains valid, but universally small prefix diagrams or shared history cannot make this representation subquadratic. The witnesses give lower bounds, not exact node counts or matching upper bounds. The final-stage obstruction survives discarding earlier roots; the history obstruction already accounts for cross-stage sharing. The family's directly certified chain solution separates this representation barrier from the complexity of full OBST.

5.29 Positive affine normalization does not remove the size barrier

A more permissive representation could share blocks that differ by a positive scale as well as an additive shift. The next result gives the same size obstruction under that equivalence, using finite integer weights with polynomial binary encoding.

Lemma 172 (Separating quadratic scores)**.** *Let $n\geq 2$, $A=16n^2+1$, $e_t=t+2nt^2$ for $1\leq t\leq n$, and $w_i=A^{e_{n-i}}$ for $0\leq i<n$. For integer vectors u,v,u',v' whose coordinates have magnitude at most $n-1$, equality*

$$(u\cdot w)(v'\cdot w)=(u'\cdot w)(v\cdot w)$$

implies the polynomial identity $(u\cdot X)(v'\cdot X)=(u'\cdot X)(v\cdot X)$. If u,v are linearly independent and $u'\neq 0$, this forces $u'=\lambda u$ and $v'=\lambda v$ for some nonzero rational λ.

Proof. All unordered sums e_a+e_b are different. Equality with e_c+e_d implies

$$a+b-c-d=-2n(a^2+b^2-c^2-d^2).$$

The left side has absolute value at most $2n-2$, so both sides vanish. Equality of sums and sums of squares gives $ab=cd$, hence equality of the unordered pairs.

Every coefficient in the difference of the two quadratic polynomials has magnitude at most $4(n-1)^2$. Substitution of the w_i sends distinct monomials to distinct powers of A. If the substituted polynomial were nonzero, its highest nonzero term would dominate all lower terms, since $4(n-1)^2 < A-1$. Thus scalar equality implies polynomial identity. For the last assertion, the independent linear forms $U = u \cdot X$ and $V = v \cdot X$ are coprime over the rationals. From $UV' = U'V$, U divides U'. Both are nonzero linear forms, so $U' = \lambda U$; cancellation gives $V' = \lambda V$. □

Theorem 173 (Size under affine block equivalence)**.** *Use the weights of Lemma 172, and put $p_i = w_i - 1$ and $q_j = 1$. Let D_k^{aff} count equivalence classes of nonnull, depth-tagged dyadic block interiors in the k-key prefix, where functions may share whenever $f = \alpha g + \beta$ for $\alpha > 0$. Let $\mathcal{H}^{\mathrm{aff}}$ count their union across all prefixes, allowing the same equivalence across stages. With N, Q, d_0, L_k as in Theorem 171,*

$$D_k^{\mathrm{aff}} \geq L_k, \qquad D_n^{\mathrm{aff}} \geq \binom{N+2}{3} = \Omega(n^3), \tag{178}$$

$$\mathcal{H}^{\mathrm{aff}} \geq (n-2)\binom{N+2}{3} - \binom{N+3}{4} = \Omega(n^4). \tag{179}$$

All original weights are positive. The unique optimum is the right chain and its necessary height is n. The largest input weight has $O(n^3 \log(n+1))$ bits, and the complete input has $O(n^4 \log(n+1))$ bits.

Proof. Consecutive normalized weights differ by a factor at least A. Thus $n \sum_{j>i} w_j < w_i$, proving the chain and necessary height by suffix dominance. The same inequality gives lexicographic optimization of valuations. The exponent and bit-length bounds follow directly from $e_n = 2n^3 + n$ and $\log A = O(\log(n+1))$.

Use the witnesses (d, s, r, k) and three interior addresses from (173)–(174). All their feasibility and valuation identities depend only on lexicographic dominance, so they hold for these weights. Zero-pad the valuation vectors to n coordinates and write

$$u = v(t_+) - v(t_-), \qquad z = v(t_*) - v(t_-).$$

The first nonzero coordinate of u is r and equals one. Also $u_{k-1} = 0$, whereas the last nonzero coordinate of z is $k-1$ and equals $-\nu_2(Q) \neq 0$. Thus u, z are linearly independent. Their coordinates have magnitude at most $n-1$, and $z_r = s - d$.

For the actual block function f, define its three-entry invariant

$$R(f) = \frac{f(t_*) - f(t_-)}{f(t_+) - f(t_-)} = \frac{z \cdot w}{u \cdot w}. \tag{180}$$

The denominator is nonzero by dominance. This ratio is unchanged by every positive affine transformation. If two same-depth witnesses have equal ratios, cross multiplication and Lemma 172 give $u' = \lambda u$ and $z' = \lambda z$. The first nonzero coefficient of each u is one, forcing equal r and $\lambda = 1$. The last nonzero coordinate of z then recovers k, and its rth coordinate recovers s. Consequently every witness remains distinct under affine equivalence, including across stages. The counts in (178)–(179) are the same counts as before. □

The result concerns the coarsest exact positive affine equivalence, without assuming an implementation or charging for canonicalization. It also holds for blocks that include their first address: whole-block equivalence would imply equality of the three strictly interior invariants. Thus adding

a scalar multiplier to normalized references cannot supply a universally small diagram. The same family still has a linear arithmetic-work chain construction and certificate; bit-operation costs are additional. A useful replacement must share more than affine copies of fixed dyadic blocks, avoid these prefix functions, or exploit additional input structure.

5.30 Lazy prefix evaluation with a rearrangement bound

One can avoid materializing a prefix function by evaluating only the predecessor states requested by its skyline recurrence. The following relaxation supplies exact lower bounds for that search. Its failure theorem identifies an ordering constraint that the relaxation loses.

For normalized nonnegative weights $w_0, \dots, w_{n-1}$ and residual gaps $g_0, \dots, g_n$, with $g_0 = g_n = 0$, recall that the reward of slots $s_0 < \cdots < s_{k-1} = b$ is

$$R = \sum_{i=0}^{k-1} w_i v(s_i) + \sum_{i=1}^{k-1} g_i \min\{v(s_{i-1}), v(s_i)\}, \qquad v(s) = \nu_2(s).$$

Its maximum is $-f_k(b)$; feasibility means $k \le b < 2^h$. Put $a_i = 2w_i + g_i + g_{i+1}$, and let $T_t(r)$ be the sum of the t largest values among $a_0, \dots, a_{r-1}$, with $T_0(r) = 0$.

Lemma 174 (Mixed-prefix rearrangement bound)**.** *For every feasible nonempty prefix,*

$$U_2(k,b) = (2w_{k-1} + g_{k-1})v(b) + \sum_{d=1}^{h-1} T_{\min\{k-1, \lfloor (b-1)/2^d \rfloor\}}(k-1), \tag{181}$$

$$2f_k(b) \ge -U_2(k,b).$$

After $O(n \log(n+1))$ preprocessing and space, each bound can be computed in $O(h \log(n+1))$ arithmetic and dictionary operations.

Proof. Use $2\min(x,y) \le x + y$ to distribute each residual gap to its two endpoints. This gives

$$2R \le \sum_{i=0}^{k-2} a_i v(s_i) + (2w_{k-1} + g_{k-1})v(b).$$

The final coefficient excludes g_k, which is absent from this prefix. For the other $k-1$ keys, drop the ordering constraint and choose slots from $[1, b-1]$. Nonnegative coefficients permit choosing the largest available valuations. An exchange of an inverted pair changes the dot product by $(a - a')(v - v') \ge 0$, so pairing coefficients and valuations in the same decreasing order is optimal for this relaxation. Exactly $\lfloor (b-1)/2^d \rfloor$ available slots have valuation at least d. Summing these threshold contributions gives (181), including the case $k = 1$.

Sort and compress the coefficient values. Maintain a persistent binary count-and-sum tree after each prefix insertion. This uses $O(n \log(n+1))$ nodes and work. Descending through the larger-value child first computes any $T_t(r)$ in $O(\log(n+1))$ work, taking an entire child's sum when its count fits the remaining quota. There are at most $h - 1$ such queries per bound. Rational coefficients require no rounding; comparisons with a cost incumbent are made after doubling that incumbent. □

The specified search. For $k \geq 2$, Lemma 40 gives

$$f_k(b) = \min_{\substack{a \in S(b) \\ a \geq k-1}} \{f_{k-1}(a) + c_k(a,b)\}, \qquad c_k(a,b) = -w_{k-1}v(b) - g_{k-1}\min\{v(a), v(b)\}. \tag{182}$$

The base is $f_1(b) = -w_0 v(b)$. Add a sentinel state $(n+1, 2^h)$, with zero key and gap reward, whose value is the minimum of f_n. At an expanded state, form its at most h feasible skyline candidates and their lower bounds $-U_2(k-1,a)/2 + c_k(a,b)$. Process candidates in increasing lower-bound order, memoizing exact states. A candidate is discarded when its lower bound is at least the current feasible incumbent. Once that comparison holds, all remaining candidates may be discarded together. Store an attaining predecessor with each exact value. An explicit stack suffices because every call decreases k. Traceback and sorting-free ancestor repair produce the original-weight tree.

If V states are expanded, including the sentinel, and Q distinct bounds are computed, then $Q \leq Vh$ and complete work is

$$O\big(n\log(n+1) + Qh\log(n+1) + Vh\log(h+1) + nh\big). \tag{183}$$

Candidate enumeration costs $O(Vh)$, candidate sorting costs $O(Vh\log(h+1))$, and each distinct bound has the cost of Lemma 174. Storage is $O(n\log(n+1) + Q + V + nh)$, including the stack's at most n candidate lists and full repair. These bounds count exact arithmetic and dictionary operations; operand bit costs are additional.

For unrestricted construction, choose a certified sufficient height $h \leq n$. A fixed $8n^2$ budget on the counted search work, followed by ordinary Knuth if exhausted, bounds full work by $O(n^2)$. This includes unsuccessful search, height selection, normalization, and full tree reconstruction. The next theorem shows why the unbudgeted search itself has no polynomial guarantee from this bound.

Theorem 175 (Exponential obstruction to rearrangement pruning)**.** *For $n \geq 6$, let $p_i = (n+1)^{n-i} - 1$ and $q_j = 1$, and use their necessary optimal height $h = n$. The completed unbudgeted search above expands at least*

$$\sum_{r=1}^{n-3} \binom{n-2}{r} = 2^{n-2} - 2 = \Omega(2^n) \tag{184}$$

distinct actual-prefix states. The same lower bound holds with any candidate processing order, if pruning uses only the stated bound and a feasible incumbent for the current state. Memoization does not remove these states. The original input is positive and has $O(n^2\log(n+1))$ bits in total.

Proof. Common-gap normalization gives $w_i = (n+1)^{n-i}$, zero residual gaps, and cost shift n. The unique optimum is the right chain by Theorem 167, so $h = n$ is necessary. The largest weight has $O(n\log(n+1))$ bits. For every index r,

$$w_r > (n-1)\sum_{i>r} w_i. \tag{185}$$

Consequently weighted reward strictly respects lexicographic order of valuation vectors with coordinates in $[0, n-1]$.

First consider a candidate child $f_m(a)$ satisfying

$$m \geq 4, \qquad a > 3 \cdot 2^{n-2}. \tag{186}$$

The first three keys are free in its relaxation. The three highest valuations among slots before a are $n-1, n-2, n-2$, at positions $2^{n-1}, 2^{n-2}, 3 \cdot 2^{n-2}$. Since the weights strictly decrease, these valuations are assigned in that order to w_0, w_1, w_2.

No increasing slot sequence has this initial valuation triple. A first valuation $n-1$ forces slot 2^{n-1}. A following valuation $n-2$ then forces slot $3 \cdot 2^{n-2}$, after which no valuation-$n-2$ slot remains. Thus every actual prefix valuation vector is lexicographically smaller within its first three coordinates. At the first difference r, its loss of at least w_r exceeds all possible gains at later coordinates by (185).

This argument compares the candidate's relaxed bound with the optimum of its *parent*, as required for pruning. For an ordinary parent, append its forced final key to the relaxed reward vector; this is exactly adding the transition reward, because all residual gaps vanish. All coordinates after the first difference, including the appended coordinate, still lie in $[0, n-1]$. The relaxed reward is therefore strictly greater than the reward of every feasible parent prefix. For the root sentinel, no coordinate is appended and the same comparison applies to all n keys. Equivalently, the candidate lower bound plus its transition cost is strictly below the true parent optimum. Every feasible incumbent is at least that optimum, so this candidate can never be pruned by the stated rule, independently of processing order.

Now choose any nonempty proper subset $B \subsetneq \{0, \dots, n-3\}$. Start at $(k, b) = (n+1, 2^n)$ and process its elements d in descending order, following

$$(k, b) \longmapsto (k-1, b-2^d). \tag{187}$$

Previously subtracted powers have larger exponents, so b is divisible by 2^{d+1}. Hence $b - 2^d$ is precisely the leading-bit prefix of $b-1$ ending at bit d: it is a genuine skyline predecessor. The sum x of all subtracted powers is less than 2^{n-2}, so every child endpoint $2^n - x$ exceeds $3 \cdot 2^{n-2}$. Its key count is at least four, since $|B| \le n-3$, and no greater than its endpoint, proving feasibility and (186). Every edge must therefore be processed unless its child has already been evaluated and cached. Induction down these paths forces evaluation of all their endpoints

$$\left(n - |B| + 1,\ 2^n - \sum_{d \in B} 2^d \right).$$

Distinct subsets give distinct states, all with at least four keys, so each is expanded at least once. Counting all nonempty proper subsets gives (184). □

The family retains its linear arithmetic-work chain constructor and certificate. The obstruction concerns this particular recurrence with this particular relaxation. It pinpoints the information needed by a stronger pruning bound: the order in which high valuations can be attained. Exact finite controls and complete-construction operation counts appear in Appendix T.

5.31 The size and rotation structure of a coarse optimal family

The coarse certificate in Section 2.30 suggests replacing an ambiguous payload by all intervals and root choices reachable through coarse optima. The following exact counts delimit that representation, even when all coarse weights are positive and their total mass is linear.

For fixed coarse weights, define the *reachable optimal-interval grammar* as follows. Start with $[0, n)$. At each reached nonempty interval $[i, j)$, include every root r attaining its unrestricted coarse optimum, and reach both children $[i, r)$ and $[r+1, j)$. Identify repeated intervals. A root triple is (i, j, r), not a complete tree. By optimal substructure, this grammar contains exactly the intervals

and root triples occurring in full coarse optima: each reached choice can replace the corresponding optimal subtree without changing the full cost.

Theorem 176 (Quadratic intervals and cubic root choices)**.** *Let $B = 2^k$, $k \geq 1$, and $n = 6B - 1$. Give every coarse key and every coarse gap weight one. The reachable optimal-interval grammar contains the $(B+1)^2$ distinct intervals*

$$[B + u,\, 2B - 1 + u + v), \qquad 0 \leq u, v \leq B. \tag{188}$$

The number of its root triples on just these intervals is exactly

$$(B + 1)(B/2 + 1)^2. \tag{189}$$

Consequently the explicit reachable grammar has $\Theta(n^2)$ nonempty intervals and $\Theta(n^3)$ root triples in the worst case, at coarse total mass $2n + 1$.

Proof. Uniform mixed cost is $2I + 3n$, so the coarse optima are exactly the trees of minimum internal path length. Take a perfect upper skeleton with $4B - 1$ keys, and expand $2B$ of its $4B$ bottom gaps. Every resulting tree fills all levels through depth $k+1$ and has its remaining $2B$ keys at depth $k + 2$; hence it has minimum internal path length.

The four subtrees at depth two each begin as a perfect $(B - 1)$-key tree with B expandable bottom gaps. Let their expansion counts be $x_0, x_1, x_2, x_3 \in [0, B]$, with sum $2B$. Set $x_0 = u$, $x_1 = v$. The remaining sum $2B - u - v$ is always in $[0, 2B]$, so choose $x_2 = \min\{B, 2B - u - v\}$ and $x_3 = 2B - u - v - x_2$. The second subtree then has exactly the interval in (188). Its start determines u, and its length determines v, proving distinctness and full-tree reachability.

Write its start as $i = B + u$. Its length is $B - 1 + v$. Every minimum-path-length tree of that size has a perfect $(B - 1)$-key upper skeleton and v bottom expansions. If t expansions lie in the left half, its root is $i + B/2 - 1 + t$, where precisely

$$\max\{0, v - B/2\} \leq t \leq \min\{v, B/2\}.$$

Every such t is realizable, and replacing the subtree preserves full optimality. Thus there are $\min\{v, B - v\} + 1$ roots on that interval. Summing first over v and then over u gives

$$(B + 1) \sum_{v=0}^{B} (\min\{v, B - v\} + 1) = (B + 1)(B/2 + 1)^2.$$

There are at most $n(n+1)/2$ nonempty intervals and $n(n+1)(n+2)/6$ interval-root triples in any BST recurrence, proving the matching orders. □

The counts concern a representation prepared from the coarse weights. They cannot be reduced by discarding coarse-optimal trees on the assumption that small perturbations will never select them. The following standard exposing-weight idea makes this precise. Onishi and Hoshi [23] study optimality regions of key-weighted BSTs and establish that every tree has a nonempty region; they also distinguish their adjacency from single rotations. We give the elementary mixed-weight consequence needed here.

Lemma 177 (Every coarse optimum can be selected uniquely)**.** *For any coarse-optimal tree T on $n \geq 2$ keys and any $\delta > 0$, there are arbitrarily small positive key perturbations, with all gap perturbations zero, satisfying (88) for which T is the unique fine-weight optimum.*

Proof. Put $A = n^2 + 1$, and give key i perturbation $e_i = \varepsilon A^{-d_T(i)}$, where $d_T(i)$ is its depth in T. In any subtree of T, let its root perturbation be w. Every proper descendant has perturbation at most w/A. The perturbation cost of the prescribed subtree is therefore at most $w + n(n-1)w/A < 2w$. Any competing subtree with a different root costs at least $2w$ just for that root key. Induction for equal roots proves that T uniquely minimizes perturbation cost among all full trees. It also minimizes coarse cost, so it uniquely minimizes their positive scaled sum. Finally $E \le n\varepsilon$, and taking $0 < \varepsilon < \delta/[n(n-1)]$ gives the certificate. The choice may be made arbitrarily small. □

For Theorem 176, every counted interval and root triple therefore occurs in a uniquely selected fine optimum for some certified perturbation of the same positive coarse input. This does not require one fine input to select all the counted triples simultaneously. Nor does it force an algorithm to materialize the grammar: a different implicit representation remains possible.

Root holes and a strict rotation trap. Small exact examples rule out two further shortcuts. For $p = (4, 1, 4)$ and $q = (1, 1, 1, 1)$, the three root-constrained optimum costs are $24, 25, 24$. Knowing the smallest and largest optimal roots therefore does not imply that all roots between them are optimal.

For a near-uniform rotation example, write $r(L, R)$ for a tree rooted at zero-based key r, and omit the children of a leaf. Let

$$T = 1\big(0,\, 3(2, 4)\big), \qquad U = 2\big(1(0, \varnothing),\, 3(\varnothing, 4)\big).$$

Give every coarse key and gap weight one, and take fine weights

$$p = (D, D+1, D+1, D, D), \qquad q = (D, D, D, D, D, D), \qquad D > 8. \tag{190}$$

Here $E = 2$ and $(n-1)E = 8 < D$, so the coarse-family certificate holds. The coarse optimum is 27. Direct depth sums give

$$C(T) = 27D + 4, \qquad C(U) = 27D + 3.$$

The four single rotations from T, respectively at edges $(1, 3)$, $(3, 2)$, $(3, 4)$ and $(1, 0)$, have costs

$$27D + 5, \qquad 29D + 3, \qquad 29D + 5, \qquad 33D + 6.$$

Thus T is a strict local minimum under every single rotation, while U is better. The global optimum is exactly $27D + 3$: among the six coarse optima, the perturbation costs are $4, 4, 4, 3, 3, 5$, and the certificate excludes every other tree. The six trees correspond to choosing two of the four bottom gaps of the perfect three-key upper skeleton. Their coarse-cost-preserving rotation graph has components of sizes two and four: the two root-$1/3$ trees form one edge, while the four root-2 trees independently choose each two-key child's orientation. Rotating the root of a root-2 tree leaves the minimum-path-length family. After dividing (190) by D, these are arbitrarily small perturbations of strictly positive uniform weights.

Descendant promotions combined with bottom-gap exchange. Strict local minima also occur under the combined richer move set. For nine keys, start with the perfect seven-key upper skeleton and expand bottom gaps 6 and 7, numbered from zero. Give every outcome coarse weight one and choose fine weights $p_i = D + e_i$, $q_j = D + f_j$, where

$$e = (2, 5, 4, 4, 5, 2, 0, 4, 4),$$
$$f = (3, 4, 4, 6, 8, 7, 1, 1, 7, 2), \qquad D > 584.$$

The total error is 73, so the certificate holds. All 28 coarse optima come from expanding two of the eight bottom gaps and have cost 59. The selected tree has fine cost $59D + 219$, whereas expanding gaps 0 and 5 gives the global optimum $59D + 212$. Consider promoting any descendant to the root of any existing subtree, or exchanging one expanded bottom gap for one unexpanded gap. Among the resulting trees, the fourteen distinct neighbors still in the coarse family all cost at least $59D + 224$ by direct depth sums. Every remaining promotion leaves the coarse family and is more expensive by the certificate. Thus the selected tree is a strict local minimum under the combined move set. Exact neighbor values and independent full-tree checks appear in Appendix A.28.

The theorem obstructs explicit coarse-family expansion; the five-key example obstructs descent using only single rotations. Neither is a lower bound for arbitrary full-OBST algorithms. Independent checks and complete original-weight controls appear in Appendix A.27.

6 Core-size tightness and further questions

Theorem 6 rules out an $O(n \operatorname{polylog} n)$ bound for all intermediate cores at linear integer mass. The remaining size question is the $\sqrt{\log n}$ factor between this lower bound and $O(n\sqrt{WL})$ when $W = n$. A stronger upper bound must respect the explicit odd-ramp density formula, or a stronger lower bound must force additional nonzero cells. The family arose from the adversarial search; its theorem follows from the deletion lemma and the closed-form density, independently of the search outcomes.

The aggregate bound of Theorem 8 already removes one logarithmic factor from construction time by charging individual outcomes across geometrically spaced layer boundaries. It leaves the per-layer $\sqrt{\log n}$ gap above unresolved. Theorem 142 shows why simply removing the height factor from each tile charge cannot close it: even one weight unit can cause a logarithmic number of positive entries in one tile, although Corollary 143 bounds its entire layer core linearly.

The bound concerns an explicit density representation. The unrestricted subquadratic construction problem remains open, as does the possibility of computing with a more compact representation of these structured dense cores. The quadratic ruler example already shows why representation size alone cannot establish a general algorithmic lower bound.

6.1 A boundary term in the local support charge

The remaining gap suggests the following open local hypothesis. For an absolute constant A, every accepted early-layer tile T of side s and inclusive outcome span J_T satisfies

$$K(T) \le A(s + W(J_T)) \qquad \text{for nonnegative integer input weights.} \tag{191}$$

Here $K(T)$ counts positive densities, rather than their sum. The side term accommodates the known one-key obstruction. Integer normalization is essential: rescaling an input preserves support while changing mass. The hypothesis remains unproved. The following one-key bound explains its compatibility with the logarithmic local obstruction.

Lemma 178 (Single-key square support)**.** *With one positive key and zero gap weights, any $s \times s$ square of ordinary density cells lying wholly on one side of the capacity line has at most $1 + \lfloor \log_2 s \rfloor \le s$ positive entries. The square need not be maximal or dyadically aligned. The logarithmic dependence has sharp leading coefficient one.*

Proof. Use the ruler-pair classification in Section 5.9. Each positive density corresponds to consecutive multiples $u < v$ of 2^r, for some $1 \le r < h$. Below capacity the mapping is $i = x - u$,

$j = c + x + 1 - v$; above it the mapping is $i = x + 1 - v$, $j = c + x - u$. For a square wholly off capacity, the possible u and v values occupy ordered disjoint integer intervals I, J, each with at most s values, and gap $g = \min J - \max I \geq 2$. Clipping to the allowed ruler range only shortens these intervals.

Every contributing pair spans a common point between I and J. At each scale, intervals between consecutive multiples have disjoint interiors, so at most one pair at that scale contributes. Their lengths lie between g and $g + 2s - 2$, whose ratio is at most s. There are at most $1 + \lfloor \log_2 s \rfloor$ powers of two in that range. This also handles $s = 1$. Multiplying the positive key weight preserves support. Finally Theorem 142 has $s = 2^{h-2}$ and $K(T) = h - 3 = \log_2 s - 1$, proving sharpness of the leading logarithmic term. □

Lemma 179 (Exact single-gap densities)**.** *Fix $h \geq 1$, $C = 2^h$, $c = C - 1$, and put a unit weight at gap x, with all other weights zero. Ordinary off-capacity densities above $j - i = c$ vanish. Below capacity put $g = c-(j-i) > 0$. The density is one exactly in the following cases, and zero otherwise:*

1. *$x = i$ or $x = j + 1$, and $g + 1$ is a power of two;*
2. *$i < x \leq j$, $g + 2 = 2^r$, and $x - i - 1 \equiv 2^{r-1} \pmod{2^r}$.*

Proof. For an interval $[i, j)$ containing the gap with $k = j - i \leq c$, put $u = x - i$ and $v = u + c - k = x + c - j$. Its maximal gap reward $F(u, v)$ is the largest a for which $[u, v]$ contains an aligned integer block $[z, z + 2^a - 1]$, with $2^a \mid z$. Its cost is $h - F(u, v)$. Indeed, the block represents 2^a consecutive external slots of a collapsed perfect subtree, whose $2^a - 1$ internal slots are omitted. The u preceding keys fit exactly when $z \geq u$; the $k - u$ following keys fit exactly when $z + 2^a - 1 \leq u + c - k = v$. The mixed-slot deletion repair gives a tree with no greater gap depth. Conversely, a gap at depth $h - a$ in an embedded height-h tree occupies such an aligned block. This also covers empty intervals, where $[u, v] = [0, c]$ and $F = h$.

Write $F = \sum_{a=1}^{h} I_a$, where I_a indicates that an aligned block of length $d = 2^a$ fits. For an interior gap the density is

$$F(u, v) + F(u - 1, v - 1) - F(u, v - 1) - F(u - 1, v).$$

For I_a this difference is -1 when the outer interval $[u-1, v]$ is exactly one aligned d-block, $+1$ when it is exactly two consecutive d-blocks, and zero otherwise. To check every endpoint case, removing the right endpoint changes I_a only if that endpoint finishes a block and no earlier complete block is present. Moving the left endpoint from $u - 1$ to u changes this event only if the removed endpoint starts that same block or its immediate predecessor. These give the one-block and two-block cases respectively. The outer interval has length $g+2$. At $g+2 = 2^r$, the positive contribution from level $r - 1$ cancels the negative contribution from level r when $u - 1 \equiv 0 \pmod{2^r}$; otherwise it survives precisely when $u - 1 \equiv 2^{r-1} \pmod{2^r}$.

For $x = i$ or $x = j+1$, the endpoint gap cost is $h - \lfloor \log_2(g+1) \rfloor$. The two contributing interval costs give density $\lfloor \log_2(g + 1) \rfloor - \lfloor \log_2 g \rfloor$. Gaps outside the four intervals contribute zero. Above capacity, every gap has comparison length h, whose interval sums have zero density. □

Lemma 180 (Universal unit-outcome support envelope)**.** *Define $\mathcal{E}_h$ on ordinary off-capacity cells by*

$$\begin{aligned} j - i < c: &\quad g = c - (j - i) > 0, \quad g + 1 \text{ or } g + 2 \text{ is a power of two;} \\ j - i > c: &\quad g = (j - i) - c > 0, \quad g + 1 \text{ is a power of two.} \end{aligned}$$

The support of every possible single positive key or gap is contained in $\mathcal{E}_h$. Every dyadically aligned square of side s, wholly in the ordinary region and on one side of capacity, contains at most $2s$

envelope cells; below capacity the bound is $2s-1$. *Maximality is unnecessary. The leading coefficient two is sharp for the union over all unit outcomes.*

Proof. The preceding lemma handles gaps. The key ruler-pair classification in Section 5.9 gives capacity distance $2^r - 1$. Scaling a single outcome preserves its support.

The claim is immediate for $s = 1$. Otherwise write the square as $[a, a+s) \times [b, b+s)$, with a, b multiples of s. Successive interval lengths have multiplicities $1, 2, \ldots, s, \ldots, 2, 1$. Below capacity, the smallest gap is $g_0 = C - (b - a + s) > 0$. Since $b - a \geq s$, we have $C \geq 4s$, so g_0 is a positive multiple of s. The gap range is $[g_0, g_0 + 2s - 2]$. A contributing power d with $g = d - 1$ or $d - 2$ exceeds s and is a multiple of s. Only $d = g_0 + s$ or $d = g_0 + 2s$ can contribute. They cannot both be powers of two: the first is at least $2s$ and their difference is s. The first contributes diagonals of lengths s and $s - 1$; the second contributes only the final corner. This proves the bound $2s - 1$.

Above capacity, $g_0 = b - a - s - C + 2$, and only $g = d - 1$ contributes. If $C \geq s$, write $g_0 = ms + 2$; positivity forces $m \geq 0$. For $m \geq 1$, contributing powers are at least $2s$, so a window of width $2s - 2$ contains at most two, each contributing at most s cells. For $m = 0$, powers $d \leq s$ contribute $d - 2$ cells each, and $d = 2s$ contributes two. Their sum is $2s - 2\log_2 s \leq 2s$. If $C < s$, write $g_0 = ms - C + 2$. Since $2 \leq C \leq s/2$, positivity forces $m \geq 1$. Contributing powers are at least s, and again at most two fit in the distance window. Restricting a padded grid to arbitrary n only removes cells.

For sharpness take $s \geq 2$, $n = C = 4s$, and $T = [0, s) \times [2s, 3s)$. Its envelope diagonals $j - i = 2s$ and $2s+1$ contain s and $s-1$ cells. A cell on the first is realized by a unit key at one-based position $i + 1$; a cell on the second by a unit gap at $x = i + s + 1$, using Lemma 179. The parent meets capacity, so the tile is accepted and maximal. This attains $2s - 1$ for the union over different inputs. □

For early layers $h < L$, the envelope uses $O(L)$ diagonals and therefore $O(nL)$ cells per layer, or $O(nL^2)$ over all early layers. Theorem 155 shows that support outside this envelope still requires accounting for joint outcomes across scales.

Lemma 181 (Conditional improvement from a boundary charge)**.** *If* (191) *holds uniformly, then the aggregate completed core size satisfies*

$$S = O_A(nH + (n + W)L^2 + n\sqrt{W} + HWL).$$

At $W = O(n)$ *this is* $O(n^{3/2})$*, and the existing Monge construction would take* $O(n^{3/2}\log(n+1))$ *arithmetic operations, including full reconstruction.*

Proof. At early layer $h < L$, put $c = 2^h - 1$. The established tile count and inclusive-position overlap bounds give at most $24n/s$ side-s tiles and total span mass at most $60W(1 + c/s)$. Combining the hypothesis with the area bound yields

$$K_{h,s} \leq \min\{24ns,\ A[24n + 60W(1 + c/s)]\}.$$

Separate the constant-per-scale term from Wc/s, and sum the geometric minimum over dyadic scales. Including the exceptional and completion diagonals gives

$$K_h = O_A((n + W)L + \sqrt{nWc}).$$

Now $\sum_{h<L}\sqrt{c_h} = O(\sqrt{n})$, so the early layers contribute $O_A((n+W)L^2 + n\sqrt{W})$. Late layers retain their proved bound $n + LW$, giving the claimed aggregate expression. At linear mass $H = O(L)$, and $nL^2 = O(n^{3/2})$. Product, conversion, and succinct reconstruction costs are $O(S\log(n+1))$. □

This conditional core scale matches the odd-ramp lower bound. Establishing or refuting (191) is therefore a direct local route to deciding the remaining logarithmic gap; the main theorem continues to use the proved height-dependent charge.

A Independent verification and operation counts

The theoretical statements above do not depend on these finite tests. Independent cubic recurrences, literal tree enumeration, direct original-cost evaluation, and operation counters provide falsification controls.

The per-outcome overlap audit checks 766 padded dyadic partitions, 4508 scales, and 1,891,752 position/scale incidences, including sizes immediately below and above powers of two. An independent all-root recurrence checks 290 mixed-weight inputs: 1977 original/padded layer equalities, 452,644 local residual bounds, and 28,013 weighted tiles. It examines 685,551 dense diagnostic states and 8,636,426 root candidates. The tests include all binary inputs through three keys, zero mass, uniform and random mixed weights, geometric weights, and mass at either endpoint gap. The executed source and operation counts are recorded in `aggregate_core_audit.json`. These dense verification operations are separate from the sparse construction's theoretical work bound.

Reproduction code and source-hashed records are available in the accompanying repository. The filenames below identify those records; they are not definitions needed by the proofs.

A.1 Implemented condensed constructors

The compact path is now implemented in Python and independently in C++. The product algorithm is the published compression, balanced core split, staircase merge, and decompression method of [14]. We credit that primitive separately from the OBST reduction. A matrix stores its top row, left column, and positive density triples. Two linear bucket passes canonicalize the triples. Boundary extraction uses one-dimensional sweeps; the product merge uses adjacent-entry recomputation along its staircase. Each relevant core row or column is scanned only a constant number of times per merge. None of these operations enumerates the full interval table.

Exact early stopping. The constructors compare consecutive completed layers by their top rows and canonical core triples; the left column is fixed. Equality is therefore equality of the whole table. A fixed point satisfies the ordinary OBST recurrence with zero diagonal. Induction on interval length proves that this recurrence has a unique solution, namely C. Stopping at such a fixed point is exact. Equality of the single full-interval cost is not used. Retained layer witnesses reconstruct a complete optimum: any unfinished nonempty terminal would contribute at least $M > HW$, exceeding the cost of an optimal tree of height at most H. All charges are nonnegative. Reconstruction uses an explicit stack and visits every original key.

Implementation costs and arithmetic. Both implementations retain ordinary integer coordinate maps and recursive witness nodes. Their witness storage is bounded by $O(H(n+K)\log(n+1))$ words, where $K = O(n + \min\{n^2, n\sqrt{WL}\})$ bounds the completed cores. This can exceed the succinct storage bound in Theorem 1 by a logarithmic factor. Compression bounds recursive dimensions by core size, and the core count halves along a recursive branch, so the time bound is preserved. No succinct rank/select implementation is claimed.

The selected perturbation sweep certified amplitude 8 around 1,000 at $n = 256$, amplitude 4 at $n = 512$, and amplitude 1 at $n = 1,024$; larger tested amplitudes at those sizes failed. These are

seeded observations, not universal thresholds. Ordinary random inputs failed the certificate. Sizes 384, 768, and 1,000 deliberately skip the shortcut because their minimum-I shape space exceeds the logarithmic width budget.

A.2 Exact bounded-excess construction by dyadic intervals

Validation, failed precursor, and prior work. The implementation passed 1,200 restricted-optimum comparisons against all shapes through eight keys with budgets $0, \ldots, 4$, and 3,000 full mixed-weight checks against cubic DP. It separately checks direct tree costs, both residual-bound implementations, and exact fallback. An additional 104 family checks include zeros, geometric weights, spikes, and key-only/gap-only controls. The seeded size sweep is retained. A precursor substituting ordered depth bounds into the old certificate passed 480 exhaustive checks but rescued none of 24 near-uniform fixtures at its 32-iteration budget. This negative experiment did not prove optimal multiplier search useless; its cost and the failed perfect-tree hypothesis motivated expanding the candidate class.

On the 1,023-key periodic fixture with weights 16 and 17, the excess-two DP finds cost 312,154 and excludes other shapes with bound 312,185. It uses 27,570 states, 137,835 transition checks, 6,052 sorting comparisons, and 1,022 merge comparisons, versus Knuth's 523,776 states and 1,046,532 root candidates. At $n = 1{,}024$, a seeded amplitude-four perturbation uses 48,003 states and 240,000 transition checks, plus 13,784 sorting and 1,264 merge comparisons. It certifies cost 19,477,996.

The dyadic/prefix-code viewpoint and sequential frontier DP are established ideas. Becker [8] gives a sequential mixed-weight DP for a near-minimal *height* restriction in quadratic time. The band above uses excess *total unweighted path length*, a different restriction; it does not improve Becker's stated problem. Priority of this particular potential, band bound, and exclusion combination remains unestablished.

A.3 A split-height construction covering every input size

Exact verification and operation evidence. The harness passed 450 exhaustive weighted height-class checks through eight keys, 44 exhaustive tall-path minima, 2,000 cubic-checked mixed inputs with complete fallback (917 successful certificates), and 104 additional adversarial-family inputs. An independent all-root height recurrence checked 520 candidates and 261,482 stored short-table entries; the same 520 cases also compared SMAWK with a retained divide-and-conquer matrix search. The larger sweep has 36 fixtures at three height budgets each. Every near-uniform fixture certified, including previously skipped sizes 384, 768, 1,000, and 1,536. All random, geometric, and gap-spike fixtures failed the certificate, sometimes despite an optimal restricted candidate.

At $n = 1{,}000$, the minimum-height candidate and full optimum are both 18,973,413, below the exclusion bound 18,974,193. The optimizer uses 117,453 short states, 220,483 short root candidates, 84,937 genuine matrix queries, 10,268 padding queries, 16,016 prefix scans, and 5,630 coarse transitions. The bound adds 10,648 sorting and 1,335 merge comparisons, plus 1,000 contractions; all keys are reconstructed. Knuth uses 500,500 states and 1,002,162 root candidates. These distinct operation types are reported separately, without treating them as equal-cost instructions.

A.4 A variable interval frontier for exact full construction

Independent verification. The harness passed 960 weighted exhaustive-shape comparisons through eight keys on arbitrary nondecreasing contours, including infeasible classes. It checked 914 interval costs independently and 90,193 top minima by brute search. Another 2,400 full solves,

using both modes on 1,200 inputs through 40 keys, matched cubic DP and reconstructed costs. All 24,600 internal subtrees in the cubic-optimal trees satisfied Lemma 2. Families include zeros, ties, geometric variants, spikes, and key-only and gap-only controls. A further 56 fixtures through 2,048 keys matched Knuth in both modes, including ordinary random inputs that failed the preceding height certificates.

At 2,048 random keys, the adaptive version chose depth six and returned cost 21,053,524. It used 309,261 interval cells and 616,642 root candidates, versus Knuth's 2,098,176 cells and 4,215,527 candidates. Additional work included 24,035 contour mass checks across seven parameter depths, 1,245,431 genuine matrix queries, 149,319 padding queries, 131,136 scans of each row/column type, and 107,859 coarse transitions. The uniform version stored 337,550 cells. On log-uniform weights the corresponding cell counts were 308,487 versus 431,951; on a single heavy key they were 1,048,576 versus 1,573,376. Geometric inputs selected the full table. All operation categories and parameter levels are retained separately.

A.5 Dominant-gap separators and local mass frontiers

Verification and complete work accounting. The harness exhaustively checks 22,140 weight assignments in $\{0, 1, 2\}$ through four keys against every tree shape. It checks 3,211 qualifying gap certificates and their cost identities, 2,036 endpoint certificates, and 380 key certificates. A further 3,280 full solves, using both frontier and Knuth terminal strategies on 1,640 inputs through 40 keys, match independent cubic costs and direct tree evaluation. Inputs include empty, zero, tied, geometric, spike, key-only and gap-only controls. Both strategies and the direct global frontier also match Knuth on 40 fixtures through 2,048 keys.

At $n = 2048$, the direct and local frontier interval counts are:

Fixture	direct frontier	local frontier	terminal components
Random	308,115	308,115	1
Near-uniform	309,290	309,290	1
Geometric	2,098,176	0	0
Heavy key	1,048,576	153,660	2
Heavy gap	1,047,552	153,582	2
Random background, heavy gap	1,047,552	153,362	2
Hierarchical gaps	522,242	22,333	64

Cell counts do not include all work. On hierarchical gaps, 63 gap separators require 254 endpoint checks, 6,142 index-build comparisons and 1,476 range query comparisons. Terminal solves use 42,885 root candidates, 39,284 genuine top-matrix queries, 6,261 padding queries, 8,122 contour mass checks, 6,952 scans of each row/column type, and 1,792 coarse transitions. Their 1,922 keys are reconstructed and relabeled. On the heavy-gap fixture, 305,118 terminal root candidates and 1,214,092 genuine top queries accompany the reduced interval table. All counter categories are retained; metadata such as local selected depths and actual heights are summed over terminals, not reported as a global tree height. Geometric input uses 2,048 endpoint separators and 4,095 endpoint checks, with no index or terminal work.

A.6 Joint heavy-outcome cores

Independent checks and operation counts. Every nonempty selected-outcome subset and every tree shape through four keys were examined. In all 1,937 non-core-closed cases a core-closed tree has componentwise no-greater selected depths and at least one strict improvement. On 22,140

inputs with weights in $\{0, 1, 100\}$, all 17,054 qualifying nontrivial certificates match the optimum over every shape and the composition identity. Another 2,640 full solves on 1,320 inputs through 32 keys, with both kernels, match independent cubic DP and reconstructed costs. On 520 small inputs the threshold sweep matches direct enumeration of every threshold and payload partition. Both kernels also match Knuth and the previous separator constructor on 30 larger fixtures through 2,048 keys.

At $n = 2048$, the following cell counts include both core and payload solves, or the complete fallback when certification fails:

Fixture	previous separator	joint core/frontier	core keys
Random	309,201	309,201	—
Near-uniform	309,290	309,290	—
Two heavy gaps, unit background	2,098,176	10	4
Two heavy gaps, random background	2,098,176	306,506	4
45 heavy gaps	304,486	23,965	90
45 moderate gaps	304,486	23,695	90
45 mixed heavy outcomes	305,421	24,016	68

The moderate-gap fixture sets each selected weight to exactly $(H + L)W_0 = 42{,}395{,}178$, with $H = 9$, $L = 12$, and $W_0 = 2{,}018{,}818$. The old size-based condition would require $(90 + 12)W_0$ and fails. Work includes 63,549 sorting comparisons, 2,047 Fibonacci additions, 3,024 height checks, 1,958 activations, 1,912 unions, and 1,511 find steps. Kernels use 45,365 root candidates, 116,338 genuine top queries, 23,904 padding queries, 12,336 contour checks, and 17,488 scans each for rows and columns. All 2,048 keys are reconstructed. Local kernel metadata such as heights and chosen depths are sums across calls, not global tree heights. The uniform two-gap case still pays selection, 2,044 payload-node constructions and relabelings, and ten core DP cells; it does not perform only ten total operations.

A.7 Joint heavy-outcome cores

Full construction beyond the quadratic-table scale. On the unit-background two-heavy-gap family, full trees were constructed at $n = 512, 2048, 8192, 32768, 65536$. Each exact cost matched an independent cubic solve of the four-key instance

$$p = (1, 1, 1, 1), \qquad q = (1, n^3, 2n - 7, n^3, 1),$$

plus $2I_*(n - 4) + 3(n - 4)$ for the uniform payload, and direct evaluation of the full reconstructed tree. The uniform closed form was separately checked against cubic DP through 16 keys. At $n = 65536$, the exact cost is 1,407,374,885,846,941. The construction used ten core DP cells and fifteen core root candidates. It also used 196,799 sorting comparisons, 65,535 Fibonacci additions, 65,535 height checks, 65,532 activations, 65,531 unions, 65,529 find steps, and 196,609 uniformity comparisons. All 65,532 payload keys were constructed and relabeled, and the four core keys were reconstructed.

A.8 Residual-cost certificates for heavy-outcome cores

Independent exact checks. On all 22,140 assignments in $\{0, 1, 4\}$ through four keys, every one of 17,054 nontrivial threshold cases passed the excluded-class lower-bound check against all tree shapes. There were 13,669 passing witness tests and 15,181 passing exact-candidate tests, including 1,512 cases rescued by refinement. Another 14,912 excluded-tree checks used arbitrary

selected subsets, weights in $\{1, 3\}$, and up to three keys. These latter selections need not be weight thresholds.

Both kernel choices and both refinement settings passed 6,560 full solves on 1,640 inputs through 40 keys against independent cubic DP and direct tree evaluation. All 1,640 candidate portfolios matched direct enumeration of the two representatives in each core-size band. Forty-eight larger fixtures and nine additional fixtures with independently varying positive key and gap weights matched Knuth through 2,048 keys. The varying backgrounds are integers in $[1, 1000]$ and their selected outcomes have weight $4000n$; the unit-background success rows below use weight $4n$. These are finite tests, not an assertion that every such random instance passes certification.

At $n = 2048$, selected operation categories are:

Fixture	Previous intervals	New intervals	Protected proof cells
Two linear-weight gaps, unit background	568,350	10	10
45 linear-weight gaps, unit background	304,486	1,394	4,095
45 mixed outcomes, unit background	305,421	1,250	2,346
Two gaps, varying background	951,851	307,541	30,941
45 gaps, varying background	304,486	23,757	22,519
45 mixed outcomes, varying background	305,421	24,163	27,558
Random control	308,860	308,860	33,380

New intervals count all executed full-weight kernel cells, including fallback, but exclude the separately displayed protected-core proof cells. The proof column sums all attempted cores. Selected-state metadata is not counted again as executed cells. Sorting, profile scans, lower-bound contractions, witness construction, kernel matrix work, and reconstruction have separate raw counters; this table is not a total instruction count. The random control fails all ten witnesses and pays additional proof work before fallback.

A.9 Residual-cost certificates for heavy-outcome cores

Full large constructions and a remaining certificate gap. For every $n = 2^h$, $3 \le h \le 16$, the full $\theta = 4n$ construction matches Equation (120), direct evaluation of the entire output tree, and an independent four-key cubic core plus the uniform payload formula. At $n = 65,536$, the optimum is $3,604,381$. The proof core and full-weight core each use ten cells and fifteen root candidates. Other counts include 196,799 selection-sort comparisons, 131,204 bound-sort comparisons, 65,536 bound contractions, 65,535 two-queue comparisons, 65,532 activations, 65,531 unions, and 65,529 find steps. All 65,536 witness keys are constructed and evaluated, then all output payload and core keys are constructed. The lower bound is $3,604,482$, giving a certificate margin of 101. The previous method was only planned at this size: it rejects the core and selects 146,498,625 frontier cells at depth 11.

Failure remains informative. At $n = 2048$ and selected gap weight $2n$, the exact core-closed cost $71,611$ equals the unrestricted optimum, but the exclusion lower bound is only $67,586$; even refinement cannot certify it. At weight n, the closed candidate costs $61,371$ while the unrestricted optimum is $59,332$. Thus some failures reflect a weak certificate and others reflect a genuinely suboptimal proposed structure. Improving the bound must preserve this distinction.

Raw results and source hashes are in `core_exclusion*.json` under `obst/results`. An initial full-harness failure was a metadata bug: list-valued fallback fields were passed to arithmetic counter update. Direct assignment fixed it, and the final full harness was rerun. A completed single-threshold experiment was superseded after it missed several useful cores; the published selector and every final result use the bounded portfolio. The general subquadratic objective remains unresolved.

A.10 Coupled two-gap skeleton certificates

Verification and measured limits Independent forest enumeration passed 11,880 checks. Exhaustive small weight inputs numbered 24,188, with 51,654 pair-exclusion checks and separate checks for both first and tail lower bounds. Among these are 3,288 exact closed candidates accepted by the coupled bound but rejected by B_{basic}. Complete construction passed 6,560 comparisons with independent cubic DP, covering both payload kernels and refinement modes. Larger tests at $n = 128, 512, 2048$ include random, near-uniform, geometric, single-spike, adjacent-pair, interior-pair, and varying-weight inputs. Neither favorable examples nor these exhaustive finite checks replace the preceding proofs. Raw counts and source hashes are stored in `pair_skeleton*.json`.

The nonuniform-family harness checks 31 placement/size combinations, including clustered, spread, and random placements; 28 are compared with Knuth and eight with cubic DP. All satisfy the exact residual formula and the proved certificate margins. Fourteen unit-background cases extend through $n = 65,536$. At $n = 2048$ with unit keys and selected gap weight $2n$, the earlier bound $67,586$ is replaced by $71,619$, which certifies the exact cost $71,611$. At gap weight n, the closed candidate remains suboptimal and is correctly rejected.

At $n = 16,384$ the seeded nonuniform instance has 1,023 doubled key weights. Full construction returns cost $767,240$, with witness $768,946$, first bound $780,208$, and tail bound $795,651$. Its 7,758,912 short states, 3,922,465 top states, and 20 total core states give 11,681,397 DP states versus 134,225,920 in a full Knuth table. Parameter-selection metadata is not counted a second time. There are also 42,351,013 top-matrix queries, 15,502,473 short-root candidates, 261,181 forest sorting comparisons, 49,143 forest merges, and 256,673 contour mass checks. The complete tree is reconstructed and its cost evaluated directly. The largest instance is certified by the theorem and construction, not compared with an independently executed quadratic table.

The chief unresolved issue is how to retain useful path-union information when no constant-size core can be certified. The present method does not bound the number of necessary boundary states for arbitrary mixed input. Its general worst-case construction remains quadratic.

A.11 An ordered depth-chain lower bound

Preliminary strength and cost. At $n = 256$, normalized gaps $(\text{OPT} - \text{LB})/W$ for the chain bound versus order-free Huffman were 0.15003 versus 0.22080 on a random mixed instance, and 0.00873 versus 0.49294 on an alternating instance. However, 300 ascent iterations used 1,082,368 key-depth cells including initialization, compared with 32,896 interval states for Knuth. The unique-root variant uses two passes and still leaves 254 of 256 potentially improving roots on the random instance, and 255 on the alternating instance. Only two survive on the gap-spike instance. These results identify a lower-bound component and its limitations; they do not establish a faster general exact solver.

A.12 Exact depth ordering and an arbitrary-window obstruction

Independent verification. The harness enumerated all 873,612 sequences in $\{1, \ldots, n\}^n$ for $1 \le n \le 7$. The characterization accepted exactly the independently generated BST depth sequences: respectively 1, 2, 5, 14, 42, 132, and 429. The stack recognizer agreed with direct interval scans on every sequence. Width3 and exact flow already admit four invalid sequences at $n = 6$ and 34 at $n = 7$. The positive family was checked at $t = 16$ for all 127 widths $2 \le w \le 128$: exact costs, window validity, zero flow, and the tree-cost lower bound all passed. Knuth and direct tree evaluation checked every instance; independent cubic DP also checked $w \le 24$. Another 120 cases

combine $2 \le w \le 16$ with forced prefix depths $1 \le D \le 8$. Cubic DP, Knuth, direct costs, and inspection of the optimal prefix verify the exact additive identity and preserved strict gap.

A.13 Endpoint responses: insufficiency and representation size

Independent checks and remaining question. The exact-arithmetic harness checks 875 response queries against cubic DP on 175 inputs through seven keys. Rational half-plane clipping checks exposed pieces; 725 retained planes with positive endpoint weights satisfy the Fibonacci bound. The geometric family passes 408 hinge queries, including thresholds and neighboring integers; 1,496 two-parameter probes through 33 keys; 204 full cubic comparisons; 30 all-shape checks that every optimum has the prescribed depth pair; and 680 strict endpoint inequalities. The product checks execute 887,808 Knuth root candidates. Another 304 queries expose all outer-root pieces, and 384 queries test arbitrary parameters and the global transition threshold. All 2,184 factored complete constructions agree with Knuth and direct evaluation, and construct exactly n keys. Full-quadrant envelopes through nine keys have 5, 10, 17, and 26 pieces, agreeing with the proved formula. The factored constructions use 14,492 threshold comparisons, 8,736 outer-minimum comparisons, and construct 52,232 keys in total; these categories exclude the independently executed reference solvers.

The collision search found an initial surrounding context with outer weights 16; reducing them to seven preserves the one-unit optimum difference. The published certificate independently checks domination of discarded planes and enumerates both full eight-key tree sets. It does not depend on the clipping implementation or sampled parameters. Code, raw results, and the harness source hash are in `boundary_response.py/json`.

Legal-domain collision and signed recovery. The independent recovery audit checks 800 integer inputs: all binary mixed vectors through four keys and seeded inputs through 24 keys, including geometric, uniform, gap-only, key-only, and zero weights. It compares 3968 signed value queries against a fresh all-root recurrence using 1,926,832 root candidates, with negative endpoint weights permitted. Literal shape evaluation supplies 107,608 further signed cost controls. The boundary-path classification has 19,184 shape controls and 2792 unique-chain checks. All recovered gap vectors agree exactly, and each instance checks the query-count and rational-denominator bounds.

The whole legal quadrant is certified by two complete intercept tables and 18 exact domination checks. Independent enumeration evaluates all 2860 ordered trees across the two eight-key contexts and agrees with the root-conditioned costs. The signed absolute probe $(0, -1)$ is checked by the original recurrence. Twelve larger recoveries reach 65,536 keys using the separately proved chain-line oracle. Random and uniform instances at that size use 65,536 value queries each; the all-zero instance uses three. These larger rows verify complete recovery from the chain representation; they have no independent large signed-DP oracle. The result record retains counts, the exact collision, verification scopes, and five source hashes. The recovery theorem bounds calls to its specified oracle and does not provide a faster general OBST solver.

A.14 A core-sparse Monge reduction with terminal penalties

Independent evidence. The condensed matrix harness passed 1,600 random rectangular products, checking 251,779 output values and exact leftmost witnesses by enumeration, and 1,600 arbitrary submatrix selections. Sparse distance-matrix products at dimensions 128, 512, and 2048 each retain seven nonzero output core entries from seven entries in each input. Core visits remain 673 in these three checks, while dimension scans grow. This exercises nontrivial sparse output, not just products with empty cores.

The complete Python constructor passed 1,134 solves, including 682 exhaustive binary-weight cases through four keys (including two empty-key inputs), 838 cubic checks, and 32 fixed-budget controls that disable early stopping. On 78 layers of the small controls, all 3,367 interval values and leftmost witnesses agree with explicit all-root enumeration. Full tree audits check 15,544 reconstructed keys. Geometric and folded weights through 64 keys remain adversarial controls; the geometric input requires all 64 layers.

The C++ address/undefined-behavior sanitizer run passed 1,278 complete solves, including 826 independent cubic checks, 12 Python compact comparisons, random weights through 48 keys, geometric/folded inputs through 64 keys, and all-zero and single-active-gap controls. It checked 18,248 reconstructed keys. The release run passed 1,362 solves and 88,904 reconstructed keys, including 1,302 independent Python Knuth comparisons. All release cases also compare against the separately implemented C++ Knuth constructor. The independent parent-array audit verifies inorder validity, unique key coverage, and direct mixed weighted cost. Three malformed/out-of-domain inputs are explicitly rejected by each binary. No sanitizer error was reported. Results include source hashes, compiler flags, and seeds.

A.15 A centered core bound for large integer baselines

Independent tests and remaining limitations The dense falsification harness `centered_layers.py` passed 1,100 complete solves, 824 cubic comparisons, 3,317 all-root cells, 1,538 baseline all-shape checks, 99,087 baseline-density checks, 685,288 tile checks, and 2,327 scale checks. It explicitly computed 4,153,053 dense states and 8,223,340 root candidates and reconstructed 11,656 keys. Baselines include 2^{256} and 2^{200} independently for keys and gaps. Zeros, concentrated gap mass, and geometric weights remain in the suite.

At 256 keys, three large-baseline fixtures have deviations 501, 505, and 504, normalized height budgets 14, 14, and 12, and maximum interior cores 6,428, 6,386, and 6,351. The earlier numerical-mass budgets were all 256. These are budget comparisons, not measured speedups: the original constructor already stops at exact fixed points. The 64-key geometric control still has $H = 64$ and all 2,016 interior density cells nonzero.

The actual condensed variant `centered_obst.py` passed 999 complete solves, 790 cubic comparisons, 32 fixed-budget controls, 1,001 intermediate values and leftmost witnesses, and 12 original-compact controls. It checked 9,992 reconstructed keys. Its large-baseline 256-key fixtures finish in ten layers with completed cores 6,624, 6,540, and 6,496. The earlier solver and its source-hashed compiled performance controls remain unchanged.

A.16 Ruler patterns and a dense-core representation barrier

Exact checks, counts, and controls The dense `ruler_layers.py` diagnostic checked 921 layer solves, 70,100 weighted-subsequence interval identities, 22,437 range maxima, 20,432 all-root cells, and 2,168 all-shape layers. It explicitly computed 5,708,220 layer states and 11,108,348 candi-

dates. Its separate full-table controls used 1,422,592 states and 2,692,718 candidates, with 96 cubic comparisons. These are diagnostic costs, not an implementation of a new fast layer constructor.

The family audit checked 68 sizes through 512, 221,284 interval/formula equalities, 202,127 instances of (143), 179,486 instances of (144), 2,974 enumerated shapes, and 13 cubic controls. At $n = 512$, all 130,816 ordinary core cells and all 129,795 valid fourth-difference cells were nonzero. The theorem uses only its explicitly proved quadratic subset. Formula and direct tree-cost checks continue through 65,536 keys without dense interval tables.

The full exchange/gate/fallback constructor passed 1,210 solves, 826 cubic comparisons, 658 checks against the prior minimum-path-length shape DP, and 386 independent enumeration checks over 950 minimum-path-length shapes. It certified 349 inputs; 512 failed the exchange condition and 349 passed that condition but failed the global gate. All fallback costs were checked. Both orientations and nonmonotone compensated key fixtures are included, alongside ordinary random and geometric failures.

A.17 Exact transfers for joint key/gap signatures

Independent checks and complete work counts The identity harness preceded the implementation. It checked all forty boundary-count intervals through block length ten, 1,365 direct costs using independent slot-expanded tree shapes, and 228 block path-independence identities. All 120 feasible short boundary/count reconstructions were checked. Another 1,000 forward/backward transfer vectors were compared with explicit word enumeration, including negative profile values and unreachable entries. The oracle does not reuse the production deque.

The final source-hashed M4 suite passes 1,023 standalone full solves, 778 cubic comparisons, 1,023 unrestricted shape-DP controls, 762 independent shape enumerations, and 1,023 independent checks of multiplicity sorting. There are 366 certificates and 657 gate rejections, including 411 in preflight. Another 1,023 portfolio solves return the same exact costs: 912 select signatures and 111 select the affine route. Eight large portfolio controls agree with the standalone constructor. Inputs include zeros, ties, large integers, geometric weights, arbitrary block values, and one-key blocks. Both scaling families at 1536 keys also match Knuth and the unrestricted shape optimizer.

Table 1 includes every block transfer performed during traceback, not just the initial forward sweep. The four-block 98,304-key solve also scans 2,228,224 source positions, pushes 311,317 deque entries, and performs 311,283 deque value comparisons, 262,146 traceback split checks, 98,304 residual contractions, and 110,590 bound-queue comparisons. Its certificate sorts thirteen multiplicity records using 26 comparisons and expands all 196,609 input weights. The sixteen-block solve uses forty-nine records, 167 sorting comparisons, 2,207,873 deque pushes, 2,415,139 deque value comparisons, and 524,310 traceback split checks. Both construct and directly evaluate all 98,304 keys. These are distinct operation categories, not a machine-instruction total. Maximum profile width counts ranks and is not a measured peak-memory byte count.

A.18 Global alternating-gap bounds and linear profile construction

Independent evidence and full-construction counts The exact word/tree harness preceded the production constructor. Recognition matches literal word enumeration on all 1,022 nonempty binary patterns through nine keys; 865 eligible patterns additionally match independent minimum-path-length tree enumeration. A separate exhaustive oracle checks 13,834 parity/tree inequalities and 69,170 mixed-weight/tree inequalities through nine keys, including exact equality of each size/parity minimum with the gap formula. The perfect-size counterexample is retained.

The full suite passes 3,157 standalone solves against Knuth, including 2,847 cubic controls and

Keys	Blocks	Window queries	Potential terms	Max. profile width
1,536	4	40,968	10,242	1,024
1,536	16	188,552	47,138	1,024
6,144	4	163,848	40,962	4,096
6,144	16	753,800	188,450	4,096
24,576	4	655,368	163,842	16,384
24,576	16	3,014,792	753,698	16,384
98,304	4	2,621,448	655,362	65,536
98,304	16	12,058,760	3,014,690	65,536

Table 1: Actual full-constructor transfer counts on certified non-affine mixed signatures, including linear-space traceback recomputation. Larger-than-1536 cases use the proved certificate and direct output-cost checks, rather than executing the dense comparison tables.

1,325 class-candidate comparisons against the unrestricted shape DP. Another 3,157 routed solves agree. There are 1,201 accepted full solves: 1,083 use the global profile equality, 116 use the older mixed gate on perfect shapes, and two are empty instances. The remaining 1,956 route to exact fallback: 86 have unsupported key alphabets, 600 unsupported gaps, 241 unsupported high counts, 905 fail profile attainment, and 124 fail the perfect-shape mixed gate. Zeros, ties, huge integers, random, geometric, near-uniform, arbitrary binary patterns with and without a baseline, and earlier affine inputs are included.

Both scaling families now have no added baseline. At 393,216 keys, a nonperiodic witness-generated pattern has 262,111 joint-signature runs but only 196,305 interval transitions; strict alternation has 393,216 runs and 196,608 transitions. They require 212,793 and 262,144 membership queries respectively, and one traceback step per stored high boundary. Each orders two multiplicity records, expands 393,216 key weights, constructs every key, and directly checks the full mixed cost. Neither uses a residual-Huffman contraction or a dense shape state. These are operation categories, not machine-instruction totals or measured peak bytes. Ten scaling portfolio controls agree; the two 1536-key cases also match Knuth and the unrestricted shape DP. Larger cases use the proved global bound and direct output-cost verification.

A.19 Threshold runs and a linear Cartesian certificate

Independent evidence and complete operation counts The threshold probe preceded the production constructor. Across all 1022 binary gap patterns through eight keys it checks 862118 indicator/tree inequalities and equality with the exhaustive optimum for every pattern. It checks 20560 weighted tree inequalities and key equality characterizations, plus 1200 mixed cubic-DP controls.

The full constructor harness checks 3000 cases against independent cubic and Knuth recurrences, including 2610 literal exhaustive controls, and another 3000 portfolio solves. The Cartesian attempt accepts 885, rejects 927 key profiles, and rejects 1188 gap profiles. The harness also checks 21844 gap vectors against explicit threshold and sorted-rank sums, 36 positive multilevel fixtures against Knuth, six larger family instances, and a 20000-key linear-height tree. Twenty independent selection scaling controls cover random, ascending, descending, duplicate, and organ-pipe weights through 100000 gaps.

At $n = 327679$, there are 327679 distinct key weights and 65538 gap levels. The exact full cost is 253331707448885247. The attempt uses 655322 Cartesian comparisons, 327679 key visits, 327680 gap visits, 327680 gap-bound scan items, and reconstructs all 327679 keys. The checked

extrema formula avoids selection on this family. There are no interval-DP cells. For the general selection controls at $N = 100000$, the sum of outer rank-level sizes is 231070; all recursive selection, partition, copy, sum, and comparison counts are separately retained in `threshold_complete.json`. These counters describe logical work, not measured peak bytes or machine instructions.

A.20 Shifted alphabetic certificates beyond key-capacity equality

Independent checks, scaling, and limitations The independent probe precedes the optimized constructor: 960 cubic controls, 3629 scalar-transfer checks, and 37500 literal tree inequalities. It certifies 138 candidates, including 24 strictly above the old bound. The constructor checks all 9840 ternary gap vectors through seven keys against cubic values and the literal compatible-pair implementation, including identical ordered gap depths. Another 960 full mixed solves match cubic and Knuth, including 560 exhaustive controls: 336 certificates and 624 fallbacks, with all 279 constant-key cases accepted. Random, near-uniform, geometric, binary, and zero inputs are included. Eight family sizes through 895 keys match Knuth; the four smallest also match cubic. A 1199-key skewed tree tests iterative traversal.

Five large constructions reach $n = 229375$, with 196608 distinct key weights, capacity deficit 32768, and exact full cost 2029411090432. The independent explicit family tree, constructed tree, and bound agree. The largest run uses 229375 compatible merges, 458749 priority pushes and pops, 6885341 priority comparisons, 5136371 heap-item comparisons, 1473827 key-sort comparisons, and 5021690 meld calls. It reconstructs every key and uses no interval-DP cells. The solver receives only p, q.

A.21 Endpoint prices and the limit of additive transfers

Independent evidence and measured costs The independent harness precedes the optimizer. It checks 2052 literal transfer identities and exact cost-difference ranks through eight keys: rank $2n - 2$, nullity three. For 60 three- and four-key instances, an independent arrangement of all alphabetic-tree and key-rank breakpoints evaluates 5111 rational points; every maximum agrees with the price search. On 600 mixed cases through 64 keys, 1712 individual oracle values match independent cubic values, and small cases also match literal trees. All 600 bounded full solves match cubic, Knuth, and direct cost: 220 certify, including 152 previous rejections, and 380 fall back. The uncapped diagnostic certifies 228, rescues 160, and completes every maximum within ten calls in this suite. There are 358 strict bound gaps and 600 clamped-transfer monotonicity controls.

Nineteen exact reduction controls produce the four-key obstruction. A further 6000 seeded positive cases find no nonintegral price maximum; this is not an integrality theorem. Intermediate prices are fractional: the obstruction search queries $y = 1/3$ and obtains $F = 68/3$. The code uses exact rationals and does not assume integral maxima.

A.22 Root/nonroot coupling beyond the transfer barrier

Independent controls, regressions, and a retained conjecture The independent case-partition harness precedes the optimized constructor. Across 840 mixed instances through eight keys, it checks 246480 literal conditioned inequalities. The bound strictly improves $K + A$ on 373 cases and is exact on 324. Random, near-uniform, geometric, and zero/binary inputs are included.

The normalized full implementation matches cubic, Knuth, and reconstructed original cost on 1040 instances through 64 keys; 720 also match literal tree enumeration. There are 800 independent bound controls and six additional family checks. The route certifies 505, including all 240 constant-size base cases, and falls back on 535. It exceeds the completed maximum transfer bound in 84 cases and rescues 11 price rejections, but loses 47 price certificates. These finite counts are not distributional guarantees, and the fixed endpoint choice is not universally superior to price optimization.

A separate diagnostic retains an unproved marked-boundary event family. It inserts a barrier at one weighted interior key, permits ordinary compatible merges on either side, and charges the key weight when a pair first consumes that barrier. Greedily taking the cheapest event fails: for $p = (0, 4, 0)$, $q = (8, 10, 2, 18)$, it returns 82 instead of 80, with strict primary choices. Inverting a unit uniform transfer gives the positive instance $p = (1, 5, 1)$, $q = (7, 9, 1, 17)$, with costs 79 and 77 respectively.

The alternative probe tries the cheapest crossing pair after every prefix of ordinary merges, completes compatible merging, and evaluates reconstructed trees with the original positional key cost. It matches all 17769 checked instances, including 15769 literal-tree controls; 4134 controls also enumerate every crossing choice without changing the best result. This is a conjectural candidate family, not an exact solver theorem. There are 20800 invalid depth reconstructions across the attempts, which are rejected. Merge charges are not assumed to equal positional costs.

Its candidate costs are not unimodal in the activation prefix. With gaps $(17, 83, 93, 11)$ and weight 92 on key 1, the ordinary merges cost 100 and 104; activation before, between, or after them gives costs 500,681,500. Thus a single-valley search is unjustified even with positive gaps and strict ordinary priorities. The full probe records 6759 nonunimodal sequences, 116 greedy failures, and 65 charge/cost disagreements. No subquadratic complexity is claimed for the probe. The proved root/nonroot certificate does not depend on this conjecture.

A.23 Exact checks and preliminary measurements

The code uses nonnegative integer weights, an independent cubic recurrence, Knuth's recurrence, and direct evaluation of reconstructed trees. The basic harness additionally enumerates all tree shapes for small sizes. CPU experiments ran on the sponsor's M4 Mac mini. The reported quantities are algorithmic counts, not inferred asymptotic bounds from elapsed time.

- Single-key contraction: 22,143 exhaustive weight assignments in $\{0, 1, 2\}$ through $n = 4$, and 6,000 seeded random instances through $n = 20$. Every cost and reconstructed tree matched the cubic oracle.
- Block theorem: 19,486 qualifying interval contractions across 4,500 seeded instances through $n = 16$ matched independent cubic costs.
- Block algorithm: 3,000 random instances through $n = 20$, including fixed and adaptive thresholds, matched the cubic oracle and tree evaluation. Additional small cases check threshold selection against every threshold.

In the heavy-separator instance all gaps and ordinary keys have integer weights in $\{1, 2, 3\}$, and 22 randomly selected keys have weight 10^6. The block algorithm retains exactly 22 keys. Its state reduction is about 12.7-fold, but this is not a claim of a 12.7-fold runtime speedup. On ordinary random mixed weights the gain is small. The single-key pass contracts every key for the tested sequence $p_i = 4^i, q_i = 4^i$, but not for most of the ratio-two sequence. Geometric integer inputs also illustrate why bit complexity must be distinguished from arithmetic-operation complexity.

A.24 Additional shape and contraction controls

The recursive constructor passed 3,000 random mixed inputs through thirty keys against cubic recurrence and direct tree evaluation. The minimum-path- length optimizer matched literal shape enumeration on 360 weighted inputs through eight keys. Its full fallback variants matched 3,000 further inputs. The affine-corridor audit includes independent candidate, regret, and complete construction checks; its parameter-dependent state bound is proved in the main text. Source-hashed records retain the individual operation categories.

A.25 Additional recurrence and certificate controls

Reference checks and implementation boundary. The dense audit `penalty_layers.py` passed 1,092 complete solves: 680 exhaustive binary-weight inputs through four keys, 384 random inputs through 32 keys, and 28 size/family fixtures through 256 keys. These include zeros, near-uniform, alternating, varying mixed, and geometric weights. All complete values and trees match Knuth and direct cost evaluation. There are 7,972 independent cubic interval checks, 8,048 all-root cell checks, 792 explicit padded-product checks, 792 all-shape truncated-layer checks, and 139,992 verified dyadic tiles. The geometric family correctly needs all 64 layers at 64 keys; no logarithmic-in-n claim is made for its exponentially large mass.

At 256 keys, the maximum interior core over all layers was 3,857 for mixed weights in $\{1, 2\}$, 6,196 for weights in $\{100, 101, 102\}$, and 5,008 for mixed weights in $\{1, \ldots, 16\}$, against 32,640 possible interior positions. These are representation counts, not runtime speedups. The audit actually computed 6,318,862 dense states and examined 12,381,033 root candidates across its full suite. It constructed 11,912 keys. Source-hashed raw data is in `penalty_layers.json`.

The lexicographic oracle matched 400 exhaustive shape-profile comparisons through eight keys. Uncapped search matched exhaustive evaluation of all allowed integer prices on 1,200 inputs. On 3,000 additional full-OBST inputs through thirty keys, both capped fallback versions matched cubic DP and direct tree costs; 87 previously unsuccessful unrestricted certificates were recovered. A separate seeded sweep recovered the $n = 1,024$, amplitude 2 input with two additional oracle calls, and the $n = 2,048$, amplitude 1 input with three. Larger tested perturbations usually exceeded the profile ceiling. This applies standard concave-price optimization to the certificate; it is not a general subquadratic result.

In the seeded 2,048-key case, the candidate costs 43,032,446; the bound rises from 43,031,347 at $\delta = 0$ to 43,032,459 at $\delta = 125$. The three extra passes use 6,144 contractions and 6,141 pair comparisons, in addition to 4,096 shape states, 6,142 transitions, 20,784 sorting comparisons, and 2,752 original merge comparisons. Knuth uses 2,098,176 interval states and 4,198,710 root candidates on that input.

Twelve full constructions reach $n = 262143$, with 262141 distinct key weights, 262144 distinct gap weights, and cost 92530765201355. The best preceding transfer bound is 91912294498251, a deficit of 618470703104. The largest run uses three alphabetic calls, 524285 compatible merges, 13894100 priority comparisons, 17365446 heap-item comparisons, and 1482337 key-sort comparisons. All scans, heap events, rebasing, and output visits are recorded; there are no interval-DP cells. Seven sizes through 511 keys match Knuth and three match cubic. Larger sizes match a separately constructed perfect tree and the closed form. Six smaller sizes additionally complete the exact endpoint price search and verify the full-class deficit independently.

Both oracles passed 50 comparisons against exhaustive length-sequence minimization, 5,000 arbitrary-multiplier bound checks against cubic OBST, and 1,050 direct comparisons of the full and shortened oracles. Forced-root marginals were additionally checked against exact root-constrained costs.

The initial probe checks 1092 alphabetic controls, 2400 mixed controls, 616500 literal tree evaluations, and 42563 pair evaluations: 2385 candidates are exact and 15 are suboptimal. Exhausting key multisets and ordered gaps over $\{0, 1\}$ and $\{1, 2\}$ through seven keys gives 7172 cases, including 14 binary failures and no $\{1, 2\}$ failures; the latter finite observation is not a theorem. There are 2036 constant-key controls, 6918 profile/depth identities, and 1421 transfer controls. All 8192 integer corners of $1000(p, q) \pm 1$ retain a gap at least 996. Padding is checked by Knuth through 256 keys (largest weight 732 bits), with cubic checks through 16.

All 512 integer corners of $100(p, q) \pm 1$ are checked; the smallest observed gap is 94, above the proved lower bound 56. With all nine weights distinct, the example

$$p = (996, 997, 1998, 999), \qquad q = (1000, 2001, 1002, 1003, 1004)$$

has full optimum 23999 and maximal transfer bound 23004. A constant supporting plane certifies that maximum in one call.

A.26 Coarse optimal families and complete fine-weight refinement

The independent audit checks 716 coarse inputs: all 680 binary mixed inputs through four keys, and 36 seeded inputs at $n = 5, 8, 12$. All 7540 interval minimum/maximum optimal-root pairs agree with a separate all-root recurrence. Independent optimal-tree counts agree with every uniqueness decision; 311 coarse optima are unique. The audit then checks 1432 complete signed-rational refinements against the original all-root optimum, original-weight tree traversal, and coarse optimum. Every fine payload cell count equals $\sum_j m_j(m_j + 1)/2$.

Six further controls exercise automatic scale selection; four exercise certificate rejection. Three zero-coarse-mass controls preserve the full optimal family, including optima taller than a balanced sufficient height. Twenty queries reuse four prepared partitions, agree with independent all-root optima, and perform no new Monge products. The record `coarse_face_audit.json` retains its seed, counts and eight executed-source hashes.

The scale audit constructs eight unstructured lattice perturbations. Coarse weights are independently drawn from $\{0, \dots, 7\}$; a large integer scale and distinct position perturbations make all $2n+1$ original weights different. Automatic scale selection accepts at $m = 7$ in each retained row. Complete construction includes both coarse orientations, the unrestricted-child closure products, partition extraction, and every fine payload. Selected exact counts are:

n	Trial	Forced keys	Largest payload	Fine DP cells	Core visits
32	0	32	0	0	553,436
32	1	32	0	0	476,696
64	0	56	3	15	1,910,066
64	1	62	2	3	1,937,568
128	0	106	11	96	6,453,968
128	1	0	128	8256	7,087,828
256	0	183	28	772	20,585,886
256	1	202	8	168	19,925,218

Fine DP cells and core visits are separate operation categories. Fewer payload cells do not imply less total work than ordinary Knuth at these sizes. In particular, the second 128-key input has global coarse ambiguity. No measured exponent or distributional bound on payload sizes is claimed.

Four ordinary mixed inputs at the same sizes reject all sixteen candidate scales and use Knuth. Three non-perfect uniform coarse inputs at $n = 32, 64, 128$ retain the entire payload table, as predicted by the explicit root-ambiguity example. Twelve further full queries reuse four perfect-size uniform preparations at $n = 31, 63, 127, 255$ and require neither fine DP cells nor new products. These uniform bases already admit direct linear construction. All 27 outputs agree with independent unrestricted Knuth and a separate original-weight tree traversal. The record `coarse_face_scale.json` retains every input, separate preparation and reference counters, all failed routes, exact accepted scales and error magnitudes, seed and eight source hashes. Bit-operation costs are separate from these arithmetic counts.

A.27 Reachable coarse-family grammar and rotation audit

The harness first compares the entire reachable interval and root-triple sets with literal optimal-tree enumeration on forty inputs through eight keys. It evaluates 10,275 literal tree costs. Twenty-seven separate perturbations select each uniform coarse optimum uniquely, checked against every competing small tree. Four exact scales verify the five-key strict rotation trap, its full optimum and all four neighboring costs. The three-key root-hole example is checked by an independent all-root recurrence.

For $B = 2, 4, 8, 16, 32$, a separate all-root computation builds the complete coarse-optimal grammar. Every interval in (188) is checked for reachability and for its entire optimal-root set. All 1,493 selected intervals also receive an explicit full-tree witness, whose original coarse cost equals the independent optimum. Selected counts are:

B	n	Selected intervals	Selected root triples
2	11	9	12
4	23	25	45
8	47	81	225
16	95	289	1377
32	191	1089	9537

For each size, a shuffled permutation of $1, \ldots, 2n+1$ gives distinct positive perturbations of all original key and gap weights. With $D = 2n \sum e_o$, the certificate holds. Independent Knuth construction and a direct full-tree cost traversal agree on all five fine inputs; each returned tree is also coarse-optimal. These are complete original-weight controls, not a new fast constructor.

The direct reference computations use 24,972 interval states and 1,349,726 root candidates. Their reachable graphs contain a total of 9,829 intervals and 47,070 root triples across the fixtures. Those are verification operations. Larger rows through $B = 2^{20}$ evaluate the proved formulas only, with no graph enumeration. The record `coarse_face_structure.json` retains the seed, all five original fine inputs, full costs, reference counts, exact scale controls and both executed-source hashes.

A.28 Adaptive coarse-grammar construction audit

The independent harness establishes the two-tree family before importing the new constructor. Direct all-root recurrences verify its complete reachable grammar at $m = 2, 4, 8, 16, 32$; literal tree enumeration checks the two smallest sizes. The persistent entry index is tested on 2,869 entries of separately generated condensed Monge matrices.

For 96 coarse inputs through twelve keys, including all-zero inputs, the prepared interval and root-triple sets equal those of an independent all-root recurrence. Fifty-six of these inputs also receive literal optimal-tree checks, evaluating 5,000 tree costs. Each preparation serves three independently perturbed rational fine inputs: all 288 full outputs match the independent original-weight optimum and remain coarse-optimal. These errors have both signs where nonnegativity permits. Ninety-six zero-budget preparations verify exact fallback, and the three-key root-hole fixture verifies that the missing middle root is excluded. Separate ordinary-input controls test automatic scale rejection. The direct reference computations total 20,900 states and 224,951 root candidates.

The two-tree scaling rows run complete generic preparation through $n = 383$. Each has the proved $V = n+1$, $A = n+2$, $Q = 4(n+1)/3$. Three full rational queries per preparation test generic refinement and the direct linear constructor, including the exact difference (96). Independent full Knuth controls cover the first six sizes, through $n = 191$; the largest row uses the proved two-tree classification and direct traversal of every original outcome. Additional perturbations deterministically select each of the two roots. All seven direct-constructor rejection controls fall back correctly.

n	Preparation core visits	Index nodes	Fine root candidates
5	6,834	47	7
11	40,522	187	13
23	174,784	632	25
47	602,090	1,827	49
95	1,893,664	4,948	97
191	5,557,466	12,965	193
383	15,476,304	32,590	385

Preparation includes both orientations, both unrestricted closure products and index construction. Core visits are one instrumentation category, not total instructions or a speed claim. Each generic fine query evaluates $n+1$ states and $n+2$ roots, checks $2n+1$ error terms and writes n keys. The direct constructor checks the same number of errors, sums $4m$ choice terms and writes the complete tree.

Three unstructured integer bases at $n = 32, 64, 128$ have respectively $V = 32, 65, 137$ states and $A = 32, 66, 143$ rules. Two rational fine queries per base match independent Knuth construction. Four additional full runs include automatic scale selection, generic preparation and output for the two-tree family. On the dense uniform control $n = 6B - 1$, the default budget finishes at $B = 2$ and exhausts at $B = 4, 8, 16$; every full output still agrees with Knuth. The retained record `coarse_grammar_audit.json` includes the seed, fine inputs, separate preparation/query counters, rejection work, verification scopes and eight executed-source hashes.

Combined-exchange control. For the nine-key example in Section 5.31, the 28 minimum-IPL shapes are checked against all 4,862 full shapes. The fourteen distinct coarse-preserving neighbors, combining every descendant promotion with one bottom-gap exchange, have error costs

$$224, 226, 227, 228, 229, 230, 230, 230, 231, 231, 234, 235, 238, 239.$$

The selected local tree has error cost 219, and the minimum among the 28 coarse optima is 212. At $D = 585$, full enumeration and Knuth give costs 34,734 and 34,727 for the candidate and optimum. Every descendant promotion, including those leaving the coarse family, is also checked directly. The source-hashed `coarse_exchange_control.json` retains the weights, candidate tree and every neighbor. This finite counterexample motivated optimizing the retained grammar directly.

A.29 Coarse-regret screening and root-range index audit

The harness first checks the signed allowance using independent all-root recurrences and minimum-height attaining trees, before importing the new constructor. It checks 540 optimal-subtree inclusions. Sixty generic finite Monge matrices of orders $n+1$, $1 \le n \le 15$, have independently generated signed boundaries and nonnegative densities. Dense minima verify 130,020 indexed root-range entries and 17,940 threshold enumerations. The hierarchy checks the per-depth $2K$ density charge, product-core bound, index-node bound, total inner dimensions and exactly $2n - 1$ range nodes.

The complete constructor audit checks 108 original-weight instances, 1,980 reached-interval root sets and, independently, all 1,980 threshold formulas using direct sums over interval outcomes. Each complete result matches an independent full optimum; 72 controls additionally evaluate 2,352 literal tree costs. Ninety-six of the 108 supplied coarse-base/scale pairs fail the earlier small-error certificate, yet screened refinement is exact. This statement concerns those supplied pairs, not every scale an automatic selector might try. Another 108 controls exhaust a zero expansion budget and return the independent Knuth optimum.

Automatic construction is checked on random mixed, near-uniform and geometric weights at $n = 16, 32, 64, 128, 256$. The first two families complete screening at each size; the geometric family exhausts the budget at each size and returns the exact fallback result. Four dense uniform-coarse-face controls complete at $n = 23$ and exhaust at $n = 47, 95, 191$. Five rational inputs, three unrelated inputs reusing one preparation, and twenty uniform or empty controls are also checked. Original-weight traversal verifies each returned full tree. The small independent recurrences evaluate 4,632 states and 15,500 root candidates.

A separate matched experiment uses positive random weights with up to 256 bits, at $n = 32, 64, 128, 256$, and compares four precisions on each identical input. Every row matches an independent full Knuth optimum. The record includes complete inputs, exact scales, original optimum costs, all failed work, and separate preparation and query counters. Table 2 reports selected counters; P is preparation core visits, Q is query range-minimum calls, and F is fallback root candidates. Dashes indicate that screened-grammar expansion was interrupted. These event categories are not a common unit of machine work, and V, A exclude preparation.

n	b	V	A	K	P	Q	F
32	1	–	–	116	172498	1685	1039
32	16	–	–	252	317835	2481	1039
32	36	127	187	262	326638	1183	0
32	216	33	34	263	328840	246	0
64	1	–	–	268	624306	4095	4034
64	16	–	–	653	1176520	5682	4034
64	49	–	–	666	1205482	7069	4034
64	343	104	112	679	1226437	988	0
128	1	–	–	654	2153460	9430	16062
128	16	–	–	1802	4749757	12460	16062
128	64	–	–	1880	4985710	17705	16062
128	512	301	333	1910	5056080	3349	0
256	1	–	–	1611	6335828	21299	66070
256	16	–	–	4805	16041939	27430	66070
256	81	–	–	5214	17327199	41114	66070
256	729	1029	1311	5269	17521921	14244	0

Table 2: Matched exact constructions for wide original weights. Precision $b = L^3$ completes screening at every tested size; smaller precisions often exhaust the counted budget. Preparation remains substantial.

At $n = 256, b = 729$, the screened grammar has 1,029 intervals and 1,311 roots, but preparation alone visits 17,521,921 core entries; the independent Knuth reference tests 66,070 root candidates. These counts establish neither a practical speedup nor an asymptotic exponent. The work bounds in Section 2.32 follow from the density-charge and regret proofs. All seventeen executed-source hash entries in the two records are checked against the retained sources. The records and reproduction commands accompany the manuscript; the theorem statements and proofs require no access to implementation files.

A.30 Implicit ordered references and nonlinear corridor audit

The interval-oracle harness establishes its claims with independent all-root tables before importing the optimized implementation. It checks 7,344 full root-cost sequences for the stated unimodality, 7,344 direct shape formulas, 756 literal interval-tree costs and 680 instances of (145). The optimized index then matches 17,424 interval values, 44,064 root-range minima and 44,064 complete threshold sets, including both orientations and partial valid-root ranges. Each threshold query also satisfies its binary-search call bound.

Full refinement checks 144 arbitrary original-weight queries, 54 literal full-tree controls and 144 zero-budget fallbacks. Two invalid coarse inputs are rejected, and three rational coarse inputs are reflected and reused for unrelated rational originals. Twenty-four larger full constructions at $n = 23, 47, 95, 191, 383, 767$ match independent Knuth. The six sparse-bump originals fail both

ordered exchange conditions, yet complete exact screening. For example, at $n = 767$ the coarse index stores 7,670 dyadic entries; the sparse-bump query reaches 959 intervals and 1,067 roots and makes 3,482 root-cost calls. The ordinary and densely perturbed failure cases retain all interrupted work before exact fallback. Preparation uses neither a Monge product nor a core list.

The nonlinear-corridor harness separately constructs literal expansion words before importing its solver. It verifies 1,638 exchange-regret inequalities and 530 optimal-prefix inclusions. The implemented class optimizer then matches 128 literal class optima and their state bounds. Full original construction passes 108 independent cubic controls, 84 literal full-tree controls, eighteen rational controls and twenty direct uniform controls. Forty-two scaling controls through $n = 1{,}535$ match independent Knuth: twenty-eight curved, irregular and reflected bounded-error cases satisfy the nonlinear-family guarantee; ordinary random and geometric cases exercise the unrestricted fallback.

Case	n	Retained states	Admitted transitions	Proved state cap
Curved, unit error	1535	26173	51320	53185
Curved, error two	1535	54437	107848	137205
Irregular increments	1535	17845	34664	33065
Reflected curve	1535	24384	47742	48635
Large curved	6143	183788	363478	372593
Large curved	24575	1007790	1999194	1965885
Large curved	98303	5295604	10525670	10026753

Table 3: Complete nonlinear-corridor constructions. Rows through $n = 1{,}535$ have independent Knuth controls. The three larger rows use the proved certificate and full original-cost traversal; no large Knuth run is asserted.

The larger curved instances use

$$f_i = i + \lfloor i^2/n \rfloor, \quad B = 2n \sum_i f_i + 20(n+1)^2,$$

with key error -1 at $i \equiv 0 \pmod 3$ and $+1$ elsewhere, and gap error -1 at odd positions and $+1$ at even positions. At $n = 98{,}303$, the fitted errors are $E_p = 32{,}767$ and $E_q = 98{,}304$, so $D = 622{,}576$. The forward affine fit from the earlier implementation has state upper bound 1,073,807,361 on the same input; this is a preflight bound, not an executed state count. The nonlinear corridor actually retains 5,295,604 states. The full Knuth table would contain 4,831,789,056 nonempty intervals, a formula-only comparison. These measurements check the implementation; the asymptotic guarantee is proved in Corollary 75.

A separate dense diagnostic covers eight residual patterns at seventeen sizes through $n = 512$, always with a baseline forcing minimum IPL on every interval. It rejects the proposed height-only final-core bound. The previous ruler theorem already gives the qualitative obstruction; the ramp observations motivate the exact maximum-support formula in Corollary 138. Dense diagnostic work and every failed route remain in the records. All three records include executed-source hashes and exact counters. The constructor and oracle bounds include fitting, preprocessing and reconstruction and exclude operand bit costs.

A.31 Range certificates and sharp box-threshold audit

The independent harness was written before the range-based constructor. It enumerates 6,917 trees through nine keys to check both shortening-capacity bounds, including 435 perfect-size gap

charges. It checks every joint key/gap box vertex at three and seven keys, as well as separate key-only and gap-only boxes: 33,304 exact all-root comparisons at the stated threshold. The explicit shifted-perfect construction is checked at fifteen sizes through $n = 65{,}535$, for three key/gap range pairs and below, at and above the threshold, giving 135 exact cost-difference identities. Fifty-four small boundary cases also have independent Knuth controls and test the full constructor's direct or fallback decision.

Full construction is compared with 576 independent cubic optima, including 96 rational cases checked by exact scaling to integer reference inputs, and 144 literal full-tree controls. Ninety-six fixtures separately satisfy the general range guarantee. Twenty-eight larger inputs through $n = 1{,}535$ are compared with independent Knuth. Fourteen curved or reflected inputs are certified by the new range condition although all fail the earlier $(n-1)E_0$ baseline gate. Ordinary random and geometric inputs retain all rejection work before exact fallback.

The larger curved family uses

$$f_i = i + \lfloor i^2/n \rfloor, \qquad B = n(f_{n-1} + 4) + 2,$$

with key errors -1 at multiples of three and $+1$ elsewhere, and gap errors -1 at odd indices and $+1$ at even indices. Three full constructions at $n = 6{,}143$, 24,575 and 98,303 are checked by the proved range certificate and direct original-weight traversal. These large rows have no quadratic-reference comparison. The corridor states and transitions are unchanged by subtracting the common baseline from the earlier curved family; all fitting, certificate and output counts are recorded separately. The certificate proof, rather than the measured state counts, supplies Corollary 79. The completed record retains exact inputs or formulas and eight executed-source hashes.

B Adversarial core search and the odd-ramp proof audit

The search fixes nonnegative integer total mass at $W = n$ and maximizes

$$\frac{\max_{0 \le h \le H} \delta(E_h)}{n\sqrt{W \lceil \log_2(n+1) \rceil}}.$$

The core includes the n finite-completion diagonal entries. All layer costs and density counts use exact integers. A whole-table fixed point, when reached, justifies repeating the same count for subsequent layers. Single-unit and larger mass transfers, swaps, and reversals preserve the mass constraint. Simulated annealing affects only search acceptance; floating point is not used to compute costs or cores.

There are 48 starts over twelve sizes from 16 through 1,024, including non-power-of-two sizes, and 24 additional concentrated starts at six sizes. They perform 145,155 candidate evaluations in total. The first batch uses spread mixed weights, key-only weights, and uniform initial keys; the second uses spaced heavy keys, spaced ramps, clustered equal weights, and random sparse positions. Each batch first checks 232 complete layer vectors against the independent recurrence, with all-root and literal-tree checks on small inputs; selected winners receive further comparisons. Search budgets, seeds, full winning vectors, all layer counts, and operation counters are recorded. The table reports the best result at each displayed size across these starts.

$n = W$	Largest completed core	Normalized ratio
16	69	0.482152
32	185	0.417226
64	463	0.341792
128	1162	0.283691
256	2724	0.221680
512	6470	0.176604
1024	14582	0.134175

A separate diagnostic evaluates eight explicit patterns at fifteen sizes, for 120 complete layer vectors through 4,096 keys. These include a contiguous ramp, a contiguous random block, a spaced ramp, isolated key and gap mass, binary blocks, ruler-weight sampling, and random unit placements. The contiguous ramp gives 13,461, 36,276, and 98,168 completed core entries at 1,024, 2,048, and 4,096 keys, respectively. These measurements motivated the odd-ramp family with exact mass $W = n$ in Theorem 6. The proof uses the separately specified odd weights, not a fitted formula for the diagnostic's initial ramp.

The decreasing ratios in the first table alone would have been misleading as evidence for a universal near-linear bound. The actual conclusion follows from the theorem's explicit density coordinates. Search supplies candidate patterns and opportunities for falsification; it supplies neither an upper bound nor an asymptotic tightness certificate.

Independent proof controls. The new mathematical audit checks 2,652 straight and 2,652 bent rotation identities, 894 dyadic partitions through $n = 129$, and 4,277 scale bounds. The odd-ramp audit compares its subsequence recurrence with 445 literal subset enumerations and checks 388 perfect-slot embeddings against an all-root interval recurrence. There are 42,748 independent checks of the rightmost-zero formula and 17,688 checks of the closed-form density against subsequence DP. At $n = 1024$, all 1,896 cells in the proved set also match the original penalty-layer recurrence. That original-layer audit and the smaller controls evaluate 4,736,120 dense states and 8,852,066 root candidates. They are diagnostics, not operations of the sparse construction algorithm.

$n = W$	Proved positive cells	Subsequence DP	Original-layer DP
1024	1896	checked	checked
4096	15792	checked	—
16384	128736	formula only	—
65536	1039296	formula only	—

The last two rows evaluate the proved coordinate count and densities; they are not additional independent full-layer computations. The lower bound holds for every $n = 2^{2t}$, $t \geq 5$, by its proof. These records are in `core_size_search*.json`, `core_size_patterns.json`, `core_size_lemmas.json`, and `ramp_core_tightness.json`. Each records hashes of the exact sources used.

C Local tile charging: probe and exact one-key audit

The tile-mass probe evaluates mixed random, sparse, uniform, geometric, and concentrated inputs, followed by fixed-mass local mutations. Its objective is the ratio of a maximal tile's density mass to the mass of its own span. The 1199 evaluations include 123 independent all-root controls. They examine 19,699,184 dense states, 38,539,843 root candidates, and 1,169,332 tiles. The largest

observed ratio is four at $n = 256$, $h = 7$, with one key carrying all the weight. Scaling that weight to one leads to Theorem 142; the search alone does not establish the unbounded ratio.

The separate formula audit checks every interval value and every interior density of the explicit family through $n = 1024$. It uses all-root recurrences through $n = 64$ and monotone-root recurrences at larger sizes. There are 700,024 interval-formula checks, 697,992 density-formula checks, and 1023 penalty-line checks. The exact completed core sizes are:

n	h	Completed core	Tile support	Tile mass
16	3	33	0	0
32	4	71	1	1
64	5	149	2	2
128	6	307	3	3
256	7	625	4	4
512	8	1263	5	5
1024	9	2541	6	6

The general one-key corollary is additionally checked on 1316 cases: every key position through $n = 16$, two positive masses, and all tested layers, using literal all-root recurrence. These contribute 88,170 additional density checks. Together the formula harness records 6,387,678 dense diagnostic states and 12,975,174 root candidates. Larger rows through $h = 30$ evaluate only the proved formulas and are explicitly labeled as such. They are not independent dense solves.

The executed sources and exact operation records are retained in `dyadic_mass_probe.json` and `single_key_tile.json`. The proofs in Section 5.9 establish the asymptotic local obstruction and global linear bound independently of these finite checks.

D Independent mixed-slot construction audit

The reconstruction procedure is checked on all 32,902 nonempty slot subsets of perfect trees through height four. Every surviving key has comparison length no larger than its selected slot's length, and every gap satisfies the neighboring-key identity.

The constructor has 717 independently checked complete solves, including 473 literal slot-minimum and literal tree-minimum controls. Its independent height recurrence examines 558,638 interval states and 15,547,880 root candidates, including the smaller scale controls. The positive four-key example in Section 2.8 has full optimum 121; literal minimum-internal-path-length enumeration gives 218. The small constructor runs record 82,568 slot states, 145,171 suffix-level updates, and 69,863 lower-level scans. Preprocessing, traceback, array initialization, node deletion, and output have separate counters.

The complementary above-capacity formula is checked on 23,920 intervals against full mixed penalty layers, and 18,000 ordinary densities agree with their zero-gap counterparts. Those independent layer computations examine 184,056 states and 1,300,792 root candidates.

There are 53 scale rows through perfect capacity 4095, using positive mixed weights with baselines 2^{60} on keys and 2^{61} on gaps. Twenty-three smaller rows also use the independent all-root recurrence; the larger rows check reconstructed height, direct original cost, and the slot objective, with their independent-reference flag set to false. At height twelve the operation counts are:

n	r	Slot states	Suffix updates	Lower-level scans
4095	0	4095	8177	4083
4094	1	8188	16351	8165
4093	2	12279	24522	12246
4091	4	20455	40852	20402
4087	8	36783	73468	36694
4079	16	69343	138516	69190

Each row tabulates 4095 valuations, builds 4095 perfect-tree nodes, deletes exactly r nodes, and outputs all n original keys. The stated asymptotic bound follows from the valuation-sum proof, independently of these rows. The executed source hashes and operation records are in `mixed_slots.json`.

E Binary-weight search and exact gap-core audit

The binary diagnostic performs 84 searches: seven power-of-two sizes, three placements of positive weights, and four initial patterns (random, alternating, centered, and prefix). Mutations exchange a zero and a one; occasional reversals preserve that invariant. Mixed runs have mass n; key-only and gap-only runs have mass $n/2$. The objective is the maximum completed core across all terminal-penalty layers, including the completion diagonal. Whole-table fixed points allow identical remaining layers to be reported without recomputation.

The search evaluates 117,084 candidates, recording 40,476,345,560 dense states, 80,519,888,452 root candidates, and 40,301,399,592 density evaluations. These are diagnostic operations, not sparse-constructor costs. An independent Python reference checks 81 vectors, including every winning vector through $n = 128$; its small cases include all-root and exhaustive-shape controls. The largest value among the four starts in each placement is:

n	mixed	keys only	gaps only	proved gap upper bound
16	69	50	44	44
32	182	137	100	100
64	465	341	228	228
128	1151	861	516	516
256	2762	2118	1156	1156
512	6473	5032	2564	2564
1024	14701	11571	5635	5636

The last column follows from Theorem 12; it is an upper bound over every fixed-mass binary gap input and all layers. The mixed and key-only columns support no asymptotic assertion.

A separate all-root recurrence checks 2,189 inputs: all binary gap vectors through eight keys, all vectors over $\{0, 1, 3\}$ through five keys, seeded nonbinary vectors, four search winners, and four explicit alternating family members. It checks 5,282 mass identities, 32,899 one-missing-slot formulas, 22,335 capacity densities, 157,129 vanishing above-capacity densities, and 3,976 whole-triangle identities. The explicit family contributes 10,524 independently evaluated density formulas. Total reference work is 492,579 states and 8,892,512 root candidates. The binary bound is checked on 60 eligible layers. Exact weights, seeds, operation counts, and source hashes are retained in `binary_core_search.json` and `gap_core_identity.json`.

F Mixed boundary-mass audit

An independent all-root recurrence checks 308 inputs: every mixed binary vector through three keys, seeded integer vectors through sixteen keys, and alternating families through $n = 128$ with four baseline pairs. The audit checks 756 mixed mass identities, 6,656 one-missing-slot formulas, 3,376 densities immediately below capacity, 136,680 above-capacity densities, and 80 closed family mass formulas. Reference work is 369,396 states and 13,048,220 root candidates. The source hashes and exact family rows are in `mixed_boundary_mass.json`. These checks accompany the proofs in Section 5.10; the asymptotic conclusions follow from the displayed identities.

G Capacity-band construction audit

The independent all-root reference covers 272 mixed inputs: exhaustive binary weights through three keys, seeded integer vectors, and geometric weights through 32 keys. Across widths $0, 1, 2, 4, 8, n$ where distinct, it checks 253,303 retained values, 1,287 complete reconstructed trees, and 4,100 per-layer candidate bounds. Reference work is 94,179 states and 924,277 root candidates. The band solver uses 199,627 states and 366,637 root candidates, with initialization recorded separately.

Sixteen scale rows use near-uniform strictly positive key and gap weights, heights six through twelve, and widths $0, 1, 4, 16$. Each full instance has $n = 2^h - 1 - d$, so positive widths include nonperfect sizes. These larger rows verify reconstructed original-weight cost and operation counters; their independent-reference flags are false. At height twelve:

n	d	base entries	band states	root candidates
4095	0	4096	40974	40974
4094	1	8189	118792	196622
4091	4	20450	343803	702361
4079	16	69224	1174685	2621548

Each row additionally scans n input positions and outputs all n keys. These counts cover all retained intervals across all twelve layers, rather than one slot-DP instance. The asymptotic bound follows from band closure and telescoping root ranges in Theorem 13. Exact rows and executed-source hashes are in `capacity_band.json`.

H Capacity interaction audit

The independent all-root audit includes every mixed binary vector through five keys, seeded integer inputs through 32 keys, and unit-input controls. It checks 28,259 capacity identities, 6,416 telescoping totals, 5,856 subadditivity bounds, and 5,856 width-one interaction inequalities. Its 3,091 vectors use 371,731 states and 3,302,476 candidate roots. It also retains the four-key wider-band counterexample from Section 5.11.

The separate sharp-family audit checks heights two through six, including 28 independently solved unit inputs at the smaller sizes. It verifies five alternating-family formulas, three direct interaction differences, and five uniform-key wider-tail formulas, using 194,152 states and 6,131,120 root candidates. Larger rows through height 128 are closed-form evaluations only. These records are `capacity_interaction.json` and `capacity_interaction_sharp.json`.

An initial optimized dense diagnostic investigated the proposed all-width allowance, maximizing $[T_d(w) - \sum_x w_x T_d(e_x)]/(hW)$ over every band width and early layer. It uses 15 fixed-mass searches

through 256 keys, with 1,200 mutations each and three starting patterns per size. Structured inputs include uniform keys, uniform gaps, mixed weights, alternating keys, odd ramps, and seeded random weights. Independent all-root controls check 150 optimized vectors. Including unit-profile construction, structured evaluation uses 157,388,626 states and 313,030,555 root candidates; search uses 1,667,827,872 states and 3,310,832,256 candidates. The largest observed ratio is 1223/1536, attained by uniform keys. The sparse family in Theorem 146 subsequently disproves this allowance. Exact weights, seeds, counters, and executed-source hashes are in `core_interaction_search.json`.

I Sparse interaction falsification and formula audit

The one-sided all-root diagnostic checks 2,956 vectors and 91,370 inequalities, using 1,770,275 states and 28,876,151 root candidates. It finds the ten-key above-capacity counterexample in Section 5.11.

The subsequent sparse diagnostic evaluates 19,614 inputs, including every unit pair on keys and gaps through 64 keys, structured unequal-weight pairs, and sparse random inputs through 512 keys. Its optimized recurrence includes unit-profile construction: 4,587,178,568 states, 9,062,247,119 root candidates, and 19,197,336 band evaluations. Independent all-root controls verify the four retained winners through 64 keys. The observed ratios exceed one and grow with input size; these finite violations motivated the explicit family of Theorem 146.

The family audit checks the combined input and both unit inputs against independent all-root recurrences through height seven. It verifies 123,585 eligible density cells, 920 band entries, and three complete family identities, using 870,624 states and 66,536,256 root candidates. Literal rank choices verify 32,424 local table entries through height twelve, with 415,042 feasible rank pairs counted. Larger rows through height 128 evaluate only the proved closed forms. These are separately labeled in `sparse_interaction_barrier.json`. The other records are `one_sided_interaction.json` and `sparse_interaction_search.json`; all retain executed-source hashes.

J Rank-reward stability and expanded tail diagnostics

The rank-template audit checks 432 small inputs. Canonical maximizing sequences are independently compared with 27,717 literal subsets over 3777 intervals. It verifies 6817 coordinate monotonicity inequalities and 15,930 single-rank perturbation bounds. Ruler clipping is checked on 75 larger fixtures through 256 keys, giving 45,840 inequalities; 540 upper densities are compared with independent all-root mixed layers. The generic subsequence work totals 286,337,285 matching candidates and 1,350,475 stored interval scores. The record is `rank_reward_tail.json`.

The periodic obstruction has a separate audit through $h = 7$. It checks 491 boundary entries, 20 family identities, and 11,036 upper density cells using 2,429,031 subsequence candidates. At $h = 3$, 48,620 literal subsets independently check all 55 eligible interval scores. Larger rows through $h = 128$ evaluate only the proved closed form. These modes are labeled in `periodic_tail_barrier.json`.

The original total-tail target $T_{h,d} = O(W(L + c_h/d))$ is tested separately. A sparse diagnostic evaluates 23,183 inputs through 1024 keys, including all unit pairs on keys and gaps through 64 keys, unequal pairs, phase-changing combs, ramps, and random small supports. Its largest exact ratio to $W(L + c_h/d)$ is 485/168. The optimized recurrence records 17,157,958,100 states, 33,808,959,382 root candidates, and 47,383,400 band evaluations. Five retained winners through 128 keys have independent all-root controls. A second diagnostic evaluates 462 repeated sparse clusters through 2048 keys; its maximum is 2828/1107, with 3,107,692,416 states, 5,649,382,012 root candidates,

and 3,997,440 band evaluations. Its two small retained winners also have independent all-root controls. These finite maxima do not prove an upper bound. The records `sparse_tail_search.json` and `cluster_tail_search.json` retain exact weights, objectives, operation counts, and executed-source hashes.

K Alternating-tail matching and sharp-scale audit

The matching audit compares generic rank-subsequence scores with an independent cardinality-matching recurrence on 2887 inputs. It checks 147,191 endpoint inequalities and constructs 41,517 augmenting paths in 18,703 packings, verifying every resulting matching, removed weight, and repair inequality. There are 58,142 global tail-bound checks. The inputs include all binary vectors through nine keys at every admissible odd capacity, and random integer weights through 128 keys. Sixteen cluster instances through $C = 128$ check (25), 1045 endpoint formulas, and 32 terminal scores. Eleven spaced-maximum families check the additive scale. Larger cluster rows through $h = 128$ use only the proved formula. The record is `alternating_tail_bound.json`.

A separate compiled probe tests the wider clipped-ruler candidate $U_d/[P(t + (c + 1)/d)]$. It evaluates 25,683 inputs and records 84,957,303 interval scores, 4,737,828,651 candidate transitions, and 82,044,166 upper densities. All 451 small control vectors agree with the generic subsequence recurrence. Its maximum observed ratio is 321/292, on a height-seven template with two unit keys. This disproves coefficient one in that inequality but supplies no asymptotic upper estimate. The exact weights, seed, counts, and source hashes are retained in `periodic_tail_probe.json`.

L Rank-weighted repair certificates and exact barriers

The paired canonical diagnostics each evaluate 7225 inputs, including all ternary interior vectors on ten positions and structured, random, and sparse inputs through 64 positions. Each run has 6976 independent generic-subsequence controls, 224,622 endpoint checks, and 26,402 global certificate checks. The leftmost run explicitly constructs 1,385,304 repair paths and finds 13,185 losses strictly above the true loss; the rightmost run constructs 546,117 paths and finds 7087 strict losses. Every path is checked for feasibility, cardinality, and its full signed retained-rank accounting. The records are `rank_matching_repair.json` and `rightmost_rank_repair.json`.

The separate exact-family audit checks 3786 zero-loss equivalences on 306 inputs, and 16 instances of Lemma 149, including weights above 2^{64}, against 160 literally enumerated old matchings and independent shortened scores. For Theorem 150, heights three through ten give 4088 explicit rightmost boundary witnesses and repair-loss formulas, 225 independent scalar-DP witness controls, and eight complete certificate formulas. Every optimal starting matching is enumerated at both height-three thresholds and the largest height-four threshold, giving 1023 checks of tie-independent loss; the two height-three optimum sets are also checked by literal enumeration without a forced-prefix assumption. There are 32,352 explicitly checked repair paths in this family audit.

Full generic subsequence tables independently give zero true tails through height five. Eight explicit constructions check the exact component counts of Corollary 151. Larger rows at heights 16, 32, 64, and 128 evaluate only the proved formulas. The independent scalar controls use 41,799,787 transitions; the generic controls use 225,793 candidate transitions and 6637 scores. Exact inputs, operation counts, and executed-source hashes are retained in `rank_repair_barrier.json`. The finite runs audit the proofs; the unbounded ratios follow from the displayed families.

M Weighted cluster tails and input sensitivity diagnostics

For integer weights and an integer template in $[0, t]$, a right input-key derivative is computed exactly by maximizing the pair (ordinary score, reward at the perturbed key) lexicographically in each interval. An independent evaluation uses

$$D_x U_d(p) = U_d((t+1)p + e_x) - (t+1)U_d(p).$$

A nonzero primary-score gap is at least one; scaling by $t+1$ prevents any slope reward, at most t, from closing it. Among primary maximizers, the perturbation selects the largest slope. Taking the signed tail sum preserves the identity. No finite-difference rounding is involved.

The weighted-cluster audit compares generic subsequence and independent cardinality-matching tables on 81 inputs, including all ternary vectors on a three-key cluster, unequal weights, zero minima, and ties. It checks 3652 endpoint formulas, 162 terminal scores, 169 coordinate derivatives, and 47 joint increments. The generic recurrence uses 44,724,022 candidate transitions and 308,926 scores; the matching recurrence uses 28,460,181 transitions and 187,123 scores. Larger formula-only rows are explicitly labeled. The record is `weighted_cluster_tail.json`.

A separate full-ruler diagnostic checks 21,020 fixed-mass inputs through 256 positions and records 3,119,522,724 candidate transitions, 20,512,788 scores, and 19,108,544 density derivatives. All 29 independent scaled perturbation controls agree with its lexicographic score/slope recurrence. The maximum derivative ratio to $h - 1 + 2^h/d$ is 416/129, on 128 positions. This finite value implies no asymptotic bound or obstruction for the full ruler. A preliminary canonical-witness probe checks 5616 Euler identities on 608 inputs and computes 11 exact directional controls. At ties its canonical coefficients can differ from right derivatives; these quantities remain distinct in the records `key_sensitivity_search.json` and `key_tail_sensitivity.json`. All records retain exact inputs, operation counts, seeds, and executed-source hashes.

N Threshold falsification and boundary-tile diagnostics

The nested-threshold probe evaluates 14,432 inputs and 35,722 width comparisons, including exhaustive ternary interior weights on ten positions. It records 1852 upper-relation violations and 4003 reverse-relation violations. The counterexample in Lemma 153 is separately checked by literal enumeration of 876 subsequences for 36 interval scores, comparison with the generic subsequence recurrence, and three original all-root terminal-layer controls. The records are `nested_tail_probe.json` and `nested_tail_example.json`.

The broad boundary-plus-mass diagnostic evaluates 21,436 inputs, with structured inputs through 1024 keys, non-power-of-two sizes, sparse unit outcomes, and fixed-mass mutations over keys, gaps, and mixed outcomes. It scans 87,706,185 accepted tiles. Its largest observed ratio $K(T)/(s+W(J_T))$ is 121/136, on a gap-only input. All 93 small independent all-root controls agree with its layerwise maxima; these controls enumerate 15,301 tiles independently. The compiled run records 3,240,243,236 states, 6,584,091,125 root candidates, and 3,217,996,956 density evaluations. The record is `tile_boundary_probe.json`.

The maximum-over-all-tiles objective often settles on a small tile. A second search therefore restricts its objective to capacity-adjacent tiles with $s \geq 4$ and $c \geq 4s$, preserving total mass $n/4$ or n. It evaluates 37,224 inputs through 512 keys and visits 212,592,240 tiles; the largest eligible ratio is 11/14. Six retained small winner tiles agree with independent all-root layers. It records 8,185,066,752 states, 16,269,275,134 root candidates, and 8,131,873,536 density evaluations. The record is `tile_capacity_support.json`. Both searches are finite falsification diagnostics; neither proves (191).

The one-key square audit checks 1,038,774 maximal squares over all key positions at selected sizes through 128 keys, including non-power-of-two inputs. It also checks 1300 arbitrary squares, including non-dyadic sides, and compares 155 full support sets against independent all-root layers. The proof of Lemma 178 uses only the ruler-pair classification and interval geometry. The record is `one_key_square.json`. All records retain exact inputs, operation counts, seeds where used, and executed-source hashes.

O Full-ruler clipping sharpness and directed support searches

The audit of Theorem 18 enumerates feasible ranks of the two positive keys on 16 parameter pairs, through 1024 keys. It checks 3544 compatible score decompositions, 3544 coarse score mappings, and 16 parity sums. Twenty generic subsequence controls independently check full and clipped tails through 128 keys, using 2,370,068 score transitions and 20,424 densities. Twelve original all-root controls through 64 keys include nonzero gap weights and a key-weight scaling factor of seven; they use 1,473,440 root candidates and 71,664 states. Literal enumeration checks 11,520 subsequences for 18 boundary scores.

The sparse-rank boundary counts in Lemma 154 have 28 independent two-key rank controls against the generic subsequence diagnostic, plus 840 literal subsequences for 14 scores on the smallest instance. Across both families, the rank audit uses 174,756 one-rank candidates and 433,864 two-rank candidates for 14,276 scores. Larger formula-only rows are marked separately. The records are `ruler_clipping_sharp.json` and `spaced_rank_tail.json`.

A preceding joint input/template search evaluates 22,212 inputs through 128 positions. Its largest observed ratio to $P(\max v + \sum v/d)$ is 16/17. The explicit family in Lemma 154 instead has unbounded ratio. The search records 865,917,888 score transitions and 9,089,940 densities; three retained small tails have 34,560 literal-subsequence checks for 54 scores. The record is `template_tail_probe.json`.

A separate fixed-side, variable-integer-mass search attacks (191). It evaluates 40,557 inputs through 256 keys and visits 121,969,824 accepted tiles. Its maximum observed ratio is 136/149, at side 32 with 136 positive entries and local mass 117. Fifteen retained small winner tiles agree with independent all-root layers, using 2,266,528 candidates and 109,936 states. The compiled search records 2,806,155,040 states, 5,480,712,131 root candidates, and 2,774,094,048 density evaluations. These finite tests leave the local support hypothesis open. The record is `tile_support_frontier.json`. All records retain exact inputs, operation counts, and executed-source hashes; seeded searches also retain their seeds.

P Unit-support geometry and off-envelope interactions

The single-gap density formula has 274,066 independent ordinary-density controls, using 3,079,878 all-root candidates and 307,625 states. The geometric audit checks 78,240 maximal dyadic tiles, 688 literal envelope counts, 248 single-key support sets, and 1222 arbitrary squares against the weaker $8s$ bound. The largest observed envelope-to-side ratio is 511/256, consistent with the proved $2s-1$ family. The record is `unit_support_envelope.json`.

A directed two-outcome search excludes every envelope cell from its support objective. It evaluates 819 inputs through 2048 keys and scans 22,696,956 tiles. Its counts are 3,760,248,528 states, 7,001,021,069 root candidates, and 3,755,113,776 density evaluations. Six retained small winner tiles agree with independent all-root layers, using 718,432 candidates and 35,984 states.

This search suggested the explicit family; the asymptotic obstruction follows from Theorem 155. The record is `interaction_support_probe.json`.

The explicit-family audit checks $s = 8, 16, \ldots, 1024$, hence $n = 64, 128, \ldots, 8192$. It verifies 36 density-one witnesses and 144 four-corner score formulas by independent feasible-rank enumeration, using 23,808 single-rank and 2,090,940 pair-rank candidates. At $n \leq 128$, twelve scores also agree with a generic subsequence recurrence, using 302,446 transitions and 2806 score entries; three densities agree with original all-root layers, using 2,375,360 root candidates and 59,936 states. Larger cases use feasible-rank enumeration, rather than a dense all-root computation. The record is `unit_envelope_obstruction.json`. All three records retain executed-source SHA-256 manifests. Verification counts do not enter the proofs of the gap formula, geometric envelope, or interaction family.

Q Pair amplification and exact binary dilation

The fixed-tile amplification search evaluates 2470 inputs through 1024 keys, including translated pairs, shared endpoints, unequal weights, and local integer-weight mutations. It records 1,858,339,360 states, 3,315,818,765 root candidates, and 6,790,720 density evaluations. Six retained tiles have independent all-root controls using 7,126,080 candidates and 179,808 states. The largest observed ratio $K(T)/(s + W(J_T))$ is 12/25; no upper bound follows from this finite maximum. The record is `pair_amplification_probe.json`.

The independent binary-dilation audit checks saturation on 1536 cases using 21,588 literal subsequences, with arbitrary nonnegative rank rewards. The two scope counterexamples use twelve additional literal subsequences. Forty dilation instances give 6142 score controls and 4392 density controls from separate generic subsequence recurrences, using 416,654 transitions and 8958 score entries. Original all-root layers with arbitrary nonzero gap weights check 2286 entries and 1434 densities, using 398,656 root candidates and 38,304 states. All 42 positive controlled fine densities are outside the unit-outcome envelope. Rounded endpoints inside zero blocks are included. The record is `binary_dilation_audit.json`.

A larger original-layer audit checks 27 replicated pair, alternating, and three-block patterns through 1024 keys. Every instance preserves the coarse tile's core cardinality and density mass; for $d > 1$, every positive fine density is off-envelope. This run uses 22,874,208 states, 40,616,815 root candidates, and 84,672 density evaluations. It uses monotone roots; the preceding audit supplies independent all-root controls at smaller sizes. The record is `binary_dilation_family.json`. All three records retain executed-source SHA-256 manifests. The exact identity and support transport follow from Section 5.19, independently of these finite checks.

R Plateau cores and rank-threshold compatibility

The endpoint-cut search evaluates 1960 inputs through 1024 keys, including replicated coarse patterns, unequal blocks, positive endpoint fringes, and integer-weight mutations. It records 1,707,458,304 states, 3,241,769,152 root candidates, and 6,204,928 density evaluations. Fourteen retained tiles have independent all-root controls using 16,627,520 candidates and 419,552 states. The largest observed ratio $K(T)/(s + W(J_T))$ is 20/37. This finite maximum proves no upper bound. The record is `endpoint_amplification_probe.json`.

The one-block audit checks 5561 complete above-capacity support sets against the original all-root recurrence, including nonzero gap weights and positive endpoint cuts, through 64 keys. Its 1,143,360 density controls include 635,883 positive-endpoint zero identities. The original recurrence

uses 33,319,148 root candidates and 2,723,869 states. Independent literal visibility enumeration checks 193,533 pairs; 23,375 block rewards agree with both direct ruler sums and (26). Constructed support sets include the exact values, not only cardinalities. The local square corollary has 50,032 controls, exhaustive through twelve keys and sampled at larger sizes. Twelve larger sharp-family constructions reach 32,768 keys; these use the linear constructor rather than the original recurrence. At the largest size it computes 16,383 ruler values, performs 16,382 sliding updates, pushes 16,383 and pops 16,369 stack indices, considers 32,738 pairs, and outputs 16,356 densities. The record is `plateau_core_audit.json`.

The complementary below-capacity construction. The independent audit covers 1025 block inputs: every contiguous block through 17 keys, selected cases through 64 keys, rational common weights, and heights through twice the binary length of n. It compares 8812 complete cores with the original all-root recurrence, checking 1,070,212 ordinary densities, 88,033 top-sum interval values, and 222,434 local tiles. The top-sum identity also has 336 arbitrary-template controls with literal subset enumeration. The original references use 18,032,986 root candidates and 1,556,996 states. Sixteen larger sparse constructions reach 65,536 keys and enumerate every layer through twice the binary length. Their checks use the explicit coordinate identities, without a large original all-root reference. Counts and final source hashes are in `plateau_below_audit.json`.

The separate fixed-tile search maximizes $K(T) - W(J_T)$, using variable integer mass with key-only, gap-only, and mixed mutations on both sides of capacity. It evaluates 34,692 inputs through 640 keys, using 22,971,840 tile-density checks and 11,439,157,710 optimized root candidates. Twenty-seven small winners agree with independent all-root layers, using 9,000,132 candidates. The maximum observed normalized excess is $39/64$, from $K = 63$, $W = 24$, $s = 64$. The record `tile_excess_probe.json` retains full winners, counts, seed, and source hashes. These finite results do not prove the proposed $K = O(s + W)$ bound; Theorem 26 has an independent proof.

The rank-threshold discovery probe evaluates 65,454 inputs using 15,176,940 subsequence transitions. Its first obstruction has 22 keys, full reward seven, and threshold optima $(5, 2, 1)$. Independent literal enumeration checks all 170,544 length-15 subsequences of that input. The explicit family in Theorem 158 and the smaller example are checked by all 32,752 single deletions through 16,384 keys, evaluating 425,838 retained positive ranks and 5,176,333 threshold comparisons. For $n \leq 128$, 22 independent generic subsequence controls use 145,386 transitions and 66 score entries; four original all-root controls with nonzero gaps use 2,812,064 root candidates and 73,456 states. The larger rows enumerate every deletion and are not original all-root runs. The record is `rank_threshold_obstruction.json`, which also hashes the raw search artifact. All three audit records retain executed-source SHA-256 manifests. The theorems follow from their proofs independently of these finite checks.

S Finite band products and complete two-block cores

The generic band audit checks 1332 inputs, including exhaustive binary sequences through $N = 4$ and signed integer sequences through $N = 16$. It compares 28,024 finite product entries against 2,098,072 literal pair choices, including infeasible completion entries, and checks 23,821 feasible rewards. A further 3996 coarsened-core and 11,808 coarsened-product entry controls test the local aggregation in Lemma 46. These checks test exact values and the finite penalties, not only the support constants.

The interior audit checks 576 support sets on 168 mixed inputs through 32 keys against original all-root layers, with 5690 densities and 672 literal subsequence intervals from 20,517 choices. Its original recurrence uses 1,345,248 root candidates and 146,592 states. Twenty-four larger condensed constructions reach 16,384 keys, with 661 additional square controls. Those rows are constructor checks rather than original all-root runs. The record is `band_dominance_audit.json`.

The complete audit checks 952 mixed inputs through 64 keys, including all 792 binary words with exactly two positive runs through ten keys and 160 seeded larger inputs with common positive key weight three. Nonzero gap weights are included. All 2953 complete above-capacity support sets and 334,032 densities agree with the independent all-root recurrence, which uses 12,507,215 root candidates and 784,506 states. Fourteen larger full constructions reach 8192 keys and include 864 local square controls. For two unit keys at that size, the constructor evaluates 31,017 distinct density candidates and outputs 3219 positive densities. It performs 26,527 range queries and 15,631 condensed-matrix value queries; the record separately counts the reused sparse-product operations. These larger rows do not have original all-root references. The record is `two_block_core_audit.json`. Both records retain executed-source SHA-256 manifests. The bounds follow from the finite product, visibility, and endpoint-case proofs, independently of the finite audits.

T Finite rank transfers and dense-core audit

The audit exhausts 363 inputs over key weights $\{0, 1, 3\}$ through five keys. It compares 23,232 finite transfer entries, including infeasible completion entries, with independent dense products using 459,456 candidate additions. All 11,385 feasible entries also agree with a separate subsequence DP and 52,245 literal subsequences. Thirty-two rational-weight inputs add 1048 exact entry controls.

For 70 mixed inputs through 32 keys, including geometric, increasing, near-uniform, constant, and random key weights with nonzero gap weights, 1172 original all-root layer scores agree with (160). Separately, 32 many-block inputs with varied positive-block lengths have 128 original-layer support-set comparisons and 3544 interior density controls. Together these original all-root controls use 987,840 root candidates and 104,460 states.

One hundred chains of signed unary functions supply 3850 feasible-score controls against an independent state DP. The explicit width-one family in Theorem 163 has 1493 formula entries checked against both that DP and condensed products, and 341 density-one square cells. Six weighted-ramp constructions through $c = 255$ check 33,138 transfer formula entries and 32,136 densities from Theorem 161. The four smaller rows, through $c = 63$, also have 2130 independent subsequence-DP entry controls. The two larger rows use the condensed product and the closed form, without that additional DP control. At $c = 255$, $r = 128$, the strict feasible strip has 12,097 positive entries and the side-64 square has 2048. The product has 127 chain multiplications, whose total input-core count is 206,530; its internal multiplication operations are recorded separately.

The record is `rank_transfer_audit.json`. It preserves exact operation counts and executed-source SHA-256 manifests. These finite checks verify the implementations and identities at the recorded sizes; the dense families and construction bounds follow from the proofs.

Slot-window products and full prescribed-height trees. The independent slot-window audit exhausts 363 inputs over $\{0, 1, 3\}$ through five keys. All 23,232 finite product entries, including infeasible completion entries, agree with separately evaluated dense products using 459,456 candidate additions. The 2946 feasible entries also agree with an independent weighted-subsequence DP and 20,340 literal subsequences. Twenty-five rational single-run cases, including runs longer than capacity, check another 6820 finite factor entries.

One hundred mixed inputs through 24 keys use constant internal gaps, arbitrary endpoint gaps, and uniform, geometric, zero, or random keys. The complete constructor agrees with 200 original all-root height optima; every output tree has independently checked inorder, height, and original mixed cost. The references use 235,160 root candidates and 34,270 states. The audit also checks 220 insufficient-capacity rejections and the explicit rejection of unequal internal gaps.

Nine checkerboard squares through side 512 verify 349,524 four-corner transition densities against (158). Eight larger uniform or three-run constructions through 8192 keys include the sparse product, witness traceback, full tree repair, and independent output-tree cost and height. Uniform rows additionally match the closed minimum-IPL optimum; the larger three-run rows have no independent dense optimum reference. At 8192 keys and height 14, the three-run construction has 61,431 input core entries, two run-product multiplications, and a final core of 20,147 entries. Leaf traceback scans the 16,383 slots twice in total, once for buckets and once for output. The record `slot_run_audit.json` contains exact operation counts and seven executed-source hashes. The theoretical bounds and signed obstruction follow from the proofs above.

Vector construction with arbitrary mixed gaps. The subsequent audit exhausts all 680 mixed binary inputs through four keys. Literal enumeration checks 7880 tree-cost normalization

identities. Independent prefix enumeration checks 12,496 state values and their attaining predecessors using 53,824 selected-slot sequences. Forty-eight finite forced-end factors supply 15,600 entry comparisons with the prior condensed window factor and 14,256 four-corner Monge controls, including infeasible entries and rational weights.

Another 84 mixed inputs through 32 keys include geometric weights, unequal gaps, rational weights, long key runs and isolated gap changes. Together the two input groups give 1528 comparisons with the original all-root height optimum, each with independent output-tree order, height and original mixed cost, plus 1184 insufficient-capacity checks. The reference recurrence uses 828,560 root candidates and 107,736 states.

Twelve larger complete constructions through 8192 keys cover uniform keys, three key runs, and three key runs with exceptional internal and endpoint gaps. Uniform rows additionally match the closed minimum-IPL optimum; the other larger rows have output-tree checks but no independent large optimum oracle. At 8192 keys and height 14, the case with three exceptional internal gaps has five transformed key runs, six long-run passes and five singleton passes. It makes 752,788 finite run-entry queries, processes 61,437 singleton states, and stores 180,224 predecessor slots. The record `slot_vector_audit.json` retains the separate exact counters and eleven executed-source hashes. SMAWK is the published primitive cited in Theorem 30; the full construction bound follows from its finite Monge interface and the valuation-sum amortization.

Sufficient unrestricted height and full vector construction. The height audit exhausts 22,140 mixed inputs over $\{0, 1, 9\}$ through four keys. A further 72 inputs through eight keys bring the literal shape count to 323,604, including 30,955 optimal shapes. The audit computes μ and z independently from every interval. It checks 152,072 sorted marked-set cuts and 790,245 positive-subtree depth assertions across the optimal shapes. Separately, 287,760 arbitrary marked subsets through three keys check the disjoint-charge bound without choosing the largest weights.

There are 6454 complete constructor controls against shape or original all-root optima, including rational scaling, geometric weights, isolated positive outcomes and zero-mass intervals; three empty inputs are checked separately. The all-root references use 1,357,542 root candidates. Twelve additional full vector constructions through 384 keys use three key runs and moderate or arbitrarily large exceptional key/gap weights. Every one matches an independent ordinary Knuth optimum and has its original tree cost, order and height checked. These twelve runs deliberately force the vector kernel: at these sizes the work selector would choose Knuth.

For the 384-key row with one 2^{192} key and one 2^{128} gap, the untrimmed sufficient height is 278; marking those two outcomes gives height 18, with remaining mass 767 and $\mu = 1$. The full forced kernel makes 12,489,698 finite run-entry queries, processes 524,018 singleton states, and stores 1,835,008 predecessor slots. This verifies the construction and height theorem, and does not establish a practical speed advantage at that size. A separate 1024-key geometric input checks iterative reconstruction of the complete rightmost-root chain against its endpoint-dominance optimum $2^{1025} - 1026$.

Twelve height-only plans with two exceptional outcomes and background keys in $\{1, 2, 3\}$ give heights 20, 23, 26 and 29 at sizes 1024, 4096, 16,384 and 65,536. Exceptional exponents 64/80 and 4096/4112 give the same budgets; no full construction is run at these sizes. A separate all-root audit checks 145,424 intervals across 1552 original or globally normalized inputs, using 2,551,952 root candidates and retaining minimum height among tied optimal costs. Rational and large weights are included. The records `trimmed_height_audit.json` and `trimmed_interval_audit.json` retain separate operation counters, reference scopes, and seven and three executed-source hashes, respectively.

Valuation-class records and sparse full-tree reconstruction. The record audit first checks every nonempty slot subset through height three: 135 subsets and 461 selected-key depth inequalities. A separate 6231-window control checks exact highest-valuation slot selection and earliest ties through height five, without scanning in the constructor.

All 680 binary mixed inputs through four keys, together with 60 further mixed, rational, geometric and run-structured inputs through 32 keys, give 1540 full constructions at prescribed heights. Every one matches both the dense prefix-vector optimum and an independent original all-root height recurrence. The audit checks all 226,280 prefix values and their attaining compressed predecessors, plus exact equality of the retained breakpoint sets at 6778 stage boundaries. At small heights, 53,824 literal slot selections provide a third prefix-value control. The original all-root reference uses 881,564 root candidates.

Eight larger complete constructions through 1024 keys cover a zero-weight middle run and genuinely mixed residual gaps. Their selected sufficient heights certify unrestricted optimality, and all match independent ordinary Knuth optima and direct output-tree cost, order and height checks. At 1024 keys, the zero-middle row uses height 25, implicit capacity 33,554,431, and 93,615 retained states across three stages. It makes 8,405,286 finite factor queries and 201,726,864 threshold-term evaluations; the repaired ancestor tree has 1034 vertices. These counts are distinct operation categories, rather than a wall-time comparison.

Three more full constructions keep 16 uniform keys while prescribing heights 16, 32 and 64. They all return cost 54, matching the closed minimum-internal-path-length optimum. At height 64 the implicit capacity is $2^{64} - 1$, while the constructor retains 15,120 prefix states, queries 302,820 destination rows, and constructs only 16 ancestor vertices. These rows check the absence of perfect-tree expansion; the integer complexity bound follows from the score-range proof, not from their measured sizes. The record `slot_records_audit.json` retains separate operation counts, reference scopes and seven executed-source hashes.

The marked-spectrum audit checks 60 inputs through five keys by enumerating 1428 slot selections, including rational marked weights and integer unmarked weights. Six further complete unrestricted mixed constructions through 192 keys place one exceptional key and one exceptional gap among three integer background runs. Gap exponents 64 and 512, paired with key exponents 80 and 528, all match independent ordinary Knuth optima. At 192 keys, both exponent choices select height 17 and retain the same 45,729 records across five stages; sparse repair constructs exactly 192 vertices. These equal counts are an observed control, while (41) proves the magnitude-independent bound. The separate `slot_marked_records_audit.json` retains six source hashes.

Exact exponential prefix-record support. The universal envelope in Lemma 165 has three independent controls. Earliest valuation-preserving embeddings are checked on all 32,902 nonempty slot subsets at heights one through four, covering 246,221 selected positions. Seventy-two arbitrary mixed rational inputs, including large exceptional keys, check 12,424 retained records against the envelope and compare 18,088 prefix values with the dense mixed recurrence; every complete output tree is independently checked. Finally, for each height $h = 1, \dots, 6$ and every key count $1 \le n \le 2^h - 1$, geometric weights of base $h + 1$ attain the envelope at all prefixes: 2,667 stage sets and 97,155 dense states are checked, including $k > h$.

The positive mixed family in Theorem 167 is checked by a dense subsequence recurrence that retains every ending slot, independently of the compressed kernel. Across sizes 2 through 16, all 1,965,265 prefix values and attaining compressed predecessors agree. The same states check the popcount support test and the exact strict-increase locations of prefix maxima. All 135 complete record sets agree with the binomial formula. A separate enumeration checks 65,519 explicit subset-

of-scales witnesses. Through five keys, 208,376 literal slot selections provide another reference; 2055 literal BST shapes through eight keys verify the unique full optimum. Nine complete record-kernel trees through ten keys agree with that optimum.

At 16 keys, the middle prefix with eight keys has exactly 39,195 records; the final prefix has 65,520. The 135 measured stage counts also satisfy the summed identity in (168). Thirty further ascending, descending and shuffled magnitude-order diagnostics through 12 keys retain their counts without implying a universal upper bound. Formula evaluations at sizes 32 through 1024 are labeled separately; no exponential prefix enumeration is run at those sizes.

Six complete constructions of the same family by the existing affine solver reach 1024 keys. Every result matches its closed unique-chain cost and has all output links, order, depths and original mixed cost checked. Independent Knuth references are used only through 256 keys. At 1024 keys, the affine constructor uses exactly 2047 root candidates and returns the complete chain; the largest input weight has 10,242 bits. This checks the separation between explicit-record size and full-problem difficulty. The record `slot_record_barrier.json` retains exact counters, reference scopes, and eight executed-source hashes. No timing supports the theorem.

Implicit mixed prefix diagrams. The additive-diagram constructor is compared with a direct all-predecessor prefix recurrence on 72,880 entries, including infeasible positions. All 1392 complete small trees agree with an independent original all-root height recurrence and have every key, gap, order constraint, and depth checked. The inputs include all 680 binary mixed instances through four keys at two heights, plus rational and geometric controls through 24 keys. The direct prefix reference evaluates 486,336 predecessor candidates; these are reference operations, separate from constructor work.

Five complete geometric constructions use the general diagram solver at necessary height $h = n$, for $n = 8, 16, 24, 32, 48$. Each agrees with the closed unique-chain optimum and independent ordinary Knuth. At 48 keys the final function has 45,391 reachable diagram nodes, while its explicit record count is $2^{48} - 48$. Construction creates 3,069,517 nonterminal nodes, keeps 1,145,477 nodes reachable from all prefix roots, and makes 3,282,551 minimum calls, 2,290,986 addition calls, and 2,200,220 prefix calls. The large difference between final and intermediate sizes is retained explicitly. A 64-key scale run was interrupted and supplies no completed row or complexity evidence. The source-hashed record is `slot_dag_audit.json`; no timing is used.

A separate prefix-operation probe checks 108 signed functions with arbitrary infeasibility patterns at heights one through nine, comparing all 12,264 positions and checking the stated linear-in-hD operation bounds. The monotone-minimum family is checked for $m = 1, \ldots, 12$: every normalized middle-level node is counted, and the total is exactly $3 \cdot 2^m - 2$. There are 45,744 address controls of both operands and their minimum, exhaustive through $m = 6$ and sampled at the proved kink boundaries thereafter. The formula has the independent proof in Theorem 168.

Nine additional unrestricted mixed constructions use certified sufficient heights and match ordinary Knuth plus complete output-tree checks. They include random and near-uniform keys and gaps through 64 keys, and geometric keys with nonconstant, nonaffine gaps through 16 keys. The 64-key random row has height 11, 413 final nodes, and 187,957 total created nodes including its terminal; the near-uniform row has height 10, 230 final nodes, and 50,714 total nodes. At 16 keys the nonaffine geometric row has height 16, 1283 final nodes, and 199,887 total nodes. These complete controls establish correctness on the displayed inputs; the measured sizes supply no universal graph-size bound. The record `slot_dag_probe.json` retains input weights, operation counters, reference counts, and source hashes.

Actual-prefix size search and skyline traceback. A directed search uses exact dense mixed prefix values, independently normalizes every dyadic block, and maximizes the largest reachable diagram across all stages. It evaluates 204 inputs at $n = h = 6, 8, 10, 12, 14, 16$, using shuffled geometric keys, geometric gaps, and local coefficient or swap mutations. Every full optimum agrees with ordinary Knuth. The largest sizes found are respectively $58, 157, 310, 548, 1335, 2479$; the corresponding final sizes are $49, 81, 140, 311, 341, 1920$. Each winner has every prefix entry independently compared with the implicit constructor. Height n is sufficient on these inputs; the search does not certify that it is necessary. These finite sizes imply no growth exponent. The record is `slot_dag_size_search.json`.

A second search uses positive mixed inputs with necessary height n. For $A = 8(n+1)^2$, choose coefficients $\alpha_i, \beta_j \in \{1, \dots, n\}$ and set

$$p_i = \alpha_i A^{n-i}, \qquad q_j = \beta_j A^{\max(0,n-j-1)} \quad (0 \le i < n,\ 0 \le j \le n).$$

For a suffix beginning at i, put $r = n - i$ and let R be the mass of all its outcomes other than p_i. Then $R < 2nA^r/(A-1) + n$, so $nR/p_i < 3n^2/(A-1) < 1$. The right chain has suffix cost at most $p_i + nR$, while a different root incurs at least $2p_i$. Recursion forces the unique right chain and hence height n. This argument applies to every allowed coefficient mutation. At the same six sizes, 48 inputs are evaluated. The largest found sizes, also the winners' final sizes, are $48, 123, 266, 495, 829, 1283$. Every candidate has an independent Knuth optimum check; each winning complete skyline construction agrees with the closed chain cost and has all links, gaps, order constraints, and depths checked. The source-hashed record is `slot_dag_chain_search.json`.

The separate skyline audit checks 42,144 prefix values against direct all-predecessor minimization, 168,576 predecessor queries against literal candidate scans, and 185,856 endpoint-dominance inequalities. It includes all 680 binary mixed inputs through four keys and 24 rational inputs through 12 keys, totaling 704 complete all-root optimum and tree checks. Queries use zero, integer, and rational nonnegative gap prices, and include infeasible prefixes and the global endpoint sentinel. Six complete constructions on the unconstrained search winners agree with Knuth and all independently counted stage sizes. Their slot traceback makes exactly n^2 path steps at $h = n$ and creates zero diagram nodes, as predicted by Theorem 41. The record `slot_dag_skyline_audit.json` retains operation counts, executed source hashes, and the hash of its consumed search record.

Scalar-carry forward construction. The independent forward audit checks every one of 60,336 prefix entries against direct predecessor minimization, including infeasible positions. All 680 binary mixed inputs through four keys are checked at both the minimum feasible height and one larger height; 24 rational inputs through 12 keys add nonintegral controls. All 1384 complete trees agree with the original all-root height recurrence and have their links, order, depths, and original key/gap costs independently verified.

The same harness separately constructs the inclusive prefix-minimum diagram and checks 15,993 expanded carry contexts. For each actual non-pruned dyadic block it verifies that the inclusive-prefix minimum is the input block minimum, and that the normalized first-address value is exactly the relative carry, including infinity. Every executed context is covered, and distinct contexts have distinct input/prefix node pairs. This checks the earlier, weaker $O(hD^2)$ state bound by node-pair injection. The sharper reset-path proof of Theorem 43 applies to the same unchanged constructor and is audited separately below.

Twelve further full constructions reuse the retained winners of both size searches. Every stage has exactly the independently counted diagram size, and every complete optimum agrees with Knuth and output-tree checks. On the 16-key unconstrained winner, the scalar-carry constructor

creates 14,124 nodes versus 144,500 in the previous construction of that same input; it expands 3829 carry contexts. On the 16-key necessary-chain winner, the respective counts are 19,090 and 199,603 nodes, with 10,110 carry contexts. All twelve new constructions make zero calls to generic pointwise diagram minimum, addition, or prefix operations. The record `slot_dag_forward_audit.json` retains exact counters, nine executed source hashes, and hashes of both consumed search records.

A separate scale audit constructs positive mixed inputs from the strict-dominance chain family above, with random coefficients and necessary height $h = n$. Each full tree agrees with its closed unique-chain cost and ordinary Knuth; all links, depths, keys, and gaps are checked. Its exact counts are:

n	final nodes	all created nodes	history nodes	carry contexts
24	4928	106914	65358	58124
32	12410	358192	215203	197331
48	45391	1939134	1145477	1081655
64	111668	6355399	3722796	3567441

The all-created count includes the terminal; history counts nodes reachable from all retained prefix roots. At 64 keys there are also 38,536,277 first-address replacement visits, retained rather than hidden by the smaller final diagram. No timing or measured exponent supports the bound. The record `slot_dag_forward_scale.json` retains every original weight, eight source hashes, and all operation categories. These inputs are distinct from the earlier interrupted 64-key experiment.

Reset paths and exact generic sharpness. A separate audit checks the reset-path cover of Lemma 42 on 1416 updates, examining all 34,186 expanded states and 18,513 reset edges. For each update it reconstructs inherited and reset edges, checks at most one inherited child, checks that each parent input node determines a single reset seed, and verifies that the seed paths cover exactly the expanded states. It checks both the unweighted and depth-weighted bounds, every node-assembly and replacement count in (52), the recursive call identity, and the summary bounds. No generic pointwise minimum, addition, or prefix call is made.

There are 148 arbitrary-diagram controls, including signed rational and infinite entries, with four compact inputs at heights 16, 32, 64 and 128. They test the counting lemma only. Separately, 372 complete mixed trees are verified: all 168 binary inputs through three keys at two heights, 24 rational inputs through 16 keys, and the twelve retained size-search winners. The first 360 have original all-root height references; the winners have independent Knuth references and independently counted stage sizes. Every complete tree is checked against its original weights. The record `slot_dag_reset_audit.json` retains all counters, nine executed-source hashes, and both consumed search-record hashes.

The sharpness audit checks the exact closed formulas of Theorem 169 for $1 \le m \le 16$ and for $m \in \{24, 32, 48, 64\}$. It examines 14,700 expanded contexts and checks 7085 input values, exhaustively through $m = 6$ and with labeled samples thereafter. At $m = 64$, the 129-node input at height 128 causes exactly 6240 expanded contexts and 316,096 node-assembly calls, with the same number of replacement visits. Every row also checks the explicit endpoint-dominance violation. These are general-diagram operation controls, not complete OBST constructions. The record `slot_dag_reset_sharp.json` retains seven source hashes. The closed formulas supply the asymptotic tightness independently of these finite checks.

Punctured diagrams and complete mixed construction. The new representation is checked against all 60,336 direct prefix entries and 94,640 direct predecessor queries with integer and rational gap prices. All 1384 small complete trees agree with the original all-root height recurrence: every

binary input through four keys at two feasible heights, plus 24 rational inputs through 12 keys. Every output tree's original key/gap cost, order, links, and depths are checked.

On 96 arbitrary signed/rational/infinite diagrams, conversion agrees with independent dense canonicalization and all 6024 nonzero-address values. These inputs check representation and operation counts only. Twelve retained adversarial mixed inputs additionally have every prefix compared by canonical equality with conversion of the earlier ordinary-diagram constructor. Their complete trees agree with independent Knuth optima. Across these controls and the generic sharp family, 5456 updates check all 59,358 expanded states and 26,108 reset edges, exact reset-path coverage, $Z + T$ assembly calls, $2Z + 1$ recursive calls, and the summary bounds. The sharp family checks Corollary 170 through $m = 64$, including its exclusion from the OBST domain. At $m = 64$ its 128-node input requires 6240 expanded states and exactly 6303 assembly calls.

The sorting-free repair is separately checked on every one of 32,902 nonempty slot subsets through height four. Its complete output tree equals the retained sorting-based repair in every case; the ancestor traversal makes exactly two visits per collected vertex. This checks the replacement of sorting by traversal independently of the optimizer. The record `slot_punctured_audit.json` retains all counts, ten source hashes, and both consumed size-search hashes.

A separate full-construction audit uses the identical necessary-height mixed chain inputs of the earlier scale audit. Each constructed tree matches the unique-chain formula, independent Knuth, and all original weights. Exact counts for the punctured representation are:

$n = h$	final nodes	all created nodes	assembly calls
24	3054	42092	58424
32	7806	143956	197859
48	29143	795874	1082831
64	72235	2636579	3569521

The same 64-key input previously created 6,355,399 ordinary nodes and performed 38,536,341 assembly calls and 38,536,277 first-address replacement visits, including the initial ordinary point diagram in the assembly count. The new constructor performs no first-address replacements. These are matched exact operation counts; the theorem supplies the bound.

Four additional arbitrary mixed integer inputs at 32, 64, 128 and 256 keys use the proved sufficient height selector and agree with independent Knuth and complete-tree checks. The selected heights are 11, 13, 14 and 16. At 256 keys the final punctured diagram has 460 nodes; complete construction creates 73,226 nodes and makes 114,633 assembly calls. All intermediate work is retained, and these measurements assert no all-stage size bound. The record `slot_punctured_scale.json` retains every original weight, six source hashes, and the consumed earlier scale-record hash.

Actual-prefix cubic and shared-history quartic witnesses. The audit for Theorem 171 first compares every greedy ending-slot reward with an independent dense subsequence recurrence for $2 \le n \le 12$, giving 89,749 exact entry comparisons. Eight complete constructions at $n = 12, 13, 16, 17, 24, 25, 32, 48$ then check all three entries of every witness at every prefix. There are 507,376 witnessed nodes and 1,522,128 entry comparisons. Each witness also checks the first and last nonzero coefficient positions and the coefficient identifying s. Actual canonical node identities are distinct across all prefix roots, so the history test includes full cross-stage sharing. Odd sizes check the even probe offset $Q = 2\lceil n/2 \rceil$ separately from $Q = n$.

Every full output agrees with independent Knuth, the closed chain cost, and original-weight tree checks; all 187 suffix dominance certificates hold. The finite node counts and proved lower bounds are:

n	final nodes	final lower bound	shared history	history lower bound
12	287	56	2044	434
13	379	84	2912	714
16	767	165	7242	1815
17	937	220	9429	2585
24	3054	969	42092	16473
25	3488	1140	50176	20235
32	7806	2600	143956	60450
48	29143	11480	795874	404670

The record `slot_punctured_barrier.json` retains every stage count, all operation counts and seven source hashes. Rows from 64 through 4096 keys evaluate only the proved formulas; they are not enumerations or full constructions. The lower-bound proof is independent of these measurements.

Positive affine block sharing. The audit for Theorem 173 checks all 43,941 unordered quadratic exponents at selected sizes through $n = 256$ for uniqueness. It then compares 18,213 greedy scores with independent dense subsequence values at $2 \le n \le 10$. Five complete constructions at $n = 8, 12, 13, 16, 17$ check every witness's three entries against the actual prefix function: 5567 distinct affine invariants and 16,701 exact entry comparisons. Every invariant is also checked after two nontrivial positive affine transformations, giving 11,134 exact rational controls. Distinctness is checked across all prefix stages, and the coefficient tests verify linear independence and recovery of r, s, k.

All five original-weight trees agree with ordinary Knuth and the closed chain cost, with 66 strict suffix certificates. The largest normalized weight has 10,322, 38,740, 50,247, 98,499 and 119,841 bits in the five rows, respectively. These finite bit lengths are reported separately from arithmetic-operation counts. The record `slot_affine_barrier.json` retains each exponent sequence, stage counts, exact operation counts and eight source hashes. The existing additive solver supplies the tested function values; no affine-normalized solver is claimed or required for the equivalence-class lower bound.

Lazy search, valid bounds, and forced exponential states. The audit for Lemma 174 exhausts the 680 binary mixed inputs through four keys and adds 48 seeded rational inputs at $n = 5, 8, 12$. At two heights per input it checks 55,744 bound entries against dense exact prefix values and independently sorted dot products. It checks all 14,245 evaluated prefix states against fresh all-predecessor recurrences and all 1456 height-constrained optima against the original all-root recurrence. Another 2184 complete constructions use budgets zero, seventeen, and the default $8n^2$. All original-weight tree costs match unrestricted Knuth. Of these, 2174 use fallback and ten finish the lazy search; every exhaustion occurs at exactly one counted operation beyond its budget. The record `slot_lazy_audit.json` retains these counts, seed and eight source hashes.

The matched scale audit retains 22 complete constructions with independent Knuth controls. Five uniform rows at $n = 32, 64, 128, 256, 512$ finish the lazy search, expanding exactly n states each. Their counted search work is 5290, 12,756, 30,263, 87,117 and 199,099 operations, respectively. All ten bounded and binary mixed rows at those sizes reach the default budget and then construct the optimum with Knuth; their unsuccessful search work is retained. The chain family uses an explicit diagnostic budget of two million search operations. Its completed and budget-exhausted rows are:

n	Necessary h	Expanded states	Search work	Completion route
6	6	67	4221	Lazy
8	8	382	32,013	Lazy
10	10	2037	211,825	Lazy
12	12	10,216	1,267,315	Lazy
14	14	12,587	2,000,001	Knuth fallback
16	16	8689	2,000,001	Knuth fallback
18	18	6554	2,000,001	Knuth fallback

The record `slot_lazy_scale.json` retains every original input, height, budget, search and fallback counter, separate reference counter, seed and eight source hashes. These finite counts are not a fitted exponent or a speed claim. Uniform trees already have a linear constructor.

A separate unbudgeted audit for Theorem 175 completes $n = 6, 8, 10, 12$, checking all 14, 62, 254 and 1022 distinct states in (184). Their canonical paths contain 14, 62, 254 and 1022 distinct edges. Every edge is checked for skyline membership, feasibility, strict comparison of its candidate bound with the exact parent optimum, and presence of its child in the completed search cache. The full runs cache 68, 383, 2038 and 10,217 exact states, including base states. Post-search bound comparisons add operations to this audit's counters; the scale table above reports construction search work alone. The record `slot_lazy_barrier.json` retains all counters and eight source hashes. Its larger rows, through $n = 256$, evaluate the proved subset-count formula only. The theorem follows from the explicit paths and strict dominance, independently of these finite checks.

U Unimodal domination and certified-envelope audit

The audit exhausts 920 unimodal words over $\{0, 1, 2\}$ through eight keys and checks all 136,802 nonempty subsequences for componentwise domination by the constructed contiguous block. Its independent definition rejects 8920 nonunimodal words. Another 1589 nonnegative reward-template controls, from 12,864 literal subsequences, check the optimization identity without assuming ruler rewards.

On 236 mixed inputs through 32 keys, including monotone, constant, isolated peak, geometric, irregular unimodal and rational weights, the original all-root recurrence checks 31,438 interval values and contiguous witnesses, 560 complete above-capacity support sets, and 24,032 densities. All 99,398 eligible small square controls satisfy $2s - 1$. The original recurrence uses 565,705 root candidates and 65,081 states.

The arbitrary-input audit includes every binary word at nine and ten keys, plus 100 random mixed inputs through 32 keys, totaling 1636 inputs. It independently checks 101,300 interval-unimodality decisions, 200,892 upper-cost inequalities, 91,143 certified exact interval values, and 130,408 envelope densities. The original recurrence checks 38,221 exact certified densities and 4988 certified support sets, using 2,345,440 root candidates and 435,040 states. The two signed-correction examples have eight literal score controls from 253 subsequences, separate from those recurrence checks.

Fourteen larger sparse constructions reach 65,536 keys. They have 1168 sampled literal ruler-window controls and 2780 constructed local-square controls; five single-key rows also match the sharp closed-form count. They do not have original all-root layer references at those sizes. At $n = 65,536$ both fixtures perform 917,538 ruler-window updates and 983,026 range-tree build-node operations. The isolated-peak input outputs 65,294 core entries across all layers, and the irregular rational mountain outputs 1201. All entries are constructed without scanning interval pairs.

The record is `unimodal_above_audit.json`. It retains exact operation counts, both signed-correction witnesses, verification scopes, and the final executed-source SHA-256 manifest. The support and work bounds follow from contiguous domination, the ruler recurrence and visibility geometry.

V Valley domination, implicit queries, and dense-square audit

The independent audit exhausts 920 valley words over $\{0, 1, 2\}$ through eight keys. It checks all 136,802 nonempty subsequences for componentwise domination by the selected end blocks, and 6198 arbitrary nonnegative reward vectors. An independent definition rejects 8920 nonvalley words.

On 256 mixed inputs through 64 keys, the original all-root recurrence checks 56,076 above-capacity interval values and their two-block witnesses. Inputs include monotone and constant words, geometric and irregular valleys, rational weights, and nonzero gap weights. The original recurrence uses 1,396,993 root candidates and 113,799 states. The explicit dense family and its normalized rational version have 672 square cells checked against that recurrence, at sides four, eight, and sixteen. Each square also has the proved exact support count $s^2/2$.

Fourteen larger implicit constructions reach 65,536 keys. Their checks comprise 896 sampled literal largest-key selections, 896 literal ruler-prefix sums at arbitrary lengths, and 640 sampled dense-formula cells. These are separate from the original all-root controls; no original all-root layer is computed at the large sizes. At $n = 65,536$, each construction performs 1,835,076 ruler-window updates and stores 1,966,150 window entries across the forward and reversed arrays, including their zero bases. No interval table or dense core is constructed.

The record `valley_above_audit.json` retains operation counts and the executed-source SHA-256 manifest. The theoretical work bounds follow from the two-wing binary search and ruler-prefix recurrence; the core lower bound follows from (163) and its residue count.

W Valley support charges and sparse-core construction audit

The audit uses 1513 small inputs: all 1461 ternary valley words through nine keys, 48 varied mixed inputs through 32 keys, and four integer or normalized rational dense-family inputs. Literal sorted sums check 73,082 auxiliary densities, 32,810 two-threshold formulas, and 4267 complete auxiliary support sets. The same number of constructed ruler support sets agree with dense evaluation of the independently audited query representation. Every eligible small square gives a boundary-mass control, totaling 198,321. Integer inputs additionally check the support count against that mass.

Original all-root mixed layers, separately from those query controls, check 49,278 densities and 2644 complete original support sets on 972 inputs. They use 1,006,805 root candidates and 163,435 states. The nonvalley word p^+ in Section 5.23 has a separate original control, with 495 root candidates and 135 states, confirming original density one and auxiliary density zero.

Twelve larger constructions through 16,384 keys enumerate all above-capacity cores on binary end blocks and bounded integer valleys. Their 3264 sampled literal density controls are kept separate from the original all-root controls. At the largest size the binary input emits 4084 original entries from 8178 auxiliary events; the bounded valley emits 16,477 from 32,995. Both preprocessings perform 393,276 ruler-window updates. Five complete dense-family constructions also check 5456 cells against (163), through side 64. The record `valley_core_audit.json` retains all operation counts, scopes, the nonvalley witness, and executed-source hashes. The support and work bounds follow from the charge and event-enumeration proofs.

X Periodic phase construction and full-tree audit

The independent audit checks 1042 small inputs: all mixed binary vectors through four keys, every binary key/gap period through length three at several complete and incomplete repetition lengths, and seeded geometric, zero, rational, and final-gap-defect inputs through 48 keys. Literal period enumeration checks every detected joint period. Independent all-root tables check 62,376 original interval values and 54,672 leftmost roots. Extended all-root tables additionally check all 38,756 synthetic phase states used in the cyclic cancellation, including intervals crossing the input endpoint. These references use 1,993,994 root candidates and 154,391 states. Separate ordered-tree enumeration evaluates 7882 shapes on the small exhaustive cases.

Twenty larger matched instances through 512 keys use an independent ordinary Knuth recurrence, which examines 1,744,227 candidates. Every reconstructed tree is traversed independently, checking inorder placement, all child links, all key and gap depths, and weighted cost. Sixteen further complete constructions reach 262,144 keys, with periods one, three, and five and both zero and positive gap patterns. At that size the period-five mixed input uses exactly 2,621,435 root candidates, and the period-three input uses 1,572,861, as predicted by (76). The all-zero case reconstructs the full leftmost-optimal chain iteratively. These large rows verify complete construction, exact operation counts, and independently evaluated tree costs; they do not claim an independent quadratic optimality reference at that size.

The result record retains optimal costs, reconstruction counts, verification scopes, and executed-source hashes. No timing or fitted exponent is used. The complexity and optimality claims follow from the exact quotient (74) and cyclic amortization (76), both of which include the synthetic phases.

Y Boundary extension and full mixed-tree audit

The implementation first checks Wilber's LWS primitive against a fresh dense recurrence: 768 instances through 64 vertices, with signed finite row and column offsets and quadratic separation costs, use 549,120 literal arc candidates. All values and leftmost predecessors agree. The implementation follows the two-search batching algorithm in Wilber's Figure 1. Invalid cells internal to SMAWK use an ordered tag above all finite values; valid arc queries remain finite. Each batch's comparisons therefore equal those of a sufficiently large finite completion. The implementation records batch widths and checks that their sum is at most three times the number of vertices; this is the potential argument from Wilber's linear bound.

The full mixed audit checks 980 small inputs: all binary mixed vectors through four keys and seeded periodic-middle inputs through 27 keys, including rational weights and boundary weights as large as 10^{40}. Explicit middle selection and automatic selection are checked separately. Independent all-root tables use 351,256 candidates and 49,492 states, verifying 57,416 original interval values and 44,230 stored root witnesses. Another 6944 rightmost-root controls test the new-column reduction directly against the original tables, independently of the periodic solver. Literal substring enumeration checks 3324 longest-middle decisions using 162,564 candidate substrings. Small ordered-tree enumeration evaluates 7880 shapes. Every reconstructed tree is independently traversed to check ordering, links, key and gap depths, and weighted cost.

Sixteen matched inputs through 512 keys use the ordinary uncompressed Knuth recurrence, examining 1,847,784 candidates. Twelve larger complete automatic constructions reach 65,537 keys, with periods one, three, and five and one or seven arbitrary fringe keys. At that size, the one-key right fringe and period-one middle have exactly 65,537 new boundary states, 131,071 central root candidates, and 1,309,476 valid LWS transition queries. With period five and seven fringe keys, the corresponding counts are 458,738, 655,295, and 8,264,895. The automatic scan performs 1,113,976 signature comparisons on each of these largest inputs. These are operation counts; central root candidates and LWS transition queries are different operations.

The large rows verify complete construction and independently evaluated tree costs; they have no separate quadratic optimality oracle. The work bound comes from Lemma 56 and the exact state identity (80). The result record retains the seed, costs, operation counts, verification scopes, and six executed-source hashes. No timing or fitted exponent enters the theorem.

Z Affine translation and finite-domain construction audit

Before running the quotient solver, the harness checks the shape-cost identity on 6250 literal ordered trees through seven keys. The transformations include positive scales below and above one and offsets of both signs. The full audit then checks 1063 small mixed inputs: all binary key/gap vectors through four keys, seeded affine inputs through 32 keys, and 21 finite inputs whose hypothetical continuations develop negative weights. Explicit witnesses include periods close to the input length, testing the terminal diagonals as well as the complete phase cycles.

Independent all-root tables examine 640,661 candidates and 74,877 states. They check 84,584 original interval values, all 74,877 leftmost roots, and 28,534 stored phase states. Ordered-tree enumeration separately checks 7882 small optimal costs. Recognition has an independent reference: solving the original two-variable linear equations checks 8644 period-feasibility decisions, and 860 automatic inputs verify the smallest admissible period. This reference does not use the production adjacent-sum test. Another 2916 checks validate returned parameter witnesses directly.

Sixteen matched inputs through 512 keys use the ordinary uncompressed Knuth recurrence, examining 1,393,110 candidates. Nine larger arithmetic-drift constructions reach 65,537 keys, with periods one, three, and five. At that size the period-one construction examines exactly 131,073 root candidates; the period-five case examines 655,325 phase candidates and 18 terminal candidates. The candidate bounds and exact state count are checked against (85) and (86). Two further exact geometric constructions reach 4095 keys; their integers grow with the input length, so their counts concern arithmetic work.

Every output is independently traversed to check key order, child links, key and gap depths, and weighted cost. The eleven largest construction rows have no separate quadratic optimality reference; their optimality guarantee is Theorem 60. The result record retains the seed, operation counts, exact costs, per-row reference scopes, and four source hashes. No timing, fitted exponent, or infinite positive extension enters the proof.

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*Dedication

Glory first be to The Holy One, Blessed be He (הַקָּדוֹשׁ בָּרוּךְ הוּא), from whom all good things come. To Him we attribute the ability that has provided for all good to come and be demonstrated anywhere in our work, and declare ourselves unworthy servants. If in any way we have erred, or our work is unsatisfactory, we ask mercy and kindness, for our effort is good and earnest.